

Pharmacology AIIMS 2017

Question 1:

How much drug will remain at the end of 6 hours when a patient is given 200 mg drug, 75 mg excreted in 90 min and if the drug follows first-order kinetics?

- a) 50 mg
- b) 30.5 mg
- c) 12.5 mg
- d) 6.25 mg

Question 2:

What is the most bactericidal anti leprotic drug amongst the given drugs?

- a) Amoxicillin
- b) Ciprofloxacin
- c) Ofloxacin
- d) Erythromycin

Question 3:

Which of the following is not true about P-glycoprotein?

- a) Pgp stands for permeability glycoprotein
- b) Also known as CD243
- c) Also known as ABCB1
- d) Also known as MDR2

Question 4:

For which of the following drugs, is absorption increased after a fatty meal?

- a) Griseofulvin
- b) Amphotericin B

- c) Ampicillin
- d) Aspirin

Question 5:

A patient with Prinzmetal angina is administered isosorbide mononitrate and is noticed to have symptomatic relief. What is the mechanism of action of nitrate in this case?

- a) Endothelium independent coronary vasodilation
- b) Reduced cardiac contractility
- c) Increased left ventricular end diastolic volume
- d) Decreased diastolic perfusion pressure

Question 6:

The drug of choice in anaphylactic shock is?

- a) Adrenaline 0.5 mg 1/1000 IM
- b) Adrenaline 1 mg 1/10000 IV
- c) Atropine 3 mg
- d) Adenosine 6 mg followed by 12 mg IV

Question 7:

What instruction do you give to a mother who is lactating for drug use?

- a) No advice as most of drug are not present in breast milk
- b) Give longer half-life drugs
- c) Tell the mother to feed when it is least efficacious
- d) To feed just before the next dose, when the plasma concentration of the drug would be the least

Question 8:

In a case of acute exacerbation of asthma, salbutamol was given, however, no improvement was noticed. Intravenous corticosteroids and aminophylline were given and the condition improved. What is the probable mechanism by which the corticosteroids would have improved the condition?

- a) Corticosteroids increase bronchial responsiveness to salbutamol
- b) Corticosteroids cause direct bronchodilatation when used with xanthines
- c) Corticosteroids indirectly increase the effect of xanthines on adenosine receptor
- d) Corticosteroids increase mucociliary clearance

Question 9:

Which of the following opioids is a partial agonist at mu and a full agonist at kappa receptors?

- a) Pentazocine
- b) Buprenorphine
- c) Tramadol
- d) Fentanyl

Question 10:

The drug of choice for insomnia in blind people is

- a) Latanoprostene
- b) Zerviate
- c) Tasimelteon
- d) Ramelteon

Question 11:

Dofetilide belongs to which class of anti-arrhythmic drugs?

- a) Class I
- b) Class II
- c) Class III
- d) Class IV

Question 12:

What is the mechanism of action of vancomycin?

- a) Cell wall synthesis inhibition
- b) Cell membrane inhibition
- c) Peptide synthesis inhibition
- d) 50s ribosome inhibition

Question 13:

The following table gives BP values before and after administration of drug 'X'. What would be the probable mechanism of action of drug 'X'?

DRUG X		
SBP	120	150
DBP	80	68
HR	72	86

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- a) α_1 agonist and β_1 agonist
- b) M2 agonist and M3 agonist
- c) α_1 antagonist and β_2 agonist
- d) β_1 agonist and β_2 agonist

Question 14:

A patient requires Ceftriaxone 180 mg. You have a 2 ml syringe with 10 divisions per ml. The vial contains 500 mg/5ml of ceftriaxone. How many divisions in the 2 ml syringe will you fill to give 180 mg ceftriaxone?

- a) 18
- b) 1.8
- c) 20

d) 2

Question 15:

The drug that decreases the size of prostate is:

- a) Tamsulosin
- b) Sildenafil
- c) Finasteride
- d) Flutamide

Question 16:

Which one of the following is a placebo?

- a) Herbal medication with no effect known
- b) Physiotherapy
- c) Sham surgery
- d) Cognitive behavioural therapy

Question 17:

A young male developed 5 episodes of loose stools 2 hours after eating in a restaurant. He is afebrile and has mild dehydration. What should be the treatment?

- a) Ciprofloxacin and tinidazole
- b) Only ciprofloxacin
- c) Only ORS
- d) Ciprofloxacin, tinidazole and ORS

Question 18:

Atropine is indicated in all of the following poisonings except _____.

- a) Baygon
- b) TIK 20
- c) Parathion

d) Endrin

Question 19:

27-year-old female came for treatment of infertility to OPD. Bromocriptine was prescribed. What could be the possible reason?

- a) Hyperprolactinemia**
- b) PCOD**
- c) Hypogonadotrophic hypogonadism**
- d) PID**

Question 20:

Which is a mood stabiliser with an anti-suicidal role?

- a) Carbamazepine**
- b) Lamotrigine**
- c) Lithium**
- d) Valproate**

Question 21:

Which of the following is not an idiosyncratic reaction of carbamazepine?

- a) Blurred vision**
- b) Agranulocytosis**
- c) Rash**
- d) Steven Johnson Syndrome**

Question 22:

In a middle-aged diabetic patient with chronic kidney disease stage 5, which of the following drugs can be given safely without altering the dose?

- a) Vildagliptin**
- b) Linagliptin**

- c) Saxagliptin
- d) Sitagliptin

Question 23:

Which of the following drugs can be used as an antidote for fibrinolytic therapy?

- a) Protamine
- b) Streptokinase
- c) Heparin
- d) ϵ -Aminocaproic acid

Question 24:

A known epileptic patient on levetiracetam has been seizure-free for 2 years. But he develops agitation and anger issues interfering with his day-to-day activities as a result of the drug intake. Which of the following measures is to be taken next?

- a) Slowly taper the drug over next 6 months
- b) Continue levetiracetam since a 5 year seizure free interval is needed
- c) Stop levetiracetam and start on a different anti-epileptic
- d) Discontinue the drug as he is seizure free

Question 25:

A drug was given IV at a rate of 1.6mg/min. The clearance of the drug is 640ml/min. The half life of the drug is 1.8 hour. Calculate the plasma concentration of the drug at steady state.

- a) 2.88mg/ml
- b) 0.004mg/ml
- c) 3.55mg/ml
- d) 0.002mg/ml

Question 26:

Which of the following disease-modifying anti rheumatoid drug increases extracellular adenosine?

- a) Methotrexate
- b) Lefunamide
- c) Sulphasalazine
- d) Hydroxychloroquine

Question 27:

Composition of Lente insulin is:

- a) 70% amorphous 30% crystalline
- b) 70% crystalline 30% amorphous
- c) Only 70% amorphous
- d) Other name of NPH insulin

Question 28:

Mechanism of action of Oseltamivir as an antiviral agent is:

- a) Inhibition of M2 receptor
- b) Neuraminidase inhibition
- c) Inhibition of RNA dependent DNA polymerase
- d) Apoptosis of infected cells

Question 29:

A male patient who is suffering from a psychotic disorder is on antipsychotic, antidepressant and antihypertensive treatment. He suddenly developed tachycardia and chest tightness. On evaluation he had metabolic acidosis. ECG showed widened QRS complex and prolonged QT interval. Which of the following drugs is responsible for this?

- a) Amitriptyline
- b) Amlodipine
- c) Chlorthalidone
- d) Enalapril

Question 30:

You and your friend plan for a mountain trip for the day after. What medication would you advise him to help with his motion sickness?

- a) Ranitidine tablet before start of trip
- b) Dimenhydrinate before start of trip
- c) Scopolamine patch the night before trip
- d) Omeprazole the night before and before start of trip

Question 31:

A patient was treated for septicemia with norepinephrine infusion. Which of the following best explains the benefit of this drug?

- a) Bradycardia
- b) Tachycardia
- c) Increases urine output
- d) Increases mean arterial blood pressure

Question 32:

Which of the following suggests storage of drugs in tissue?

- a) Excretion in saliva
- b) Excretion in urine
- c) Large volume of distribution
- d) Small volume of distribution

Answer Key

Question No.	Correct Option
1	b
2	c
3	d
4	a
5	a

6	a
7	d
8	a
9	a
10	c
11	c
12	a
13	d
14	a
15	c
16	c
17	c
18	d
19	a
20	c
21	a
22	b
23	d
24	a
25	d
26	a
27	b
28	b
29	a
30	c
31	d
32	c

Detailed Explanations

Solution to Question 1:

In first order kinetics, a constant fraction of the drug is excreted in a given time.

Since 75 mg of 200 mg is excreted in 90 minutes, the fraction of drug excreted in each 90 mins is $75/200$, which is $3/8$. Thus, the fraction of the drug that remains in the body is $5/8$.

There will be 4 cycles of excretion of the drug (6 hours = 360 mins, and $360/90 = 4$). Thus, with a starting dose of 200mg, the drug remaining in the body after 4 hours will be:

$$200 \times \frac{5}{8} \times \frac{5}{8} \times \frac{5}{8} \times \frac{5}{8} = 30.5\text{mg}$$

Alternate way to calculate:

75 mg of 200 mg is excreted in 90 minutes, thus $75/200 = 0.375$ is eliminated

After 90 minutes; $200\text{ mg} - 75\text{ mg} = 125\text{ mg}$ remains.

The second 90 minutes (after 180 minutes) = $125 \times 0.375 = 46.875\text{ mg}$ is eliminated, and $125 - 46.875 = 78.125\text{ mg}$ remains

The third 90 minutes (after 270 minutes) = $78.125 \times 0.375 = 29.297\text{ mg}$ is eliminated, and $78.125 - 29.297 = 48.828\text{ mg}$ remains

The fourth 90 minutes (after 360 minutes) = $48.828 \times 0.375 = 18.310\text{ mg}$ is eliminated, $48.828 - 18.310 = 30.518\text{ mg}$ remains after 6 hours.

Solution to Question 2:

Ofloxacin is the most bactericidal anti leprotic drug amongst the given drugs.

It exhibits high activity against *M. leprae*, and it is being used in alternative multidrug therapy (MDT) regimens in rifampicin-resistant cases. As a component of MDT, it has been found to hasten the bacteriological and clinical response.

Among the macrolides, clarithromycin has activity against leprosy bacilli and is used for alternative MDT in cases resistant to both rifampicin and fluoroquinolones. Azithromycin shows activity against *Mycobacterium avium* complex but not lepra bacilli.

Note: Rifampicin is the most potent bactericidal anti leprotic drug.

Solution to Question 3:

P-glycoprotein is also known as multidrug resistance protein 1 or MDR1 (not MDR2). It is also denoted by cluster of differentiation 243 (CD243).

P-glycoprotein (P-gp) is one of the first members of the ATP-binding cassette (ABC) transporter, which acts as an efflux membrane transporter. It is responsible for limiting cellular uptake and the distribution of xenobiotics and toxic substances. It is associated with multidrug resistance (MDR).

- Increased P-gp expression: Reduces the absorption of drugs that are substrates for P-gp
- Decreased P-gp expression: Leads to supratherapeutic plasma concentrations and drug toxicity

Solution to Question 4:

The absorption of griseofulvin is increased after a fatty meal.

Note: Fatty meals increase the rate of absorption of efavirenz to such an extent as to cause toxicity. Hence, efavirenz is recommended to be taken on an empty stomach.

Drug absorption reduced /delayed by food	Drug absorption increased by food
AmpicillinAspirinAtenololAzithromycinCaptoprilIndinavirIsoniazidRifampinSucralfateTetracyclineDoxycycline	AtovaquoneCarbamazepineChlorothiazideCefuroximeDiazepamGanciclovirLithiumMetoprololRitonavirHydralazineLumefantrine

Solution to Question 5:

The mechanism of action of nitrates in Prinzmetal angina is endothelium-independent coronary vasodilatation.

Prinzmetal angina occurs due to vasospasm of the coronary arteries. Nitrates relax the smooth muscles of epicardial coronary arteries and relieve coronary artery spasm.

In the case of stable angina, the primary mechanism of action of nitrates is preferential dilatation of veins (by increasing cGMP). Venous dilatation leads to a decrease in the venous return to the heart (preload). This decreases the ventricular end-diastolic volume and thereby a decrease in myocardial oxygen consumption. The diastolic perfusion pressure (aortic blood pressure in diastole - left ventricular pressure in diastole) is increased.

Solution to Question 6:

Adrenaline 0.5 mg 1/1000 by intramuscular route (IM) is the drug of choice for treatment of anaphylactic shock. Hence it is the answer here.

In case the patient develops life threatening condition, adrenaline 0.25 mg IV is given.

Note : This is a controversial question. The following inference from journal may help to clarify the concept.

- As vasodilatation is the main pathological change early in anaphylaxis, this enables the subcutaneous or intramuscular absorption of adrenaline to be rapid and effective.
- Thus, when the disease is treated early and is progressing slowly, or venous access is difficult or the patient is unmonitored, intramuscular adrenaline has advantages in terms of safety and is usually effective.
- Later, when intravascular volume is depleted and shock occurs with a life threatening situation, or there is severe dyspnoea or airway compromise, the intravenous route is necessary to achieve optimal absorption.

Solution to Question 7:

Ideally, the mother should feed just before the next dose, when the plasma concentration of the drug would be the least.

- The drugs are present in breast milk in variable amount but lesser than in plasma of mother.
- Breastfeeding is advised at least 3-4 hours after drug intake so that plasma concentration is lesser and hence the concentration in the breast milk is also less.

Solution to Question 8:

Corticosteroids improved the condition by increasing the airway responsiveness to salbutamol.

Mechanism of action of corticosteroids in an acute asthma attack:

- Increases responsiveness to β -2 agonists
- Decreases mucous secretion
- Decreases inflammatory cell response.

Steroids increase the transcription of β -2 receptor genes and hence increases responsiveness to β -2 agonists.

β -2 agonists increase the effect of steroids by enhancing the action of glucocorticoid receptors.

Theophylline increases the effects of steroids by activating the enzyme histone deacetylase which is the target for steroids as well.

Note: As per recent GINA 2021 guidelines, aminophyllines are no longer preferred in the management of acute severe asthma due to their low efficacy and poor safety profile.

Solution to Question 9:

Pentazocine is a partial agonist at mu and a full agonist at kappa receptors.

Analgesia produced by pentazocine is predominantly mediated by the κ receptor.

Pentazocine is a mixed agonist-antagonist opioid agent. It is a full agonist at the κ receptor and a partial agonist or weak antagonist at the μ receptor

Note: Buprenorphine is a partial agonist at mu receptors but an antagonist at kappa receptors

Solution to Question 10:

The drug of choice for insomnia in blind people is tasimelteon.

Option A - Latanoprostene is used in primary open-angle glaucoma.

Option B - Zerviate is a cetirizine ophthalmic solution used in allergic conjunctivitis.

Option D - Ramelteon is a synthetic tricyclic analog of melatonin used in sleep onset difficulty insomnia.

Solution to Question 11:

Dofetilide belongs to class III of anti-arrhythmic drugs.

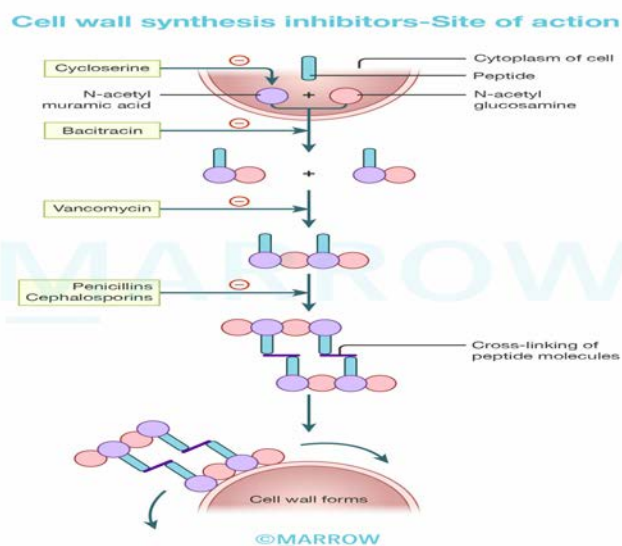
Solution to Question 12:

The mechanism of action of vancomycin is cell wall synthesis inhibition.

The bacterial cell wall has long chains of peptidoglycan and is cross-linked to each other by the action of the enzyme transglycosylase. Vancomycin is a glycopeptide group of antibiotics that binds to the D-alanyl-D-alanine terminus of the synthesized peptidoglycan. This inhibits the transglycosylase enzyme and prevents elongation and cross-linking of peptidoglycan. This leads to the weakening of the cell wall causing lysis.

It exhibits a time-dependant or concentration-independent killing of bacteria. It primarily acts on Gram +ve organisms which are resistant to other antibiotics such as methicillin-resistant staphylococcus and ampicillin resistant enterococci.

Lipoglycopeptides are a new generation of glycopeptide congeners which include telavancin, dalbavancin, and oritavancin. Dalbavancin and Oritavancin are administered as a single intravenous dose for skin and soft-tissue infections.



Solution to Question 13:

The probable mechanism of action of drug 'X' is β_1 and β_2 agonism.

According to the given table, systolic BP and heart rate is increased due to β_1 agonistic effect, and diastolic BP is decreased by vasodilatation due to β_2 action.

Option A: Action on α_1 causes $\uparrow$ DBP

Option B: Action on M_2 causes $\downarrow$ HR and action on M_3 causes vasodilatation and reflex increase in HR. Hence there is no net effect on heart rate

Option C: α_1 antagonism causes vasodilatation and $\downarrow$ DBP. β_2 action causes vasodilatation and DBP $\downarrow$

Solution to Question 14:

- 500 mg of drug is in 5 ml and hence 100 mg in 1ml. Thus 180 mg in 1.8 ml
- 10 divisions for each ml
- 1 ml = 10 divisions
- Therefore, 1.8 ml = $10 \times 1.8 = 18$ Divisions

Solution to Question 15:

Finasteride reduces prostatic size.

Finasteride:

- It is a competitive inhibitor of 5α -reductase enzyme, thus inhibiting the conversion of testosterone into its active form DHT(Dihydrotestosterone) which is responsible for androgenic action in Prostate gland.
- It produces effect on long term use (nearly 6 months) by reducing prostatic epithelial cell proliferation thereby reducing the size of prostate and improves urine flow.

Tamsulosin is an α_1 Selective Antagonist which dilates prostatic urethra and hence produces immediate symptom relief in BPH. It has no effect on reducing size of the prostate.

Sildenafil acts by inhibiting phosphodiesterase-5 (PDE-5) and is used in the treatment of erectile dysfunction.

Flutamide is a Nonsteroidal Androgen Receptor antagonist with no hormonal activity. They are used for treatment of metastatic prostatic cancer in conjunction with GnRH agonists.

Solution to Question 16:

Sham surgery (placebo surgery) is a faked surgical intervention that omits the step thought to be therapeutically necessary.

Note: While option A may be considered a placebo due to there being no known effects, it is not certain as to whether all possible effects (known and unknown) have been considered. Hence,

weighing options A and C, the better option among the two would be C as sham surgery is a widely used placebo.

Solution to Question 17:

It is a case of acute diarrhea in an afebrile, stable patient which can be managed with ORS alone.

Management of diarrhea:

- Mild-moderate dehydration - ORS.
- Severe dehydration- IV fluids are given.
- Afebrile patient - No antibiotics are given.
- Febrile patient - empirical antibiotics. If only bacterial etiology is suspected, ciprofloxacin is given. If giardiasis is suspected, metronidazole or tinidazole are recommended.

Solution to Question 18:

Endrin is an organochloride pesticide that causes neurotoxicity. There is no antidote available and only symptomatic treatment is done.

Parathion and TIK20 are organophosphates and baygon is a combination of organophosphate and carbamate. For all of these, atropine is the drug of choice.

Solution to Question 19:

The patient is likely to be suffering from hyperprolactinemia.

Prolactin inhibits the secretion of GnRH, which leads to the decreased secretion of LH and FSH resulting in infertility. Bromocriptine acts on Dopamine D2 receptors to reduce prolactin secretion.

Solution to Question 20:

Lithium is a mood stabilizer with an antisuicidal role.

Maintenance treatment with lithium has shown to decrease the rate of suicide in mood disorders. Lithium was found to be better than other mood stabilizers such as carbamazepine, lamotrigine, and valproate.

Lithium is also used in recurrent depression with a cyclic pattern is controlled by either lithium or imipramine. It is also used to augment standard antidepressant response in acute major depression in those patients who have had an inadequate response to monotherapy.

Solution to Question 21:

Blurred vision is not an idiosyncratic reaction of carbamazepine.

Idiosyncratic blood dyscrasias with carbamazepine, include fatal cases of aplastic anaemia and agranulocytosis. The most common idiosyncratic reaction is an erythematous skin rash. It can rarely cause Steven Johnson Syndrome.

Solution to Question 22:

Linagliptin can be given safely without altering the dose for this patient.

Linagliptin is a dipeptidyl-peptidase-4 (DPP-4) inhibitor and is eliminated by the hepatobiliary route. Hence no dosage adjustment is needed in renal failure.

It is used in type 2 diabetes as monotherapy or as combination therapy with other drugs when metformin is not tolerated or contraindicated.

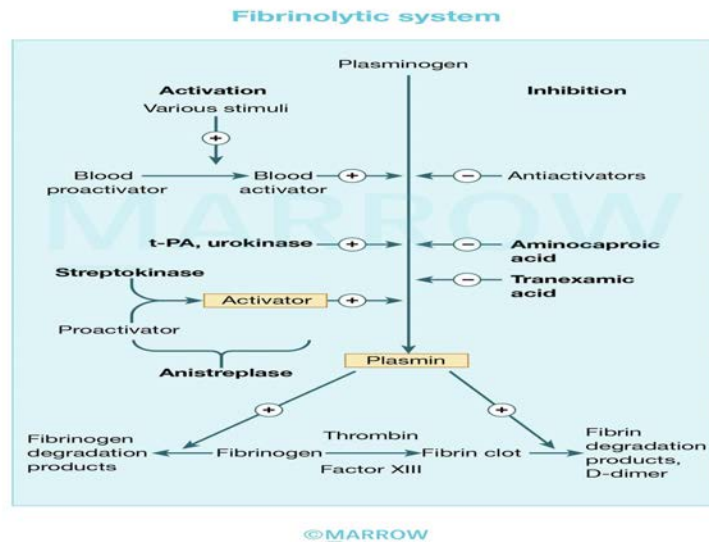
Other DPP-4 inhibitors like sitagliptin, vildagliptin, saxagliptin, alogliptin show renal elimination.

Solution to Question 23:

ϵ -Aminocaproic acid can be used as an antidote for fibrinolytic therapy.

It is a lysine analog that binds competitively to plasminogen, blocking plasminogen from binding to fibrin and the subsequent conversion to plasmin. This activity subsequently results in the inhibition of fibrinolysis.

It should not be given unless hemorrhage is life-threatening because it inhibits intrinsic fibrinolytic activity and can precipitate DIC. Fresh frozen plasma and cryoprecipitate may also be used to replenish fibrin and clotting factors.



Solution to Question 24:

The mood changes observed in the above patient must be due to an adverse effect of levetiracetam. Abrupt stoppage of the drug can result in rebound seizures. So the best management will be to slowly taper the drug over the next 6 months.

Adverse effects of levetiracetam:

- Somnolence
- Asthenia
- Ataxia
- Dizziness.
- Mood and behavioral changes
- Psychotic reactions (rare)

Solution to Question 25:

Plasma steady-state concentration is achieved when the amount of drug being administered equals the amount of drug being excreted from the body.

The amount of drug being administered = 1.6mg/min

Amount of drug being excreted = Clearance (CL) x Plasma concentration of the drug (C)

Clearance = 640ml/min

Therefore,

1.6mg/min = 640ml /min (clearance) x Plasma concentration of the drug

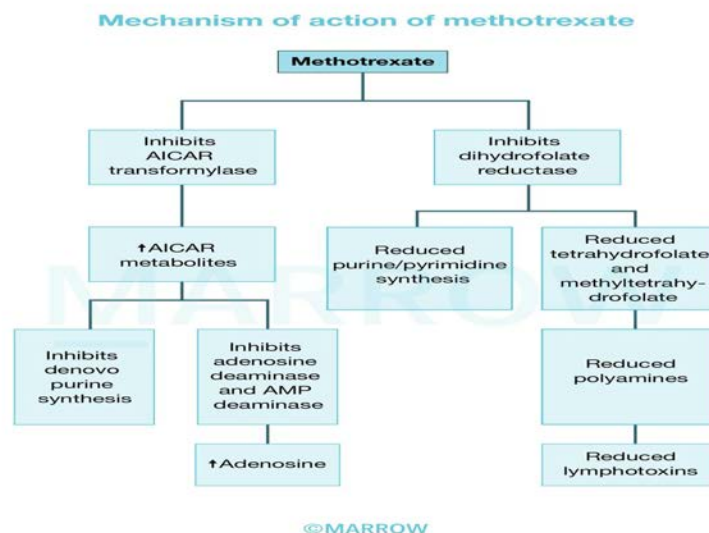
Hence,

Plasma concentration of the drug = $1.6 \text{ mg/min} / 640 \text{ ml/min} = 0.002 \text{ mg/ml}$.

Solution to Question 26:

Methotrexate is the disease-modifying anti rheumatoid drug that increases extracellular adenosine. Methotrexate's principal mechanism of action relates to inhibition of amino-imidazole carboxamide ribonucleotide (AICAR) transformylase and thymidylate synthetase.

AICAR accumulates intracellularly, competitively inhibits AMP deaminase, leading to an accumulation of AMP. The AMP is released and converted extracellularly to adenosine, which is a potent inhibitor of inflammation. As a result, the inflammatory functions of neutrophils, macrophages, dendritic cells, and lymphocytes are suppressed.



Solution to Question 27:

Lente insulin is an intermediate-acting insulin and is a combination of Ultralente and Semilente in the ratio of 7:3

Two types of insulin zinc suspensions have been produced. The one with large particles is crystalline and practically insoluble in water (Ultralente). It is long-acting. The other has smaller particles and is amorphous (Semilente), is short-acting. Their 7:3 ratio mixture is called 'Lente insulin' and is intermediate-acting.

Solution to Question 28:

Oseltamivir and zanamivir are neuraminidase inhibitors.

They competitively and reversibly interact with the enzyme sites to inhibit viral neuraminidase activity. This interferes with the release of the progeny from infected host cells, halting the spread of infection. Early administration is crucial.

Oseltamivir is given at a dose of 75 mg, twice a day for 5 days for the treatment of Influenza A and B. For prophylaxis, it is given at a dose of 75mg, once per day and continued for the duration of exposure.

Solution to Question 29:

The given clinical scenario is suggestive of torsades de pointes. Only amitriptyline causes this condition, among the above drugs.

Torsades de pointes (French: twisting of points) is a life-threatening form of polymorphic ventricular tachycardia with rapid asynchronous complexes and an undulating baseline on ECG.



Solution to Question 30:

In the above case, a scopolamine patch the night before the trip should be advised.

Treatment for motion sickness :

- Scopolamine patch- preferred, most effective.
- Do not give to children <10 years of age.
- Place patch behind the ears at least 4 hours before travel and replace every 72 hours as needed.
- Transdermal patch – Most effective
- Oral dose - 0.4–0.6 mg one hour before travel and then TID may be recommended.

- Cinnarizine – 30mg, 2 hours prior to travel, and 15mg TID as required. Not as effective as scopolamine. Moderately effective.
 - Promethazine
 - Other drugs (minimally effective):
 - Cyclizine
 - Dimenhydrinate
 - Diphenhydramine
 - Meclizine
- PPI's / ranitidine / traditional ginger – all are ineffective.

Solution to Question 31:

The benefit of using noradrenaline in septicemia is that it increases the mean arterial blood pressure. It acts as a powerful peripheral vasoconstrictor and inotrope.

Noradrenaline causes a rise in both systolic BP and diastolic BP, resulting in a rise in the mean arterial blood pressure. This increases the organ perfusion in patients in septic shock or cardiogenic shock.

Another advantage of noradrenaline is that it does so without causing tachycardia, which is often a troublesome event with other inotropes in critical care.

Options A and B: The rise in mean arterial pressure causes reflex bradycardia which is not a major contributor to the treatment of septic shock.

Option C: Low-dose dopamine infusions and furosemide are used to increase urine output in critically ill patients.

Solution to Question 32:

Storage of drug in tissue suggests large volume distribution.

Drugs with small volume of distribution are confined to vascular component.

Drugs with large volume of distribution are in compartments like adipose tissue, muscle, bone.

Volume of distribution depends on:

- Lipid solubility
- pKa value of a drug
- Degree of plasma protein binding
- presence of tissue-specific transporters
- differences in regional blood flow.

Volume (V) = Dose administered I.V. / Plasma concentration

e.g., when 1000mg of drug is injected I.V. and plasma concentration of drug is 50mg/L,

- $V=1000/50$

- $V=20L$

Pharmacology AIIMS 2018

Question 1:

Which of the following is not a side effect of salbutamol?

- a) Hypoglycemia
- b) Hypokalemia
- c) Tremors
- d) Tachycardia

Question 2:

Which of the following drugs should be sold only on prescription from a registered medical practitioner?

- a) Schedule M drugs
- b) Schedule H drugs
- c) Schedule J drugs
- d) Schedule X drugs

Question 3:

The loading dose of a drug is governed primarily by:

- a) Clearance
- b) Volume of distribution
- c) Plasma half life
- d) Bioavailability

Question 4:

Therapeutic index is used for assessing which of the following factors?

- a) Safety
- b) Efficacy

- c) Potency
- d) Toxicity

Question 5:

Which of the following drugs causes flagellate pigmentation?

- a) Bleomycin
- b) Doxorubicin
- c) Vincristine
- d) Minocycline

Question 6:

Which is the anticoagulant of choice for the management of cancer-associated thromboembolism?

- a) Unfractionated Heparin
- b) Warfarin
- c) Low molecular weight heparin
- d) Dabigatran

Question 7:

Which of the following new drugs has been approved for the treatment of amyotrophic lateral sclerosis?

- a) Edaravone
- b) Piracetam
- c) Dabrafenib
- d) Trametinib

Question 8:

What is the drug of choice for scrub typhus during pregnancy and otherwise respectively?

- a) Chloramphenicol, doxycycline

- b) Doxycycline, doxycycline
- c) Doxycycline, azithromycin
- d) Azithromycin, doxycycline

Question 9:

Which of the following is an example of a GPCR?

- a) Muscarinic
- b) Nicotinic
- c) GABA-A
- d) Insulin

Question 10:

The package description of a drug states that 500mg of paracetamol is present in each unit of the tablet formulation, but in reality, each unit contains only 200mg. What is the terminology used to refer to these kinds of drugs?

- a) Misbranded drug
- b) Adulterated drugs
- c) Spurious drugs
- d) Counterfeit drugs

Question 11:

A patient presented to the casualty with signs of dehydration due to severe vomiting. The intern gave the patient a parenteral medication, following which the patient developed grimacing and abnormal posturing. What could have been the inciting drug for this reaction?

- a) Atropine
- b) Ranitidine
- c) Metoclopramide
- d) Domperidone

Question 12:

Identify the wrongly matched pair of antifungals with its mechanism of action:

- a) Ketoconazole : Lanosterol-14-demethylase inhibition
- b) Flucytosine : Inhibits microtubule synthesis and thus hinders mitosis
- c) Terbinafine : Squalene epoxidase inhibition
- d) Amphotericin B : Sequesters ergosterol

Question 13:

Which of the following parameters is used to assess the extent of drug absorption in a plasma concentration vs. time graph?

- a) T_{max}
- b) C_{max}
- c) T_{1/2}
- d) Area under the curve

Question 14:

Which of the following routes of administration will not bypass hepatic first-pass metabolism?

- a) Sublingual
- b) Oral
- c) Intravenous
- d) Transdermal

Question 15:

For the action of heparin as an anticoagulant, availability of which of the following chemical substances is mandatory?

- a) Thrombin
- b) Plasmin
- c) Antithrombin-III
- d) Thrombomodulin

Question 16:

What is the bioavailability of a drug administered intravenously?

- a) 10%
- b) 100%
- c) 50%
- d) Variable

Question 17:

A drug solution containing 500 mg/5 ml of the drug is given to you. You are provided with a 2 ml syringe (having 10 divisions for each ml). How many divisions of the 2 ml syringe need to be administered to deliver 180 mg of the drug to a patient?

- a) 1.8
- b) 10
- c) 14
- d) 18

Question 18:

Which of the following is gametocidal to all the species causing malaria?

- a) Primaquine
- b) Chloroquine
- c) Quinine
- d) None

Question 19:

A drug label suggests storage in a 'cool' place. What is the temperature at which this drug is to be stored?

- a) - 2 degree Celsius
- b) 0 degree Celsius
- c) 2-6 degree Celsius
- d) 8-15 degree Celsius

Question 20:

A patient with AIDS was treated with zidovudine, lamivudine, indinavir, ketoconazole and cotrimoxazole. He developed nephrolithiasis, central obesity, hyperlipidemia and insulin resistance. Which of the drugs could have been the cause?

- a) Ketoconazole
- b) Indinavir
- c) Lamivudine
- d) Zidovudine

Question 21:

If carbamazepine is added to ongoing valproate therapy, what adverse effect is risked?

- a) Pancreatitis
- b) Thrombocytopenia
- c) Hepatotoxicity
- d) Hyperammonemia

Question 22:

What is the mechanism of action of dapson, considering that it is used in the treatment of leprosy, fungal infections and, also for immunomodulatory actions?

- a) Cell wall synthesis inhibitor
- b) Folic acid synthesis inhibitor
- c) Protein synthesis inhibitor
- d) Cell membrane inhibitor

Question 23:

What is the mechanism of action of sodium nitroprusside?

- a) Increased cAMP
- b) Stimulation of guanylyl cyclase
- c) Calcium channel blockage

- d) K⁺ channel opener

Question 24:

Levothyroxine/liothyronine acts by binding to thyroid receptors. Thyroid hormone receptors belong to which of the following class of receptors?

- a) Transmembrane receptor linked to tyrosine kinase
- b) Nuclear receptor
- c) Ion channel receptor
- d) G protein coupled receptor

Question 25:

The Nobel Prize in Physiology or Medicine, 2018 was awarded to James P. Allison and Tasuku Honjo for which of the following discoveries?

- a) Discovery of mechanisms of autophagy
- b) Discoveries of molecular mechanisms controlling the circadian rhythm
- c) Discovery of cancer therapy by inhibition of negative immune regulation
- d) Discovery that mature cells can be reprogrammed to become pluripotent

Question 26:

Which of the following drugs does not require dose reduction in patients with renal failure?

- a) Gentamicin
- b) Vancomycin
- c) Amphotericin B
- d) Doxycycline

Question 27:

Which of the following drugs does not increase the risk of bleeding in patients undergoing warfarin therapy?

- a) Cimetidine
- b) Carbamazepine

- c) Isoniazid
- d) Amiodarone

Question 28:

Which of the following is used in acute mountain sickness?

- a) Dexamethasone
- b) Digoxin
- c) Diltiazem
- d) Acetazolamide

Question 29:

Which of the following anti-tubercular drugs is associated with ophthalmic toxicity?

- a) Ethambutol
- b) Isoniazid
- c) Kanamycin
- d) Rifampin

Question 30:

A patient with diabetic macular edema developed glaucoma. Which of the following drugs would you consider giving last?

- a) Beta blocker
- b) Alpha agonist
- c) Pilocarpine
- d) Prostaglandin analogues

Question 31:

What is the mechanism by which acetylcholine causes a decrease in the heart rate?

- a) Delayed diastolic depolarization
- b) Increase in plateau

- c) Decrease in preload
- d) Increase in afterload

Question 32:

Diamox is given to a patient with glaucoma. It is a non-competitive inhibitor of the carbonic anhydrase enzyme. Which of the following would be the effect of this drug?

- a) Decrease in V_{max}
- b) Decrease in K_m
- c) Decrease in both K_m and V_{max}
- d) No change in V_{max}

Question 33:

Low apparent volume of distribution of drug indicates

- a) Low $t_{1/2}$
- b) Low bioavailability
- c) Low efficacy
- d) Not extensively distributed to tissues

Question 34:

An infant had a serum bilirubin level of 33 mg/dL. The mother might have taken which of the following drugs during the third trimester for this to occur?

- a) Ampicillin
- b) Co-trimoxazole
- c) Azithromycin
- d) Chloramphenicol

Question 35:

Lithium usage in pregnancy can lead to which of the following teratogenic effects?

- a) Cardiac defects

- b) Facial defects**
- c) CNS defects**
- d) Urogenital defects**

Answer Key

Question No.	Correct Option
1	a
2	b
3	b
4	a
5	a
6	c
7	a
8	d
9	a
10	a
11	c
12	b
13	d
14	b
15	c
16	b
17	d
18	a
19	d
20	b
21	c
22	b
23	b
24	b
25	c
26	d
27	b
28	d

29	a
30	d
31	a
32	a
33	d
34	b
35	a

Detailed Explanations

Solution to Question 1:

Hypoglycemia is not a side effect of salbutamol. It causes hyperglycemia.

The side effects of salbutamol are due to the stimulation of extrapulmonary β_2 receptors. They are more common after oral or intravenous administration. The side effects seen are:

- Muscle tremor
- Tachycardia
- Hypokalemia
- Restlessness
- Hypoxemia
- Increase in free fatty acid, insulin, glucose, pyruvate, and lactate

Solution to Question 2:

Schedule H drugs should be sold only if the prescription is from a registered medical practitioner (RMP).

Important schedules under the “drugs and cosmetics act”:

- C and C1: Biological and special products
- F and F1: Bacterial vaccines
- G: Drugs that must be labelled with the word “caution” should not be taken without medical supervision
- J: List of diseases and ailments which a drug may not claim to prevent or cure
- M: Good Manufacturing Practices (GMP)
- P: Expiry period of drug formulations
- W: Drugs marketed under generic name only

- X: Psychotropic drugs requiring a special license for manufacture and sale
- Y: Requirements and guidelines on clinical trials, import, and manufacture of new drugs.

Solution to Question 3:

The volume of distribution of the drug determines the loading dose of the drug.

Loading Dose = Vol. of distribution x Desired steady state plasma concentration.

When the time to reach a steady-state is long, as it is for drugs with long half-lives, it may be desirable to administer a loading dose that promptly raises the concentration of drug in plasma to the target concentration.

Option A: Clearance is used to determine the maintenance dose.

Maintenance dose = Clearance x Desired Steady-state plasma concentration.

Options C and D: Plasma half-life primarily determines the frequency of dosing, while bioavailability determines the route of administration. Though bioavailability may affect the dose necessary, it will not be the primary factor considered during the calculation of the loading dose.

Solution to Question 4:

Therapeutic index is used to assess the safety of the drug administered.

Therapeutic index = TD_{50}/ED_{50}

- TD_{50} (Median toxic dose) = Dose required to produce a toxic effect in 50% of animals tested.
- ED_{50} (Median effect dose) = Dose required to produce a therapeutic effect in 50% of the individuals.

Option B: Efficacy is defined as the maximum effect produced by the drug. It expresses the effect produced by the receptors when bound by a certain drug.

Option C: Potency refers to the concentration (EC_{50}) or dose (ED_{50}) required to produce 50% of maximum effect where EC_{50} is the concentration of agonist resulting in a half-maximal response.

Option D: Median toxic dose is the dose required to produce a toxic effect in 50% of animals tested. Lethal dose is the dose required to kill 50% of animals tested.

Solution to Question 5:

Flagellate hyperpigmentation/flagellate erythema is a characteristic cutaneous reaction to treatment with bleomycin.

Clinical features include multiple linear streaks (erythematous or hyperpigmented) at sites of trauma and generalized pruritus.



Solution to Question 6:

The anticoagulant of choice for the management of cancer-associated thromboembolism is low-molecular-weight heparin (LMWH).

It is the preferred agent because of more bioavailability, lack of need for anticoagulant monitoring, and a long $t_{1/2}$.

Venous thromboembolism (including deep vein thrombosis, pulmonary embolism, and splanchnic vein thrombosis), is a frequent complication of cancer and is the second-leading cause of death in cancer patients receiving chemotherapy.

Based on the results from the Comparison of Low-molecular-weight heparin vs. Oral anticoagulants for long-term anticoagulation in cancer patients with venous Thromboembolism (CLOT) Trial, all recent guidelines now advise LMWH for cancer-associated thromboembolism.

However, unfractionated heparin (UFH) can be used in patients with renal failure, and fondaparinux can be considered in patients with heparin-induced thrombocytopenia (HIT).

Solution to Question 7:

Edaravone was approved by the FDA for the treatment of amyotrophic lateral sclerosis (ALS).

It has free radical scavenging properties that may reduce oxidative stress. The exact mechanism of action is unknown. It is to be administered as an IV infusion with an initial treatment cycle of daily dosing for 14 days, followed by a 14-day drug-free period. Subsequent treatment cycles consist of dosing on 10 of 14 days, followed by a 14 day drug-free period.

The major adverse effect of edaravone is hypersensitivity because the infusion contains sodium bisulfite. Other adverse effects include bruising, gait disorder, and headache.

Tafinlar (dabrafenib) and Mekinist (trametinib), are both indicated as monotherapy in BRAF V600E/K (BRAF V600E mutation-positive) melanoma. They are approved for combined administration, for the treatment of anaplastic thyroid cancer (ATC) that cannot be removed by surgery or that has spread to other parts of the body (metastatic).

Piracetam is a nootropic agent (a drug that enhances cognitive function) that is used off-label to enhance memory. It is also used to treat myoclonus.

Solution to Question 8:

The drug of choice for a case of scrub typhus during pregnancy is azithromycin. For other patients the drug of choice is doxycycline.

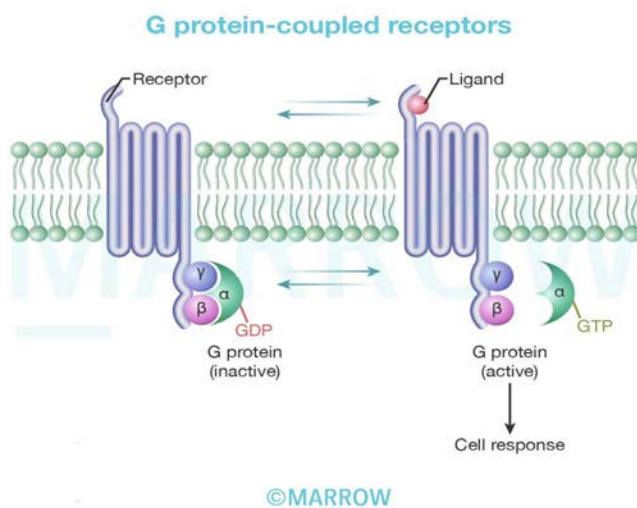
An alternative treatment option in pregnant women and patients allergic to doxycycline is chloramphenicol.

Solution to Question 9:

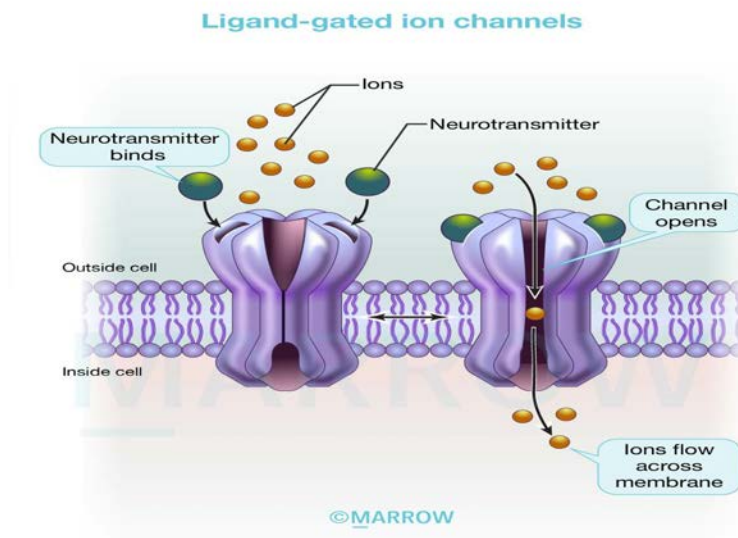
Muscarinic receptors are G-protein-coupled receptors among the given options.

M1, M2, M3, M4, and M5 are the muscarinic Ach receptors.

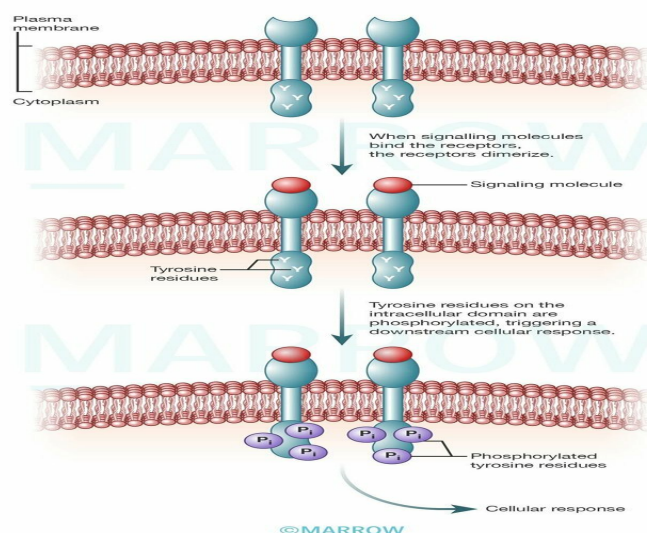
- M1 and M3 act through Gq protein and activate IP₃/DAG causing increased cytosolic calcium.
- M2 and M4 act through Gi protein and inhibit adenylyl cyclase causing increased potassium conductance.



Nicotinic receptors and GABA-A are examples of ion channel coupled receptors.



Insulin receptor is a tyrosine kinase receptor.



Solution to Question 10:

A drug stating that 500mg of paracetamol is present in each tablet, but contains only 200mg is a misbranded drug, according to the CDSCO (Central drugs standard control organization).

The drug and cosmetics act initially passed in 1940 and amended in 2016 provides a definition of misbranded, spurious, and adulterated drugs.

A drug is considered misbranded if it fulfills any of the following three conditions:

- If the appearance of the drug has been altered to conceal physical damage or make it appear of greater therapeutic value than it actually is (or)
- If it is not labeled in the prescribed manner (or)

- If the label or container of the drug makes a false or misleading claim about the drug. In this case, the packet states that there is 500 mg of paracetamol present as opposed to the 200 mg that is actually present.

A drug is considered spurious if it fulfills the following conditions:

- If the drug is imported under a name that belongs to another drug (or)
- If the drug is an imitation of another drug with the intent to deceive (or)
- If the label depicts a manufacturer that is either fictitious or a real company with no ties to the product (or)
- If it has been substituted wholly or in part by another drug or substance.

A drug is considered adulterated if it fulfills the following conditions:

- If it consists wholly or in part of any filthy, putrid, poisonous, or decomposed substance (or)
- If it has been manufactured under unsanitary conditions (or)
- If it bears a coloring other than the one prescribed (or)
- If it has been mixed with any substance that may reduce its quality or strength.

*Note: The term counterfeit drugs is a broad term and is used by the WHO. This may also be conceived to include misbranded, spurious, or adulterated drugs. However, the term misbranded drugs would be a more appropriate answer here since it is likely that the reference used is from CDSCO.

Solution to Question 11:

The given clinical scenario describes the adverse effects characteristic of metoclopramide. The most common adverse effects of metoclopramide involve the central nervous system because it crosses the blood-brain barrier (BBB) and blocks the central D₂ receptors.

The adverse effects include:

- Restlessness
- Drowsiness
- Insomnia
- Anxiety
- Agitation
- Extrapyramidal effects - acute reaction with high doses and long-term therapy. (due to central dopamine receptor blockade)
- Dystonias
- Akathisia
- Parkinsonian features
- Tardive dyskinesia

Adverse events are more common among the elderly. Long-term use should be avoided unless absolutely necessary.

Solution to Question 12:

Flucytosine acts by inhibition of fungal DNA synthesis.

It is a pyrimidine antimetabolite. After uptake into fungal cells, fungal cytosine deaminase activates it to 5-fluorouracil, which is an anticancer drug. This is further converted into metabolites that inhibit thymidylate synthase which is necessary for DNA synthesis. Thus fungal DNA synthesis is inhibited. Mammalian cells are spared as they lack cytosine deaminase enzyme.

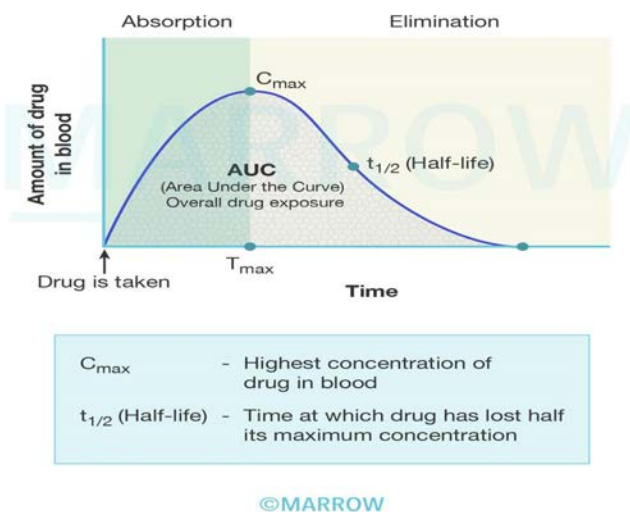
Flucytosine has clinically useful activity against (3C):

- Cryptococcus
- Candida species
- Agents causing chromoblastomycosis

Solution to Question 13:

The area under the curve of plasma concentration vs. time curve gives the extent of absorption, which is calculated by the trapezoidal rule.

The following image shows a plasma concentration vs. time graph:



Two drugs may have the same rate of absorption but a different extent of absorption or bioavailability.

The factors upon which the extent of drug absorption depends are:

- Lipophilicity - If too lipophilic (e.g., acyclovir), the drug is not soluble enough to cross the water layer adjacent to the cell.
- Hydrophilicity - If too hydrophilic (e.g., atenolol), the drug cannot cross the lipid cell membrane.

Solution to Question 14:

Hepatic first-pass metabolism occurs in drugs administered via the oral route.

When drugs are given orally, they are absorbed in the intestine and reach the liver through the portal vein. During this process, the drug is acted on by enzymes of the liver and intestine. Additionally, the liver may excrete part of the drug in the bile, further reducing the bioavailability. This is called the first-pass metabolism.

In all the other routes, the drug directly enters systemic circulation without reaching the liver.

Note: Bioavailability is the fraction of an administered dose of unchanged drug that reaches the systemic circulation. Hence, the bioavailability of a drug administered intravenously (I.V) is 100%. It is neither subjected to the first-pass metabolism as in the oral route nor does it depend on local factors as in subcutaneous, intramuscular, or transdermal.

Solution to Question 15:

Antithrombin-III is mandatory for the action of heparin as an anticoagulant. Reduced antithrombin levels can lead to resistance.

Heparin resistance refers to failure to achieve a therapeutic aPTT with daily doses of heparin of 35,000 units or more.

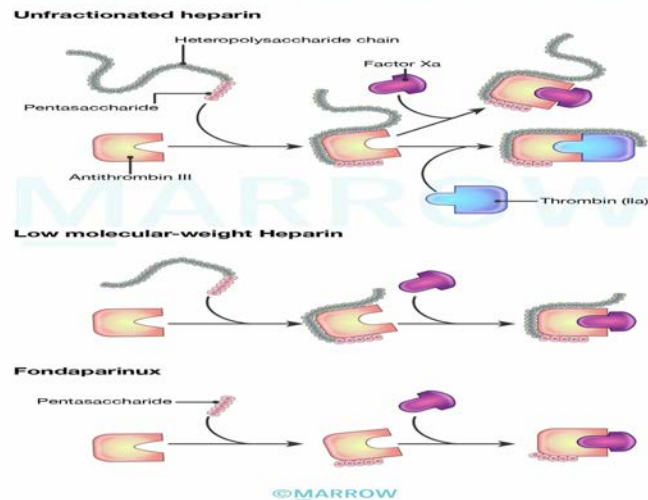
In true heparin resistance, there is an actual blunting of heparin's anticoagulant effect. This occurs due to:

- Low ATIII levels (most common cause)
- Increased heparin-binding plasma proteins (acute phase reactants)
- Increased heparin clearance (e.g. due to splenomegaly in liver disease)

Low ATIII levels could be due to decreased synthesis (hereditary, hepatic failure), increased consumption (DIC, MAHA, or in a bypass circuit) or increased loss (nephrotic syndrome).

The dose of heparin has to be increased to achieve therapeutic aPTT. Severe antithrombin deficiency may require antithrombin concentrate as well.

Heparin, LMWH, and Fondaparinux - the structure and their actions



Solution to Question 16:

The bioavailability of a drug administered intravenously (I.V) is 100%.

It is neither subjected to the first-pass metabolism as in oral route nor does it depend on local factors as in subcutaneous, intramuscular, or transdermal. In IV route, the drug is directly delivered to the systemic circulation and hence has the maximum availability for action at the intended site.

Bioavailability is the fraction of an administered dose of unchanged drug that reaches the systemic circulation. Hence, by definition, when a medication is administered intravenously, its bioavailability is 100%.

Intrathecal administration does not have a bioavailability of 100%.

Solution to Question 17:

Concentration of drug solution given = $500/5$ ml or 100 mg/ml

-> 100 mg of drug in 1ml

i.e 180 mg of drug in "X" ml

-> $X = 180/100 = 1.8$ ml

So, 1.8 ml of the drug with 180 mg of the drug is to be given to the patient.

In the 2 ml syringe provided, each ml has 10 divisions i.e. each division corresponds to 0.1 ml.

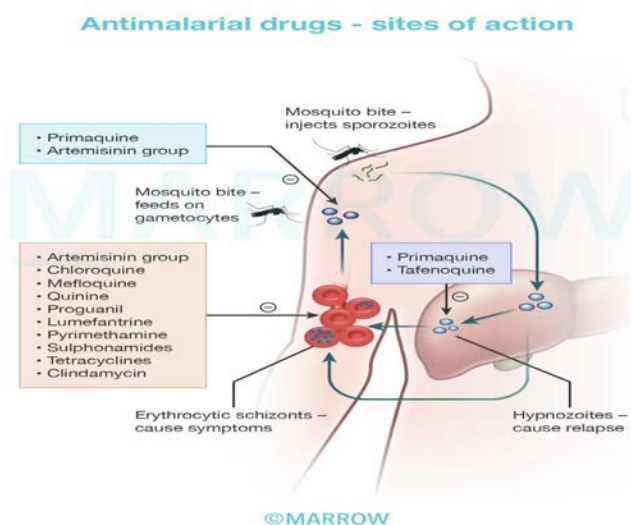
So, for delivering 1.8 ml, $1.8/0.1 = 18$ divisions need to be administered.

Solution to Question 18:

Primaquine is gametocidal to all species causing malaria. It inhibits transmission of malaria.

The drugs with clinically important gametocidal activity (against all species) are:

- High activity:
- Artemisininins
- Primaquine
- Low to moderate activity:
- Chloroquine
- Quinine
- Atovaquone/proguanil
- No activity:
- Mefloquine
- Pyrimethamine
- Sulfadoxine
- Tetracycline



Solution to Question 19:

Storing at a cool place means to store at 8-15 degrees C.

Examples for drugs to be stored under cool temperature include insulin, nicorandil.

Label instructions	Temperature
Store frozen	-200C
Keep cool	8-150C
Store at room temperature	15-250C

Solution to Question 20:

Indinavir, a protease inhibitor causes the above given adverse effects.

Patients being treated with protease inhibitors (e.g., indinavir) develop a syndrome involving the derangement of lipid metabolism. This causes changes in body fat distribution. Type 2 diabetes mellitus has also been reported. This is called HIV/HAART-associated lipodystrophy.

It is typically seen after 6–12 months of protease inhibitor therapy.

Solution to Question 21:

If carbamazepine is added to ongoing valproate therapy, it can result in hepatotoxicity.

A combination of valproate and carbamazepine increases hepatotoxicity. Valproic acid may inhibit carbamazepine clearance and increase steady-state carbamazepine blood levels.

Solution to Question 22:

The mechanism of action of dapsone is inhibition of folic acid synthesis.

Dapsone [4,4'-diaminodiphenylsulphone (DDS)] is a structural analog of para-aminobenzoic acid (PABA) and a competitive inhibitor of dihydropteroate synthase in the folate pathway. It is weakly bactericidal and can cause mild hemolysis, and rarely anemia or psychosis.

DDS syndrome is a rare side effect of dapsone therapy in leprosy patients which starts after 5-6 weeks of starting dapsone and leads to exfoliative dermatitis, hepatosplenomegaly, lymphadenopathy, fever and hepatitis and may be fatal. It is also called 'five-week dermatitis'.

Solution to Question 23:

Sodium nitroprusside acts by stimulation of guanylyl cyclase.

Sodium nitroprusside releases nitric oxide which in turn stimulates guanyl cyclase. This activates cGMP secondary messenger which results in the relaxation of the smooth muscle.

Solution to Question 24:

Thyroid hormone receptors belong to the class of nuclear receptors.

Nuclear receptors are a family of ligand-regulated transcription factors that are activated by various lipid-soluble signals, including retinoic acid, thyroid hormone, and steroid hormones. They primarily function as transcription factors and directly interact with the genomic DNA to regulate gene expression.

Nuclear receptors can either remain anchored in the cytoplasm via chaperone proteins until freed by ligand-binding (e.g., steroid and vitamin D receptors) or can reside in the nucleus bound to specific DNA response elements even in the absence of ligand (e.g., thyroid hormone, PPAR γ , and retinoic acid receptors).

Solution to Question 25:

The Nobel prize in Physiology or Medicine, 2018 was awarded to James P. Allison and Tasuku Honjo for the discovery of cancer therapy by inhibition of negative immune regulation.

Immune checkpoint inhibitors inhibit proteins that keep immune responses to cancer cells in check. When these proteins are blocked, the brake on the immune system is released and T cells are able to kill cancer cells more efficiently.

The Nobel prize 2018 in Physiology or Medicine was awarded to James P. Allison and Tasuku Honjo for their breakthrough work on unleashing the body's immune system to attack cancer, that has led to an entirely new class of these drugs and brought lasting remissions to many patients who had run out of options.

Solution to Question 26:

Doxycycline does not require dose reduction in renal failure. Doxycycline, minocycline, and glycylicyclines like tigecycline have fewer renal side effects.

Antibiotics causing acute kidney injury are as follows:

- Acute tubular necrosis:
- Amphotericin B
- Vancomycin
- Acyclovir
- Foscarnet
- Tenofovir
- Acute interstitial nephritis:
- Penicillin

- Cephalosporins
- Quinolones
- Rifampicin

Solution to Question 27:

Carbamazepine does not increase the risk of bleeding in patients undergoing warfarin therapy.

Warfarin is mainly metabolized by the CYP2C9 enzyme. Drugs that inhibit CYP2C9 enzyme reduce the metabolism of warfarin and thus, enhance the anticoagulant effect of warfarin and increase the risk of bleeding.

Drugs causing CYP2C9 inhibition:

- Amiodarone
- Azole antifungals
- Cimetidine
- Clopidogrel
- Cotrimoxazole
- Disulfiram
- Isoniazid
- Metronidazole

Solution to Question 28:

Acetazolamide is used in the treatment of acute mountain sickness for symptomatic relief. It can also be used for prophylaxis.

Solution to Question 29:

Ethambutol is associated with ophthalmic toxicity.

Optic neuritis resulting in decreased visual acuity and loss of ability to differentiate red from green is the most serious adverse effect of ethambutol. It is a dose-dependent adverse effect that can be unilateral or bilateral.

Tests of visual acuity and red-green discrimination before the start of therapy and periodically thereafter are recommended.

Solution to Question 30:

Prostaglandin analogs are the last drugs to be considered in this patient since macular edema is one of their side effects.

Adverse effects of PGF 2alpha analogs are:

- Blurring of vision
- Macular edema (especially in aphakic patients)
- Eyelash hypertrichosis
- Iris pigmentation (Heterochromia iridis)
- Peri-orbital fat depletion

Solution to Question 31:

The mechanism by which acetylcholine causes a decrease in the heart rate is by decreasing the diastolic depolarization.

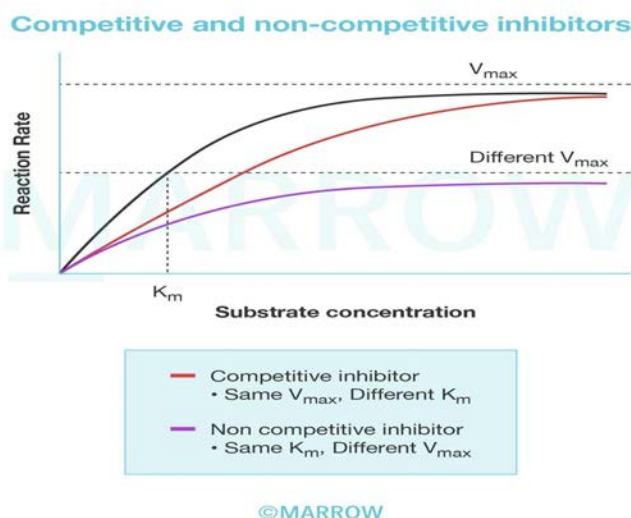
The acetylcholine released from the vagal nerve acts on the M2 receptor ($\beta\gamma$ subunit of G protein) on the SA node. This opens a special set of potassium channels causing an efflux of potassium (I_{KACh}) which hyperpolarizes the SA node (I_h). The resultant current (I_h) slows the diastolic depolarization thereby causing bradycardia.

Solution to Question 32:

Non-competitive inhibitors cause a decrease in V_{max} .

In non-competitive inhibition:

- The inhibitor binds to the adjacent site and not the catalytic site of the enzyme
- The catalytic property of the enzyme is reduced and hence V_{max} is decreased
- The affinity of the enzyme is not altered and hence K_m is unaltered



Solution to Question 33:

Low apparent volume of distribution of a drug indicates that it is not extensively distributed in the tissues.

Apparent Volume of distribution (V_d) = dose administered I.V / plasma concentration E.g: If a 500 mg of drug A is given i.v (100% bioavailability) and the plasma concentration reached is 10 mg/liter then the apparent volume of distribution (V_d) is $500/10 = 50$ liters.

But the normal blood volume is about 5 liters. A high V_d is due to accumulation of drug A in extravascular tissues resulting in low plasma concentration. Therefore, the apparent volume of distribution is an imaginary volume throughout which the drug is distributed if the plasma concentration remains the same throughout.

Option A: $t_{1/2}$ relationship with volume of distribution (V_d) and clearance (Cl) is given by the formula $t_{1/2} = 0.693 \times V_d / Cl$. $t_{1/2}$ is directly proportional to V_d but it is also inversely proportional to clearance. If a drug has a low volume of distribution but clearance is very low, $t_{1/2}$ can still be high. So low V_d doesn't always imply low $t_{1/2}$. So the best answer among the given options is D.

Option B and C: Low V_d does not indicate low bioavailability or low efficacy.

Solution to Question 34:

Cotrimoxazole is a combination drug consisting of trimethoprim and sulfamethoxazole, of which sulfamethoxazole is a sulfonamide. Sulphonamides displace bilirubin from plasma albumin, leading to elevation of serum bilirubin levels.

These drugs can cross the placenta and are secreted in milk and can therefore cause kernicterus in newborn infants if they are given to pregnant women near term or during lactation.

Solution to Question 35:

Lithium use in pregnancy can lead to cardiac defects.

Ebstein's anomaly is the specific cardiac defect with lithium use in pregnancy.

In Ebstein's anomaly, the radiological sign seen is a box-shaped heart.



Pharmacology AIIMS 2019 (May & Nov)

Question 1:

Which of the following is not an adverse effect of salbutamol?

- a) Tachycardia
- b) Tolerance
- c) Hypokalemia
- d) Hypoglycemia

Question 2:

Propranolol is drug of choice for:

- a) Ulcerated infantile hemangioma
- b) Lymphangioma circumscriptum
- c) Capillary malformation
- d) Pyogenic granuloma

Question 3:

Clostridium difficile associated diarrhea occurs maximally due to:

- a) Lincosamides
- b) Tetracyclines
- c) Carbapenems
- d) Macrolide

Question 4:

A 53 year old man is found to have stage 2 hypertension without any co morbidities. Which drug will you start?

- a) Furosemide
- b) Chlorthalidone

- c) Spironolactone
- d) Triamterene

Question 5:

Which of the following is a schedule X drug?

- a) Halothane
- b) Thalidomide
- c) Ketamine
- d) Colistin

Question 6:

LMWH is given by which route?

- a) Intravenous
- b) Subcutaneous
- c) Intramuscular
- d) Intradermal

Question 7:

A patient presents to the emergency department with diarrhea, sweating, excessive lacrimation, and salivation. His pulse rate is 55/min and his plasma cholinesterase level was 50U/mL. Which of the following can be used to treat this patient?

- a) Neostigmine
- b) Atropine
- c) Epinephrine
- d) Benztropine

Question 8:

What is pegloticase used for?

- a) Chronic gout

- b) Rheumatoid arthritis
- c) Psoriatic arthritis
- d) Paralytic ileus

Question 9:

Which of the following GPCRs does not act via the K⁺ channel?

- a) Angiotensin 1 receptor
- b) Muscarinic M2 receptor
- c) Dopamine D2 receptor
- d) Serotonin 5 HT1 receptor

Question 10:

Which of the following is not used in heart failure?

- a) Trimetazidine
- b) Sacubitril
- c) Metoprolol
- d) Nesiritide

Question 11:

Which of the following causes an increase in ACh release from the presynaptic neuron?

- a) Potassium channel blocker
- b) Sodium channel blocker
- c) Calcium channel blocker
- d) Chlorine channel blocker

Question 12:

Apart from mu-opioid receptors, on which other receptor does tramadol act?

- a) Anticholinergic
- b) 5HT_{1a} inhibition

- c) Serotonergic and norepinephrine reuptake inhibition
- d) Antihistaminic

Question 13:

In which of the following conditions is non-pulsatile dosing of GnRH agonist not used?

- a) Male infertility
- b) Endometriosis
- c) Precocious puberty
- d) Prostate cancer

Question 14:

Which of the following cannot be used to decrease intraocular pressure in glaucoma?

- a) Dexamethasone
- b) Acetazolamide
- c) Apraclonidine
- d) Mannitol

Question 15:

Prucalopride is a _____.

- a) 5HT_{2a} agonist
- b) 5HT_{2b} agonist
- c) 5HT₄ antagonist
- d) 5HT₄ agonist

Question 16:

Z track technique is used for?

- a) Administration of nicotine patches
- b) Monitoring of carbamazepine therapy
- c) Administration of long acting depot antipsychotics

- d) Monitoring of lithium therapy

Question 17:

Tafenoquine is used in which of the following conditions?

- a) Radical cure of plasmodium vivax
- b) Prophylaxis of malaria in pregnancy
- c) Treatment of severe falciparum malaria
- d) Treatment of malaria in children <2 years

Question 18:

Which is not given in Wilson's disease?

- a) Zinc
- b) Penicillamine
- c) Calcium citrate
- d) Trientine

Question 19:

What dose of epinephrine is used in anaphylactic shock?

- a) 1:10000 I.V. - 10 ml
- b) 1:1000 I.M. - 0.5 ml
- c) 1:1000 I.M. - 1 ml
- d) 1:1000 I.V. - 0.5 ml

Question 20:

Which of the following is true about tedizolid?

- a) Active against gram positive organisms
- b) Poor oral bioavailability
- c) Active against anaerobes
- d) Causes peripheral neuropathy

Answer Key

Question No.	Correct Option
1	d
2	a
3	a
4	b
5	c
6	b
7	b
8	a
9	a
10	a
11	a
12	c
13	a
14	a
15	d
16	c
17	a
18	c
19	b
20	a

Detailed Explanations

Solution to Question 1:

Salbutamol in large doses can result in hyperglycemia, not hypoglycemia.

Option A: Tachycardia due to reflex sympathetic stimulation due to vasodilator action on β_2 receptors and direct effect on cardiac β_1 receptors (β_2 selectivity is lost at high concentrations)

Option B: Tolerance occurs due to desensitization on long-term use

Option C: Hypokalemia due to direct effect on skeletal muscle uptake of K^+ ions

Solution to Question 2:

Propranolol is the drug of choice in the management of infantile hemangioma complicated by ulceration or in symptomatic patients.

It acts by causing vasoconstriction, inducing apoptosis and inhibition of angiogenesis.

Solution to Question 3:

Clostridium difficile-associated diarrhea is mostly due to lincosamides such as clindamycin

Antibiotic-associated colitis is caused by Clostridium difficile infection due to the destruction of the normal gut microbial flora by the use of antibiotics. The antibiotics most frequently implicated include clindamycin, fluoroquinolones and broad-spectrum penicillins, and cephalosporins

Clostridium difficile infection occurs frequently in hospitals and nursing homes where antimicrobial use is high and the environment is contaminated by C. difficile spores.

Although all antibiotics can cause the condition, the most commonly implicated are:

- Clindamycin
- Third-generation cephalosporins (cefotaxime, ceftriaxone, cefuroxime, and ceftazidime)
- Fluoroquinolones (ciprofloxacin, levofloxacin, and moxifloxacin)
- Broad-spectrum penicillins

Solution to Question 4:

Chlorthalidone is one of the first choice drugs in the treatment of hypertension without any comorbidities.

First-choice drugs used in the treatment of hypertension in patients without any comorbidities include:

- Thiazide diuretics: Chlorthalidone is usually preferred to hydrochlorothiazide.
- Calcium channel blockers: Extended-release, long-acting dihydropyridines are preferred.
- ACE inhibitors: Captopril is preferred for the initiation of therapy. Enalapril and ramipril twice daily may also be used.

In the presence of any associated conditions or comorbidities, the first choice drugs for the treatment of hypertension would be as follows:

Conditions associated with hypertension	First choice drugs for hypertension
Diabetes Chronic kidney disease with proteinuria	Inhibitors of renin-angiotensin system.
Angina	Beta-blockers or calcium channel blockers
Heart failure	ACE inhibitors and alpha+beta blockers such as carvedilol
Benign prostatic hyperplasia	α_1 Blockers
Pregnancy	Labetalol
Atrial fibrillation	Diltiazem and verapamil

Solution to Question 5:

Ketamine is a schedule X drug.

According to The Drugs and Cosmetics Act-1945,

Schedule X drugs are those that have high abuse liability and are to be sold with a valid prescription only.

A prescription in duplicate, separate license requirement, meticulous storage, and dispensing records must be maintained by the chemist.

Option B: Thalidomide is a schedule H drug.

Option D: Colistin is a schedule P drug.

Solution to Question 6:

LMWH is given by the subcutaneous route.

Heparin, low molecular weight heparin (LMWH), and fondaparinux are not absorbed through GI mucosa and must be given parenterally.

LMWH and fondaparinux are given subcutaneously.

Heparin can be given through intravenous infusion and subcutaneous routes.

Solution to Question 7:

The clinical features are suggestive of organophosphate poisoning and IV atropine is the treatment of choice.

The organophosphates commonly used as pesticides are anticholinesterases. Toxicity occurs by excessive inhalation of vapors, ingestion, or percutaneous absorption. It inhibits the

cholinesterase activity which results in an increase of acetylcholine, leading to cholinergic symptoms.

The predominant initial signs are of muscarinic excess which includes: (DUMBBELSS)

- Diarrhea
- Urination
- Miosis
- Bronchospasm
- Bradycardia
- Emesis
- Lacrimation
- Salivation
- Sweating

This may be followed by cognitive disturbances, convulsions, and neuromuscular blockade.

Treatment:

- Maintenance of vital signs particularly respiration
- Removal of clothing and washing of the skin
- IV atropine 2- 5 mg. If IV administration is not possible, then 2 mg IM every 5-10 minutes (till muscarinic symptoms disappear)
- Pralidoxime, a cholinesterase reactivator can also be used for treatment

Solution to Question 8:

Pegloticase is used in the treatment of chronic gout.

Pegloticase is a pegylated uricase (urate oxidase). It catalyzes the enzymatic oxidation of uric acid into allantoin, which is a more soluble and inactive metabolite.

It is used in the treatment of severe, refractory chronic gout or in cases where other urate-lowering therapies are contraindicated.

Solution to Question 9:

Angiotensin 1 receptor acts via Ca^{2+} channels. It does not act through second messengers via the K^{+} channel.

G-protein coupled receptors activate inwardly rectifying potassium channels and increase the permeability of the potassium ions causing hyperpolarisation of the cell. They are called G protein-coupled inwardly-rectifying potassium channels (GIRKs). These receptors include:

- M2 muscarinic

- A1 adenosine
- $\alpha 2$ adrenergic
- D2 dopamine
- 5HT₁ serotonin
- Somatostatin receptors.

Solution to Question 10:

Trimetazidine is not used in heart failure.

Trimetazidine is a newer antianginal drug that is a pFOX inhibitor which partially inhibits the fatty acid oxidation pathway in the myocardium. This decreases the oxygen requirement of the heart.

Nesiritide (natriuretic peptide), sacubitril (neprilysin inhibitor) and metoprolol (beta-blocker) are used in the treatment of heart failure.

Solution to Question 11:

Potassium channel blockers cause an increase in ACh release from the presynaptic neuron.

Presynaptic potassium channel block promotes the activation of voltage-gated calcium channels. This leads to a release of acetylcholine in the neuromuscular junction.

Aminopyridines (K⁺ channel blockers) are used in the treatment of Lambert-Eaton myasthenic syndrome which helps to improve muscle strength.

Solution to Question 12:

Tramadol inhibits the reuptake of serotonin and norepinephrine, apart from its action on the mu-opioid receptor.

It is a synthetic codeine analog that is a mu-opioid receptor agonist.

Tramadol is a centrally acting analgesic used in the treatment of mild to moderate pain.

Solution to Question 13:

Pulsatile dose of GnRH agonist is used in the treatment of male infertility.

Pulsatile administration of GnRH agonists results in increased FSH and LH secretion and is used to treat male and female infertility.

Non-pulsatile or continuous dose of GnRH agonist results in suppression and decreased concentrations of gonadotropins LH and FSH. They are used in:

- Endometriosis
- Uterine fibroids
- Central precocious puberty
- Ca prostate
- Ca ovary and breast
- Controlled ovary stimulation

Solution to Question 14:

Dexamethasone is a long-acting glucocorticoid and is not used to decrease the intraocular pressure in glaucoma.

Acetazolamide, apraclonidine, and mannitol are used in lowering intraocular pressure in glaucoma.

Solution to Question 15:

Prucalopride is a specific 5HT₄ receptor agonist which increases bowel movements.

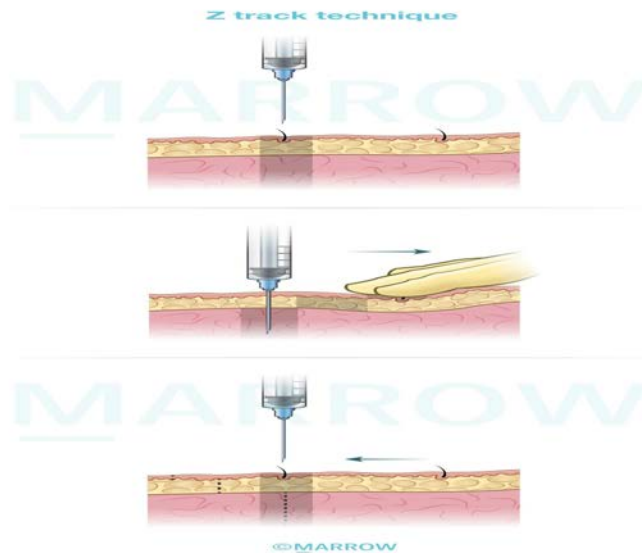
It is approved for use in women with chronic constipation where other laxatives have failed to provide relief.

Solution to Question 16:

The Z-track technique is used for the intramuscular administration of long-acting depot antipsychotics.

Advantages:

- The 'Z-track' prevents leakage of the medication into the subcutaneous tissue and skin staining through the tract to the surface of the skin
- Pain intensity is also reduced



Solution to Question 17:

Tafenoquine is used in the radical cure of *Plasmodium vivax*. Total eradication of plasmodium species which includes clinical curative and hypnozoites phase are called radical cure.

Primaquine and tafenoquine (derivative of primaquine) have very high activity against the latent tissue forms (hypnozoites) of the species. It is primarily used for the chemoprophylaxis and radical cure of *P. vivax* and *P. ovale* infections.

Solution to Question 18:

Calcium citrate is not used in the treatment of Wilson's disease.

The drugs used in the management of Wilson's disease are:

Chelating agents: These drugs bind to copper to form complexes that get excreted in the urine. D-penicillamine and trientine are first-line drugs in symptomatic patients with hepatic diseases.

- D-penicillamine: It is the recommended drug due to its availability and cost-effectiveness as compared to trientine. Since it is an antagonist to pyridoxine, supplementation of pyridoxine is required. It can precipitate various adverse effects including aplastic anemia, myasthenia gravis, and membranous nephropathy. It can also worsen neurological symptoms
- Trientine: It is administered in those who cannot tolerate D-penicillamine
- Tetrathiomolybdate: It has been identified to be effective in patients with neuropsychiatric manifestation

Elemental zinc:

- Zinc blocks the intestinal absorption of copper by inducing metallothionein (endogenous chelator in enterocytes)

- It is commonly used as a maintenance therapy along with chelating agents. It can also be used as first-line therapy in pregnancy

Solution to Question 19:

Epinephrine (1:1000 solution) 0.5 ml via intramuscular route is preferred in the treatment of anaphylactic shock.

For intravenous use, 1:1000 adrenaline must be further diluted to 1:10,000 and should be used very cautiously as slow bolus.

There is a much greater risk of causing harmful side effects by either inappropriate dosage or misdiagnosis of anaphylaxis when using IV adrenaline. Hence, IM injection is preferred over intravenous (IV) bolus because it is faster in many situations and safer (lower risk of cardiovascular complications, such as severe hypertension and ventricular arrhythmias).

For intramuscular, subcutaneous, and inhalational (nebulization) routes of administration, 1:1000 dilution can be used.

In cardiac arrest, the 1:1000 adrenaline must be diluted to 1:10,000 for IV use. The recommended dose of epinephrine is 1mg = 10 mL of 1:10,000 solution IV administered every 3 to 5 minutes during resuscitation as per the CPR guidelines.

Epinephrine is also used in the treatment of hypotension, to induce local vasoconstriction to reduce the diffusion of local anaesthetics and to induce bronchodilation in asthma and COPD.

Solution to Question 20:

Tedizolid is active against gram-positive organisms.

Tedizolid belongs to the class of oxazolidinones. It has poor activity against most gram-negative bacteria and anaerobes. It inhibits protein synthesis by binding to the P site of the 50S ribosomal subunit.

It is used for the treatment of skin and soft tissue infections, respiratory tract infections caused by streptococci, and *S. aureus* including MRSA.

Common adverse effects include myelosuppression leading to anemia, leukopenia, thrombocytopenia, and pancytopenia.

Pharmacology AIIMS 2020 (May and Nov INI-CET)

Question 1:

All of the following are the uses of enteric-coated drugs except:

- a) Protect the acid labile drug from gastric acidity
- b) Protect the gastric mucosa from irritant effects of certain drugs
- c) Prolongs the half life of the drug
- d) Deliver drugs distal to stomach

Question 2:

A patient who was on antiepileptics for the management of seizure disorder started having visual disturbance. Evaluation reveals the contraction of the visual field. The drug responsible is likely to be?

- a) Phenobarbitone
- b) Vigabatrin
- c) Ethosuximide
- d) Levetiracetam

Question 3:

If a patient who is on metoprolol is started on verapamil, which of the following side effects would be seen?

- a) Bradycardia with AV block
- b) Torsades de pointes
- c) Tachycardia
- d) Ventricular fibrillation

Question 4:

The response on stimulation of muscarinic receptor is:

- a) Erection

- b) Ejaculation
- c) Bronchodilation
- d) Increased myocardial contractility

Question 5:

Rosuvastatin should not be combined with which of the following drugs?

- a) Oxycodon
- b) Clarithromycin
- c) Adalimumab
- d) Rivoraxaban

Question 6:

Which of the following is not a prodrug?

- a) Proguanil
- b) Sulindac
- c) Primidone
- d) Flurouracil

Question 7:

Which of the following topical antifungal has both anti-inflammatory effects and anti-pruritic action?

- a) Ketoconazole
- b) Sertaconazole
- c) Luliconazole
- d) Terbinafine

Question 8:

A patient is started on lithium therapy. Which of the following side effects should the patient be asked to watch out for to monitor for toxicity?

- a) Malaise and sore throat
- b) Vertigo and Diarrhoea
- c) Fever
- d) Breathlessness

Question 9:

A psychiatric patient was started on lithium for the management of bipolar disorder. To monitor the blood lithium levels, at what time interval should the blood sample be collected after the administration of lithium tablets?

- a) After 24 hours
- b) After 12 hours
- c) After 6 hours
- d) Immediately after taking tablets

Question 10:

Which of the following is a second line anti-tubercular drug?

- a) Streptomycin
- b) Ethionamide
- c) Acyclovir
- d) Vancomycin

Question 11:

Which of the following anti-tubercular drug is not bactericidal?

- a) Isoniazid
- b) Rifapentin
- c) Thioacetazone
- d) Kanamycin

Question 12:

Which of the following drugs do not need dose adjustment in renal failure?

- a) Sitagliptin
- b) Saxagliptin
- c) Vildagliptin
- d) Linagliptin

Question 13:

The antidote for fibrinolytic therapy is?

- a) Urokinase
- b) Dalteparin
- c) Tranexamic acid
- d) Alteptase

Question 14:

Which of the following is least likely to be useful in a patient with severe COVID-19?

- a) Low molecular weight heparin
- b) Oxygen
- c) Favipiravir
- d) Steroids

Question 15:

If codiene is coming to the market in combination with dextromethorphan as an anti-tussive, this combination _____

- a) Should not be used due to risk of addiction
- b) Can be used as both have different mechanism of action
- c) Should not be used as it is irrational
- d) Can be used to treat cough effectively

Question 16:

The normality of 2M HCl is ____ as 1M H₂SO₄.

- a) Half
- b) Twice
- c) Same
- d) 4 times

Question 17:

An elderly woman on treatment for migraine presents with chest pain. She has a previous history of angina/MI. Which of the following mechanism of the prescribed drug for migraine might have caused this presentation?

- a) 5-HT antagonist
- b) 5-HT agonist
- c) Calcium Channel Blocker action
- d) Nitric oxide mediated vasodilation

Question 18:

A patient presented with pelvic fracture and head trauma after a road traffic accident. He was given IV morphine for pain relief following which he developed respiratory difficulty. What could be the most likely cause?

- a) Pelvic Fracture
- b) CO₂ retention due to opioid use
- c) Cerebral Edema
- d) Pulmonary embolism

Question 19:

Which of the following antidepressants causes the least sexual dysfunction?

- a) Fluoxetine
- b) Venlafaxine
- c) Mirtazapine
- d) Imipramine

Question 20:

Which of the following statements is true about disulfiram?

- a) When used with alcohol, the patients feels drowsy, with chest discomfort and respiratory depression
- b) It is an anticraving drug
- c) It can be initiated even in minimally motivated person
- d) Its action persists up to 2-3 days after stopping the drug

Question 21:

Which of the following drugs promotes bone formation?

- a) Calcitonin
- b) Raloxifene
- c) Bisphosphonates
- d) Teriparatide

Question 22:

A known asthmatic patient with fever and sore throat was prescribed an antibiotic. Two days later he presents with a maculopapular rash. Which of the following drugs might have caused this?

- a) Prednisolone
- b) Salbutamol
- c) Fexofenadine
- d) Amoxicillin

Question 23:

What is the most common drawback of the given device?



- a) Bacterial contamination
- b) Does not provide multiple doses
- c) Drug deposition in the oropharynx
- d) Not portable

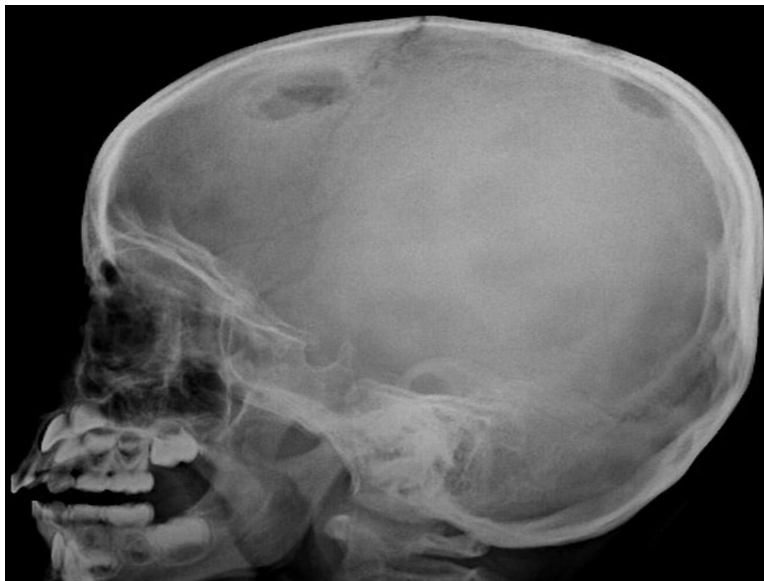
Question 24:

Which of the following statements is correct for a patient on high dose of penicillin and gentamicin?

- a) TDM is done with penicillin because of narrow therapeutic index
- b) TDM is done with gentamicin because of narrow therapeutic index
- c) TDM is done with both because of dangerous side effects
- d) Both drugs do not need monitoring

Question 25:

A boy presents with pallor and lymphadenopathy, the X-ray of his skull is given below. Which of the following drug is not used for the management of the suspected diagnosis?



- a) Cyclosporine
- b) Methotrexate
- c) Vinblastine
- d) Vincristine

Question 26:

A patient was treated with TCA for a major depressive episode in the hospital for 4 weeks. What is the main concern while considering his discharge from the hospital?

- a) Need for drug monitoring
- b) ECG monitoring for arrhythmias
- c) Suicidal risk
- d) Sedation being the side effect

Question 27:

Which of the following drugs exacerbates myasthenia and is usually avoided in a patient with myasthenia gravis? 1. Lidocaine 2. Digene 3. Meropenem 4. Phenytoin

- a) 1 and 2
- b) 2 and 3
- c) 1 and 3
- d) 1 only

Question 28:

Which of the following drugs are used for absence seizures? A. Carbamazepine B. Valproate C. Ethosuximide D. Gabapentin

- a) A, B and C
- b) B and C only
- c) B only
- d) A, B and D

Question 29:

Match the following dilution of adrenaline and its use: A. 1:1000 B. 1:10000 C. 1:100000 1. Cardiac arrest 2. Anaphylaxis 3. Local anesthesia

- a) A-1 B-2 C-3
- b) A-2 B-1 C-3
- c) A-3 B-2 C-1
- d) A-1 B-3 C-2

Question 30:

Match the following bacteria with the drug of choice used for each.

- a) A-1, B-3, C-4, D-2
- b) A-2, B-1, C-4, D-3
- c) A-3, B-4, C-1, D-2
- d) A-4, B-1, C-2, D-3

Question 31:

Which of the following drug reduces mortality in COVID-19 patients? A. Dexamethasone B. Remdesivir C. Favipiravir

- a) B and C
- b) A and B
- c) Only A

d) A, B , C

Question 32:

Which of the following statements are true?

- a) A and C
- b) A, B and C
- c) A and D
- d) B, C and D

Question 33:

Which of the following is correct about the drugs used in Gout?

- a) A, B & C
- b) A, B, C & D
- c) A & B
- d) C & D

Answer Key

Question No.	Correct Option
1	c
2	b
3	a
4	a
5	b
6	c
7	b
8	b
9	b
10	b
11	c
12	d

13	c
14	c
15	c
16	c
17	b
18	b
19	c
20	a
21	d
22	d
23	c
24	b
25	d
26	c
27	a
28	b
29	b
30	b
31	c
32	b
33	a

Detailed Explanations

Solution to Question 1:

Enteric coating of the drug does not prolong the half-life of the drug.

The half-life of a drug depends on the volume of distribution and rate of clearance from the blood. It is given by $t_{1/2} = 0.693 \times \text{Volume of distribution} / \text{Clearance}$.

The enteric coating can be done for the following reasons:

- Protection of active pharmaceutical ingredients, from the acidic environment of the stomach (e.g. enzymes and certain antibiotics).
- To prevent gastric distress or nausea from a drug due to irritation (e.g. sodium salicylate).
- For the delivery of drugs that are optimally absorbed in the small intestine to their primary absorption site in their most concentrated form.
- To provide a delayed-release component for repeat action.
- Required for minimizing the first-pass metabolism of the drug

Solution to Question 2:

The given patient has developed a contraction of the visual field which is a side effect of vigabatrin.

Solution to Question 3:

The patient will be prone to developing bradycardia and AV block.

Both β -blockers and non-dihydropyridine calcium blockers have a propensity to slow down SA Node function and AV conduction. A combination of these drugs increases the risk further and is contraindicated. In contrast, propranolol has safely been used with the dihydropyridine group (nifedipine) of calcium channel blockers.

Solution to Question 4:

Erection is mediated by muscarinic receptors.

Erection results from parasympathetic action which is mediated via muscarinic/nicotinic receptors, whereas sympathetic action is mediated by adrenergic receptors.

Solution to Question 5:

Rosuvastatin should not be combined with clarithromycin.

Clarithromycin is a macrolide antibiotic and it can cause drug interactions because of its ability to inhibit organic anion-transporting polypeptides 1B1 and 1B3 (OATP1B1 and OATP1B3) which are involved in the excretion of statins like rosuvastatin and pravastatin.

Individuals taking rosuvastatin, pravastatin, or fluvastatin with clarithromycin had an increased risk of hospitalization for acute kidney injury, an increased risk of hyperkalemia and an increased risk of all cause mortality.

Solution to Question 6:

Among the given options primidone is not a prodrug.

Prodrugs are drugs that are in their pharmacologically inactive form. They are converted to their active forms by metabolism.

The prodrug may offer advantages over the active form in being more stable, having better bioavailability, or fewer side effects and toxicity. Some prodrugs are activated selectively at the

site of action. The given options are examples of prodrugs:

- Sulindac is a sulfoxide prodrug used in inflammatory diseases including osteoarthritis, rheumatoid arthritis, acute gouty arthritis, ankylosing spondylitis.
- Proguanil is a prodrug and its active metabolite is cycloguanil. It is used in the treatment and prevention of malaria.
- Fluro-uracil is a prodrug having the active metabolite as fluorouridine monophosphate

Solution to Question 7:

Sertaconazole has both anti-inflammatory effects and anti-pruritic action.

Sertaconazole inhibits the release of pro-inflammatory cytokines from activated immune cells.

Solution to Question 8:

Vertigo and diarrhoea are known side-effects of lithium therapy. Lithium therapy can cause side effects at both therapeutic and toxic levels.

Known significant side effects of lithium therapy include

- Polyuria-polydipsia
- Dry mouth
- Leukocytosis Headache
- Hand tremors
- Confusion
- Decreased memory
- Muscle weakness
- Hyperreflexia
- Muscle twitch
- Vertigo
- ECG changes
- Nausea & vomiting
- Diarrhoea

Other options:

No evidence of significant association with sore throat, fever or breathlessness has been documented with lithium therapy

Solution to Question 9:

The blood sample for serum lithium estimation is to be collected after a time interval of 10-12 hours from the previous dose of lithium.

Lithium is used in the management of bipolar disorder, mania, and has anti-suicidal properties. It has a narrow therapeutic index, and blood levels in excess of this may cause symptoms of toxicity. Therapeutic drug monitoring is done by measuring trough lithium levels 12 hours after the previous dose, at least 5 days after the initiation of the treatment with lithium.

Solution to Question 10:

Ethionamide is a second line anti-tubercular drug.

Ethionamide is structurally related to isoniazid and acts by inhibiting the mycolic acid synthesis. It is effective against bacilli resistant to other drugs and has been proved effective in infections due to atypical mycobacteria. It is effective against both intracellular as well as extracellular organisms.

The first line anti-tubercular drugs include: Isoniazid, Rifampicin, Pyrazinamide, Ethambutol

The second line drugs include fluoroquinolones, ethionamide, capreomycin, kanamycin, amikacin, cycloserine, thioacetazone.

Note: This is a recall question when ethionamide is a second-line among the given options, and streptomycin was a first line injectable drug.

Solution to Question 11:

Thioacetazone is not a bactericidal drug, it is bacteriostatic.

Thioacetazone is never used alone. It is only used as a combination with first-line TB medications, such as isoniazid and rifampicin. It is used primarily to prevent the development of bacteria that are resistant to first-line drugs and to treat patients infected with drug-resistant TB.

It should never be used in HIV patients as it can cause severe and fatal skin reactions. Side-effects include gastrointestinal disturbances, blurring of vision, hemolytic anemia and urticaria.

Isoniazid is a first-line bactericidal drug that is active against both intracellular and extracellular bacilli.

Rifapentin is a rifampin congener having a similar mechanism of action. It is a bactericidal drug having a longer half-life than rifampin.

Kanamycin is bactericidal and is active against bacilli resistant to streptomycin, INH, and cycloserine.

Solution to Question 12:

Linagliptin does not require dose adjustment in patients with renal failure.

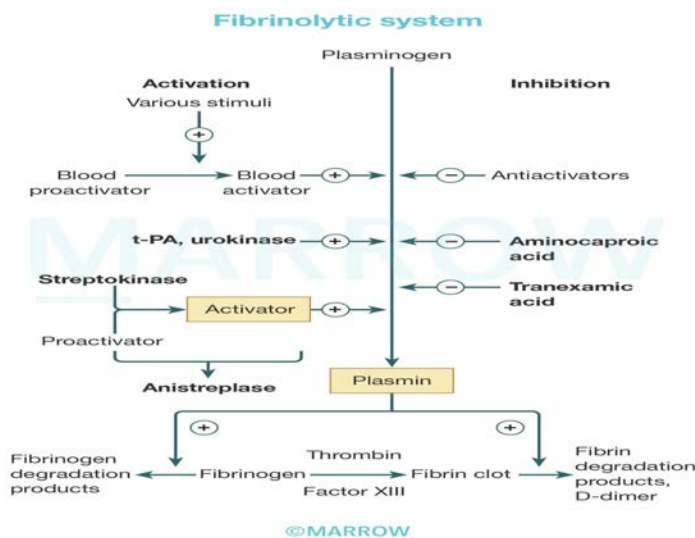
Linagliptin is a dipeptidyl-peptidase-4 (DPP-4) inhibitor. It is used in type 2 diabetes as monotherapy or in combination therapy when metformin is not tolerated or contraindicated. It binds extensively to plasma proteins and is cleared primarily by the hepatobiliary system. Hence, no dosage adjustment is needed in renal failure.

Other DPP-4 inhibitors (sitagliptin, vildagliptin, saxagliptin, alogliptin) show renal elimination and therefore require dose reduction in renal failure.

Solution to Question 13:

The antidote for fibrinolytic therapy is tranexamic acid.

Tranexamic acid is a lysine analogue that competitively bind to lysine binding sites on plasminogen and plasmin, and block their interactions with fibrin and thereby inhibiting fibrinolysis. It is given intravenously in trauma resuscitation, massive hemorrhage, and in women with postpartum hemorrhage.



Other options:

Option A (Urokinase) and D (Alteplase) are fibrinolytic drugs.

Option B: Dalteparin is low molecular weight heparin.

Solution to Question 14:

Favipiravir (inhibits viral RNA polymerase) was the first oral drug approved in India for restricted emergency use for the treatment of mild to moderate COVID-19 patients. It was not approved for

severe COVID-19.

Note: This is a question from INICET 2020, during which time Favipiravir had gained approval from DCGI for mild to moderate COVID. The current clinical management protocol for COVID-19 does not recommend Favipiravir.

Solution to Question 15:

It is an irrational drug combination and should not be given.

Codeine is an opioid analgesic having specific antitussive action while dextromethorphan is a centrally active NMDA receptor antagonist that may be effective as an antitussive. Both are centrally acting antitussives, and do not provide any added advantage when combined. Hence it is an irrational drug combination.

Rational fixed-dose combination drugs for the treatment of cough usually contain:

- A centrally acting anti-tussive (either codeine or dextromethorphan; not both), with
- Antihistamines, decongestants and expectorants.

Option A - Both of these drugs carry a risk of addiction, but it is not the reason to not give the drug. The main reason is that the combination is irrational.

Options B & D - Both codeine and dextromethorphan have different mechanisms of action and can treat cough effectively. However, the combination of two centrally acting antitussives does not provide any added advantage compared to using one drug alone. Hence, the combination is irrational.

Solution to Question 16:

The normality of 2M HCl is the same as 1M H₂SO₄.

Normality of a solution can be calculated from its molarity by multiplying it by its valence. Valence is the number of protons a salt releases in aqueous solution.

Here Valence of HCl is 1 while H₂SO₄ is 2.

Hence Normality of HCl = 2M x 1 = 2 N

similarly Normality of H₂SO₄ = 1M x 2 = 2N

Normality is defined as the number of gram equivalents dissolved per liter of solution. Where gram equivalents are the number of moles of a solute multiplied by its valence

Molarity is defined as the number of moles of a solute dissolved per liter of solution.

Solution to Question 17:

The patient may have been given sumatriptan, a 5-HT_{1b/1d} receptor agonist which causes vasoconstriction and used in migraine. Thus, sumatriptan will worsen angina by causing coronary vasoconstriction.

Angina will be worsened if the O₂ demand of the heart is increased or there is reduced blood flow to the already ischemic areas.

Solution to Question 18:

The most likely cause of respiratory difficulty in the given clinical scenario is CO₂ retention due to opioid use, as the patient was administered morphine after head trauma.

Head injury is considered as a contraindication to morphine use because:

- Morphine retains CO₂ that increases intracranial pressure (ICP) and adds to the already increased ICP.
- Therapeutic doses can result in marked respiratory depression.
- Interferes with the assessment of progress by causing miosis, vomiting, and altered mentation.

Solution to Question 19:

Among the given options mirtazapine causes the least sexual dysfunction.

Mirtazapine and mianserin block histamine H₁ receptors and also have high antagonistic activity against 5HT_{2A}, 5HT_{2C}, and 5HT₃ receptors. Sexual dysfunction is not a known side-effect of mirtazapine.

Due to its sedative effect, mirtazapine is used in depression with insomnia.

Option A: Fluoxetine is a selective serotonin reuptake inhibitor (SSRI) that can cause side effects like GI disturbances, sexual dysfunctions, may increase the risk of suicidal thoughts or behavior, risk of serotonin syndrome with MAOIs, and some CYP interactions. Vilazodone is an SSRI not associated with sexual dysfunction or weight gain.

Option B: Serotonin-norepinephrine reuptake inhibitors like venlafaxine have side effects that include nausea and dizziness, may increase the risk of suicidal thoughts or behavior, and can also cause sexual dysfunctions.

Option D: Tricyclic antidepressants like imipramine have side effects like orthostatic hypotension, weight gain, GI disturbances, sexual dysfunction, seizures, and irregular heartbeats.

The various drugs classes used in depression and anxiety are listed below

Class of drugs	Examples
Selective serotonin reuptake inhibitors (SSRI)	FluoxetineFluvoxamineParoxetineSertralineCitalopram
Serotonin-norepinephrine reuptake inhibitors	Venlafaxine, desvenlafaxineDuloxetineMilnacipran

Class of drugs	Examples
Tricyclic antidepressants	Amitriptyline Imipramine Nor tripyline Desipramine Amoxa pine
Atypical antipsychotics	Aripiprazole Olanzapine Queti apine Risperidone
Monoamine oxidase inhibitors (MAO)	Isocarboxazid Selegiline Phene lazine
Atypical antidepressants	Bupropion Trazodone Mirtaza pine Mianserin

Solution to Question 20:

Disulfiram is used in behavioural modification technique for alcohol de-addiction. The principle of action is aversive conditioning.

Disulfiram is an irreversible inhibitor of both cytosolic and mitochondrial forms of aldehyde dehydrogenase. Ethanol consumption during disulfiram therapy can lead to increased concentrations of acetaldehyde. Increased acetaldehyde levels can cause flushing, pulsating headache, throbbing pain in head and neck, nausea, vomiting, rashes, respiratory difficulties, chest pain, autonomic manifestations and other aversive reactions.

Disulfiram can only be given to highly motivated individuals. Disulfiram should not be administered until the person has abstained from alcohol for at least 12 hours. The disulfiram-alcohol reaction may occur as long as 14 days after the last dose of disulfiram.

Disulfiram like reaction with alcohol ingestion is also seen with the following drugs:

- Metronidazole
- Sulfonamides like chlorpropamide
- Cephalosporins like cefoperazone, cefamandole

Other modalities of treatment:

- Pharmacotherapy: Typically 5-10 days, treatment for detoxification: Benzodiazepine and chlordiazepoxide are the drugs of choice.
- Rehabilitation programs: Typically 28-day programs, learn about relapse prevention and identification of triggers.
- Maintenance phase: Baclofen, Acamprosate, Naltrexone, Topiramate, SSRI, Ondansetron
- Psychotherapy: Group therapy such as Alcoholics Anonymous

Solution to Question 21:

Teriparatide increases bone formation especially when given in low doses and intermittently.

Teriparatide is a recombinant preparation of parathyroid hormone.

The pharmacologic activity of teriparatide, which is similar to the physiologic activity of PTH, includes stimulating osteoblast function, increasing gastrointestinal calcium absorption, and increasing renal tubular reabsorption of calcium.

Although it may be more effective than bisphosphonates and estrogen, it has to be administered subcutaneously every day. Hence it is indicated mostly in severe osteoporosis.

Teriparatide increases the risk of osteosarcoma and it is contraindicated in conditions that cause high risk of osteosarcoma such as:

- Paget disease
- Prior h/o of radiotherapy
- Young adults with open epiphysis
- Unexplained elevated ALP
- Hypercalcemia

Option A: Calcitonin decreases bone resorption.

Option B: Raloxifene is a selective estrogen receptor modulator, which decreases bone resorption. It is used to prevent and treat osteoporosis in postmenopausal women.

Option C: Bisphosphonates like alendronate and etidronate decrease bone resorption.

Solution to Question 22:

Skin rashes of all types may be caused by allergy to penicillin.

Amoxicillin and ampicillin are aminopenicillins. The incidence of skin rashes appears to be highest following the use of ampicillin, at about 9%.

Amoxicillin is a close congener of ampicillin with the following features:

- Better oral absorption; food does not interfere with the absorption
- Incidence of diarrhea is less
- More active against penicillin-resistant pneumococci

Solution to Question 23:

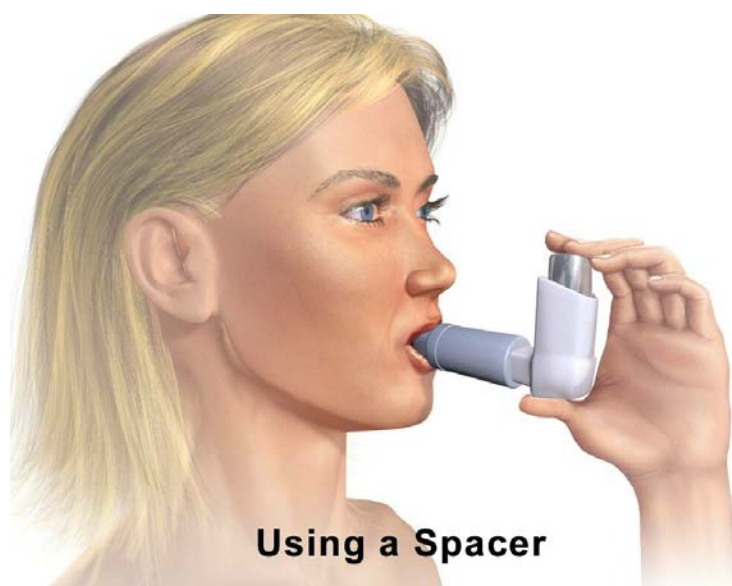
The given image shows a metered-dose inhaler (MDI). The most common side effect of this device is drug deposition in the oropharynx.

Most material inhaled using this device will be swallowed and absorbed, entering systemic circulation after undergoing the first-pass effect in the liver. Some drug will also be absorbed into the systemic circulation from the lungs.

The use of a large-volume spacer will reduce the amount of drug deposited in the oropharynx, thereby reducing the amount swallowed and absorbed from the GI tract, thus limiting the

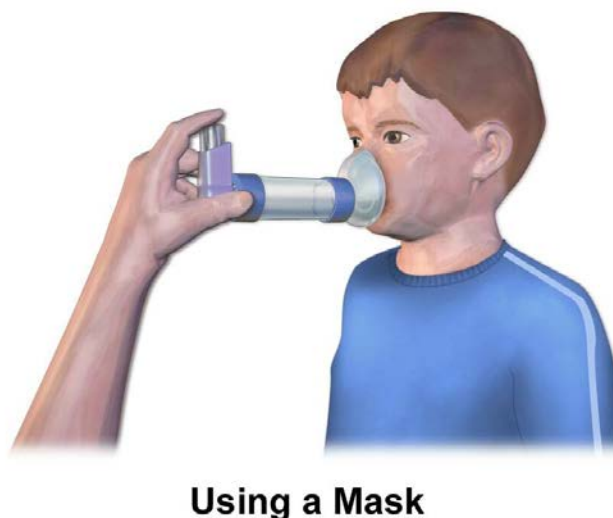
systemic effects of the drug.

The given image shows the use of spacer with MDI:



Spacers reduce the velocity and size of particles entering the upper airways. This reduces the amount of drug that impinges on the oropharynx and increases the proportion of the drug inhaled into the lower airways.

Spacer devices are useful in delivering inhaled drugs to small children, as young as 3 years of age, who are not able to use a pMDI (pressurized MDI), as seen in the image below:



Pressurized metered-dose inhalers: Drugs are propelled from a canister in the pMDI with the aid of a propellant. These devices are convenient, portable, and typically deliver 50–200 doses of the drug.

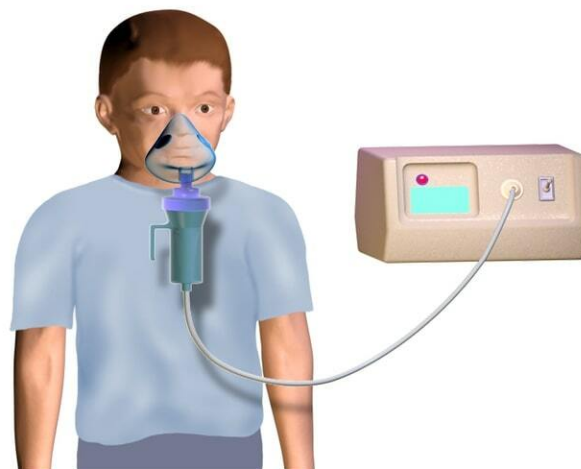
Dry powder inhalers: Drugs may also be delivered as a dry powder using devices that scatter a fine powder dispersed by air turbulence on inhalation. Children less than 7 years of age find it difficult to use a DPI. DPIs have been developed to deliver peptides and proteins, such as insulin,

systemically but have proved to be problematic because of the consistency of dosing.
The given image shows various types of dry powder inhalers:



Nebulizers: Two types of nebulizers are available. Jet nebulizers are driven by a stream of gas (air or oxygen), whereas ultrasonic nebulizers use a rapidly vibrating piezoelectric crystal and thus do not require a source of compressed gas.

The nebulized drug may be inspired during tidal breathing, and it is possible to deliver much higher doses of the drug as compared to pMDI. Nebulizers are therefore useful in treating acute exacerbations of asthma and COPD, for delivering drugs when airway obstruction is extreme (e.g., in severe COPD), for delivering inhaled drugs to infants and small children who cannot use the other inhalation devices, and for giving drugs such as antibiotics when relatively high doses must be delivered.



Using a Nebulizer

Solution to Question 24:

Gentamicin is an aminoglycoside antibiotic that requires therapeutic drug monitoring (TDM) as it has a narrow therapeutic index. It is nephrotoxic and ototoxic (predominantly vestibular effects).

The threshold for toxicity for gentamicin is the plasma concentration of 2 µg/mL, the maximum recommended for prolonged exposure.

Aminoglycoside antibiotics act by binding to the 30s ribosome and inhibiting protein synthesis. Their ability to cause misreading of DNA templates and the formation of aberrant proteins that get incorporated into bacterial cell membranes makes them bactericidal agents.

Common ototoxic drugs are:

- Antibiotics - aminoglycosides (gentamicin > tobramycin > amikacin > neomycin), erythromycin, tetracyclines.
- Chemotherapeutic agents - cisplatin, bleomycin, nitrogen mustards.
- High dose aspirin, acetaminophen, NSAID's
- Antimalarials - quinine, quinidine
- Phosphodiesterase-5 inhibitors - sildenafil

Solution to Question 25:

The given clinical scenario and the X-ray of the skull are suggestive of Langerhans Cell Histiocytosis. Vincristine is not used in the management of this condition.

Following are skeletal and radiological manifestations of Langerhan's cell histiocytosis:

Skull:

- Solitary or multiple punched-out lytic lesion without sclerotic rim.
- Double contour or beveled edge appearance (hole within a hole sign).
- Button sequestra representing residual bone.
- Geographic skull.

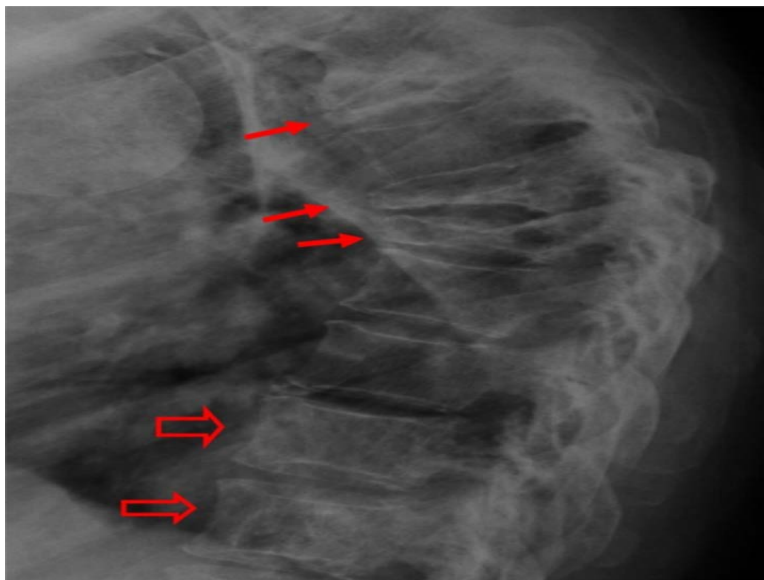
Mandible:

- Floating tooth: loss of lamina dura.

Spine:

- Vertebra Plana: Most common cause of vertebra plana in children; mostly seen in thoracic spine.

Skull in LCH, Vertebra plana:



Solution to Question 26:

Suicidal thoughts or behavior are a side effect of TCAs. Therefore, the attendants of the patient should be warned regarding this when a patient gets discharged after being treated with TCAs for a major depressive episode.

Tricyclic antidepressants (TCAs) are drugs that act by inhibiting the reuptake of noradrenaline and serotonin from the synaptic clefts.

The side effects of TCAs are due to blockade of other receptors:

- Muscarinic M receptors- constipation, dry mouth, arrhythmia
- Histamine H₁ receptors- sedation
- 5HT receptors - weight gain due to increased appetite

- Alpha₁ receptors - postural hypotension.

The various classes of drugs used in depression and anxiety are listed below:

Class of drugs	Examples
Selective serotonin reuptake inhibitors (SSRI)	Fluoxetine Fluvoxamine Paroxetine Sertraline Citalopram
Serotonin-norepinephrine reuptake inhibitors	Venlafaxine Desvenlafaxine Duloxetine Milnacipran
Tricyclic antidepressants	Amitriptyline Imipramine Nortriptyline Desipramine Amoxapine
Atypical antipsychotics	Aripiprazole Olanzapine Quetiapine Risperidone
Monoamine oxidase inhibitors (MAOI)	Isocarboxazid Selegiline Phenelzine
Atypical antidepressants	Bupropion Trazodone Mirtazapine Mianserin

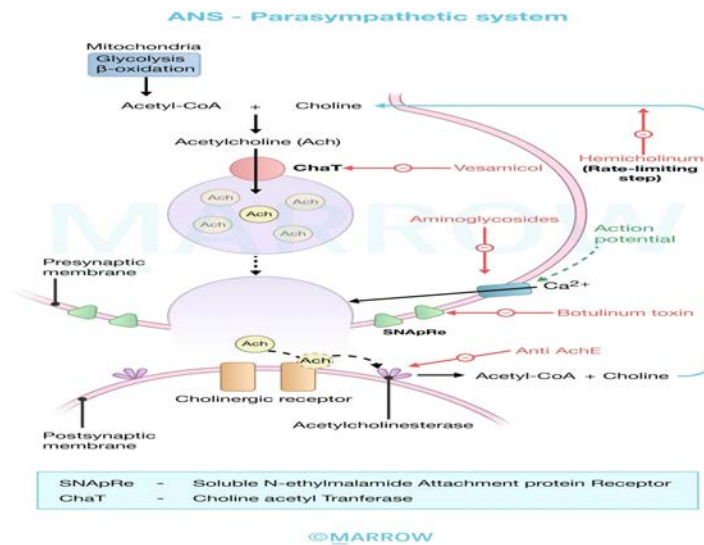
Solution to Question 27:

Lidocaine can exacerbate myasthenia gravis and should be avoided in patients with this disease. Also, digene is an antacid, that contains magnesium hydroxide as one of its active ingredients, which may exacerbate myasthenia.

Drugs that may aggravate myasthenia gravis are

- Antibiotics like aminoglycosides, quinolones, and macrolides
- Beta-blockers
- Botulinum toxin
- Local anaesthetics
- Magnesium
- Non-depolarising muscle relaxants
- Penicillamine
- Quinine derivatives

Aminoglycosides have to be avoided in a patient with myasthenia gravis. It blocks Ca²⁺ mediated exocytosis of acetylcholine from the presynaptic membrane in the neuromuscular junction. The decrease in ACh levels aggravates myasthenia gravis. Aminoglycosides can also have a direct inhibitory effect on post-synaptic acetylcholine receptors.



Solution to Question 28:

Ethosuximide and Valproate are effective for the treatment of absence seizures. However, between these 2 drugs, Ethosuximide would be a better first-line treatment choice for uncomplicated typical absence seizures followed by Valproate. Atypical absence seizures respond well to valproate.

Typical absence seizures are characterized by sudden, brief lapses of consciousness without loss of postural control. There is no postictal confusion. The following subtle motor signs may accompany the episodes

- Rapid blinking of the eyelids
- Chewing movements
- Small-amplitude clonic movements of the hands

Hyperventilation has a tendency to provoke the absence seizures.

NOTE: In a typical absence seizure, Ethosuximide is the first-line therapy for it and acts by blocking the calcium channel.

Mechanism of action of some common antiepileptics:

Mechanism of action	Antiepileptics
Sodium-channel blockers	Phenytoin Carbamazepine Oxcarbamazepine Eslicarbazepine Lamotrigine Topiramate Felbamate Valproate Rufinamide Levetiracetam
Calcium-channel blockers	Valproate Lamotrigine Ethosuximide

Mechanism of action	Antiepileptics
GABA _A receptor modulators	Benzodiazepines Phenobarbital Felbamate Primidone Topiramate Carbamazepine, Oxcarbamazepine Stiripentol Clobazam
GABA uptake inhibitors	Tiagabine Vigabatrin
NMDA receptor antagonist	Felbamate
AMPA/Kainate receptor antagonist	Phenobarbital Topiramate Perampanel
HCN channel activity enhancers	Lamotrigine
Inhibitors of brain carbonic anhydrase	Acetazolamide Zonisamide Topiramate
Potassium-channel openers	Retigabine Topiramate

Solution to Question 29:

A-2 B-1 C-3

Epinephrine (1:1000 solution) 0.5 ml I.M. is preferred in the treatment of anaphylactic shock.

In cardiac arrest, the 1:1000 adrenaline must be diluted to 1:10,000 for IV use.

Adrenaline in a concentration of 1: 50,000 to 1: 200,000 is added to local anesthetics for:

- Reducing the systemic absorption
- Prolonging the duration of action
- Providing a more bloodless field for surgery

1:1000 dilution can be used for intramuscular, subcutaneous and inhalational (nebulization) route of administration. For intravenous use, 1:1000 adrenaline must be further diluted to 1:10,000 and used cautiously.

IM injection is preferred over intravenous (IV) bolus because it is faster in many situations and safer (ie, lower risk of cardiovascular complications, such as severe hypertension and ventricular arrhythmias). There is a much greater risk of causing harmful side effects by inappropriate dosage or misdiagnosis of anaphylaxis when using IV adrenaline.

Epinephrine is also used in the treatment of hypotension, to induce local vasoconstriction to reduce the diffusion of local anaesthetics, to induce bronchodilation in asthma and COPD.

Note: . The recommended dose of epinephrine is 1mg = 10 mL of a 1:10,000 solution administered IV every 3 to 5 minutes during resuscitation as per the CPR guidelines. For nebulization, 1:1000 adrenaline is commonly used.

Solution to Question 30:

A-2, B-1, C-4, D-3

Vancomycin is Drug Of Choice for Methicillin-Resistant *Staphylococcus aureus* (MRSA) (should be used only if other drugs are resistant).

Metronidazole is the drug of choice for anaerobic infections caused by *Bacteroides*.

Uses of metronidazole include:

- Giardiasis
- Trichomoniasis
- Amoebiasis
- Bacterial vaginosis
- treatment of pseudomembranous colitis.

Meropenem is used in infections caused by ESBL producing bacteria like *Klebsiella*.

Carbapenems are not hydrolyzed by ESBL (extended-spectrum beta-lactamases). Therefore, carbapenems like imipenem, meropenem, ertapenem are resistant to hydrolysis by extended-spectrum beta-lactamases and are indicated in the treatment of infections caused by ESBL producing bacteria. ESBL is most prevalent among gram-negative organisms like *Klebsiella* and *E. coli*.

Penicillins like amoxicillin, ampicillin can be used in *H. influenzae* infections.

Solution to Question 31:

Among the given options, only dexamethasone is shown to reduce mortality in severe covid19 cases.

Patients with severe COVID-19 can develop a systemic inflammatory response that can lead to lung injury and multisystem organ dysfunction. It has been proposed that the potent anti-inflammatory effects of corticosteroids might prevent or mitigate these deleterious effects and help reduce mortality.

In severe cases, Dexamethasone is administered at doses of 6mg IV/PO once daily for 10 days or until hospital discharge. If dexamethasone is not available, then, equivalent doses of other corticosteroids, like prednisone, methylprednisone, or hydrocortisone may be used.

Other options:

- There is no evidence that favipiravir reduces mortality. It reduces duration of hospitalisation in mild to moderate disease.
- In a recent update, WHO recommended against the use of remdesivir in hospitalized patients, regardless of disease severity, as there is no evidence that it improves survival.

Solution to Question 32:

Statements A, B and C are true.

A. Morphine used in pulmonary edema

B. Pethidine used in biliary colic

C. Indomethacin is used in headache

Aspirin can also cause damage to the stomach and intestinal lining leading to the development of erosions or ulcers. Erosions may cause further complications like bleeding and perforations.

Note: Unlike Morphine, Pethidine/Mepiridine can be used for pain management in biliary colic as it has no effect on the tone of sphincter of Oddi.

Solution to Question 33:

Gout is a metabolic disease associated with hyperuricemia.

The drug of choice in an acute attack of gout is NSAIDs (Non-steroidal anti-inflammatory drugs) like indomethacin or the drug colchicine, but they are poorly tolerated and should not be used in elderly patients with chronic kidney disease.

Indomethacin is commonly used in the initial treatment of gout. In addition to inhibiting prostaglandin synthase, indomethacin and other NSAIDs also inhibit urate crystal phagocytosis

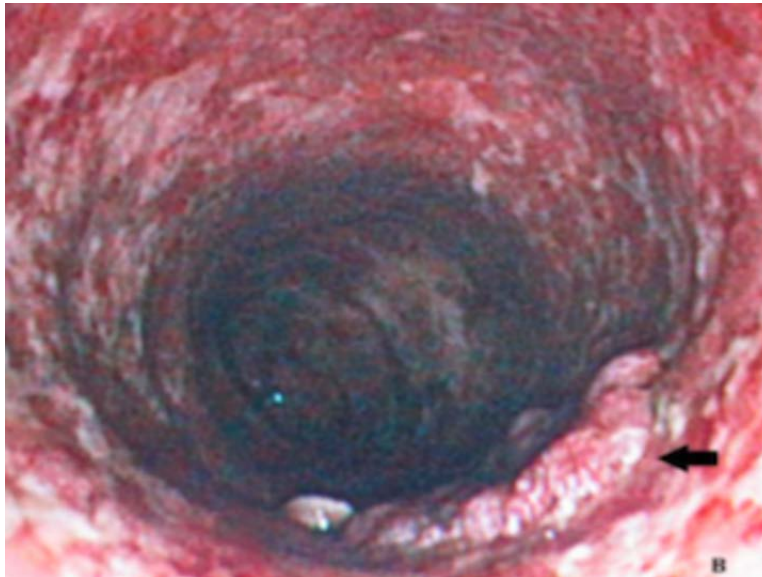
Intraarticular steroids like triamcinolone or methylprednisolone can also be given.

Aspirin is not used because it causes renal retention of uric acid at low doses. It is uricosuric only at higher doses that are poorly tolerated.

Pharmacology NEET 2018

Question 1:

A colonoscopy of an elderly patient revealed brownish pigmented lesions as shown below. He admitted to chronic laxative abuse. Which of the following laxatives is most likely to cause this condition?



- a) Bran
- b) Psyllium
- c) Senna
- d) Methylcellulose

Question 2:

A patient who is on treatment for angina, presents to you with erectile dysfunction. You decide against prescribing sildenafil because the patient is being treated with _____

- a) Calcium channel blockers
- b) Beta blockers
- c) Nitrates
- d) ACE inhibitors

Question 3:

An example of a bacteriostatic drug is _____

- a) Aminoglycoside
- b) Vancomycin
- c) Metronidazole
- d) Linezolid

Question 4:

The antiviral drug having dual antiviral activity against HIV and HBV is:

- a) Enfuvirtide
- b) Abacavir
- c) Emtricitabine
- d) Ritonavir

Question 5:

The replacement dose of hydrocortisone for chronic adrenal insufficiency is (in mg/day):

- a) 10
- b) 12
- c) 20
- d) 5

Question 6:

Which of the following is the drug of choice for scrub typhus?

- a) Doxycycline
- b) Azithromycin
- c) Chloramphenicol
- d) Ciprofloxacin

Question 7:

Which of the following drugs is associated with QT prolongation?

- a) Lignocaine
- b) Quinidine
- c) Magnesium sulfate
- d) All of the above

Question 8:

Sacubitril is a new drug approved for chronic heart failure. The mechanism of action of sacubitril is:

- a) Angiotensin II inhibitor
- b) ACE inhibitor
- c) Renin inhibitor
- d) Neutral endopeptidase inhibitor

Question 9:

Coagulation profile monitoring is needed in therapy with which of the following drugs?

- a) Fondaparinux
- b) Enoxaparin
- c) Dabigatran
- d) Lepirudin

Question 10:

Which of the following antihypertensives are not used in pregnancy?

- a) Alpha methyldopa
- b) Hydralazine
- c) Propranolol
- d) Labetalol

Question 11:

Which of the following atypical antipsychotics has the highest risk of galactorrhea?

- a) Iloperidone
- b) Risperidone
- c) Clozapine
- d) Aripiprazole

Question 12:

When the pH = pKa of a drug, which of the following statements is true?

- a) Concentration of the drug is 50% ionic and 50 % non-ionic
- b) Concentration of the drug is 90% ionic and 10% non-ionic
- c) Concentration of the drug is 75% ionic and 25 % non-ionic
- d) Concentration of the drug is 25% ionic and 75 % non-ionic

Question 13:

Corticosteroid indicated for the stimulation of fetal lung maturation in preterm labor is?

- a) Hydrocortisone
- b) Triamcinolone
- c) Methylprednisolone
- d) Betamethasone

Question 14:

What is the site of action of amphotericin B in fungi?

- a) Nucleic acid
- b) Ribosome
- c) Cell membrane
- d) Cytoplasm

Question 15:

Which of the following acts by causing both decreased bone resorption and increased bone formation?

- a) Strontium ranelate
- b) Ibandronate
- c) Teriparatide
- d) Calcitonin

Question 16:

Which is the drug used for refractory intractable rheumatic chorea?

- a) Probenecid
- b) Diazepam
- c) Haloperidol
- d) Sodium valproate

Question 17:

What is basiliximab?

- a) IL-1 receptor antagonist
- b) Anti-CD3 antibody
- c) IL-2 receptor antagonist
- d) TNF inhibitor

Question 18:

Which of the following conditions can be treated with pirenzepine?

- a) Asthma
- b) Peptic ulcer disease
- c) Glaucoma
- d) Hypertension

Question 19:

Which of the following antimicrobials is effective against an organism producing extended-spectrum beta-lactamases?

- a) Amoxicillin - clavulanic acid
- b) Penicillin
- c) Piperacillin - tazobactam
- d) Ceftriaxone

Question 20:

Which of the following inhibits the synthesis of glucocorticoid?

- a) Mifepristone
- b) Flutamide
- c) Finasteride
- d) Metyrapone

Question 21:

Isosorbide dinitrate is indicated for congestive heart failure in combination with which of the following vasodilators?

- a) Minoxidil
- b) Hydralazine
- c) Nimodipine
- d) Nitrendipine

Question 22:

What is the mechanism of action of colchicine?

- a) Inhibits synthesis of uric acid
- b) Promotes excretion of uric acid
- c) Inhibits xanthine oxidase enzyme
- d) Binds to tubulin and inhibits granulocyte migration

Question 23:

Which of the following statements is true about prasugrel?

- a) It acts as an P2 Y₁₂ agonist
- b) It is 10 times less potent than clopidogrel
- c) It is a pro drug
- d) Omeprazole interferes in its activation to an active metabolite

Question 24:

Apixaban belongs to which of the following class of drugs?

- a) Antifibrinolytic
- b) Factor Xa inhibitor
- c) Oral direct thrombin inhibitor
- d) Parenteral direct thrombin inhibitor

Answer Key

Question No.	Correct Option
1	c
2	c
3	d
4	c
5	c
6	a
7	b
8	d
9	d
10	c
11	b
12	a
13	d
14	c
15	a
16	d

17	c
18	b
19	c
20	d
21	b
22	d
23	c
24	b

Detailed Explanations

Solution to Question 1:

The above colonoscopy image depicts a brownish moiré pattern that is indicative of melanosis coli. Chronic use of anthraquinone laxatives like senna and cascara sagrada can cause melanosis coli.

Bisacodyl is also implicated in causing melanosis coli.

Solution to Question 2:

The patient would be on nitrates for angina and thus should not be prescribed sildenafil as sildenafil causes dangerous potentiation of nitrate action resulting in severe hypotension.

The combination of the two drugs has a synergistic action on the intracellular cGMP levels that can result in uninhibited vasodilation and a precipitous fall in blood pressure. Hence, the use of sildenafil is a contraindication when on nitrates.

Solution to Question 3:

Linezolid is an example of a bacteriostatic drug.

Primarily bacteriostatic drugs	Primarily bactericidal drugs
Sulfonamides Erythromycin Tetracyclines Clindamycin Chloramphenicol Linezolid Ethambutol	Penicillins, Cephalosporins Aminoglycosides Vancomycin Polypeptides Ciprofloxacin Rifampin Metronidazole Isoniazid Cotrimoxazole Pyrazinamide

Solution to Question 4:

Antiviral drug having dual antiviral activity against HIV and HBV is emtricitabine.

It is a nucleoside reverse transcriptase inhibitor similar to lamivudine. It has activity against HIV-1, HIV-2, and HBV.

Solution to Question 5:

The replacement dose of hydrocortisone for chronic adrenal insufficiency is 20 mg/day.

The treatment of chronic adrenal insufficiency (not in crisis) requires glucocorticoid replacement at a dose that replaces the physiologic daily cortisol production. This is achieved by administering an oral replacement dose of 15-25mg Hydrocortisone.

A midpoint dose of 20mg is given in 2-3 divided doses: 10mg + 5mg + 5mg.

Half the daily dose should be administered to all patients in the morning. This is to follow the normal circadian rhythm.

Solution to Question 6:

Doxycycline is the drug of choice for scrub typhus.

Azithromycin, chloramphenicol, or rifampin can be administered as alternatives in patients allergic to doxycycline and in pregnancy.

Solution to Question 7:

QT prolongation is seen with quinidine.

It causes torsades de pointes, a life-threatening form of polymorphic ventricular tachycardia.



Solution to Question 8:

Sacubitril is a neutral endopeptidase (neprilysin) inhibitor.

Sacubitril is a prodrug that is activated to sacubitrilat. It can inhibit the enzyme neprilysin (neutral endopeptidase), which is responsible for the degradation of atrial and brain natriuretic peptides (reduction of blood volume). Therefore, it increases the levels of atrial and brain natriuretic peptides and lowers blood pressure.

In 2015, the FDA approved the use of sacubitril in combination with valsartan, at a ratio of 1:1, for the treatment of chronic heart failure.

Solution to Question 9:

Coagulation profile monitoring is needed for lepirudin.

Lepirudin is a direct irreversible thrombin inhibitor and its most adverse reaction is bleeding. The risk of bleeding can be reduced by implementing an appropriate monitoring and dose adjustment strategy. Thus, coagulation profile monitoring is required for lepirudin.

However, enoxaparin, fondaparinux, and dabigatran do not bind to endothelium or many plasma proteins and thus, the response to these drugs at the usual doses is consistent from patient to patient. Hence, monitoring is not required.

Solution to Question 10:

Propranolol is a non-selective beta-blocker and is not safe for use during pregnancy.

Antihypertensives to be avoided in pregnancy:

- ACE inhibitors, ARBS
- Diuretics
- Nonselective beta-blockers
- Nitroprusside

Antihypertensives found to be safe in pregnancy:

- Hydralazine
- Methyldopa
- Dihydropyridine CCBs: should be stopped before labor as they weaken the uterine contractions
- Cardioselective beta-blockers and those with intrinsic sympathomimetic activity: may be used as a last choice

Solution to Question 11:

Among atypical antipsychotics, risperidone and paliperidone have the highest risk of galactorrhea. The adverse effect occurs due to their higher D2 blocking potency.

Solution to Question 12:

When the pH = pKa of a drug, the concentration of the drug is 50% ionic and 50% non-ionic.

pKa is defined as the pH at which a drug exists as 50% ionized (hydrophilic) and 50% unionized (lipophilic).

For e.g., phenobarbitone is an acidic drug with a pKa of 7.2. At a pH of 7.2, 50% will be unionized. In a more basic plasma (pH=7.4), a lesser fraction (around 40%) will be unionized.

Small changes in pH near pKa of a drug may cause large variations in the degree of ionization.

Solution to Question 13:

Betamethasone is indicated for the stimulation of fetal lung maturation in preterm labor.

As compared to cortisol, betamethasone binds to maternal proteins less and has less placental metabolism. This allows increased transfer to the fetus. Additionally, infants exposed to betamethasone are less likely to require postnatal resuscitation. 2 doses of 12 mg each are administered 18-24 hours apart when the delivery is anticipated before 34 weeks of gestation.

Dexamethasone, another long-acting glucocorticoid is also indicated. It has benefits such as less risk of intraventricular hemorrhage and a shorter length of stay in the NICU.

Solution to Question 14:

The site of action of amphotericin B in fungi is the cell membrane.

Amphotericin B binds to the ergosterol in the cell membrane of fungi and forms pores or channels. They increase the permeability of the membrane to ions and cause leakage of cytosolic molecules. This leads to a loss of integrity of the cell membrane and causes fungicidal effects.

Recent evidence suggests that the amphotericin B sequesters ergosterol from lipid layers, much like a sponge, resulting in fungal cell death.

Solution to Question 15:

Strontium ranelate acts by both decreased bone resorption and increased bone formation.

It is generally reserved for severe osteoporosis. Due to an increased incidence of myocardial infarction and other cardiovascular events, the use is restricted in all the approved regions.

Option B- Ibandronate which is a bisphosphonate, decreases bone resorption only.

Option C- The pharmacologic activity of teriparatide, which is similar to the physiologic activity of PTH, includes stimulating osteoblast function, increasing gastrointestinal calcium absorption, and increasing renal tubular reabsorption of calcium.

Option D- Calcitonin also decreases bone resorption.

Solution to Question 16:

The drug used for refractory intractable rheumatic chorea is sodium valproate, as it doesn't have extrapyramidal side effects.

Sydenham's chorea is characterized by involuntary, purposeless, jerky movements which mainly affect the head and upper limbs. In the tongue, it is visible as the characteristic darting movements.

Medications used to treat Sydenham chorea are:

- Mild cases - Providing a calm environment.
- Severe cases:
- Dopamine antagonists - haloperidol, pimozide
- Antiepileptic medications - sodium valproate, carbamazepine.
- Severe, refractory cases - Steroids (thought to be immune-mediated)
- IV immunoglobulins, plasmapheresis - limited studies, decreased efficacy (not currently recommended)

Solution to Question 17:

Basiliximab exhibits a high affinity towards IL-2 receptor and inhibits it. It is an anti- CD-25 monoclonal antibody.

It has a short plasma half-life of 1 week. It is used to prevent transplant rejection reactions. Its adverse effects are anaphylactic reactions & opportunistic infections.

Solution to Question 18:

Pirenzepine is used for treating gastric ulcers. It is a selective M1 anticholinergic drug. It decreases gastric acid secretions at doses that have little effect on salivation or heart rate.

Solution to Question 19:

Piperacillin-tazobactam is an antimicrobial that is effective against organisms producing extended-spectrum beta-lactamases.

Extended-spectrum beta-lactamase (ESBL) enzyme produced by bacteria causes resistance to most of the penicillins, cephalosporins (third and fourth generation), and monobactams. ESBL is most prevalent among gram-negative organisms like Klebsiella and E. coli.

ESBL can be inhibited by combining beta-lactamase inhibitors like clavulanic acid or tazobactam. Carbapenems and quinolones have inherent activity against ESBL.

Note: As piperacillin has greater activity against Klebsiella and other gram-negative organisms than amoxicillin, piperacillin-tazobactam can be considered as a better option in this question.

Solution to Question 20:

Metirapone inhibits the synthesis of glucocorticoids by inhibiting steroid 11 β -hydroxylase reversibly.

It is used to test the hypothalamic-pituitary feedback pathway in the diagnosis of Cushing syndrome.

Solution to Question 21:

Isosorbide dinitrate (ISDN) is indicated in combination with hydralazine for CHF.

Hydralazine is a direct vasodilator and its mechanism of action is not known. It also reduces nitrate tolerance.

The combination of ISDN and hydralazine showed mortality benefits in a specific group of population (African-Americans).

Solution to Question 22:

Colchicine acts by binding to tubulin and inhibits granulocyte migration.

Colchicine produces its anti-inflammatory effects by binding to the intracellular protein tubulin. This prevents the polymerization of tubulin into microtubules which is necessary for leukocyte migration and phagocytosis. Inhibition of microtubule formation prevents granulocyte migration and also arrests cell division at G₁ phase.

Solution to Question 23:

Prasugrel is a prodrug.

It acts as an irreversible ADP (P₂ Y₁₂) antagonist and is converted into its active metabolite by CYP3A4 and CYP2B6. Since omeprazole is an inhibitor of CYP2C19, it does not interfere with its activation.

It is 10-times more potent than clopidogrel and has a more rapid onset of action because of better absorption and more predictable metabolic activation.

Solution to Question 24:

Apixaban, a novel oral anticoagulant (NOAC), is a factor Xa inhibitor that is used in the treatment of VTE (venous thrombo embolism)/DVT and in its prophylaxis after major orthopedic surgeries.

Direct factor Xa inhibitors:

- Rivaroxaban
- Apixaban
- Edoxaban
- Betrixaban

Parenteral direct thrombin inhibitors:

- Bivalirudin
- Argatroban
- Desirudin

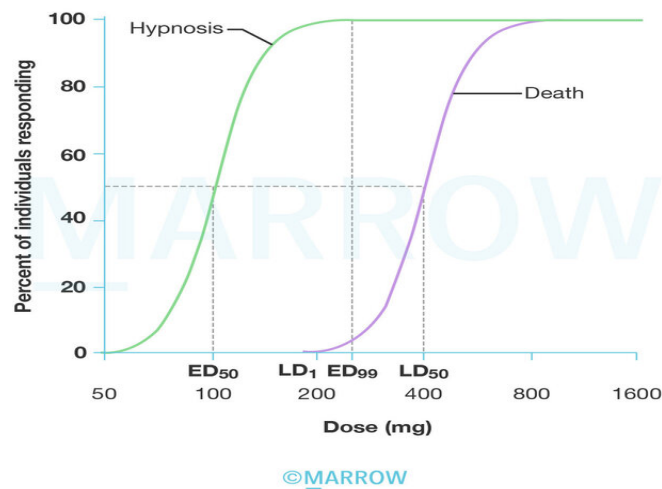
Oral direct thrombin inhibitor—Dabigatran is the only oral direct thrombin inhibitor available for clinical use.

The FDA recently approved the drug Apixaban for a new indication. The drug is now used to reduce the risk of stroke and systemic embolism in patients with nonvalvular atrial fibrillation.

Pharmacology NEET 2019

Question 1:

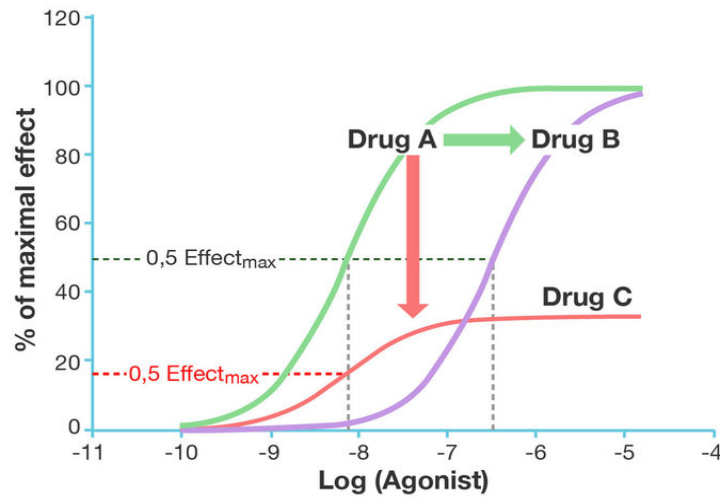
Calculate the therapeutic index of a drug whose dose-response curves are plotted as shown below:



- a) 1
- b) 2
- c) 4
- d) 8

Question 2:

Which of the following statements is true regarding the dose-response curve shown in the image below?



- a) C is competitive antagonist
- b) C is more potent than A
- c) B is more potent than A
- d) B is competitive antagonist

Question 3:

Which of the following antiglaucoma medications is unsafe in infants?

- a) Timolol
- b) Brimonidine
- c) Latanoprost
- d) Dorzolamide

Question 4:

Apparent volume of distribution of a drug is very high (6L/kg). Which of the following is true regarding the distribution of that drug?

- a) Highly bound to plasma proteins
- b) Confined to vascular compartment
- c) Sequestered in body tissues
- d) Both A and B

Question 5:

Which of the following phases of a clinical trial is also referred to as efficacy trial?

- a) Phase 1
- b) Phase 2
- c) Phase 3
- d) Phase 4

Question 6:

Which of the following is an oral factor Xa inhibitor?

- a) Bivalirudin
- b) Dabigatran
- c) Rivaroxaban
- d) Enoxaparin

Question 7:

What is the drug of choice for diphtheria carrier state?

- a) Amoxicillin
- b) Erythromycin
- c) Rifampicin
- d) Tetracycline

Question 8:

What is the mechanism of action of oseltamivir and zanamivir?

- a) DNA polymerase inhibition
- b) Protein synthesis inhibition
- c) Nucleotide analogue
- d) Neuraminidase inhibition

Question 9:

Which of the following is a synthetic PGF₂ alpha derivative used as an abortifacient?

- a) Dinoprost
- b) Dinoprostone
- c) Misoprostol
- d) Carboprost

Question 10:

Which of the following drugs causes activation of K⁺-ATP channels?

- a) Nicorandil
- b) Molsidomine
- c) Isosorbide dinitrate
- d) Nitroglycerin

Question 11:

Most effective antiemetic for chemotherapy-induced vomiting is_____.

- a) Domperidone
- b) Tegaserod
- c) Granisetron
- d) Doxylamine

Question 12:

True statement about teduglutide is :

- a) It is a GLP- 2 analogue
- b) It's half life is 6 - 8 hours
- c) It is administered orally
- d) It is recommended for patients with colorectal cancer

Question 13:

Cisatracurium is preferred over atracurium because?

- a) It has rapid onset of action
- b) It has less depressant action on heart
- c) It has short duration of action
- d) It causes less release of histamine

Question 14:

Which one of the following drugs inhibits neutrophil recruitment?

- a) Montelukast
- b) Sodium cromolyn
- c) Colchicine
- d) Febuxostat

Question 15:

Which of the following fluoroquinolones has maximum bioavailability?

- a) Ciprofloxacin
- b) Levofloxacin
- c) Gatifloxacin
- d) Norfloxacin

Question 16:

What is the mechanism of triazoles?

- a) Interferes with DNA and RNA synthesis
- b) Inhibition of ergosterol synthesis
- c) Forms pores in fungal membranes
- d) Inhibits epoxidation of squalene in fungi

Question 17:

A dermatologist wants to prescribe acitretin to a patient with psoriasis. Which of the following statements is false regarding the drug?

- a) Contraindicated in persons with hyperlipidemia
- b) It can cause increased sensitivity to sunlight
- c) Female patients should avoid getting pregnant for 6 months after therapy
- d) Acitretin has a half life of about 50 hours

Question 18:

Cilastatin is administered to prolong the action of which of the following drugs?

- a) Meropenem
- b) Imipenem
- c) Cefazolin
- d) Piperacillin

Question 19:

What is the drug used to differentiate myasthenia gravis from cholinergic crisis?

- a) Obidoxime
- b) Edrophonium
- c) Donepezil
- d) Atropine

Question 20:

Which of the following is true about pegylated-filgrastim?

- a) Duration of action is shorter than normal filgrastim
- b) Added advantage is that it can be taken orally
- c) It should not be administered in sickle cell patients
- d) It is not effective in the treatment of severe neutropenia

Question 21:

The best approach in the treatment of chronic SIADH is _____

- a) Tolvaptan

- b) Reducing fluid intake to <500 ml per day
- c) Demeclocycline
- d) Fludrocortisone

Question 22:

Intravenous mannitol is used in ____?

- a) Pulmonary edema
- b) Congestive cardiac failure
- c) Acute kidney injury with anuria
- d) Glaucoma

Question 23:

Which of the following carbapenems has increased risk of causing seizures?

- a) Meropenem
- b) Imipenem
- c) Etrapanem
- d) Doripenem

Question 24:

Which of the following helps in maintaining the patency of ductus arteriosus?

- a) PGF₂α
- b) PGE₁
- c) PGH₂
- d) PGI₂

Answer Key

Question No.	Correct Option
1	c

2	d
3	b
4	c
5	b
6	c
7	b
8	d
9	d
10	a
11	c
12	a
13	d
14	c
15	b
16	b
17	c
18	b
19	b
20	c
21	a
22	d
23	b
24	b

Detailed Explanations

Solution to Question 1:

The therapeutic index is 4.

In the above graph,

Median lethal dose (LD₅₀) = 400

Median effective dose (ED₅₀) = 100

Therapeutic index (TI) = LD₅₀/ED₅₀.

Hence,

Therapeutic index (TI) = 400/100 = 4

Clinically, the therapeutic range bound by the dose that produces the minimal therapeutic effect and the dose that produces the maximal acceptable adverse effect is more relevant.

Solution to Question 2:

In the above image, drug B is a competitive antagonist.

In competitive antagonism, the DRC is shifted to the right whereas in non-competitive antagonism the DRC is flattened.

Thus in the above image Drug B is the Competitive antagonist while drug C is the non-competitive antagonist.

Potency:

- It is defined as the amount of drug needed to produce a given response.
- A highly potent drug evokes a given response at low concentrations and vice-versa.
- The more the curve is on the left in a dose-response curve (DRC), the greater is its potency.
- Thus, in the above image, the potency is drug A > drug C > drug B.

Efficacy:

- It is defined as the maximum effect produced by the drug.
- It expresses the effect produced by the receptors when bound by a certain drug.
- In the log dose-response curve, the perpendicular height represents the drug efficacy.
- Thus, in the above image, the efficacy is drug A = drug B > drug C.

Solution to Question 3:

The use of brimonidine is unsafe in infants. It is an α -2 agonist that can cross the blood-brain barrier. This may result in CNS depression and apnoea in neonates. Therefore it is contraindicated in children less than 2 years.

α 2 agonists, through the activation of G-protein coupled receptor activation, inhibit the activity of adenylate cyclase. This reduces cAMP and hence reduces aqueous humor production by ciliary action.

Solution to Question 4:

In the present case, the volume of distribution of the drug is 6L/kg (420 L in a normal adult with reference weight 70kg) which is very high when compared to the total body water of 42L. Therefore, it can be implied that the drug is having high sequestration in tissues.

The apparent volume of distribution (Vd) is an imaginary volume throughout which the drug is distributed if the plasma concentration remains constant throughout.

$V_d = \text{Dose administered IV} / \text{plasma concentration}$

The difference in the volume of distribution based on drug type:

- Low V_d
- Restricted to the intravascular compartment
- Seen in drugs with high molecular weight (e.g., heparin) or high protein binding (e.g., warfarin - 9.8L)
- Medium V_d
- Lipid insoluble drugs (e.g., gentamicin - 20L) do not enter the cells and are distributed in the extracellular fluid volume
- Small water-soluble molecules (e.g., ethanol - 42L) are distributed in the total body water
- High V_d
- Sequestered and accumulated in the tissues
- Small lipophilic molecules (e.g., chloroquine - 13000L)

Solution to Question 5:

Efficacy and appropriate dose range is the main aim of phase 2 of clinical trials. It is evaluated on a small group of patients with the target disease.

Phase II of clinical trials involve a small group of patients (50-500). Most of the new drugs will be administered for the first time in patients with the target disease. The study is designed to obtain proof of concept (potential of the drug to treat the disease) and dose-ranging study (provides input for choosing the most appropriate dose for phase 3 studies).

Phase 3 studies are also known as pivotal studies as they establish treatment benefits.

Solution to Question 6:

Rivaroxaban is an oral direct factor Xa inhibitor. They represent a new class of oral anticoagulant drugs that do not require monitoring.

This group of drugs inhibit factor Xa in the final common pathway of clotting [conversion of Prothrombin (Factor II) to thrombin (Factor IIa)]. They are given as fixed doses and have a rapid onset of action.

Oral direct factor Xa inhibitors are indicated in:

- Prevention of embolic stroke in patients with atrial fibrillation (without valvular heart disease)
- Venous thromboembolic disease (VTE)
- Postoperative thromboprophylaxis in patients undergoing hip and knee arthroplasty

Solution to Question 7:

The drug of choice for diphtheria carrier state is erythromycin.

Erythromycin for 7 days is very effective for eradicating the diphtheria carrier state or acute infections.

Solution to Question 8:

Oseltamivir and zanamivir are neuraminidase inhibitors.

They competitively and reversibly interact with the enzyme sites to inhibit viral neuraminidase activity. This interferes with the release of the progeny from infected host cells, halting the spread of infection.

Early administration hastens the resolution of symptoms if given within 48 h of infection.

Oseltamivir is given at a dose of 75 mg, BD for 5 days for the treatment of Influenza A and B.

For prophylaxis, it is given at a dose of 75mg, OD and continued for the duration of exposure.

Solution to Question 9:

Carboprost is a 15-methyl synthetic derivative of PGF₂ alpha. It is used as an abortifacient and in refractory PPH.

Options A: Dinoprost is a natural PGF₂ alpha preparation.

Option B: Dinoprostone is natural PGE₂ preparation.

Option C: Misoprostol is a synthetic PGE₁ analog. Gemeprost is also a synthetic PGE₁ analog.

Solution to Question 10:

Nicorandil causes activation of K⁺-ATP channels. This leads to the hyperpolarization of the smooth muscle cells leading to vasodilatation.

It also has nitrate like cGMP dependent action. Nicorandil dilates both arteries and veins leading to a decrease in afterload and preload of the heart.

Molsidomine, isosorbide dinitrate, and nitroglycerin are nitrates. They do not act on K-ATP channels.

Solution to Question 11:

Among the given options, the most effective antiemetic for chemotherapy-induced vomiting is granisetron.

Granisetron is a 5HT₃ receptor antagonist. They are the most effective drugs for the treatment of chemotherapy-induced and post-operative nausea and vomiting.

Palonosetron is a second-generation 5HT₃ antagonist that has the highest affinity for 5HT₃ receptors and has the maximum potency. It is also the longest acting that makes it effective in delayed emesis associated with chemotherapy. Palonosetron has minimal/no propensity to cause QT interval prolongation.

Solution to Question 12:

Teduglutide is a GLP-2 analogue.

It has been recently approved for the treatment of short bowel syndrome. Teduglutide binds to enteric neurons and endocrine cells, stimulating the release of a number of hormones such as insulin-like growth factor, this stimulates mucosal epithelial growth and enhances fluid absorption.

The drug has a t_{1/2} of 1–2 hours and is administered subcutaneously once daily. It can potentially cause bowel cancer, therefore it is not recommended for patients with active malignancies.

Note: GLP-1 analogs are used for the treatment of Type II diabetes mellitus.

Solution to Question 13:

Cisatracurium is preferred over atracurium because it causes less release of histamine.

Other options:

Option A- Cisatracurium has a slower onset of action

Option B and C- Both atracurium and cisatracurium have similar duration of action and cardiac action.

Solution to Question 14:

Colchicine inhibits neutrophil (granulocyte) recruitment.

Colchicine acts by preventing the polymerisation of microtubules by binding to tubulin. This reduces neutrophil recruitment to inflamed tissue and also decreases the secretion of chemotactic factors by activated neutrophils, thus reducing the inflammatory response overall.

Colchicine is used in gout and Mediterranean fever.

Solution to Question 15:

Levofloxacin and ofloxacin have the maximum bioavailability among fluoroquinolones. Hence, intravenous and oral doses are the same for these drugs.

Solution to Question 16:

Triazoles act by inhibition of ergosterol synthesis.

The azole group inhibits the fungal cytochrome P450 enzyme called 14- α -sterol demethylase and results in an impairment of conversion of lanosterol to ergosterol. This results in the depletion of membrane ergosterol.

Azole antifungals are of two types:

- Imidazoles - clotrimazole, ketoconazole, luliconazole
- Triazoles - itraconazole, fluconazole, voriconazole, posaconazole and isavuconazole

Solution to Question 17:

Female patients should avoid conception for 3 years after discontinuing acitretin treatment to prevent retinoid-induced embryopathy.

Acitretin is approved for use in the cutaneous manifestations of psoriasis.

Absolute contraindications:

- Pregnant women
- Women who are planning to get pregnant
- Breastfeeding

Relative contraindications:

- Leukopenia
- Alcoholism
- Hyperlipidemia
- Hypercholesterolemia
- Hypothyroidism
- Significant hepatic or renal disease

Note: Female patients should avoid conception for 1 month after discontinuation of isotretinoin.

Solution to Question 18:

Cilastatin is administered to prolong the action of imipenem.

Imipenem is rapidly hydrolyzed by the enzyme renal dehydropeptidase. It is administered with cilastatin, a reversible inhibitor of dehydropeptidase to inhibit the breakdown and prolong the action of imipenem.

Meropenem is not hydrolyzed by renal dehydropeptidase. Cilastatin administration is not necessary with meropenem.

Solution to Question 19:

The drug used to differentiate myasthenia gravis from a cholinergic crisis is edrophonium.

Muscle weakness is a common presentation in patients with myasthenia gravis and cholinergic crisis (excessive dose of anti-ChE drugs). Intravenous injection of edrophonium improves muscular weakness in a patient with myasthenia gravis and not in a case of cholinergic crisis.

The test used for this differentiation is known as the edrophonium test.

Initially, 2 mg is administered; if improvements of symptoms do not occur within 45 seconds, an additional 8 mg can be administered. Improvement of strength is a positive test for myasthenia gravis.

Solution to Question 20:

Patients with sickle cell anemia should not receive pegylated filgrastim as it can trigger a sickle cell crisis.

Filgrastim is a recombinant human G-CSF (granulocyte colony-stimulating factor). The pegylated forms of filgrastim, pegfilgrastim, and lipegfilgrastim, are longer-acting forms. It is administered subcutaneously or by slow intravenous infusion only.

Indications of recombinant G-CSF:

- Severe neutropenia after autologous hematopoietic stem cell transplantation
- Severe neutropenia after high dose cancer chemotherapy
- Severe congenital neutropenias
- Neutropenia of patients with AIDS receiving zidovudine

Solution to Question 21:

The best approach in the treatment of chronic SIADH is the oral administration of a vaptan like tolvaptan, which is a selective V₂ receptor antagonist. This increases urinary water excretion by blocking the antidiuretic effect of AVP.

Other treatment options for chronic SIADH include:

- Fluid intake < 500 ml/day

- Demeclocycline
- Fludrocortisone

Increased thirst is a frequently observed characteristic of hyponatremia in SIADH. Fluid restriction is usually a burden to the patient. It is unreliable, impractical, and slow to work. As thirst is particularly difficult to suppress, water restriction in SIADH frequently leads to non-compliance in the long run. Vaptans would be especially suited for the treatment of these patients. Serum sodium concentrations are also corrected by vaptans and hence, vaptans substantially improve the quality of life in these patients.

Solution to Question 22:

Intravenous mannitol is used for the treatment of acute attacks of glaucoma.

Uses of mannitol are:

- To reduce cerebral edema and intracranial pressure
- Enhancement of urinary excretion of salicylates, barbiturates, bromides, and lithium
- Diagnosis of bronchial hyperreactivity (inhalation)
- Antihemolytic urological irrigation during transurethral procedures
- Treatment of acute attacks of glaucoma.

Solution to Question 23:

Imipenem is the carbapenem that is associated with an increased risk of causing seizures, especially when high doses are given to patients with CNS lesions.

Excretion of imipenem is mainly renal and dosage modification is required to avoid seizures in renal failure patients.

Solution to Question 24:

Prostaglandin E1 (PGE1) maintains the patency of ductus arteriosus (PDA). This is used in term neonates with underlying cyanotic congenital heart disease awaiting surgical intervention as they have duct-dependant circulation.

Indomethacin and Ibuprofen are the NSAID indicated for closing the ductus arteriosus in premature infants and neonates.

Pharmacology NEET 2020

Question 1:

Which of the following is an inhibitor of DNA synthesis?

- a) 6-mercaptopurine
- b) Mitomycin
- c) Actinomycin
- d) Asparaginase

Question 2:

A patient on lithium therapy developed hypertension. He was started on thiazides for hypertension. After few days, he developed coarse tremors and other symptoms suggestive of lithium toxicity. Explain the likely mechanism of this interaction.

- a) Thiazide inhibits the metabolism of lithium
- b) Thiazides act as an add on drug to lithium
- c) Thiazides increase the tubular reabsorption of lithium
- d) Thiazides cause water loss thereby increase lithium levels

Question 3:

Which of the following should not be prescribed with theophylline?

- a) Erythromycin
- b) Cefotaxime
- c) Cotrimoxazole
- d) Amoxicillin

Question 4:

Which of the following drugs gets deposited in cornea?

- a) Leflunamide
- b) Chloroquine

- c) Methotrexate
- d) Sulfasalazine

Question 5:

A patient was prescribed an antipsychotic drug following which he developed parkinsonism. Which of the following drugs can be given?

- a) Trihexiphenidyl
- b) Entacapone
- c) Rasagiline
- d) Pramipexole

Question 6:

Which of the following antiarrhythmic drugs is contraindicated in a patient with interstitial lung disease?

- a) Amiodarone
- b) Sotalol
- c) Quinidine
- d) Lignocaine

Question 7:

Which of the following drugs acts by inhibiting the late sodium current in the heart?

- a) Ivabradine
- b) Ranolazine
- c) Trimetazidine
- d) Fasudil

Question 8:

A patient with diabetes and COPD developed post-operative urinary retention. Which of the following drugs can be used for short-term treatment of the condition?

- a) Tamsulosin

- b) Bethanechol
- c) Terazosin
- d) Methacoline

Question 9:

Which of the following drugs is used for the treatment of motion sickness?

- a) Cetrizine
- b) Promethazine
- c) Misolastin
- d) Loratidine

Question 10:

Which of the following is not a prokinetic?

- a) 5HT₄ agonist
- b) Dopamine antagonist
- c) Motilin
- d) Diphenylmethane

Question 11:

Which of the following is used in the first-line management of postmenopausal osteoporosis?

- a) Estrogen
- b) Bisphosphonates
- c) Raloxifene
- d) Combined oral contraceptives

Question 12:

What is the mechanism of resistance by lactamases to β -lactam antibiotics?

- a) Alteration of penicillin binding protein
- b) Degradation of antibiotics

- c) Drug efflux
- d) Alteration in 50s ribosome structure

Question 13:

A female patient is on first-line pharmacological therapy for bipolar disorder for 6 months. She is fasting for a few days due to religious reasons and presented with seizures, coarse tremors, confusion, and weakness of limbs. Which of the following investigations should be done to help diagnose her condition?

- a) Serum potassium level
- b) Serum lithium level
- c) ECG
- d) MRI

Question 14:

Pegloticase is used in management of:

- a) Chronic gout
- b) Rheumatoid arthritis
- c) Psoriatic arthritis
- d) Calcium pyrophosphate dihydrate crystal deposition (CPPD)

Question 15:

Which of the following is an inhibitor of RNA synthesis?

- a) 5-Fluorouracil
- b) Rifampicin
- c) Nitrofurantoin
- d) Novobiocin

Question 16:

Which of the following fluoroquinolone is not given in liver disease?

- a) Ofloxacin

- b) Levofloxacin
- c) Pefloxacin
- d) Lomefloxacin

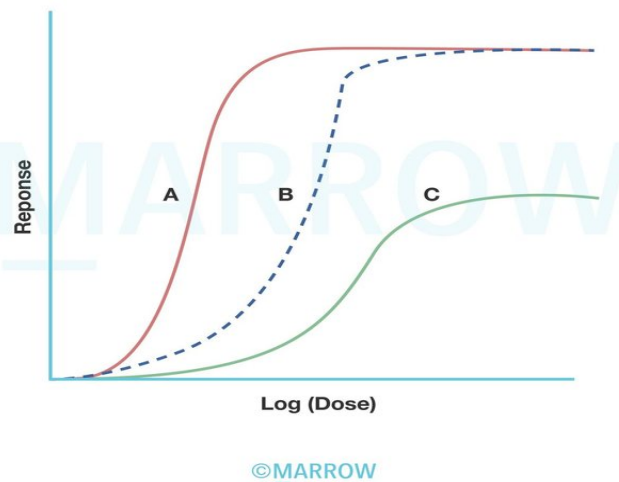
Question 17:

Theophylline causes diuresis by:

- a) Adenosine antagonism
- b) PDE 4 inhibition
- c) Beta 2 agonism
- d) PDE 3 inhibition

Question 18:

The graph below shows the relationship between the log(dose) and the response of three drugs, A, B, and C. Which of these drugs has the highest potency?



- a) A
- b) B
- c) Both A and B
- d) C

Question 19:

A patient presented with biliary colic. The intern has administered an analgesic, following which the pain got aggravated. Which of the following could be responsible for the aggravation of the patient's symptoms?

- a) Morphine
- b) Diclofenac
- c) Nefopam
- d) Etoricoxib

Question 20:

A patient with gout develops hypertension. The uric acid level was found to be elevated. Which of the following is not administered?

- a) Hydrochlorothiazide
- b) Atenolol
- c) Enalapril
- d) Prazosin

Question 21:

Voriconazole is not effective against which of the following?

- a) Candida albicans
- b) Mucormycosis
- c) Candida tropicalis
- d) Aspergillosis

Question 22:

A patient who was diagnosed with ADPKD was prescribed tolvaptan. After a few weeks, he presented with dry mouth and increased thirst. What is the likely cause?

- a) Increased free water clearance
- b) V₂ receptor agonism
- c) Increased renal cAMP levels

d) Increase in urine osmolality

Answer Key

Question No.	Correct Option
1	a
2	c
3	a
4	b
5	a
6	a
7	b
8	a
9	b
10	d
11	b
12	b
13	b
14	a
15	b
16	c
17	a
18	a
19	a
20	a
21	b
22	a

Detailed Explanations

Solution to Question 1:

6-Mercaptopurine is an inhibitor of DNA synthesis.

6-Mercaptopurine is inactive in its parent form. It must be metabolized by hypoxanthine-guanine phosphoribosyltransferase to form monophosphate nucleotide

6-thioinosinic acid. This inhibits several enzymes of de novo purine nucleotide synthesis.

Option B- Mitomycin acts as an alkylating agent and crosslinks DNA. It also generates free radicals which damage DNA.

Option C- Actinomycin causes single-strand breaks in DNA. It blocks RNA transcription as well.

Option D- L-Asparaginase causes the breakdown of asparagine and thus interferes with protein synthesis.

Solution to Question 2:

Thiazides act by increasing the tubular reabsorption of lithium.

Thiazide use is accompanied by sodium loss which is compensated in a few days by increased reabsorption of Na⁺ in the proximal tubule.

The kidney handles both lithium and Na⁺ in the same way. Thus, along with Na⁺ reabsorption, there will be an increase in the reabsorption of lithium also, resulting in decreased excretion of lithium.

Solution to Question 3:

Erythromycin should not be prescribed with theophylline.

Theophylline is mainly metabolized by CYP1A2 in the liver. Erythromycin inhibits CYP1A2 and thereby decreases the clearance of theophylline.

Factors decreasing the clearance of theophylline are:

- Drugs - erythromycin, cimetidine, ciprofloxacin, allopurinol, fluvoxamine, zileuton, zafirlukast
- Congestive heart failure
- Liver disease
- Pneumonia
- Viral infection and vaccination
- High-carbohydrate diet
- Old age

Solution to Question 4:

Chloroquine, an anti-malarial drug may get deposited in the cornea leading to corneal opacity.

Ocular toxicity of chloroquine:

- Short-term therapy - blurring of vision and diplopia, reversible on stoppage of drug.

- Long-term use (high-dose):
- Keratopathy
- Pigmentary retinopathy
- Optic atrophy

Hydroxychloroquine has lower ocular toxicity compared to chloroquine.

Drugs causing corneal opacity:

- Amiodarone
- Phenothiazine
- Indomethacin
- Chloroquine

Long-term treatment with these drugs can cause a whorl-like opacity in the corneal epithelium called cornea verticillata. It is also seen in Fabry's disease.

Solution to Question 5:

Trihexyphenidyl is the drug of choice for drug-induced parkinsonism. It is a central anticholinergic.

Drug-induced parkinsonism is a condition caused by certain drugs having D2 antagonistic action (typical antipsychotics). The decrease in dopaminergic activity results in an unbalanced cholinergic activity. Anticholinergics act by restoring the balance.

Solution to Question 6:

Amiodarone is contraindicated in a patient with interstitial lung disease.

The most serious adverse effect during chronic amiodarone therapy is pulmonary fibrosis, which can be rapidly progressive and fatal. Underlying lung disease, doses of 400 mg/day or more, and recent pulmonary insults such as pneumonia appear to be risk factors for this condition.

Amiodarone is a class III antiarrhythmic drug.

The side effects of amiodarone include:

- Pulmonary fibrosis
- Hypotension, depressed ventricular function
- Torsades de pointes
- Corneal microdeposits
- Hepatotoxicity
- Neuropathies
- Photosensitivity

- Thyroid dysfunction

Solution to Question 7:

Ranolazine acts by inhibiting the late sodium current in the heart.

Late sodium current (I_{Na}) facilitates entry of calcium via the sodium-calcium exchanger. Inhibition of I_{Na}, therefore, reduces the intracellular calcium concentration which in turn reduces the diastolic tension, cardiac contractility and work.

At higher doses, it may also modify the fatty acid oxidation, which improves the efficiency of cardiac oxygen utilization.

It is used in the prophylaxis of angina.

Solution to Question 8:

Tamsulosin is the drug used to relieve the symptoms in cases of postoperative urinary retention in a patient with COPD.

It is also efficacious in the treatment of benign prostatic hyperplasia (BPH) with little effect on blood pressure.

Option B: Bethanechol is the drug of choice to relieve the symptoms in cases of postoperative urinary retention. However, it can cause bronchoconstriction in patients with COPD. Hence tamsulosin is the best answer here.

Option C: Terazosin is an α_1 receptor antagonist, used in the treatment of benign prostatic hyperplasia (BPH).

Option D: Methacholine is a muscarinic agonist but acts specifically on bronchial smooth muscle causing its contraction. It is available as an inhalational agent and is used for the diagnosis of bronchial hyper-reactivity in patients who do not have clinically apparent asthma.

Solution to Question 9:

Promethazine is used for motion sickness. It is an antihistaminic agent that acts as an antagonist of H₁ receptors. It also has central anticholinergic activity that is responsible for its antiemetic action.

Solution to Question 10:

Diphenylmethane is not a prokinetic agent. Its derivative (Bisacodyl) is used as a laxative.

Prokinetic agents are the drugs that increase GI motility and transit of material in the GI tract. These agents include:

- Dopamine receptor antagonists- metoclopramide and domperidone
- Serotonin receptor agonists- cisapride and prucalopride
- Motilin
- Macrolide antibiotics mimicking motilin - erythromycin
- Miscellaneous agents:
 - Sincalide (C-terminal octapeptide of CCK)
 - Camicinal (motilin receptor agonist)
 - Relamorelin (ghrelin receptor agonist)
 - Velusetrag and naronapride (5HT₄ agonists)

Solution to Question 11:

Bisphosphonates are used in the first-line management of postmenopausal osteoporosis.

Bisphosphonates are analogs of pyrophosphates. They specifically inhibit bone resorption and thus are very effective for the treatment of postmenopausal osteoporosis.

Bisphosphonates concentrate at sites of active remodeling. They remain in the matrix until the bone is remodeled. Then, they are released in the acid environment of the resorption lacunae and induce apoptosis in osteoclasts.

Classification of bisphosphonates (mainly based on potency):

First-generation:

- Etidronate (least potent)
- Clodronate
- Tiludronate

Second-generation: 10-100 times more potent than the first generation

- Alendronate
- Pamidronate
- Ibadronate

Third-generation: 10,000 times more potent than the first generation

- Risedronate
- Zoledronate

Solution to Question 12:

Resistance by lactamases is through the degradation of antibiotics.

The bacterial cell wall is made of heteropolymeric peptidoglycan that confers mechanical stability. The β -lactam antibiotics inhibit the last step in peptidoglycan synthesis by acylating the transpeptidase via cleavage of the $-\text{CO}-\text{N}-$ bond of the β -lactam ring. These transpeptidase targets are collectively called Penicillin Binding Proteins (PBPs)

Beta-lactamases are enzymes produced by certain bacteria that inactivates β -lactam antibiotics by hydrolysis of the β -lactam ring in their structure.

The resistance to penicillins and other beta-lactam antibiotics can be conferred through one of the four mechanisms:

- Inactivation of antibiotic by lactamases
- Modification of target PBPs
- Impaired penetration of drug to target PBPs
- Antibiotic efflux

Solution to Question 13:

The clinical features given in the question are most likely suggestive of lithium toxicity, and hence serum lithium levels should be measured to confirm it.

Lithium is a mood stabilizer with a very narrow therapeutic index and is the first line of management in bipolar disorder.

The kidney handles both lithium and Na^+ in the same way via ENaC channel. Thus during fasting/dehydration, along with Na^+ reabsorption in renal tubules, there will be an increase in the reabsorption of lithium leading to lithium toxicity.

Lithium has a very narrow therapeutic index of 0.8-1.2 mEq/L. The severity of lithium toxicity is often divided into the following three grades:

A fine tremor is common in patients taking lithium at therapeutic doses. Therapeutic lithium tremor is generally symmetric, limited to the arms, and nonprogressive. The tremor from lithium toxicity is coarser and more disabling, may also affect the legs, and occurs with other symptoms of toxicity. Other adverse effects include mental confusion, hypothyroidism, nephrogenic diabetes insipidus, and edema.

As lithium is readily dialyzed, lithium overdose or toxicity is treated with hemodialysis.

*Note - Serum Na^+ levels are significant with respect to lithium toxicity, where as serum potassium levels becomes significant in digoxin toxicity. Hypokalemia increases the risk of digoxin toxicity.

Grade	Sr. lithium concentration(in mEq/L)	Symptoms
Mild symptoms	1.5-2.5	nausea, vomiting, tremor, and fatigue
Moderate intoxication	2.5-3.5	confusion, agitation, delirium, tachycardia, and hypertonia

Grade	Sr. lithium concentration(in mEq/L)	Symptoms
Severe intoxication	>3.5	coma,seizures, hyperthermia, and hypotension

Solution to Question 14:

Pegloticase is used in the treatment of chronic gout.

Pegloticase is a pegylated uricase (urate oxidase). It catalyzes the enzymatic oxidation of uric acid into allantoin, which is a more soluble and inactive metabolite.

It is used in the treatment of severe,refractory chronic gout or in cases where other urate-lowering therapies are contraindicated.

Solution to Question 15:

Rifampicin is an inhibitor of RNA synthesis. It binds to the beta subunit of DNA dependent RNA-polymerase (rpoB) to form a stable drug-enzyme complex. The binding suppresses the chain formation in RNA synthesis.

Option A: 5-Fluorouracil inhibits thymidylate synthesis.

Option C: Nitrofurantoin produces highly reactive intermediates that cause DNA damage

Option D: Novobiocin is a potent inhibitor of bacterial DNA gyrase.

Solution to Question 16:

Pefloxacin is mainly metabolized in the liver and hence avoided in patients with liver disease.

Quinolones with primarily hepatic metabolism include moxifloxacin, pefloxacin, and trovafloxacin (withdrawn from the market due to hepatotoxicity).

Quinolones primarily excreted by the kidneys include ciprofloxacin, levofloxacin, norfloxacin, lomefloxacin, and ofloxacin.

Solution to Question 17:

The diuretic effect of theophylline is due to blockade of adenosine (A₁) receptors.

Theophylline is a methylxanthine drug that acts by:

- Inhibition of phosphodiesterase enzyme (PDE). PDE₃ inhibition causes relaxation of bronchial smooth muscle and PDE₄ inhibition causes inhibition of the release of cytokines and chemokines

which further leads to a decrease in immune cell migration and activation.

- Blockade of adenosine receptors, which inhibits bronchoconstriction. However, blockade of A₁ receptors in the glomerular afferent arteriole and PCT causes diuresis.
- Activating enzyme histone deacetylase-2, which is responsible for its anti-inflammatory action

Solution to Question 18:

Drug A has the highest potency.

Potency:

- It is defined as the amount of drug needed to produce a given response. A highly potent drug evokes a given response at low concentrations and vice-versa.
- The more the curve is on the left in a dose-response curve (DRC), the greater is its potency.
- Thus, in the above image, the potency is Drug A > Drug B > Drug C.

Efficacy:

- It is defined as the maximum effect produced by the drug. It reflects the capacity of the drug to activate a receptor and produce the response.
- In the log dose-response curve, the perpendicular height represents the drug efficacy.
- Thus, in the above image, the efficacy is Drug A = Drug B > Drug C.
- Drugs having a similar pharmacological effect can have different efficacies e.g furosemide and metolazone are both diuretics but furosemide has greater efficacy than metolazone.

Solution to Question 19:

Morphine is the analgesic which could be responsible for the aggravation of the patient's symptoms.

It causes constriction of the sphincter of Oddi causing an increase in fluid pressure in the common bile duct and gall bladder.

Note: Pethidine causes lesser spasm of the sphincter of Oddi and can be used in biliary colic.

Solution to Question 20:

Hydrochlorothiazides are not given to this patient.

Gout is a metabolic disease associated with hyperuricemia. Thiazides increase direct urate reabsorption in the proximal renal tubules and are associated with elevated serum uric acid levels and may precipitate gout. Hence, hydrochlorothiazides are not used to treat hypertension in gout patients.

Solution to Question 21:

Voriconazole is not effective against mucormycosis.

Mucormycosis is treated by amphotericin B. Posaconazole can also be used. However, other azoles are not effective against this fungus.

Voriconazole is the drug of choice for the treatment of aspergillosis. It can also be used for *Candida* species (including *glabrata* and *krusei*), *Scedosporium* and *Fusarium*.

Solution to Question 22:

Tolvaptan causes increased free water clearance, which causes dehydration. Hence, the patient may present with dry mouth and increased thirst.

Tolvaptan is a vasopressin V2 receptor antagonist. Decreased binding of vasopressin to the V2 receptor in the kidney lowers the adenylyl cyclase activity leading to decreased intracellular cAMP. Decreased cAMP concentrations prevent the fusion of aquaporin 2 containing vesicles with the plasma membrane, leading to an increase in urinary water excretion (free water clearance), and a decrease in urine osmolality.

The common side effects of tolvaptan include:

- dehydration
- hypernatremia
- hypovolemia
- liver injury

Tolvaptan is used to delay disease progression in patients with ADPKD.

Pharmacology INI-CET 2021

Question 1:

Which of the following is false about the selection of essential drugs?

- a) Adequate safety profile needs to be established
- b) Cost to benefit has to be considered
- c) Fixed drug combination is preferred over single drugs
- d) Disease prevalence is considered

Question 2:

A 34-year-old man with hypertension is diagnosed with stage 4 chronic kidney disease, with eGFR of ≤ 30 . His doctor wants to prescribe a thiazide diuretic. Which of the following can be given?

- a) Chlorothiazide
- b) Chlorthalidone
- c) Metolazone
- d) Hydrochlorothiazide

Question 3:

A patient who was diagnosed with epilepsy was put on retigabine TDS. Now phenytoin is being added. Which of the following changes should be made to retigabine?

- a) Increase the dose
- b) Decrease the dose
- c) Stop the drug
- d) Stop retigabine and start on carbamazepine

Question 4:

A patient with history of ischemic stroke was started on clopidogrel. However, she had another attack of stroke after 6 months. Which of the following is likely to be responsible for

the failure of clopidogrel in this patient?

- a) Downregulation of CYP2C19
- b) Downregulation of CYP2D6
- c) Upregulation of CYP1A1
- d) Downregulation of CYP2E1

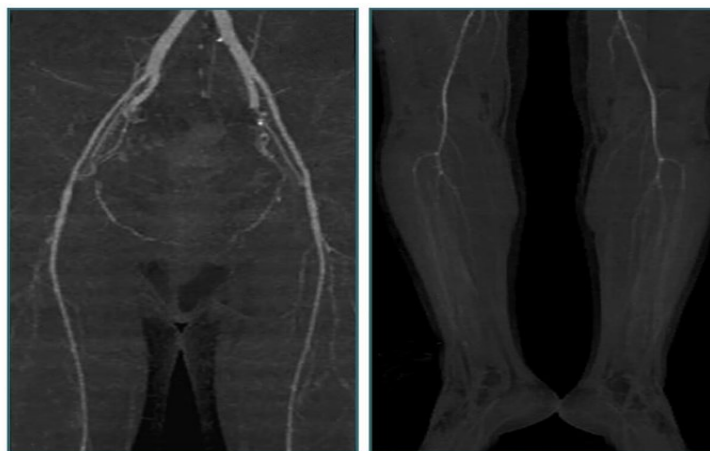
Question 5:

Which of the following can be given for long term management of obesity?

- a) Liraglutide
- b) Metformin
- c) Sibutramine
- d) Fenfluramine

Question 6:

An HIV-positive woman on ART was prescribed ergotamine for a migraine attack 4 days back. She presented with complaints that she is unable to feel her legs from the mid-thigh to her toes, for the past two days. The angiography image is given below. What is the likely diagnosis?



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- a) Atherosclerosis
- b) Ergotism

- c) Polyradicular neuropathy
- d) Herniated disc

Question 7:

Topiramate is used for

- a) Treatment of migraine
- b) Treatment of Lennox Gastaut syndrome
- c) Prophylaxis of ADHD
- d) Treatment of dementia

Question 8:

Which of the following is incorrectly matched?

- a) Andexanet alfa: Edoxaban
- b) Andexanet alfa: Rivaroxaban
- c) Idarucizumab : Dabigatran
- d) Ciraparantag: Fondaparinux

Question 9:

Inhaled Nitric oxide is approved for treatment of

- a) Labile hypertension
- b) Pulmonary hypertension
- c) Treatment of resistant migraine
- d) Resistant Parkinson's disease

Question 10:

Nivolumab is used as a checkpoint inhibitor in

- a) Retinoblastoma
- b) Medulloblastoma
- c) Pleuropulmonary blastoma

d) Hodgkin's lymphoma

Question 11:

Paradoxical bronchoconstriction seen with ipratropium bromide is due to all except

- a) Hypertonic saline
- b) EDTA
- c) Benzalkonium chloride
- d) Prejunctional M2 receptor blockade

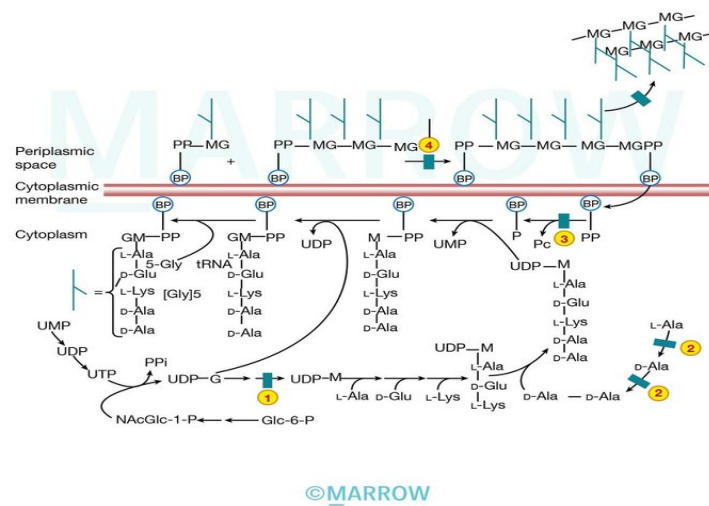
Question 12:

Antimicrobial combinations are used in all except

- a) Gonorrhea
- b) Tuberculosis
- c) Intra abdominal infections
- d) Malaria

Question 13:

Which of the following steps does Vancomycin act on?



- a) 1

- b) 2
- c) 3
- d) 4

Question 14:

What would happen to the half-life and plasma concentration of a drug which follows first-order kinetics, if the dose is doubled?

- a) Half-life doubles and plasma concentration remains the same
- b) Half-life remains the same and plasma concentration doubles
- c) Half-life and plasma concentration remains the same
- d) Half life and plasma concentration doubles

Question 15:

Which of the following is not an off-label use of risperidone?

- a) OCD
- b) Dementia
- c) PTSD
- d) Bipolar disorder

Question 16:

The elimination rate constant of a drug is 0.05/hr. What is its half-life?

- a) 20.8 hours
- b) 6.9 hours
- c) 13.8 hours
- d) 9.8 hours

Question 17:

A patient is on phenytoin for a seizure disorder. He was prescribed sucralfate 4 times a day for peptic ulcers. What should be the minimum duration of time between consumption of these drugs?

- a) 30 minutes
- b) 60 minutes
- c) 90 minutes
- d) 120 minutes

Question 18:

Category A, B, C, D, X in drug classification are indicative of which of the following?

- a) Safety in pregnancy
- b) Dose adjustment in renal failure
- c) Cost ratio from cheap to expensive
- d) As per safety or therapeutic index

Question 19:

Identify the false statement regarding suicidal inhibition.

- a) The inhibitor can bind with any site resulting in suicidal inhibition
- b) The binding of the enzyme to the substrate analogue is irreversible
- c) They are enzyme specific and used in rational drug design
- d) The inhibitor forms a product with the enzyme and the product inhibits it

Question 20:

The dose at which dopamine acts on beta 1 receptors is ____.

- a) 2-10 mcg/kg/min
- b) <2 mcg/kg/min
- c) 10-20 mcg/kg/min
- d) >20 mcg/kg/min

Question 21:

Which of the following is the incorrect statement regarding clomiphene citrate?

- a) Causes monofollicular development

- b) Inhibits negative feedback on GnRH
- c) Causes endometrial thinning
- d) Can be used with gonadotropin

Question 22:

What is the mechanism of action of remdesivir?

- a) RNA-dependent DNA polymerase inhibition
- b) RNA-dependent RNA polymerase inhibition
- c) Fusion inhibition
- d) Entry inhibition

Question 23:

Administration of which of the following drugs would increase the bioavailability of saquinavir?

- a) Ritonavir
- b) Vitamin C
- c) Cimetidine
- d) Ganciclovir

Question 24:

The table below gives the AUC of drug B alone and that of drug B when combined with another drug A. If the κ value is < 0.01 , then which of the following statements is true?

- a) Drug A decreases intestinal absorption of Drug B
- b) Drug A reduces first pass metabolism of Drug B
- c) Drug A increases the metabolism of Drug B
- d) Drug A increases renal elimination of Drug B

Question 25:

What is the mechanism of action of cyclosporine?

- a) Inhibition of calcineurin
- b) AMP kinase stimulation
- c) IMP dehydrogenase inhibition
- d) Dihydro-orotate dehydrogenase inhibition

Question 26:

Anionic and weakly acidic drugs bind to which of the following?

- a) Albumin
- b) Globulin
- c) Lipoprotein
- d) Alpha-1 glycoprotein

Question 27:

A 14-month-old boy was diagnosed with *Hemophilus influenza* meningitis. The initial treatment plan was to start on ampicillin but later was shifted to ceftriaxone. What could be the probable reason?

- a) *Hemophilus* alters penicillin binding protein sites, hence resistant to ampicillin
- b) Easier to administer IV ceftriaxone than IV ampicillin
- c) Cotrimoxazole is the drug of choice, but patient is allergic to sulfa drugs
- d) *Hemophilus* produces beta lactamase, hence resistant to ampicillin

Question 28:

A patient who is a known case of cirrhosis of the liver presents with abdominal distension and swelling of the lower limbs. The preferred diuretic for this patient would be

- a) Triamterene
- b) Eplerenone
- c) Chlorthalidone
- d) Furosemide

Question 29:

Which of the following is the most effective drug in smoking cessation?

- a) Varenicline
- b) Buspirone SR
- c) Nicotine gum
- d) Rimonabant

Question 30:

A patient who was on warfarin therapy for atrial fibrillation is prescribed a cephalosporin antibiotic for the treatment of a lower respiratory tract infection. He reports back to the hospital after having developed hemarthrosis. Which of the following cephalosporins is likely to have caused this?

- a) Cefixime
- b) Cefoperazone
- c) Ceftobiprole
- d) Ceftazidime

Question 31:

Which of the following is a novel anti-depressant?

- a) Lurasidone
- b) Vilazodone
- c) Asenapine
- d) Blonanserin

Question 32:

A 57-year-old man was brought to the hospital with complaints of dizziness and breathlessness. During his physical examination, you notice the following discolouration over his face. He is presently being treated with amiodarone, metoprolol, verapamil, lisinopril, and amlodipine. Which of the following drugs is most likely responsible?



- a) Amiodarone
- b) Amlodipine
- c) Metoprolol
- d) Verapamil

Question 33:

Which of the following is a centrally acting opioid antitussive?

- a) Ambroxol
- b) Diphenoxylate
- c) Levopropoxyphene
- d) Levorphanol

Question 34:

The anti-epileptic drugs which can also be used for the treatment of neuropathic pain are

- a) 1, 2 and 3
- b) 1, 2 and 4
- c) 1, 3 and 4
- d) 2 and 4 only

Question 35:

A breast cancer patient post-chemotherapy developed a low-grade fever with anemia and leukocytopenia ($<4000/\text{cm}^3$; DNC: 30). Which of the following antibiotics will not be used in the empirical management of this patient?

- a) Cefepime
- b) Piperacillin-Tazobactam
- c) Linezolid
- d) Meropenem

Answer Key

Question No.	Correct Option
1	c
2	c
3	a
4	a
5	a
6	b
7	b
8	d
9	b
10	d
11	a
12	a
13	d
14	b
15	d
16	c
17	d
18	a
19	a
20	a
21	a
22	b

23	a
24	b
25	a
26	a
27	d
28	b
29	a
30	b
31	b
32	a
33	c
34	b
35	c

Detailed Explanations

Solution to Question 1:

Fixed drug combinations are not preferred over single drugs while selecting essential drugs.

Criteria for selection of essential medicines as per WHO:

- There should be adequate data on the safety and efficacy of the drug.
- The drug should be available in a form where its quality including bioavailability and stability on storage is assured.
- The choice of the drug depends on the pattern of prevalent diseases, financial resources, availability of trained personnel and facilities, genetic, demographic, and environmental factors.
- The cost-benefit ratio is an important factor.
- Pharmacokinetic properties of the drug and the local manufacture and storage facilities are also considered.
- Essential drugs should preferably be single compounds.
- Fixed combinations are used only when the dosage of each component meets the requirement of the population in which it is used. It should also have the advantage in safety, therapeutic effect, adherence, and prevention of drug resistance when compared to single compounds.
- The selection of essential drugs is a continuous process and changes are to be made with developments in public health and the availability of better drugs.
- The selection should be based on rationally developed treatment guidelines.

Note:

In 1977 WHO brought the first "Model List of Essential Drugs". India made its "National Essential Drug List" in 1996 and revised the list in 2011 and renamed it as "National List of Essential Medicines". The latest edition of WHO Model List of Essential Drugs is the 21st edition published in 2019.

Solution to Question 2:

The thiazide diuretic that can be used as an antihypertensive in this patient with chronic kidney disease is metolazone.

In patients with chronic renal failure, thiazide diuretics are ineffective. But metolazone, despite being a thiazide diuretic, is effective as antihypertensives in patients with chronic renal failure.

Metolazone is useful even in severe renal failure (GFR of ~15 mL/min). It has an additive action when combined with loop diuretics like furosemide. It also has an additional proximal tubular action (thiazides mainly act on DCT). It is commonly used for edema and is occasionally used as an antihypertensive.

Diuretics in hypertension:

Thiazides are the diuretics of choice for uncomplicated hypertension.

Mechanisms of antihypertensive action:

- Decreases plasma and ECF volume by diuresis thereby decreasing cardiac output
- Even when the compensatory mechanisms restore the Na⁺ levels and plasma volume, a low BP is maintained by a reduction in total peripheral resistance.
- This reduction in total peripheral resistance is due to the persistence of a small sodium and volume deficit even after compensation. The decreased intracellular sodium is responsible for increased compliance of the vessel wall and a reduced response to constrictor stimulus.

Advantages of thiazide as an antihypertensive:

- It can be given as once a day dose and has a flat dose-response curve.
- Decreased incidence of postural hypotension and relative freedom from CNS side effects
- No tolerance
- No fluid retention
- Decreased risk of osteoporotic fractures due to its hypocalciuric action
- Low cost
- Effectiveness in isolated systolic hypertension

Thiazides are ineffective both as antihypertensive and diuretic in patients with chronic renal failure.

High ceiling diuretics: Furosemide is the prototype drug in this class and it is a weaker antihypertensive agent than thiazides. They are given in hypertension only in the presence of the following complications:

- Chronic renal failure - As thiazides are ineffective

- Coexisting refractory congestive heart failure
- Resistance to combination regimen with thiazides
- Marked fluid retention with the use of potent vasodilators.

Potassium-sparing diuretics - They are only used in combination with thiazides to prevent the loss of potassium.

Side effects of diuretics include hypokalemia, carbohydrate intolerance, dyslipidemia, and hyperuricemia.

Solution to Question 3:

The dose of retigabine should be increased when phenytoin is added because phenytoin can decrease the plasma concentration of retigabine when given together.

Retigabine is also known as ezogabine. Metabolism of the drug occurs by glucuronidation and acetylation. Its half-life is about 7-11 hours and hence requires thrice-daily dosing. When it is administered with phenytoin or carbamazepine there may be a reduction in the plasma concentration of the drug. So the dose of retigabine should be increased when given along with these drugs.

It is a potassium(K⁺) channel opener that enhances transmembrane potassium currents mediated by the KCNQ family of ion channels. It was approved as an adjunctive agent for focal onset seizures of patients above 18 years. But because of its adverse effects like retinal abnormalities and blue pigmentation, the production of the drug was discontinued by the manufacturer.

Other drug interactions of phenytoin:

It is an inducer of the enzymes CYP2C8/9, CYP3A4/5, and other CYPs. It is also an inhibitor of CYP2C9/19.

- Phenobarbitone can inhibit the metabolism of phenytoin. But the overall interaction is unpredictable as both the drug enhance each other's degradation by enzyme induction.
- Phenytoin and carbamazepine induce each other's metabolism.
- Valproate increases the level of unbound phenytoin by displacing the protein-bound form and decreasing its metabolism.
- Phenytoin increases the metabolism of steroids, oral contraceptives, doxycycline, theophylline, etc by enzyme induction.
- It produces competitive inhibition of warfarin metabolism. (Increased risk of bleeding when given together)
- Drugs like chloramphenicol, warfarin, isoniazid, and cimetidine can inhibit the metabolism of phenytoin and cause its toxicity.
- Sucralfate decreases the absorption of phenytoin by binding to it in the gastrointestinal tract.

Solution to Question 4:

Downregulation of CYP2C19 is responsible for the failure of clopidogrel in this patient.

Clopidogrel is a prodrug and its metabolic activation occurs in the liver. It is bioactivated by the CYP2C19 enzyme. If there is downregulation of this enzyme it will cause a decrease in the efficacy of clopidogrel and can lead to thrombotic events like stroke. Drugs that inhibit CYP2C19 can also decrease the antiplatelet effect of clopidogrel. Proton-pump inhibitors may cause inhibition of CYP2C19 and when they are given in combination there would be a decrease in the efficacy of clopidogrel.

Therapeutic uses: Clopidogrel is used to reduce the rate of stroke and myocardial infarction. It also decreases the rate of deaths in patients with recent myocardial infarction or stroke. Other uses are acute coronary syndrome and peripheral arterial disease.

Adverse effects: Increased risk of bleeding especially when given with an anticoagulant or aspirin. Rarely thrombotic thrombocytopenic purpura can occur.

CYP2C19: Common drugs that are metabolized include omeprazole, phenytoin, diazepam, and propranolol. Inducers of the CYP2C subfamily are rifampicin and carbamazepine. Omeprazole is an inhibitor.

Other options:

CYP2D6: Drugs metabolized include SSRIs, tricyclic antidepressants, antiarrhythmic drugs, beta-blockers, and opiates. It is inhibited by quinidine and it will prevent the formation of morphine from codeine. Hence quinidine decreases the analgesic effect of codeine.

CYP1A1/2: It metabolizes few drugs like theophylline, paracetamol, caffeine and carbamazepine. It plays an important role in the activation of procarcinogens. Inducers are rifampicin, carbamazepine, polycyclic hydrocarbons, charbroiled meat, and cigarette smoke.

CYP2E1: It is involved in the oxidation of alcohol, halothane, and minor metabolism of other drugs. It is important to note that it is involved in the formation of hepatotoxic N-acetyl benzoquinone imine from paracetamol. Chronic alcoholism is responsible for the induction of this enzyme.

Solution to Question 5:

Liraglutide is used in the long-term management of obesity.

Mechanism of action:

- Increase glucose-mediated insulin release
- Lowers the glucagon levels
- Delays gastric emptying
- Decreases feeding by stimulating GLP-1 receptors in the arcuate nucleus of the hypothalamus.

It is a GLP-1 analog used as a parenteral hypoglycemic agent and anti-obesity drug. It is administered subcutaneously. It is used as an adjuvant in patients who are not achieving glycemic control with oral agents. It can be also added with basal insulin.

It is found to decrease the risk of cardiovascular death, nonfatal myocardial infarction, and stroke in patients with type 2 diabetes and cardiovascular disease.

Adverse effects: Nausea, vomiting, diarrhea, or constipation. It should not be given in patients with a personal or family history of MEN syndrome or medullary carcinoma of the thyroid.

Other options:

Metformin: It belongs to the biguanides class of oral hypoglycemic agents. It is used as a first-line agent in the management of type 2 diabetes and it acts by reducing hepatic glucose production. It is also used to treat infertility in PCOD patients. IT is usually not associated with weight gain and in some cases, it may cause mild weight reduction.

Sibutramine: It has both serotonergic and noradrenergic action. It is also an anorectic agent with a similar mechanism of action to fenfluramine. But due to severe adverse effects, US-FDA has banned its use.

Fenfluramine: It is a serotonergic anorectic agent. It reduces food-seeking behavior due to increased serotonergic transmission in the hypothalamus. The use of is not recommended now by the US-FDA due to its severe side effects.

Solution to Question 6:

The clinical features and the CT angiography image showing diffuse symmetrical luminal narrowing of the lower limb vessels are suggestive of ergotism.

In this case, a patient on antiretroviral therapy is given ergotamine. HIV protease inhibitors used in ART like ritonavir, lopinavir, etc are potent inhibitors of the CYP3A4 enzyme. This enzyme is responsible for the metabolism of ergot compounds. Hence inhibition of this enzyme leads to decreased metabolism of ergotamine. This causes the accumulation of toxic levels of ergotamine in the patient and leads to ergotism.

Ergotamine is a partial agonist and antagonist at alpha-adrenergic, 5HT₁; and 5HT₂ receptors. It produces sustained vasoconstriction, vasomotor center depression, and visceral smooth muscle contraction.

Adverse effects: Nausea, vomiting, and abdominal pain may be seen. Vasoconstriction caused by the drug is responsible for most of the adverse effects. These are usually seen with long-term therapy with higher doses.

- It may precipitate or aggravate angina pectoris or myocardial infarction due to coronary insufficiency.
- Some patients develop cold, numb, and painful extremities.
- Leg claudication may be seen
- Gangrene of the extremities

Contraindications for ergotamine:

- Peripheral vascular disease
- Coronary artery disease
- Uncontrolled hypertension
- Sepsis

- Pregnancy
- Liver and kidney diseases

Solution to Question 7:

Topiramate is used in the treatment of Lennox-Gastaut syndrome.

Uses of topiramate:

- In newly diagnosed focal and primary generalized epilepsy in adults and children
- As monotherapy for refractory focal epilepsy and generalized tonic-clonic seizures
- Treatment of tonic-clonic seizures in patients with Lennox-Gastaut syndrome
- It is also used in the prophylaxis of migraine(not for treatment)
- It can be given as monotherapy for children of 10 years or more. But if used as adjunctive therapy it can be used in children as young as 2 years.

Mechanism of action: It reduces voltage-gated sodium currents in the cerebellar granule cells. It also activates a hyperpolarizing potassium current, enhances GABAA receptor currents, and limits the activation of the AMPA-kainate subtype of glutamate receptors. It also produces weak inhibition of the carbonic anhydrase enzyme.

Pharmacokinetics: Only 10-20% is bound to plasma proteins and the drug is largely excreted unchanged in the urine. Half-life is about 1 day. Topiramate is found to decrease the plasma concentrations of oral contraceptives, hence a higher dose of OCPs should be given along with topiramate.

Adverse effects:

- Somnolence, weight loss, fatigue, and nervousness are commonly seen
- It can precipitate renal calculi. It may be due to the weak carbonic anhydrase inhibition.
- Cognitive impairment

Solution to Question 8:

Ciraparantag is a reversal agent for all direct anticoagulants and is not useful with fondaparinux which is an indirect factor Xa inhibitor.

Ciraparantag, andexanet alfa, and idarucizumab are all reversal agents of direct anticoagulants.

Ciraparantag: It is considered as a potential reversal agent against all direct anticoagulants including dabigatran, rivaroxaban, edoxaban, apixaban, LMWH, and heparin. This drug is in the early stages of development.

Andexanet alfa: It is a recombinant analog of factor Xa and is proposed for reversal of oral factor Xa inhibitors like rivaroxaban, edoxaban, and apixaban. It rapidly reverses the action of these drugs and restores thrombin generation.

Idarucizumab: It is a licensed monoclonal antibody that specifically acts as a reversal agent for dabigatran. It has a 350 times higher affinity for dabigatran than thrombin. Hence it forms a complex with dabigatran and is excreted by the kidneys.

If a patient on direct oral anticoagulants like dabigatran, rivaroxaban, edoxaban, and apixaban presents with life-threatening bleeds and the reversal agents are not available then other options are the administration of prothrombin complex concentrate, activated prothrombin complex concentrate, or recombinant factor VII-A. Hemodialysis is useful in severe bleeding due to dabigatran.

Other reversal agents for anticoagulants:

- Protamine sulphate - Heparin
- Vitamin K - Warfarin

Note:

Andexanet alfa has received US-FDA approval in May 2018 as a reversal agent for rivaroxaban and apixaban.

Solution to Question 9:

Inhaled nitric oxide (NO) is used in the management of pulmonary hypertension.

The only FDA-approved therapeutic indication of inhaled nitric oxide is in persistent pulmonary hypertension of the newborn. It has the ability to selectively dilate pulmonary vasculature. Hence it can be used in conditions with increased pulmonary vascular resistance.

It is proposed for the:

- Long-term management of pulmonary artery hypertension (PAH) and other forms of pulmonary hypertension
- Rescue therapy for severely hypoxemic patients with or without pulmonary hypertension

Diagnostic indications:

- In heart failure patients and infants with congenital heart disease during cardiac catheterization to evaluate pulmonary vasoreactivity.
- In determining the diffusion capacity across the alveolar-capillary unit. It is considered superior to carbon dioxide in this because it has a greater affinity for hemoglobin and higher water solubility at body temperature.

Mechanism of action:

It activates soluble guanylyl cyclase and produces cGMP, which then activates cGMP-dependent protein kinase. This leads to the activation of multiple regulatory mechanisms that decrease intracellular calcium and cause vasodilation in precapillary resistance arterioles.

Adverse effects:

- Methemoglobinemia at higher concentrations.
- It can worsen left ventricular performance in patients with an already impaired left ventricular function.

Solution to Question 10:

Nivolumab is used as a checkpoint inhibitor in Hodgkin's lymphoma.

FDA approved uses of nivolumab:

- Advanced melanoma
- Advanced non-small cell lung cancer(NSCLC) that are previously treated
- Renal cell carcinoma
- Advanced head and neck cancers
- Refractory/relapsed classical Hodgkin's lymphoma

Nivolumab is a checkpoint inhibitor that blocks the interaction between PD-1 and its ligands. It is a fully human monoclonal IgG4 antibody. It has a good response rate in patients with pretreated solid tumors and hematological malignancies. Similar effectiveness is also seen with pembrolizumab which is also an anti-PD-1 monoclonal antibody.

Adverse effects:

- In melanoma patients rash is a commonly seen side effect
- fatigue, dyspnea, musculoskeletal pain, cough, decreased appetite, and constipation can be seen in advanced squamous NSCLC patients
- Pneumonitis, colitis, nephritis, hepatitis, and renal dysfunction are also seen

Important note:

The FDA on May 2021 has approved the use of nivolumab in patients having completely resected esophageal or gastroesophageal junction cancers with residual pathologic disease who have received neoadjuvant chemoradiotherapy.

Solution to Question 11:

Paradoxical bronchoconstriction with ipratropium bromide is seen with a hypotonic nebulizer solution and not hypertonic saline.

Paradoxical bronchoconstriction with ipratropium bromide is seen especially when given by a nebulizer and may be due to the following factors:

- Hypotonic nebulizer solution
- Antibacterial additives like benzalkonium chloride and EDTA
- It may be also due to prejunctional M2 receptor blockade on the cholinergic nerves present in the airway. These prejunctional M2 receptors normally inhibit acetylcholine release into the synapse. So their blockade can cause increased Ach release and bronchoconstriction.

Paradoxical bronchoconstriction is not reported with tiotropium bromide or any other long-acting muscarinic antagonists. Tiotropium is selective for M3 and M1 receptors and therefore less likely

to produce prejunctional M2 receptor blockade.

Other adverse effects of inhaled muscarinic antagonists:

- A bitter taste is seen with ipratropium
- Nebulization with ipratropium bromide may precipitate glaucoma in elderly patients. This is due to the direct effect of the drug on the eyes and can be easily prevented by using a mouthpiece instead of a facemask.
- Long-acting agents can cause dryness of the mouth.
- In elderly patients, urinary retention may be seen occasionally.

Solution to Question 12:

Gonorrhea is not managed by antimicrobial combinations. According to the latest CDC guidelines, single-agent therapy with ceftriaxone is the preferred treatment for gonococcal infections.

Single-dose of 500mg intramuscular(IM) ceftriaxone is given for gonorrhea. It is important to note that the previous guidelines that recommended dual therapy with ceftriaxone and azithromycin are no longer preferred.

Monotherapy with ceftriaxone is given when concurrent chlamydia infection is ruled out by molecular testing. If this is not done then presumptive treatment for chlamydia should be given along with the treatment of gonorrhea. According to CDC guidelines, 100mg doxycycline twice daily for 7 days is given as presumptive treatment of chlamydia in nonpregnant individuals.

Other options:

Tuberculosis: Multidrug therapy is given in tuberculosis for proper cure and to prevent the selection of drug-resistant mutants. ATT regimen is given in detail in the pearl attached.

Intraabdominal infections: It is usually polymicrobial and can result in an intraabdominal abscess or secondary peritonitis. These are managed with either single-agent or combination therapy. Broad-spectrum agents with gram-negative cover such as meropenem or piperacillin-tazobactam can be used alone. Combination therapy with metronidazole and other agents like ciprofloxacin, cefepime, etc are used as empirical agents in intraabdominal infections.

Malaria: It is also treated with antimicrobial combinations. In plasmodium vivax and ovale malaria, schizontocides like chloroquine are combined with agents that attack the hypnozoites like primaquine for radical cure. Primaquine is also given in combination with Artemisinin combined therapy(ACT) for plasmodium falciparum infections as a gametocidal agent. Treatment regimens are given in the pearl attached.

Solution to Question 13:

Vancomycin acts on step 4 in the given image.

Mechanism of action of vancomycin: It is a cell wall synthesis inhibitor. It binds to the D-Ala-D-Ala terminus of the newly formed peptidoglycan pentapeptide and inhibits the transglycosylase enzyme. This would prevent further elongation of the peptidoglycan and its cross-linking.

Vancomycin is a bactericidal agent for gram-positive organisms. It is also active against many gram-positive anaerobes like *Clostridium difficile*. Resistance against vancomycin in enterococci is due to the alternation in its binding site where D-Ala is replaced by D-Lactate.

Clinical uses:

- Parenteral vancomycin is given for bloodstream infections and endocarditis caused by MRSA.
- Vancomycin plus gentamicin is given as an alternative regimen for enterococcal endocarditis with penicillin allergy.
- In combination with other agents for suspected meningitis or meningitis due to penicillin-resistant pneumococcus.
- Oral vancomycin is given for *Clostridium difficile* colitis.

Adverse effects:

- Injection site phlebitis
- Nephrotoxicity
- "Redman" syndrome due to histamine release following infusion

Other options:

Fosfomycin inhibits step 1 in the image. It acts at a very early stage in cell wall synthesis and inhibits the cytoplasmic enzyme enol pyruvate transferase. This would block the addition of phosphoenolpyruvate to UDP-N-acetylglucosamine. This prevents the formation of UDP-N-acetylmuramic acid which is the precursor N-acetylmuramic acid. N-acetylmuramic acid is a component of the bacterial cell wall and hence cell wall synthesis is inhibited.

Cycloserine inhibits step 2 in the image. It is a structural analog of D-alanine and inhibits the enzymes alanine racemase that converts L-alanine to D-alanine and D-alanyl-D-alanine ligase. This would thereby inhibit the incorporation of D-alanine into the peptidoglycan pentapeptide. It is mainly used in the management of tuberculosis where there is resistance to the first-line agents.

Bacitracin inhibits step 3 in the image. It is a cell wall synthesis inhibitor that interferes with dephosphorylation in the cycling of the lipid carrier which transfers peptidoglycan subunits to the growing cell wall. It is only given for topical application because of its severe nephrotoxicity when given systemically.

Note:

Beta-lactam antibiotics prevent cross-linking by inhibiting the transpeptidases. (Last step in the images)

Solution to Question 14:

In this case, the half-life remains the same, and the plasma concentration doubles.

Half-life is constant in drugs following first-order kinetics irrespective of the dose as a constant fraction of the drug is eliminated per unit time.

The plasma concentration of the drug increases when the dose given is increased. In this case, as the drug dose is doubled plasma concentration is also doubled.

Solution to Question 15:

Risperidone is FDA approved for the treatment of bipolar affective disorders.

Uses of risperidone:

- Long-term management of psychosis and schizophrenia.
- Maintenance therapy in mania bipolar disorders.
- Management of irritability associated with autism.

Off-label uses of risperidone:

- Management of psychotic symptoms in delirium and dementia.
- Obsessive-compulsive disorders.
- Post-traumatic stress disorders.

Risperidone is an atypical antipsychotic that shows both D₂ and 5 HT₂ receptor blockade. It also has an affinity for alpha 1, alpha 2, and H₁ receptors. It produces a more potent D₂ blockade than clozapine. Hyperprolactinemia is an important adverse effect. It is also associated with an increased risk of stroke in elderly patients. At low doses, extrapyramidal side effects are less.

Solution to Question 16:

The half-life of the given drug is 13.8 hours.

The half-life of a drug or $t_{1/2}$ is the time taken for the plasma concentration to be reduced by 50%. The elimination rate constant or k value of a drug reflects the fraction of drug removed from the body compartment per unit of time.

The half-life of a drug and its elimination rate constant are inversely proportional to each other. They are represented by the equation:

$$k t_{1/2} = 0.693$$

$$\text{Therefore, } t_{1/2} = 0.693 / k$$

The given value for elimination rate constant k is 0.05, therefore the half-life of the given drug would be:

$$0.693 / 0.05 = 13.8$$

Therefore, the half-life of the drug would be 13.8 hours.

Solution to Question 17:

The minimum duration of time between the consumption of phenytoin and sucralfate should be 120 minutes.

Sucralfate, in an acidic environment (pH < 4) acts by producing a viscous, sticky polymer that adheres to ulcer craters and epithelial cells for up to 6 hours after a single dose. This viscous layer produced in the stomach leads to inhibition of absorption of other drugs, like phenytoin, digoxin, cimetidine, ketoconazole, and fluoroquinolone antibiotics. Therefore, sucralfate should be taken after a gap of at least 2 hours / 120 minutes after the administration of phenytoin.

Constipation is the most common side effect of sucralfate. It should be avoided in patients with renal failure who are at risk of aluminium overload. Aluminum-containing antacids should not be combined with sucralfate in these patients. As the viscous gel produced by sucralfate is sticky, it may eventually lead to bezoar formation.

It is generally used for prophylaxis of peptic ulcers, but rarely so because of its side effects and its risks outweigh its benefits.

Solution to Question 18:

Category A, B, C, D, X in drug classification are indicative of their safety in pregnancy.

The United States Food and Drug Administration (FDA) system for teratogenic potential attempts to quantify teratogenic risk from category A (safe) to category X (definite human teratogenic risk).

Category A: The possibility of fetal harm is remote and studies in women have failed to demonstrate any evidence of risk to the fetus. Examples: doxylamine, folic acid, levothyroxine.

Category B: Either animal-reproduction studies have not demonstrated a fetal risk, but there are no controlled studies in pregnant women, or animal-reproduction studies have shown an adverse effect that was not confirmed in controlled studies in pregnant women. Example: Amoxicillin, loratadine, ondansetron.

Category C: Either studies in animals have revealed adverse effects on the fetus, that is, teratogenic or embryocidal but there are no controlled studies in women, or studies in women and animals are not available. These drugs should be given only if the potential benefit justifies the potential risk to the fetus. Example: fluconazole, metoprolol, sertraline.

Category D: Though there is evidence of human fetal risk, the benefits from use in pregnant women may be acceptable despite the risk, for example, if the drug is needed in a life-threatening situation or for a disease for which safer drugs cannot be used or are ineffective. Example: lisinopril, lithium, phenytoin.

Category X: There is evidence of fetal risk based on human or animal experience and the risk of the use of the drug in pregnant women clearly outweighs any possible benefit. The drug is contraindicated in women who are or may become pregnant. Example: methotrexate, simvastatin, warfarin.

Solution to Question 19:

The false statement regarding suicidal inhibition is the inhibitor can bind with any site resulting in suicidal inhibition.

In suicidal inhibition, the inhibitor can only bind with the active site of an enzyme and is unreactive outside the confines of the enzyme's active site.

The enzyme inhibition helps to modify the rate of reactions it catalyzes and this inhibition can either be reversible or irreversible. Reversible inhibitors bind to enzymes through non-covalent bonds. Irreversible inhibitors usually bind to an enzyme through covalent bonds and do so strongly so that it can no longer dissociate.

Suicide inhibitors are a category of irreversible inhibitors also called mechanism-based inhibitors. They are specialized substrate analogs that contain a chemical group that reacts with the catalytic machinery of the target enzyme. In these types of inhibitors, the substrate analog irreversibly ties up with the binding site of an enzyme, deceiving the enzyme by concealing its functional group resulting in the formation of a highly reactive molecule, which brings about self-inactivation of the enzyme.

Suicide inhibitors bind to the enzyme by means of covalent bonds; they are enzyme-specific and unreactive outside the confines of the enzyme's active site.

An example of this would be the binding of clavulanic acid to the active site of beta-lactamases in bacteria. Clavulanic acid mimics the structure of a beta-lactam antibiotic and forms a covalent adduct with the β -lactamase active site. This leads to a rearrangement that creates a much more reactive derivative, which is subsequently attacked in its active site to irreversibly acylate the enzyme and inactivate it. Thus, amoxicillin and clavulanic acid are combined to overcome bacterial resistance to amoxicillin alone.

The development of clavulanic acid to target beta-lactamase-producing bacteria is also an example of the rational design of drugs. Rational design of drugs refers to the ability to predict the appropriate molecular structure of a drug on the basis of information about its biological receptor.

Solution to Question 20:

The dose at which dopamine acts on β -1 receptors is 2-10 mcg/kg/min.

The systemic effects of dopamine vary with its concentration in the body.

When administered at an infusion rate of 2-5 mcg/kg/min, it directly stimulates cardiac β receptors. This enhances myocardial contractility and is useful in the management of cardiogenic shock.

At doses of 5-15 mcg/kg/min and above, it stimulates α adrenergic receptors. This causes peripheral arterial and venous vasoconstriction.

At lower doses of <2 mcg/kg/min, it directly stimulates both D1 and D2 receptors. The D1 receptors in renal and mesenteric blood vessels are the most sensitive: i.v. infusion of low dose of DA dilates these vessels subsequently, causing increased renal blood flow and increased urine output.

The clinical indications of dopamine include cardiogenic and vasodilatory shock, congestive cardiac failure and symptomatic bradycardia unresponsive to atropine or pacing.

The major side effects are:

- Severe hypertension (especially when taken with nonselective β -blockers)
- Ventricular arrhythmias
- Cardiac ischemia
- Tissue ischemia/gangrene (high doses/extravasation into tissues)

Solution to Question 21:

The incorrect statement regarding clomiphene citrate is that it causes monofollicular development; rather, it causes multi follicular development.

Clomiphene is an unequal mixture of two isomers, enclomiphene and zuclomiphene. Zuclomiphene is a non-steroidal compound closely resembling an estrogen, blocks the negative inhibitory feedback of estrogens on the hypothalamus, causing a surge of gonadotropins (FSH, LH), which leads to ovulation. The continued flow of FSH encourages multiple follicular development which is relatively common.

Being an anti-estrogen, it causes thickening of cervical mucus and thinning of the endometrium.

Clomiphene citrate (CC) is used for the treatment of absent or irregular ovulation due to hypothalamic-pituitary dysfunction associated with normal levels of endogenous estradiol; the patient must be treated repeatedly until pregnancy is achieved since the normal cyclic ovulatory function does not usually resume. Clomiphene citrate has no effect on patients with ovarian or pituitary failure. Ovulation induction with clomiphene citrate and a gonadotropin like FSH increases the likelihood of multiple concurrent ovulations.

Side effects of CC include hot flushes and some nausea. However, it is usually very well tolerated.

Solution to Question 22:

The mechanism of action of remdesivir is inhibition of the enzyme RNA-dependent RNA polymerase.

Remdesivir is an adenosine nucleotide prodrug. It distributes into cells where it is metabolized to form remdesivir triphosphate (RDV-TP). It binds to the viral RNA-dependent RNA polymerase and inhibits viral replication by prematurely terminating RNA transcription.

Remdesivir (under Emergency Use Authorization) is used in the treatment of COVID-19. It may be considered in patients with moderate to severe disease, requiring supplemental oxygen, within 10 days of onset of the illness.

Dose: 200 mg IV on day 1 followed by 100 mg IV daily for 4 days (total 5 days).

The following are contraindications to the use of remdesivir:

- Renal or hepatic dysfunction (eGFR $\leq$ 30 ml/min/m²; AST/ALT $\geq$ 5 times upper limit of normal) (Not an absolute contraindication)
- Pregnant or lactating females

- Patients <12 years of age
- Patients who are not on oxygen support or in-home settings

Remdesivir can cause gastrointestinal symptoms, elevated transaminase levels, increased prothrombin time, and hypersensitivity reactions. Hence, liver function tests and prothrombin time should be obtained before and during treatment with remdesivir.

Other options:

Option A- RNA-dependent DNA Polymerase / reverse transcriptase inhibitors prevent the conversion of viral RNA into proviral DNA. They help to prevent infection of susceptible cells but do not eradicate the virus from cells that are integrated with proviral DNA. Examples: zidovudine, lamivudine, stavudine, tenofovir, didanosine, zalcitabine, and emtricitabine. These are drugs used to manage HIV infection.

Options C and D- Entry/fusion inhibitors include enfuvirtide and maraviroc. Enfuvirtide inhibits the fusion of viral and cell membranes mediated by gp41 and CD4 interactions. Maraviroc is a chemokine receptor antagonist and binds to host cell CCR5 receptor to block binding of viral gp120. They are also used to manage HIV infection.

Solution to Question 23:

Administration of ritonavir would increase the bioavailability of saquinavir.

Saquinavir is an HIV protease inhibitor that inhibits replication of both HIV-1 and HIV-2. It is metabolized predominantly by the hepatic enzyme CYP3A4. Inhibition of CYP3A4 results in an increase or a boost in the concentration of saquinavir. Ritonavir is one such drug that inhibits CYP3A4. Therefore combining it with a protease inhibitor like saquinavir results in:

- Enhanced bioavailability
- Inhibition of systemic clearance
- Inhibition of first-pass metabolism, and
- Longer half-life of the co-administered HIV protease inhibitor.

This allows for a reduction of drug dose and dosing frequency of saquinavir.

All protease inhibitors except for nelfinavir are predominantly metabolized by CYP3A4. Therefore, the administration of CYP3A4 inhibitors like ritonavir or cobicistat can increase the bioavailability of these protease inhibitors. This phenomenon is known as boosting protease inhibitors by CYP inhibitors. The dosage of ritonavir, when given as a booster drug, is 100 to 200mg once or twice daily.

Other options :

Option B: Vitamin C / Ascorbic acid is a water-soluble vitamin that can actually reduce the steady-state concentration of protease inhibitors. It increases the absorption of dietary iron.

Option C: Cimetidine is a histamine 2 receptor antagonist which is used to promote gastric and duodenal ulcer healing. It is used in the management of gastroesophageal reflux disease. It also inhibits CYPs, but only ritonavir and cobicistat are used for boosting protease inhibitors.

Option D: Ganciclovir is a guanine nucleoside analogue which has inhibitory activity against all herpes viruses, especially cytomegalovirus (CMV).

Solution to Question 24:

Drug A reduces the first-pass metabolism of Drug B.

The area under the plot of plasma concentration of a drug versus time after dosage is called the area under the curve or AUC. It can be used as a measure of the bioavailability of a drug and to calculate its clearance rate from the body.

Bioavailability refers to the extent to which an administered dose of the drug reaches its site of action; if the metabolic and excretory capacity of the intestine or liver for the drug is large, bioavailability is reduced substantially and this is called the first-pass metabolism whereas bioavailability is increased if intestinal absorption of the drug is increased.

The greater the AUC, the greater is the bioavailability of the drug.

In the question, is observed that when drug A was given along with drug B, the total AUC has increased ($820 \text{ seconds} \pm 124 \text{ seconds}$) when compared to the AUC when drug B is administered alone ($570 \text{ seconds} \pm 127 \text{ seconds}$).

The null hypothesis for the above question would be that drug A doesn't increase the AUC for drug B and for this to be true, α value should be >0.05 .

As given α value is <0.01 , it is significant and the null hypothesis is rejected, which means that drug A has indeed increased the AUC for drug B.

This indicates increased bioavailability of drug B when administered with drug A, which shows that drug A has either increased the intestinal absorption of drug B or has reduced its first-pass metabolism.

Other options:

Option A: If drug A decreased the intestinal absorption of drug B, this would mean that the AUC for drug B, when administered with drug A, would be reduced. This is due to decreased bioavailability of drug B when given with drug A.

Option C: If drug A increases the metabolism of drug B, this would lead to a shorter $t_{1/2}$ and duration of action of drug B, thereby decreasing the AUC for when drug B and drug A are given together.

Option D: If drug A increased the elimination of drug B, the AUC for drug B when given with drug A would be reduced due to reduced duration of action of drug B.

Solution to Question 25:

The mechanism of action of cyclosporine is inhibition of calcineurin activity.

Calcineurin is a calcium/calmodulin-dependent phosphatase required for the induction of T cell lymphokine production and proliferation. Cyclosporine is an immunomodulatory drug

that inhibits calcineurin activity by forming a complex with cyclophilin, a cytoplasmic-receptor protein present in target cells, to form a cyclosporine-cyclophilin complex. This drug-receptor complex specifically and competitively binds to and inhibits calcineurin, a calcium and calmodulin-dependent phosphatase. This in turn, inhibits the translocation of a family of transcription factors (NF-AT), leading to reduced transcriptional activation of cytokine genes for interleukin (IL)-2, tumour necrosis factor (TNF)-alpha, IL-3, IL-4, CD40L, granulocyte-macrophage colony-stimulating factor, and interferon-gamma. Ultimately, the proliferation of T lymphocytes is reduced, resulting in immunomodulation.

Its uses are :

- In solid organ transplantation, to treat organ rejection in kidney, liver, and heart allogeneic transplants.
- In patients with rheumatoid arthritis, when the disease has not adequately responded to methotrexate.
- It is approved to treat amyotrophic lateral sclerosis (ALS or Lou Gehrig disease) and its variants.
- In nephrotic syndrome, to treat focal segmental glomerulosclerosis not responding to corticosteroids.

Adverse effects of cyclosporine include a decrease in glomerular filtration rate (GFR) and metabolic disturbances like hypomagnesemia, hyperkalemia, dyslipidemia, gynecomastia, hypertrichosis; neurotoxicity in the form of convulsions. There is an increased risk of developing skin and lymphoproliferative malignancies in cyclosporine-treated psoriasis patients.

Other options:

Option B: AMP kinase is stimulated by metformin, a first-line orally administered drug in the management of type-2 diabetes mellitus.

Option C: IMP dehydrogenase is inhibited reversibly by the drug mycophenolate mofetil, which is indicated in the prophylaxis of transplant rejection.

Option D: Dihydro-orotate dehydrogenase is inhibited by leflunomide, a pyrimidine synthesis inhibitor indicated for the management of rheumatoid arthritis in adults.

Solution to Question 26:

Anionic and weakly acidic drugs bind to albumin.

After being distributed in the circulating blood, drugs bind to serum proteins to varying degrees. Among serum proteins, albumin and alpha-1 acid glycoprotein play important roles in protein binding for many drugs, which affects their distribution in the body.

Albumin is a major carrier of and binds to acidic drugs like phenytoin, NSAIDs. Whereas, α -1 acid glycoprotein binds basic drugs like lignocaine, disopyramide, quinidine and propranolol.

Importance of plasma protein binding:

Conditions resulting in acute phase reaction response leading to elevated levels of alpha-1 acid glycoprotein like Crohn's disease, myocardial infarction, arthritis, cancer, etc. enhance binding of basic drugs. Conditions causing hypoalbuminemia result in decreased binding sites for acidic

drugs.

When a drug is bound to the plasma proteins, it is kept in the blood, while the unbound (free) drug can produce a therapeutic effect, be metabolized, or excreted. In the majority of the drugs, this does not have a significant effect, but in drugs with a narrow therapeutic index like warfarin, it is clinically relevant as excessive free warfarin can produce toxicity.

Many drugs displace other drugs from their protein binding sites. This may cause an increase in the pharmacological effect of the displaced drug.

A drug bound to a plasma protein cannot cross membranes and this bound fraction of a drug is also not available for metabolism or glomerular filtration which can make the drug long-acting.

Other options:

Option B- Globulins are of three types - α , β , and γ -globulins. Globulins bind glucocorticoids and progesterone in the blood.

Option C- Lipoproteins like VLDL, LDL can bind anti-cancer drugs like 5-fluorouracil and doxorubicin.

Solution to Question 27:

The probable reason why the drug to be administered was changed from ampicillin to ceftriaxone is because *Hemophilus* produces beta-lactamase and hence, is resistant to ampicillin.

A considerable percentage of isolates of *Hemophilus influenzae* produce beta-lactamase, rendering them resistant to ampicillin, a beta-lactam antibiotic. However, these strains are still sensitive to second and third-generation cephalosporins.

In *H. influenzae*, the resistance to ampicillin and other β -lactam antibiotics is generally due to the production of a β -lactamase. In the case of β -lactamase-negative ampicillin-resistant strains, resistance is due to the presence of altered penicillin-binding proteins.

Ceftriaxone is a third-generation cephalosporin that is effective against *Hemophilus Influenzae* and considerably resistant to beta-lactamase. It has a very long half-life which permits once-daily dosing of the drug. Because of the efficacy of third-generation cephalosporins in the therapy of meningitis caused by *Hemophilus Influenzae* type b, ceftriaxone administered at the rate of 50 mg/kg/dose given 12th hourly should be a part of the initial empirical therapy.

Vancomycin combined with a third-generation cephalosporin like ceftriaxone is the recommended empirical antibiotic regimen in a suspected case of meningitis, beyond the neonatal period.

Solution to Question 28:

The preferred diuretic for this patient with cirrhosis presenting with abdominal distension and lower limb swelling is eplerenone.

Eplerenone is a mineralocorticoid receptor/aldosterone antagonist. High doses of mineralocorticoid receptor/aldosterone antagonists decrease total body sodium in cirrhotic patients with ascites by natriuresis.

Patients with liver cirrhosis have a decreased effective arterial blood volume because of splanchnic vasodilatation. This stimulates the renin-angiotensin-aldosterone system and results in hyperaldosteronism. This causes increased sodium and water retention and leads to the development of ascites. This also contributes to resistance to loop diuretics.

Normally, aldosterone (AL) enters the cells of the late distal tubule and collecting ducts by diffusion and binds to a specific mineralocorticoid receptor (MR) in the cytoplasm. This AL-MR complex mediates Na^+ reabsorption, which drives the secretion of K^+ and H^+ into the tubules.

Eplerenone competitively blocks this MR, preventing it from binding with aldosterone. The net effect is sodium diuresis and potassium retention. It is a potassium-sparing diuretic. Other mineralocorticoid receptor antagonists include spironolactone, canrenone, potassium canrenoate.

Some clinical uses of MR antagonists are -

- Hypertension due to primary hyperaldosteronism
- Edema associated with liver cirrhosis, congestive cardiac failure, chronic kidney disease
- They may reduce mortality in patients with myocardial infarction

Hyperkalemia is the major side effect of MR antagonists, the risk being greater in patients with renal disease or those already taking ACE inhibitors. Other important side effects are gynecomastia and decreased libido due to the anti-androgenic effect of spironolactone but eplerenone does not cause gynecomastia and decreased libido.

Other options :

Option A- Triamterene directly inhibits renal epithelial sodium channels (ENaC) in the late distal tubule and collecting duct. It is a potassium-sparing diuretic and is generally used along with other diuretics in the treatment of hypertension and edema. Triamterene is metabolized in the liver, hence triamterene toxicity may be enhanced in patients with hepatic disease. It is used in the management of Liddle syndrome.

Option C- Chlorthalidone is a thiazide-like diuretic. It inhibits Na^+ - Cl^- symporter in the early distal tubule and increases the excretion of sodium and chloride. Thiazide diuretics may be used for the treatment of hypertension, heart failure, nephrolithiasis due to idiopathic hypercalciuria, and nephrogenic diabetes insipidus.

Option D- Furosemide is a loop diuretic. It acts by inhibiting the Na^+ - K^+ - 2Cl^- symporter in the thick ascending limb of the loop of Henle resulting in increased excretion of sodium and chloride. It also has mild carbonic anhydrase inhibitory activity and also increases urinary excretion of bicarbonate and phosphate. Cirrhotic patients are often resistant to loop diuretics due to decreased secretion of the drug into tubular fluid and high aldosterone levels. Furosemide may be given along with eplerenone, but the best answer here is eplerenone since aldosterone antagonists are the main therapy for cirrhotic edema.

Solution to Question 29:

Varenicline is the most effective drug in smoking cessation.

Nicotine is the principal constituent of tobacco responsible for its addictive behaviour. One of the newest agents introduced to aid in smoking cessation is varenicline.

Varenicline works by blocking the effects of nicotine on the brain. It blocks α -4, β -2 nicotinic acetylcholine receptor subtypes and varenicline inhibits dopaminergic activation produced by smoking through their partial agonism, and decreases the craving and withdrawal syndrome that occurs with cessation attempts. It prevents nicotine stimulation of the mesolimbic dopamine system associated with nicotine addiction, thus helping to occlude the "rewarding" effect of smoking.

Varenicline is prescribed only to patients 18 years or older. Varenicline is taken orally and the dose taken is 0.5 mg once daily on days 1 to 3, increased to 0.5 mg twice daily on days 4 through 7 and 1 mg twice daily after day 8 for 11 weeks. Therapy should start one week before the target quit date. It can be continued for another 12 weeks (1 mg BD) to maintain abstinence.

The adverse effects include nausea, which can be mitigated by starting at lower doses and gradually increasing the dose. Other symptoms include disturbed sleep, sleepwalking, agitation, drowsiness, and constipation and may also increase the risk of pancreatitis. Varenicline has also been implicated in specific skin syndromes such as Stevens-Johnson syndrome, erythema multiforme, and photosensitivity. Varenicline can cause renal failure and kidney stones, thus renal function must be monitored.

The FDA has issued a warning about mood and behavioral changes associated with its use, and there is some evidence of increased cardiovascular risk.

Other options:

Option B: Buspirone sustained release is a partial agonist at 5HT receptors but affinity for D2 receptors may also be present. Its effects are slow onset, over a couple of weeks and is used in the management of generalised anxiety states.

Option C: Nicotine gum has nicotine which is an agonist at Nn type nicotinic cholinergic receptors and is also advised for smoking cessation along with psychotherapy. However, varenicline was more effective when compared with single product nicotine replacement therapy (NRT) and was similarly effective compared with combination NRT.

Option D: Rimonabant is an inverse agonist at cannabinoid receptor CB1 which decreases neurotransmitter release at GABAergic and glutamatergic synapses; its usage for smoking cessation and treat obesity was stopped after serious side effects notably severe depression and increased risk of suicide. However, it continues to be used off-label for smoking cessation.

Solution to Question 30:

The cephalosporin antibiotic most likely to have caused hemarthrosis in this patient on warfarin therapy is cefoperazone.

Cephalosporins containing a methylthiotetrazole (NMTT) group can prolong the prothrombin time. This results in clinically significant bleeding in patients already on anticoagulant therapy or having vitamin K deficiency.

This is because NMTT cephalosporins inhibit the enzyme vitamin K epoxide reductase in the liver, which is the key enzyme for vitamin K-dependent carboxylation, a posttranslational modification that is essential for the biological functions of coagulation factors.

NMTT group containing cephalosporins include latamoxef, cefmenoxime, cefoperazone, cefotetan, cefamandole.

Vitamin K epoxide reductase is also the target of anticoagulants like warfarin. Oral administration of vitamin K can prevent this problem.

Solution to Question 31:

Vilazodone is the novel antidepressant.

The antidepressant effect of vilazodone is due to its enhancement of serotonergic activity in the CNS through selective inhibition of serotonin reuptake. It binds with high affinity to the serotonin reuptake site but not to the norepinephrine or dopamine reuptake site. As a result, vilazodone potently and selectively inhibits the reuptake of serotonin.

Vilazodone is also a partial agonist at serotonergic 5-HT_{1A} receptors; the result of this activity on serotonergic transmission also plays a role in its antidepressant effect.

Thus it offers a novel combination of selective serotonin reuptake inhibition (SSRI) and serotonergic (5-HT_{1A}) receptor partial agonist activity. Because of these characteristics, vilazodone has been termed a serotonin partial agonist–reuptake inhibitor (SPARI).

Vilazodone is never combined with a monoamine oxidase (MAO) inhibitor or given within 14 days of discontinuing or initiating MAO inhibitor therapy due to the serious risk of serotonin toxicity.

Other options:

Option A: Lurasidone is a second-generation antipsychotic that is a 5-HT_{2A} and D₂ receptor antagonist. It is used in the treatment of schizophrenia and bipolar depression (as monotherapy or as an adjunct).

Option C: Asenapine is a second-generation antipsychotic which is also a 5-HT_{2A} and D₂ receptor antagonist, used in the management of schizophrenia and acute mania.

Option D: Blonanserin is a second-generation antipsychotic, which acts as an antagonist at D₂, D₃, and 5-HT_{2A} receptors. Blonanserin has a low affinity for 5-HT_{2C}, adrenergic α_1 , histamine H₁, and muscarinic M₁ receptors, but displays a relatively high affinity for 5-HT₆ receptors. It is used in the management of schizophrenia.

Solution to Question 32:

The image shows grey-blue discoloration over this patient's face and the drug responsible for this is amiodarone.

Long-term treatment with amiodarone is responsible for skin pigmentation. Elderly men are at an increased risk of developing this.

Ultraviolet light reacts with amiodarone and its metabolites to cause local vasodilatation. This increases the diffusion of amiodarone and its metabolites, resulting in their deposition in the tissues, causing pigmentation. Therefore, it is usually seen in sun-exposed areas.

Cessation of amiodarone, avoiding exposure to sunlight using sunscreen cream is recommended.

Amiodarone is an iodine-containing compound that is structurally related to thyroid hormone. It is a class III antiarrhythmic drug, which increases the duration of an action potential by blocking potassium channels. It thereby increases the time taken for repolarization. It also prolongs the refractory period and suppresses abnormal automaticity. It also has weak calcium channel blocking, sodium channel blocking, and adrenergic blocking actions.

It is used to maintain normal sinus rhythm during atrial fibrillation and prevent recurrent ventricular tachycardia.

Adverse effects of amiodarone:

- Prolonged usage of amiodarone causes interstitial lung disease.
- Heart blocks, bradycardia and QT prolongation
- Hyperthyroidism or hypothyroidism
- Corneal micro-deposits and vortex keratopathy, optic neuropathy

Drug interactions of amiodarone

- Causes digoxin toxicity by decreasing its renal clearance.
- Increases concentration of quinidine and procainamide
- Potentiates the anticoagulant effect of warfarin when given simultaneously

Solution to Question 33:

Levopropoxyphene is a centrally acting opioid antitussive.

Levopropoxyphene is an opioid derivative that is a strong and partial μ receptor agonist that helps to reduce the cough reflex and therefore, is used in the management of acute debilitating cough. The usual antitussive dose is 50–100 mg every 4 hours. It has no analgesic activity. However, it is known to cause some sedation.

Other options:

Option A: Ambroxol is derived from bromhexine. It is a mucoactive agent that helps alter the viscoelastic properties of mucus and increase mucociliary clearance, thereby providing relief in productive cough. It is commonly available as intramuscular or intravenous, lozenges, syrups, or tablets.

Option B: Diphenoxylate is an opioid agonist that is structurally similar to meperidine. It is used in the treatment of acute, chronic, and traveler's diarrhea. Upon administration, it undergoes de-esterification to difenoxin, which is its active metabolite. Its side effects include constipation, toxic megacolon, paralytic ileus and can give rise to CNS depression in children.

Option D: Levorphanol is a synthetic opioid that is also a 5HT/Norepinephrine reuptake inhibitor and an NMDA receptor antagonist which is chiefly used in the management of moderate to severe pain and has a relatively rapid onset of action. Its side effects include delirium and hallucinations.

Solution to Question 34:

The anti-epileptic drugs which can also be used for neuropathic pain are 1, 2, and 4 which are pregabalin, carbamazepine and gabapentin respectively.

Pregabalin is an anti-epileptic drug that acts by binding to $\alpha 2\delta$, a protein that is a subunit of voltage-gated calcium channels and causes a decrease in glutamate release at excitatory synapses. It is used in the management of neuropathic pain associated with diabetic peripheral neuropathy, spinal cord injury or postherpetic neuralgia.

Gabapentin is an anti-epileptic drug that, like pregabalin, acts by binding to $\alpha 2\delta$ protein. It is used in the management of neuropathic pain due to postherpetic neuralgia in adults.

The binding of gabapentin and pregabalin, also called "gabapentinoids" due to their structural similarity to GABA, to calcium channel subunit $\alpha 2\delta$ protein, is considered to mediate the anti-convulsant effects of gabapentin and pregabalin. They are used in focal onset seizures with or without progression to bilateral tonic-clonic seizures when used in addition to other anti-epileptic drugs.

Carbamazepine is an iminostilbene derivative. It blocks voltage-activated sodium channels, slowing their rate of recovery and limiting the repetitive firing of action potentials. It is used in the treatment of:

- Generalized, focal-to-bilateral and idiopathic tonic-clonic seizures
- Focal seizures
- Neuropathic pain associated with trigeminal and glossopharyngeal neuralgias.

Lamotrigine is an anti-folate agent whose mechanism of action is similar to that of carbamazepine i.e. it delays the recovery from inactivation of recombinant sodium channels. It also inhibits the release of glutamate by acting directly on the sodium channels. Therefore, when used concurrently with carbamazepine, is associated with toxicity in some patients due to increase in 10,11- epoxide of carbamazepine. It is useful in the management of:

- Focal and secondarily generalized tonic-clonic seizures in adults
- Lennox-Gastaut syndrome - a childhood disorder characterized by multiple type seizures, mental retardation, and refractoriness to anti-seizure medication.

Solution to Question 35:

Linezolid cannot be used in this patient as bone marrow suppression is a side effect of this drug and hence it can worsen the post-chemotherapy bone marrow suppression in this patient.

Linezolid can cause myelosuppression which can present as anemia, thrombocytopenia, leukopenia, and pancytopenia. It is commonly associated with thrombocytopenia and is seen 7-10 days following administration. This is usually reversed on discontinuation of the drug.

Other adverse effects of linezolid:

- It inhibits mitochondrial protein synthesis and may lead to peripheral neuropathy, optic neuritis, and lactic acidosis in patients receiving the drug for longer durations.

- It can cause serotonin syndrome when coadministered with adrenergic or serotonergic drugs like SSRIs due to its nonspecific MAO inhibition.

It is a protein synthesis inhibitor of the class oxazolidinones. It mainly acts against gram-positive organisms. It is approved for the treatment of the following conditions:

- Skin and skin structure infections by staphylococcus(Both MRSA and MSSA) and streptococcus.
- Community-acquired pneumonia due to streptococcus pneumoniae.
- Nosocomial pneumonia due to staphylococcus aureus.

Other options:

Cefepime: It is a fourth-generation antipseudomonal cephalosporin. It is useful in enterobacter and pseudomonas infections that are resistant to other cephalosporins.

Piperacillin-Tazobactam- It is a combination of piperacillin which is a ureidopenicillin and tazobactam which is a beta-lactamase inhibitor. It is useful in various infections including nosocomial infections, intraabdominal infections, pneumonia, and UTI. It is highly effective against E Coli and Klebsiella.

Meropenem: It belongs to the carbapenem group of beta-lactam antibiotics. They are more active against gram-negative organisms. Coadministration of cilastatin is not required with this drug. (Cilastatin should be coadministered with imipenem as imipenem is sensitive to renal peptidase degradation)

Pharmacology NEET 2021

Question 1:

A female patient on oral contraceptive pills (OCP) was diagnosed to have tuberculosis and started on antitubercular therapy (ATT). She was advised to use an additional barrier method of contraception. What is the reason for this advice?

- a) Teratogenic
- b) Failure of ATT
- c) Rifampicin decreases OCP metabolism
- d) Rifampicin induces OCP metabolism and cause failure

Question 2:

A 35-year-old chronic hypertensive woman visits the outpatient department for pre-conceptional counseling. Which of the following anti-hypertensives should she discontinue?

- a) Labetalol
- b) Calcium channel blockers
- c) Angiotensin-converting enzyme inhibitors
- d) Alpha methyldopa

Question 3:

A pregnant woman was prescribed indomethacin. Which of the following effects does the drug have on the fetus?

- a) Premature closure of ductus arteriosus
- b) Keeps ductus arteriosus patent
- c) Closure of fossa ovalis
- d) Premature closure of ductus venosus

Question 4:

Which of the following can be prescribed in case of an open angle glaucoma with bronchial asthma?

- a) Timolol
- b) Gemeprost
- c) Latanoprost
- d) Mannitol infusion

Question 5:

As a medical practitioner in India, how will you prescribe alprazolam to a patient?

- a) 1 mg OD before bedtime for 7 days
- b) 1/2 mg HS for 7 days
- c) 0.5 mg BD for 7 days
- d) 0.5 mg daily before bedtime for 7 days

Question 6:

A patient started on antihypertensive presented with the following picture. Which of the following drugs is the most likely cause?



- a) Lisinopril
- b) Nifedipine
- c) Atenolol

d) Clonidine

Question 7:

A patient on aspirin presented with occult blood in the stool. This is due to inhibition of _____ by aspirin

- a) Phospholipase A2**
- b) Thromboxane**
- c) Phospholipase C**
- d) Phospholipase D**

Question 8:

A 60-year-old postmenopausal female patient with a previous history of Colle's fracture came with complaints of lower backache. On evaluation, her T score was -2.5. Which of the following statements are false about the management of her condition?

- a) Teriparatide should be started before supplementing with bisphosphonates**
- b) Bisphosphonates are not given for more than a year**
- c) Calcium requirement is 1200 mg per day**
- d) Oral vitamin D3 is given along with oral calcium**

Question 9:

A patient presented with a history of multiple episodes of vomiting. He was given an intravenous drug, and later, he developed abnormal movements. Which of the following can be used to treat him?

- a) Hyoscine**
- b) Cyproheptadine**
- c) Levodopa**
- d) Benzhexol**

Question 10:

What formula is the dosing rate of a drug based on?

- a) Clearance / plasma concentration
- b) Clearance x plasma concentration
- c) Clearance x loading dose
- d) Clearance / loading dose

Question 11:

A patient on digoxin, potassium chloride, clarithromycin, atenolol, and triamterene developed atrial fibrillation with controlled ventricular rate. Serum digoxin levels were raised. Which of his medications is responsible for the same?

- a) Potassium chloride
- b) Clarithromycin
- c) Atenolol
- d) Triamterene

Question 12:

What is the mechanism of action of methotrexate in osteosarcoma?

- a) Analog of UMP
- b) Competitive inhibitor of DHFR
- c) Blocks purine synthesis
- d) Blocks pyrimidine synthesis

Question 13:

What should not be consumed by patients with bacterial vaginosis and trichomoniasis on metronidazole?

- a) Grapefruit juice
- b) Alcohol
- c) Benzodiazepine
- d) MAOI/SSRI

Question 14:

A patient with chronic kidney disease underwent a nephrectomy. Which of the following can be administered as a postoperative analgesic in this patient?

- a) Naproxen
- b) Indomethacin
- c) Diclofenac
- d) Acetaminophen

Question 15:

A patient with renal insufficiency presented with very low urine output, pedal edema, headache and BP 160/90. Which anti-hypertensive can be administered?

- a) Aliskiren
- b) Chlorthalidone
- c) Amlodipine
- d) Prazosin

Question 16:

Which of the following interleukin (IL) is inhibited by tocilizumab?

- a) IL-6
- b) IL-2
- c) IL-4
- d) IL-5

Answer Key

Question No.	Correct Option
1	d
2	c
3	a
4	c
5	d

6	a
7	b
8	b
9	d
10	b
11	b
12	b
13	b
14	d
15	b
16	a

Detailed Explanations

Solution to Question 1:

Rifampicin is a potent cytochrome P450 (CYP450) enzyme inducer. When given along with oral contraceptive pills(OCP) it would induce the metabolism of OCPs leading to contraceptive failure. This is the reason for the patient being advised to use an additional method of contraception.

Important enzyme inducers are:

- Barbiturates
- Phenytoin
- Carbamazepine
- Rifampicin
- Cigarette smoking
- Chronic alcoholism

Important drugs that inhibit CYP450 isoenzyme are:

- Macrolide antibiotics
- Azole antifungals
- Chloramphenicol
- Omeprazole
- Selective serotonin reuptake inhibitors (SSRIs)
- HIV protease inhibitors
- Cimetidine
- Ciprofloxacin

- Metronidazole

Solution to Question 2:

Angiotensin-converting enzyme (ACE) inhibitors should be discontinued in pregnancy as they can cause hypoplasia of organs, growth retardation, and fetal loss.

ACE inhibitors cause reduced perfusion and oligohydramnios leading to fetal growth retardation, calvarium maldevelopment, hypoplasia of lungs and kidneys, and limb contractures.

Antihypertensive drugs to be avoided in pregnancy:

- ACE inhibitors and Angiotensin receptor blockers(ARB) - Can cause growth retardation and fetal damage.
- Diuretics - Can decrease uteroplacental perfusion
- Nonselective beta-blockers
- Sodium nitroprusside - It is contraindicated in eclampsia.

Antihypertensive drugs that can be used in pregnancy:

- Labetalol - Combined α + β blocker
- Nifedipine - Dihydropyridine calcium channel blocker. It is useful in preeclampsia but should be stopped before labor as they weaken the uterine contractions.
- Methyldopa
- Hydralazine

Note:

Thiazide diuretics like hydrochlorothiazide are considered to be safe in pregnancy and are useful in the management of chronic hypertension in pregnant women.

Solution to Question 3:

Indomethacin can cause premature closure of ductus arteriosus if given to a pregnant woman.

Indomethacin is approved for the closure of persistent ductus arteriosus in premature infants. It is given intravenously. One of its limiting factors is the development of renal toxicity in neonates.

Indomethacin is a nonselective COX inhibitor. It has anti-inflammatory, analgesic, and antipyretic properties. It can be used in the management of conditions like rheumatoid arthritis, gouty arthritis, osteoarthritis, ankylosing spondylitis, etc. But its use is limited due to the high rate of intolerance and availability of safer agents.

Adverse effects: Diarrhea with ulcerative bowel lesions, acute pancreatitis, and hepatitis are seen. Frontal headache is the most frequent CNS effect. Dizziness, light-headedness, vertigo, and mental confusion can occur. Neutropenia, thrombocytopenia, and rarely aplastic anemia can also occur.

Note:

Prostaglandin analogs are indicated in the maintenance of patency of the ductus arteriosus.

Solution to Question 4:

Latanoprost can be given in a case of open-angle glaucoma with bronchial asthma.

Latanoprost is a PGF₂ α analog and it is not associated with systemic side effects like bronchospasm and hence can be given for asthmatic patients. Prostaglandin analogs are the first choice drugs for open-angle glaucoma.

Mechanism of action: It reduces intraocular pressure by altering the ciliary muscle tension and improving the outflow of the aqueous humor.

Advantages of latanoprost:

- Good efficacy
- Once-daily application
- Absence of systemic side effects

Disadvantage: High cost

Adverse effects:

- Blurring of vision
- Iris pigmentation
- Thickening and darkening of eyelashes
- Macular edema can occur especially in aphakic patients.

Other options:

Option A- Timolol is a non-selective beta-blocker used topically in glaucoma. It is associated with systemic effects by absorption through the nasolacrimal duct and can cause life-threatening bronchospasm in asthmatics and COPD patients. Hence they are not given in in this case.

Option B- Gemeprost is a PGE₁ analog used for the termination of early pregnancy.

Option D- Mannitol infusion is given for acute congestive glaucoma and not for open-angle glaucoma. It causes decongestion of the eye by its osmotic action.

Solution to Question 5:

The correct method of prescribing alprazolam is 0.5mg daily before bedtime for 7 days.

The drugs are prescribed in generic names legibly and preferably in capital letters. The prescription should have details to identify the patient and medication that needs to be dispensed and the therapeutic purpose. The use of abbreviations, especially Latin is discouraged. The dosage should be written in metric measure.

Guidelines for drug prescription:

The parts of an ideal prescription are as follows.

- Date
- Name and address of the doctor
- Superscription(Rx symbol)
- Information about the patient
- Name and dose of the drug prescribed(Inscription)
- Dispensing directions to the pharmacist(Subscription)
- Instructions for the patient
- Signature of the doctor

Solution to Question 6:

The image shows edema over the lips and face and its occurrence following intake of an antihypertensive is suggestive of drug-induced angioedema. Lisinopril being an ACE inhibitor is most likely to be responsible for this condition.

Angioedema with ACE inhibitors:

In very few patients(0.1%-0.5%), ACE inhibitors cause rapid swelling in the nose, throat, mouth, glottis, larynx, lips, or tongue. Once the drug is stopped symptoms improve drastically over hours. Angioedema of the intestine can occur very rarely.

ACE inhibitor-associated angioedema is a class effect, and patients who develop this adverse event should not be prescribed any other drugs within the ACE inhibitor class.

Other adverse effects of ACE inhibitors:

Hypotension - A sudden fall in blood pressure can occur following the first dose of ACE inhibitors.

Cough - It can produce a bothersome dry cough due to the accumulation of bradykinin, substance P, or prostaglandins in the lungs. The cough usually disappears within 4 days of the stoppage of the drug.

Hyperkalemia due to K⁺ retention can occur.

Acute renal failure can occur with ACE inhibitors in patients with bilateral renal artery stenosis, stenosis of the artery to a single remaining kidney, heart failure, or volume depletion due to diarrhea or diuretics.

It should be discontinued in pregnant women as it can cause developmental defects.

It can also cause maculopapular skin rash in some cases.

Other options:

Option B: Nifedipine is a calcium channel blocker. Common adverse effects include headache, flushing, and dizziness. Peripheral edema may rarely occur.

Option C: Atenolol is a beta 1 selective antagonist. Bradycardia, fatigue, cold extremities are common adverse effects. CNS side effects like nightmares and depression are less frequent with atenolol than other beta-blockers.

Option D: Clonidine is an alpha-2 adrenergic agonist. The main adverse effects of clonidine are dry mouth and sedation. Bradycardia and sexual dysfunction are other effects.

Solution to Question 7:

Aspirin causes inhibition of cyclooxygenase-1(COX-1) which would inhibit the formation of thromboxane A₂. Thromboxane A₂ is an inducer of platelet aggregation and is a potent vasoconstrictor. Hence inhibition of its formation is responsible for the bleeding complications of aspirin.

Aspirin is a noncompetitive, irreversible inhibitor of the cyclooxygenase(COX) enzyme. It is a nonselective COX inhibitor and inhibits both COX-1 and COX-2.

Uses of aspirin:

- Analgesic and antipyretic uses.
- Acute rheumatic fever
- Rheumatoid arthritis
- Post myocardial infarction and stroke due to its antiplatelet action.
- Prevention of preeclampsia

Adverse effects:

Nausea, vomiting, epigastric discomfort, and increased occult blood in stools are seen with analgesic doses(0.5-2g/day). Antiinflammatory doses(3-5g/day) can produce salicylism characterized by dizziness, tinnitus, vertigo, hyperventilation, reversible loss of hearing and vision, etc. It can cause elevation of liver enzymes and hepatic damage. Hypersensitivity reactions are also seen.

Reye's syndrome is a rare condition that can occur when salicylates like aspirin are given in children having viral infections like influenza.

Solution to Question 8:

Bisphosphonates are given for periods of more than one year (Option B is false). In this case, as the patient is having a history of osteoporotic fracture, teriparatide can be given before bisphosphonates.

Pharmacologic management of osteoporosis:

Bisphosphonates: These are pyrophosphate analogs that act by directly inhibiting bone resorption. The antiresorptive action is mainly due to its direct inhibitory effect on osteoclasts. It causes osteoclast apoptosis as well as the prevention of hydroxyapatite dissolution. Long-term bisphosphonates therapy is given for durations of 3-5 years, depending on the fracture risk of the

patient. Since they are associated with osteonecrosis of the jaw and atypical femur fractures, a drug holiday may be advised after long-term management.

Teriparatide is a recombinant PTH analog that has an anabolic effect on bone homeostasis by increasing osteoblastic activity. It increases bone mineral density and decreases the risk of fractures in postmenopausal women. Indications of teriparatide therapy include the history of osteoporotic fracture, multiple risk factors for fracture, intolerance, or failure of previous osteoporosis therapy. It is only approved for 2 years of use. Clinical trials that used teriparatide prior to bisphosphonate have been proven effective.

Denosumab is a human monoclonal antibody with a mechanism of action similar to that of osteoprotegerin (OPG). It binds to RANK Ligand (RANKL), thereby inhibiting its binding to the RANK receptor of the surface of osteoclasts. This prevents RANK receptor activation and subsequent bone resorption.

Selective estrogen receptor modulators (SERMs) - Estrogen has a protective effect that prevents accelerated bone loss. This is carried out by reducing the bone resorption effects of parathyroid hormone (PTH). Keeping in line with these effects, osteoporosis prevention in postmenopausal women has been attributed to the use of SERMs such as raloxifene. It acts as an estrogen agonist on the bone and liver.

Calcium and Vitamin D supplementation: Calcium supplementation in the elderly suppresses bone turnover and improves bone mineral density. 1200mg of calcium is daily required in adults of more than 50 years of age. Vitamin D supplementation with 400-800IU/day improves intestinal calcium absorption, suppresses bone remodeling, and improves bone mineral density.

Solution to Question 9:

Benzhexol should be used in the management of this patient. The development of abnormal movements following an anti-emetic injection suggests that the antiemetic used in this case is probably metoclopramide.

Metoclopramide is a dopamine receptor antagonist used as a prokinetic and anti-emetic agent. It can produce extrapyramidal side effects and dystonia is seen acutely following intravenous injections. Anticholinergics like benzhexol (trihexyphenidyl) are used in the management of these side effects. Benzhexol is a centrally-acting anticholinergic agent. It acts by restoring the cholinergic balance in the striatal interneurons.

Extrapyramidal side effects of metoclopramide are commonly seen in children and young adults with higher doses. Acute dystonia occurs after intravenous administration, while parkinsonian-like symptoms develop weeks after therapy. It is managed with the discontinuation of metoclopramide and with anticholinergics or antihistaminics. Irreversible tardive dyskinesia can occur with chronic treatment.

Other adverse effects:

It can cause galactorrhea by its dopamine antagonism. Methemoglobinemia can occur if used in infants.

Other options:

Option A: Hyoscine (scopolamine): It is a muscarinic receptor antagonist used in the management of motion sickness and vomiting.

Option B: Cyproheptadine is an H₁ receptor antagonist used in the management of various allergy-related conditions like allergic rhinitis, anaphylaxis, urticaria, etc.

Option C: Levodopa is the precursor of dopamine and it is given in combination with carbidopa for the management of parkinsonism.

Solution to Question 10:

The formula for drug dosing rate is clearance x plasma concentration.

The equation for dose rate given here is based on steady-state concentration. When a drug is given repeatedly at short intervals it will accumulate in the body until the elimination of the drug matches its input. So steady-state concentration is the plasma concentration at which the drug elimination and drug input balance each other.

Steady-state concentration(C_{pss}) = Dose rate / Clearance

From this equation dose rate for a target steady-state plasma concentration is calculated as

Dose rate = Clearance(CL) x target plasma concentration(C_{pss})

Solution to Question 11:

Elevated serum digoxin levels and the development of atrial fibrillation with controlled ventricular rate are suggestive of digoxin toxicity. Clarithromycin when given along with digoxin can cause digoxin toxicity.

P-glycoprotein pumps play a major role in the elimination of digoxin. Drugs that inhibit P-glycoprotein can thereby decrease the elimination of digoxin and lead to digoxin toxicity. Macrolide antibiotics like clarithromycin and erythromycin are inhibitors of the P-glycoprotein. So clarithromycin when given along with digoxin can cause digoxin toxicity.

Other drugs that inhibit P-glycoprotein are - Amiodarone, quinidine, verapamil, diltiazem, cyclosporine, itraconazole, propafenone, and flecainide.

Digoxin is one of the cardiac glycosides which is derived from the digitalis plant and used clinically.

It inhibits Na⁺/K⁺ ATPase, a membrane-bound transporter to increase intracellular sodium. As a result, there is less expulsion of calcium from the cell by sodium-calcium exchanger (NCX). Increased calcium inside the myocyte leads to an increase in cardiac contractility.

The most common cardiac manifestations of digitalis toxicity include atrioventricular junctional rhythm, premature ventricular depolarizations, ventricular bigeminy, ventricular tachycardia, and second-degree AV blockade.

Factors that increase the risk of digoxin toxicity are:

- Hypercalcemia

- Hypomagnesemia
- Hypokalemia- It increases the binding of the digoxin to the Na⁺/K⁺ ATPase pump.
- Drugs causing hypokalemia- loop and thiazide diuretics, amphotericin B, corticosteroids
- Drugs that decrease digoxin elimination (inhibitors of P-glycoprotein)
- Renal failure
- Hypothyroidism

Triamterene, atenolol, and potassium chloride cause hyperkalemia, and therefore do not cause digoxin toxicity.

Solution to Question 12:

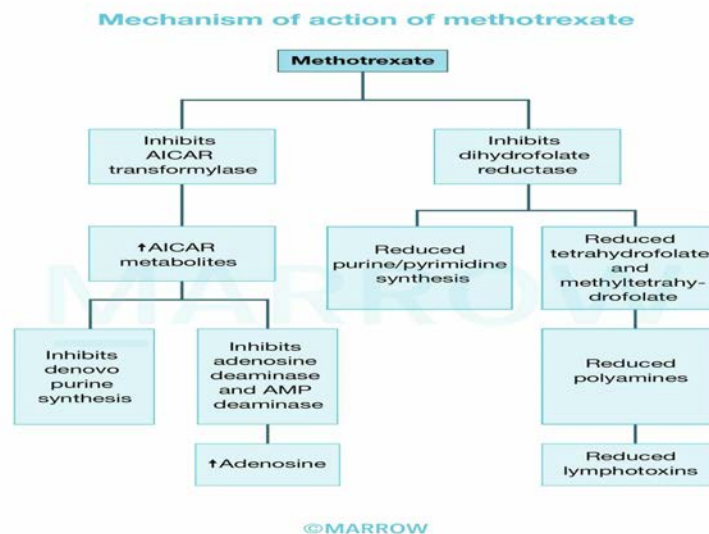
The anticancer activity of methotrexate is by inhibition of dihydrofolate reductase (DHFR).

Methotrexate belongs to the antimetabolite class of anticancer drugs. The antifolate action of the drug is responsible for its anti-cancer properties.

Mechanism of action:

It is a folic acid analog and binds to the active catalytic site of dihydrofolate reductase(DHFR). This would inhibit the formation of tetrahydrofolate(THF) which is needed for purine nucleotides, thymidylate, and the amino acids serine and methionine. This would interfere with the production of DNA, RNA, and important cellular proteins.

Low doses of methotrexate are used in the management of rheumatic diseases. It is due to the inhibition of amino-imidazole carboxamide ribonucleotide (AICAR) transformylase and thymidylate synthetase. This results in increased levels of adenosine that inhibits inflammation.



Adverse effects:

Common side effects are nausea and mucosal ulcers. Others include anemia, leukopenia, stomatitis, GI ulcerations, and alopecia. Dose-related hepatotoxicity is also seen. The GI effects and hepatotoxicity can be reduced by the use of leucovorin or folinic acid.

Solution to Question 13:

Alcohol should not be consumed by a patient on metronidazole. Metronidazole produces a disulfiram-like reaction. If the patient consumes alcohol during or within 3 days of therapy, they may experience abdominal distress, vomiting, headache, and flushing.

Disulfiram and disulfiram-like reactions:

Disulfiram inhibits the aldehyde dehydrogenase (ALDH) activity and thereby increases the concentration of acetaldehyde in the blood. When a person treated with disulfiram consumes alcohol he or she would develop symptoms of acetaldehyde poisoning. The symptoms are:

- Flushing
- Pulsating headache
- Nausea and vomiting
- Sweating
- Tachycardia
- Hypotension
- Orthostatic syncope
- Vertigo
- Blurred vision

Certain drugs can also produce disulfiram-like reactions if alcohol is consumed along with these drugs. Patients should be warned not to use alcohol along with these drugs. The drugs include:

- Metronidazole
- Tinidazole
- Cefoperazone
- Sulfonylureas

Solution to Question 14:

Acetaminophen can be given as an analgesic for the patient as it is the least nephrotoxic among the given options.

Analgesia in CKD patients:

Acetaminophen is considered a safer non-narcotic analgesic in chronic kidney disease (CKD) patients. For more severe pains opioid analgesics like tramadol or fentanyl are used.

Even though acetaminophen is considered safe, with chronic high dosage it can also become nephrotoxic.

NSAIDs like naproxen, indomethacin, diclofenac, etc are not used in CKD patients because of their direct nephrotoxic effects. In healthy normal individuals, NSAIDs have little effect on renal function and blood pressure. But in conditions where the regulatory systems are impaired as hypovolemia, congestive heart failure, hepatic cirrhosis, chronic kidney disease, or other conditions where the renin-angiotensin system is activated, renal function by the formation of prostaglandins are very important.

NSAIDs cause impairment of prostaglandin-induced inhibition of chloride reabsorption and antidiuretic hormone action. This will cause salt and water retention. It also inhibits COX-2 derived prostaglandins which regulate the renal medullary blood flow and would lead to a rise in blood pressure, increased cardiovascular thrombotic events, and cardiac failure. It can also cause electrolyte abnormalities like hyponatremia and hyperkalemia.

Solution to Question 15:

The antihypertensive that can be given in this case is chlorthalidone. The patient in this case is having renal insufficiency but there is no proteinuria mentioned in the question stem. So among the given options, the thiazide diuretic chlorthalidone can be administered as antihypertensive in this patient.

Management of hypertension in chronic kidney disease(CKD):

ACE inhibitors or ARBs are considered the first-line drugs in patients with proteinuria. These agents effectively reduce blood pressure, proteinuria and prevent kidney decline. Calcium channel blockers can be added if necessary.

For patients who do not have proteinuria, there is no established first-line therapy. One or two of the following drugs can be used:

- Thiazides/thiazide-like diuretics
- ACE inhibitors or ARBs
- Calcium channel blockers

In patients presenting with fluid retention or fluid overload, thiazide diuretics such as chlorthalidone and hydrochlorothiazide are recommended for management. In CKD stages 1 to 3 (GFR $\geq$ 30ml/min), thiazides are recommended, while loop diuretics such as bumetanide, furosemide are used in stages 4-5 (GFR $<$ 30ml/min).

Other options:

Option A: Aliskiren is a direct renin inhibitor. It effectively lowers blood pressure, reduces albuminuria, and limits left ventricular hypertrophy.

Option C: Amlodipine is a calcium channel blocker. A common side effect of amlodipine is the production of peripheral edema.

Option D: Prazosin is a selective α_1 antagonist. It is generally not used as an initial agent to treat hypertension, except in symptomatic prostatism or nightmares linked to posttraumatic stress disorder. Headache is one of the side effects of prazosin.

Solution to Question 16:

Tocilizumab inhibits IL-6.

Tocilizumab is a newer humanized antibody. It binds to the soluble, membrane-bound IL-6 receptors and inhibits IL-6 mediated signaling. IL-6 can cause T-cell activation, acute-phase protein synthesis, and stimulation of inflammatory processes in diseases like rheumatoid arthritis.

Uses:

- Rheumatoid arthritis - It is a biological DMARD and is indicated in adult patients having moderate to severe disease with inadequate response to one or more DMARDs.
- Juvenile idiopathic arthritis - It can be given for active systemic juvenile idiopathic arthritis(SJIA) or polyarticular juvenile idiopathic arthritis(PJIA) in patients of 2 years of age or more.

Adverse effects:

Common adverse effects include upper respiratory infections, hypertension, headache, and elevation of hepatic enzymes. It is also associated with serious infections including TB and opportunistic infections. Neutropenia and reduction in platelet count can also occur.

Other options:

Option B- IL-2 receptor inhibitors are daclizumab, basiliximab. They are used as immunosuppressive agents following renal and heart transplants.

Option C- IL-4 receptor inhibitor is dupilumab. They have been shown to reduce exacerbations in asthma.

Option D- IL-5 inhibitors are mepolizumab, reslizumab, benralizumab. They have been shown to reduce exacerbations in asthma.

Pharmacology INI-CET 2022

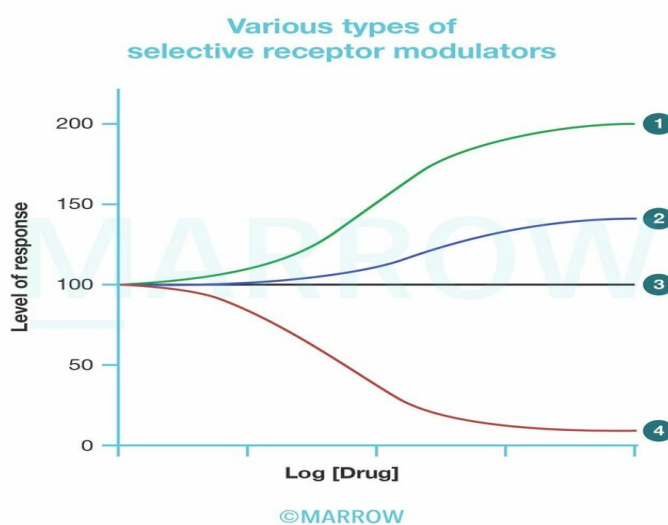
Question 1:

The extent of drug absorption is represented by:

- a) T_{max}
- b) Area under the curve
- c) C_{max}
- d) $t_{1/2}$

Question 2:

Which of the following statement is correct regarding the given graph?



- a) Drug 1 represents agonist and drug 2 represents inverse agonist
- b) Drug 3 represents agonist and drug 4 represents inverse agonist
- c) Drug 1 represents agonist and drug 4 represents inverse agonist
- d) Drug 2 represents partial agonist and drug 3 represents inverse agonist

Question 3:

Which among the following drugs is the new FDA approved immune check point inhibitor for carcinoma endometrium?

- a) Nivolumab
- b) Pembrolizumab
- c) Trastuzumab
- d) Ipilimumab

Question 4:

Which of the following is not an immune checkpoint inhibitor?

- a) Nivolumab
- b) Pembrolizumab
- c) Atezolimumab
- d) Cetuximab

Question 5:

NAPQI is the metabolite of which of the following drug toxicity?

- a) Aspirin
- b) Paracetamol
- c) Diclofenac
- d) Ketorolac

Question 6:

Identify the correct match, regarding the drug and its adverse effect.

- a) Hydralazine - heart failure
- b) Atenolol - hemolytic anemia
- c) Aliskiren - hypokalemia
- d) Verapamil - constipation

Question 7:

A lady who had undergone mastectomy for breast cancer is being treated with tamoxifen. How long should it be stopped for before she can conceive?

- a) 1 month
- b) 3 months
- c) No need to stop
- d) Can be stopped and conceived soon after stopping

Question 8:

A young boy with a history of multiple blood transfusions presents with bone pain for the past 6 months. A peripheral smear showed sickle-shaped RBCs. Which of the following drugs are not used in the treatment of his condition?

- a) Bebtelovimab
- b) Voxelotor
- c) Hydroxyurea
- d) L-glutamine

Question 9:

The half-life of letrozole is:

- a) 45 hours
- b) 70 hours
- c) 90 hours
- d) 120 hours

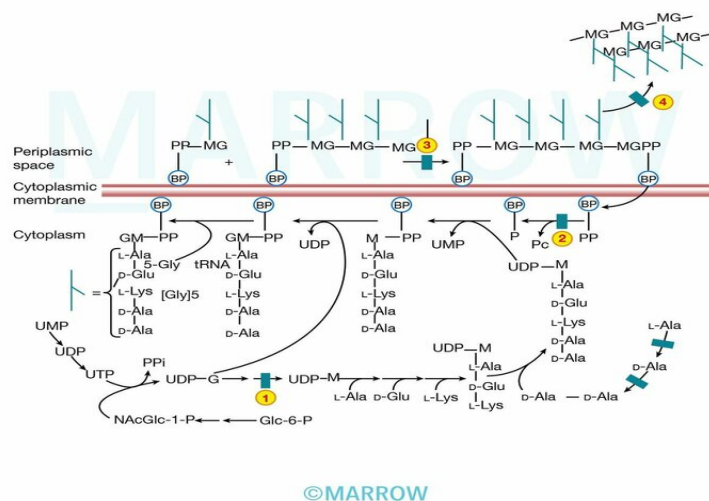
Question 10:

Identify the false statement about eicosanoids.

- a) Synthesized from arginine
- b) Synthesized from arachidonic acid
- c) Prostaglandins have 5 carbon ring
- d) Synthesized in platelets

Question 11:

The given diagram shows the steps in bacterial cell wall formation. Identify the site of action of beta-lactam antibiotics.



- a) 1
b) 2
c) 3
d) 4

Question 12:

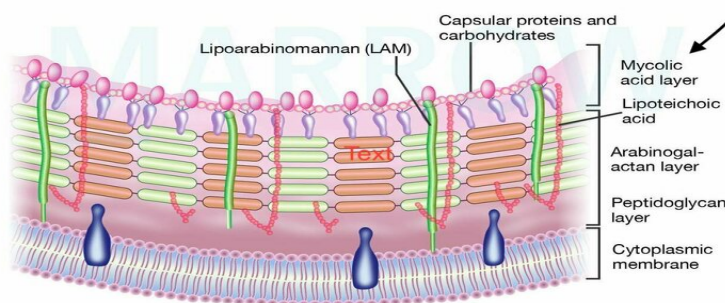
Phospholipase C acts via which of the following G protein-coupled receptors?

- a) Gs
b) Gi
c) Gq
d) Go

Question 13:

A patient was diagnosed with pulmonary tuberculosis and was started on first-line antitubercular drugs. Which of the following drug acts by inhibiting the synthesis of the marked layer shown in the image?

Mycobacterial cell wall structure



©MARROW

- a) Isoniazid
- b) Rifampicin
- c) Pyrazinamide
- d) Ethambutol

Question 14:

A man was brought to the emergency room after poisoning with an unknown substance. Muscarinic poisoning was suspected and he was treated for the same. What is the possible presenting feature which led to the diagnosis?

- a) Mydriasis
- b) Diuresis
- c) Muscle fasciculations
- d) Bradycardia

Question 15:

A 29-year-old lady came to psychiatry OPD with symptoms of hypomania. There is a past history of maniac episode. Now, she is planning to conceive. Which drug should be avoided for being highly teratogenic to the fetus?

- a) Valproate
- b) Lithium
- c) Oxcarbazepine

d) Olanzapine

Question 16:

True statement regarding the administration of 5 alpha-reductase inhibitors is/are:

- a) 1, 2, 3
- b) 1, 3, 4
- c) 1, 4
- d) 2, 3

Question 17:

The manufacturer of a drug company labels a drug as having 500 mg paracetamol. On quantitative analysis of the drug, it actually contained only 200 mg of paracetamol. What is this type of drug called?

- a) Spurious drug
- b) Adulterated drug
- c) Unethical drug
- d) Misbranded drug

Question 18:

Which of the following anti-diabetic drugs is associated with the increased risk of fractures?

- a) Rosiglitazone
- b) Liraglutide
- c) Voglibose
- d) Canagliflozin

Question 19:

The active metabolite of cyclophosphamide is:

- a) N - hydroxy cyclophosphamide
- b) 4 - hydroxy cyclophosphamide

- c) 4 - acetyl cyclophosphamide
- d) N - acetyl cyclophosphamide

Question 20:

Urinary alkalization will be most useful in toxicity of which of the following drugs?

- a) Beta blockers
- b) Calcium channel blockers
- c) Acetaminophen
- d) Aspirin

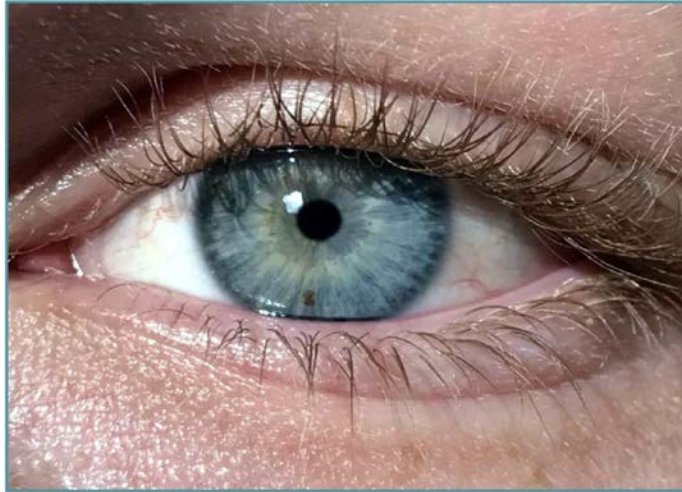
Question 21:

A hospital is conducting a study to check the efficacy of statins in multiple centers. There are 4-5 patients per center and half of the patients receive the drug and the other half receive the placebo. Neither the patients nor clinicians know which drug is being administered. What phase of the clinical trial describes the above scenario?

- a) Phase 1
- b) Phase 3
- c) Phase 2
- d) Phase 4

Question 22:

A patient who is a known addict is brought to the hospital with the intoxication of an agent. An ocular examination shows the following finding. Which of the following is true regarding this condition?



- a) This action can be reversed by phenylephrine.
- b) It cannot be reversed by levallorphan.
- c) It is caused by activation of the Edinger-Westphal nucleus.
- d) Tolerance develops due to long-term use.

Question 23:

Match the following drugs in Column A with their contraindications in Column B.

- a) A-1, B-3, C-2, D-4
- b) A-4, B-1, C-3, D-2
- c) A-2, B-4, C-1, D-3
- d) A-3, B-2, C-4, D-1

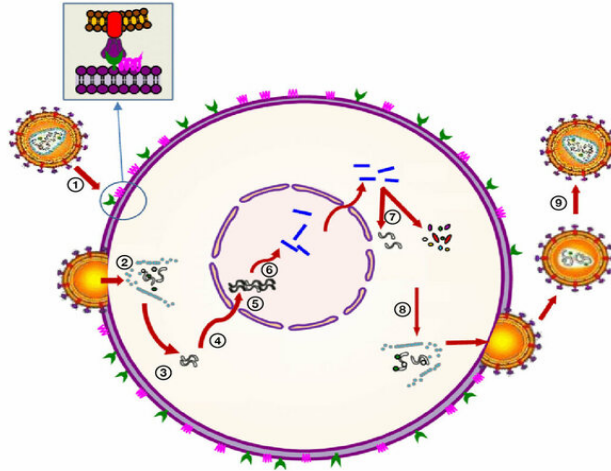
Question 24:

Which of the following statements is true regarding memantine?

- a) 1, 2, 3, 4
- b) 1, 2
- c) 3, 4
- d) 1, 2, 3

Question 25:

The image below shows the life cycle of a virus. Which of the following drugs acts at step 5?



- a) Raltegravir
- b) Enfuvirtide
- c) Tenofovir
- d) Zidovudine

Question 26:

Which of the following statements is correct regarding clinical trials?

- a) New drug application is done after phase 4
- b) Phase 4 is also called post-marketing surveillance
- c) Phase 3 is always double-blinded
- d) Phase 2 is done on healthy individuals

Question 27:

Match the following teratogenic drugs with their teratogenic effects.

- a) 1-a, 2-b, 3-c, 4-d
- b) 1-c, 2-d, 3-a, 4-b
- c) 1-d, 2-a, 3-d, 4-c

d) 1-d, 2-c, 3-b, 4-a

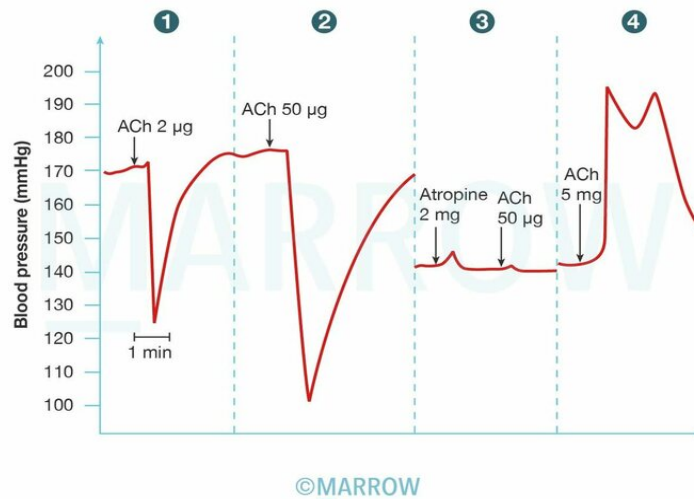
Question 28:

A young man sustained a scorpion sting 2 hours ago. On examination, his blood pressure is 108/ 60 mm Hg; HR, 72 beats/minute, and respiratory rate, 32 breaths/minute. He also complains of breathlessness. How do you manage this patient?

- a) Metoprolol
- b) Ivabradine
- c) Prazosin
- d) Frusemide

Question 29:

An experiment is being conducted on a cat spine to study the effect of acetylcholine. The blood pressure is recorded at each intervention and is shown below. What does phase 4 in the graph indicate?



- a) Muscarinic action of acetylcholine
- b) Potentiation of acetylcholine
- c) Antagonism of acetylcholine
- d) Nicotinic action of acetylcholine

Question 30:

A patient who is a known case of HIV is started on combined active antiretroviral therapy (cART). After 5 months of treatment, he develops hyperpigmentation in the palms and soles. What is the most likely drug causing this?

- a) Nelfinavir
- b) Emtricitabine
- c) Abacavir
- d) Zidovudine

Question 31:

Dorzolamide is a _____.

- a) Topically applied sulfonamide antibacterial
- b) Topical ocular carbonic anhydrase inhibitor
- c) Second-generation sulfonylurea
- d) Opioid receptor agonist

Question 32:

A patient with erectile dysfunction is given sildenafil. The penile erectile function is found to be improved after the therapy. Which of the following mediates the action of sildenafil?

- a) cAMP
- b) cGMP
- c) Calcium
- d) Phosphatidyl glycerol and inositol phosphate

Question 33:

Match the following diuretics in column A to their site of action in column B.

- a) A-2, B-3, C-1, D-4
- b) A-3, B-4, C-2, D-1
- c) A-1, B-2, C-4, D-3
- d) A-4, B-1, C-3, D-2

Question 34:

Alkalinization of urine will be helpful in the removal of _____.

- a) Weakly basic drugs
- b) Strong electrolytes
- c) Polar drugs
- d) Weakly acidic drugs

Question 35:

All of the following side effects are more commonly seen with carbamazepine than oxcarbazepine, except_____.

- a) Hypotension
- b) Hyponatremia
- c) Rash
- d) Elevated levels of gamma glutamyl transpeptidases

Question 36:

Match the following drugs with the targets of their actions.

- a) A-2, B-3 , C-1 , D-4
- b) A-4, B-3 , C-1 , D-2
- c) A-3, B-4 , C-2 , D-1
- d) A-4, B-3 , C-2 , D-1

Question 37:

A patient is brought with altered sensorium and seizures. Attender shows the laboratory report which reveals elevated transaminases. Which of the following can be used in this condition?

- a) Midazolam
- b) Diazepam
- c) Alprazolam

d) Oxazepam

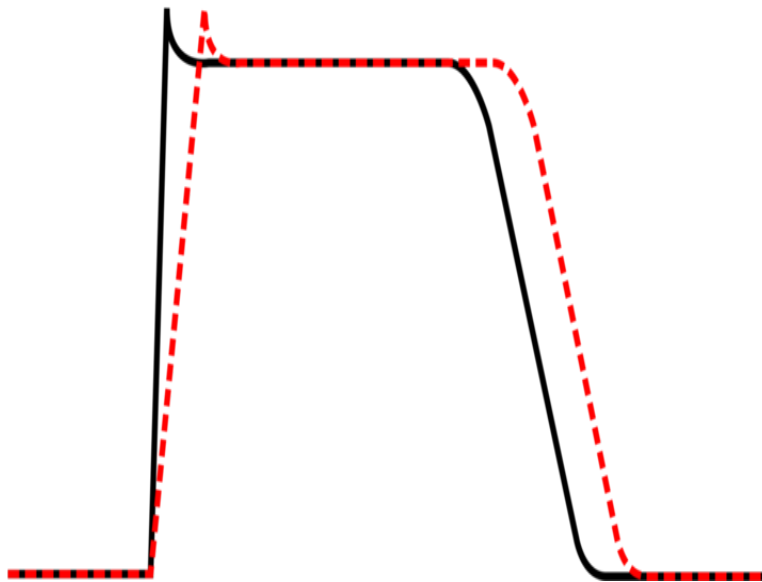
Question 38:

The lipid profile of a patient who suffered from acute coronary syndrome is given below.
What drug can be used to treat this condition?

- a) Fenofibrate 160 mg
- b) Rosuvastatin + Fenofibrate
- c) Rosuvastatin 10 mg
- d) Atorvastatin 80 mg

Question 39:

An anti-arrhythmic drug is shown to have the following effect. It is effective in both atrial and ventricular arrhythmias. Identify the drug.



- a) Tocainide
- b) Propranolol
- c) Quinidine
- d) Lignocaine

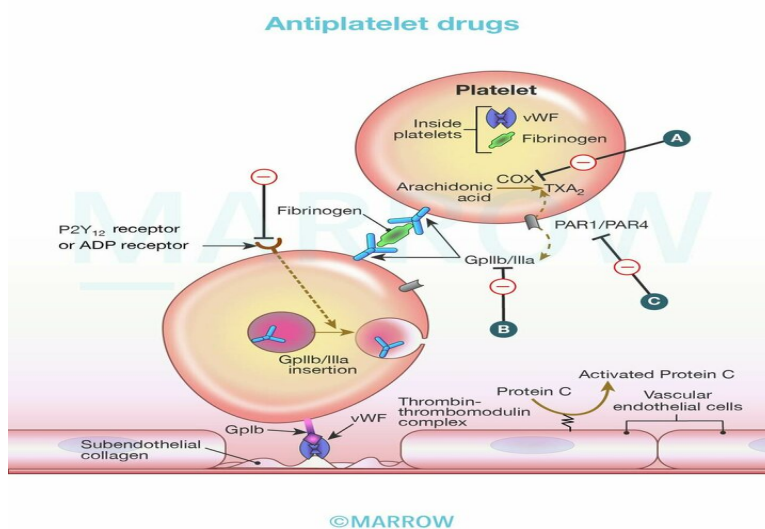
Question 40:

Which among the following is the false statement regarding statins?

- a) Although HMG-CoA reductase inhibitors substantially reduce the risk of cardiovascular events, there is mild increase in lipoprotein a (Lpa) levels.
- b) These drugs should not be stopped even in severe conditions like injury, surgery, etc.
- c) They can be given with verapamil and other enzyme inhibitors.
- d) With the long term use, there is slight increase in the incidence of type 2 diabetes mellitus.

Question 41:

Identify the drugs marked as A, B, and C.



- a) A - Aspirin, B - Vorapaxar, C - Prasugrel
- b) A - Aspirin, B - Eptifibatide, C - Vorapaxar
- c) A - Aspirin, B - Eptifibatide, C - Prasugrel
- d) A - Aspirin, B - Prasugrel, C - Eptifibatide

Question 42:

Which of the following is false regarding disulfiram?

- a) Only 3
- b) 1, and 3
- c) 1, 3, and 4
- d) 2, 3, and 4

Question 43:

G6PD can manifest with which of the following drugs?

- a) Primaquine
- b) Chloroquine
- c) Penicillin
- d) Tetracycline

Question 44:

Choose the correctly matched pairs regarding the drugs used in schizophrenia:

- a) 1,3
- b) 1,4
- c) 1,2,4
- d) 1,2,3,4

Question 45:

A patient presented with a lesion, as shown in the image below. She is an immunocompromised patient currently on HAART. What are the systemic antifungal drugs that can be added along with the HAART in this patient?



- a) Inj. Amphotericin B 4-8 weeks, followed by oral itraconazole
- b) Inj. Amphotericin B 8 weeks, followed by oral posaconazole
- c) Inj. Fluconazole, followed by oral fluconazole
- d) Inj. Amphotericin B, followed by flucytosine

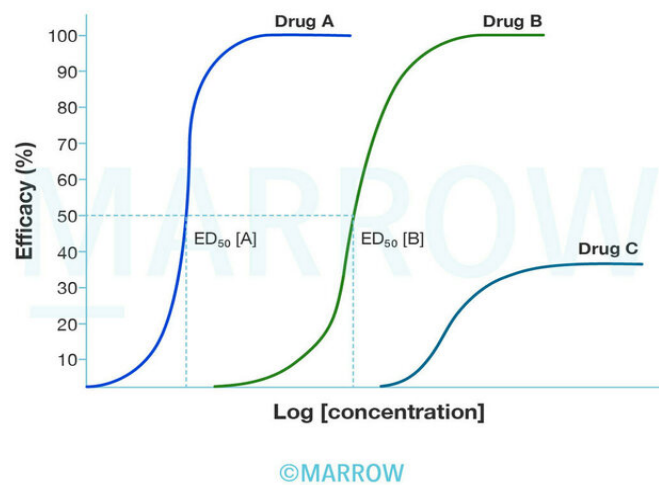
Question 46:

A patient who is on treatment for hyperlipidemia develops gallstones. What is the mechanism of action of the causative drug that was given to this patient?

- a) Inhibits HMG CoA reductase
- b) Activates PPAR alpha
- c) Binds to deoxycholic acid
- d) Decreases VLDL

Question 47:

The dose-response curve for three drugs in the same tissue is shown in the image below. Which of the following statements is true?



- a) Efficacy of drug A is more than that of drug B.
- b) Potency of drug B is more than that of drug A.
- c) Drug A & drug B are agonists.
- d) Drug C is a non-competitive antagonist.

Question 48:

Which of the following drugs is a β_2 agonist?

- a) 1, 2, 3
- b) 4, 3, 2, 1
- c) 1, 2
- d) 3, 4

Question 49:

What is the mechanism of action of sildenafil?

- a) Inhibition of 5α -reductase
- b) Inhibition of phosphodiesterase-5
- c) Release of nitrous oxide radicals
- d) Blockage of voltage-gated calcium channels

Question 50:

Choose the correct options regarding the route of administration and bioavailability.

- a) A,B,D
- b) A,C,D
- c) A and C
- d) A and D

Question 51:

A hypertensive patient on medication presents with the lab values as shown below. Which of the following drugs should be stopped?

- a) Amlodipine
- b) Metoprolol
- c) Losartan
- d) Furosemide

Question 52:

Which of the following is the most effective oral drug in smoking cessation?

- a) Varenicline
- b) Cevimeline
- c) Muscarine
- d) Atropine

Question 53:

Match the following antiarrhythmic drugs with their mechanism of action:

- a) 1-d, 2-b, 3-a, 4-c
- b) 1-a, 2-c, 3-d, 4-b
- c) 1-a, 2-d, 3-b, 4-c
- d) 1-d, 2-c, 3-a, 4-b

Question 54:

Which of the following drugs increases trabecular outflow?

- a) Timolol
- b) Netarsudil
- c) Brimonidine
- d) Acetazolamide

Question 55:

Finasteride is used to treat male-pattern baldness acts by inhibiting which of the following enzymes?

- a) Aromatase
- b) 17-hydroxylase
- c) 5-alpha reductase
- d) Phosphodiesterase

Question 56:

Rituximab is a:

- a) Polyclonal antibody which acts against CD20
- b) Polyclonal antibody which acts against CD50
- c) Monoclonal antibody which acts against CD20
- d) Monoclonal antibody which acts against CD50

Question 57:

A new drug application is filed after which phase of clinical trials?

- a) 4
- b) 3
- c) 2
- d) 1

Answer Key

Question No.	Correct Option
1	b
2	c
3	b
4	d
5	b
6	d
7	b
8	a
9	a
10	a
11	d
12	c
13	a

14	d
15	a
16	c
17	d
18	d
19	b
20	d
21	b
22	c
23	b
24	b
25	a
26	b
27	d
28	c
29	d
30	b
31	b
32	b
33	d
34	d
35	b
36	d
37	d
38	d
39	c
40	c
41	b
42	a
43	a
44	c
45	b
46	b
47	c
48	c
49	b
50	b

51	c
52	a
53	c
54	b
55	c
56	c
57	b

Detailed Explanations

Solution to Question 1:

The extent of drug absorption is represented by the area under the curve.

The rate and extent of absorption of a drug from dosage form administered by any route refer to the bioavailability of a drug. Bioavailability is determined by the concentration-time curve in blood or by its excretion in urine. The area under the blood concentration-time curve (AUC) is proportional to the dose and the extent of bioavailability of a drug.

Bioavailability is defined as the fraction of unchanged drug that reaches systemic circulation. It can be represented by the formula:

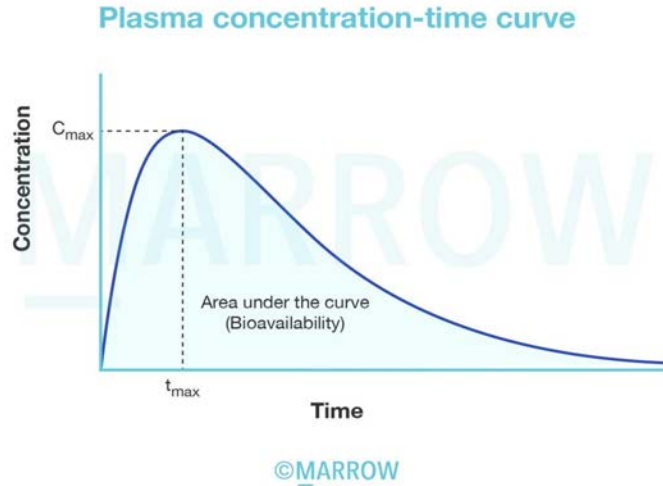
Bioavailability(F) = Quantity of drug reaching systemic circulation/ Quantity of drug administered.

Bioavailability depends on two factors:

- Absorption
- 1st pass metabolism

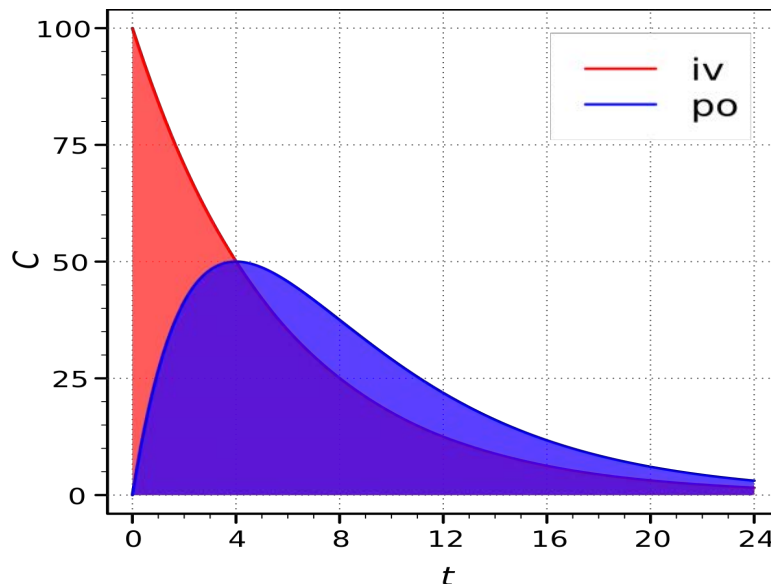
The area under the plot of plasma concentration of a drug versus time after dosage is called the area under the curve or AUC. It can be used as a measure of the bioavailability of a drug and to calculate its clearance rate from the body.

The image given below shows the AUC (bioavailability) in a plasma concentration-time curve:



Bioavailability of drug injected intravenously is 100%, but it is lower after oral ingestion because of incomplete absorption and first-pass metabolism in the intestinal wall/liver.

The image below shows the difference between the bioavailability of drugs injected intravenously(iv) and orally(PO). Note the area under the respective curves is shaded in red and blue colours which represent the absorption of the drug.



Solution to Question 2:

The correct statement is Drug 1 represents agonist and drug 4 represents inverse agonist.

Graph 1 represents the agonist: An agonist is an agent which initiates the normal physiological response when combined with a receptor.

Example: Salbutamol is a beta-2 receptor agonist.

Graph 2 represents a partial agonist: An agent which produces a submaximal or partial effect when combined with a receptor.

Example: Buprenorphine and butorphanol are partial agonists at opioid receptors.

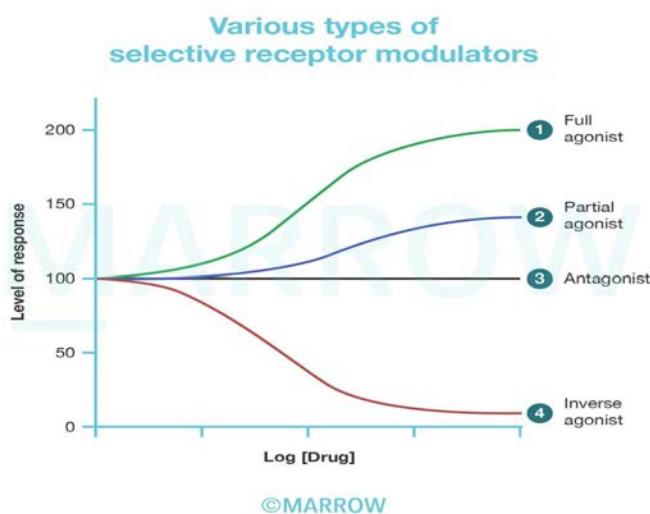
Graph 3 represents the antagonist: An agent which blocks or reduces the action of an agonist on a receptor, but does not have any effect of its own.

Example: Naltrexone and naloxone are antagonists at opioid receptors

Graph 4 represents an inverse agonist: An agent which activates a receptor to produce an effect in the opposite direction to that of the agonist.

Example: β -Carboline is an inverse agonist at benzodiazepine site of GABA-A receptor.

The image given below shows action of various types of selective receptor modulators.



Solution to Question 3:

The new FDA-approved immune checkpoint inhibitor for carcinoma endometrium is pembrolizumab.

Pembrolizumab acts by inhibiting programmed death receptor - 1 (PD-1) and was used for the treatment of advanced melanoma. Now, the FDA has approved the use of pembrolizumab for any cancer with mismatch repair (MMR) deficiency or high microsatellite instability (MSI-H).

On March 21, 2022, the food and drug administration approved pembrolizumab, as a single agent, for patients with advanced endometrial carcinoma that is microsatellite instability-high (MSI-H) or mismatch repair deficient (dMMR), as determined by an FDA-approved test.

Other options:

Option A: Nivolumab is an immune checkpoint inhibitor that acts by inhibiting programmed death receptor - 1 (PD-1) and is used for the treatment of advanced melanoma and relapsed/refractory classical Hodgkin lymphoma.

Option C: Trastuzumab is a monoclonal antibody that inhibits the HER2 receptor. It is used in the treatment of HER2-positive breast cancer.

Option D: Ipilimumab is a monoclonal antibody that binds to cytotoxic T lymphocyte-associated Protein 4 (CTLA-4) and is approved for the treatment of late-stage melanoma.

Solution to Question 4:

Cetuximab is not an immune checkpoint inhibitor. Cetuximab is a monoclonal antibody against epidermal growth factor receptor (EGFR). It is used for the treatment of colon cancer and head and neck squamous cell cancer.

Other options.

Option A: Nivolumab is an immune checkpoint inhibitor. It acts by inhibiting programmed death receptor - 1 (PD-1) and is used for the treatment of advanced melanoma and classical Hodgkin lymphoma.

Option B: Pembrolizumab is an immune checkpoint inhibitor. It acts by inhibiting programmed death receptor - 1 (PD-1) and is used for the treatment of advanced melanoma. In 2021, approval was granted for the treatment of patients with endometrial carcinoma that progressed after systemic treatment, in combination with lenvatinib.

Option C: Atezolizumab is an immune checkpoint inhibitor. It acts by inhibiting programmed death-ligand - 1 (PDL-1) and is used for the treatment of non-small cell lung cancer.

Important note:

The FDA on May 2021 approved the use of nivolumab in patients having completely resected esophageal or gastroesophageal junction cancers with residual pathologic disease who have received neoadjuvant chemoradiotherapy.

Solution to Question 5:

N-acetyl-p-benzoquinone imine (NAPQI) is a metabolite of paracetamol drug toxicity.

Paracetamol is an analgesic and antipyretic with weak anti-inflammatory action. It is mainly metabolized in the liver by conjugation with glucuronic acid and sulfate and is excreted in the urine.

At toxic doses, the glucuronide and sulfate conjugation pathway becomes saturated. The metabolites of paracetamol get converted into NAPQI. NAPQI is highly reactive and causes enzymatic dysfunction in the liver. Normally, minor amounts of NAPQI produced in our body are rapidly eliminated by conjugation with reduced glutathione (GSH) and then excreted into the urine.

Therefore, paracetamol toxicity is treated with N-acetyl cysteine (NAC), which replenishes GSH. NAC thereby helps in the excretion of NAPQI from the body. N-acetyl cysteine is an anti-oxidant and functions by detoxifying NAPQI.

Paracetamol poisoning is the most common cause of liver failure necessitating transplantation.

The maximum dose of acetaminophen that can be given is 4 g/day.

Solution to Question 6:

The correct match is verapamil causes constipation.

Verapamil is a calcium channel blocker used for the treatment of paroxysmal supraventricular tachycardia, atrial fibrillation, and atrial flutter. Bradycardia and constipation are the important side effects of verapamil.

Other options:

Option A: Hydralazine is a vasodilator used in the treatment of heart failure, and hypertensive emergencies, especially preeclampsia. The drug is used as a combination pill containing isosorbide dinitrate (BiDil) in the treatment of heart failure. The important side effect of hydralazine is drug-induced lupus.

Option B: Atenolol is a cardioselective beta-1 blocker used in the treatment of angina and hypertension. Atenolol doesn't cause hemolytic anemia.

Option C: Aliskiren is a direct renin inhibitor used as an alternative antihypertensive drug for those who do not respond or do not tolerate 1st line drugs. The side effect of aliskiren is hyperkalemia and not hypokalemia.

Solution to Question 7:

In a post-mastectomy patient undergoing therapy with tamoxifen, it should be stopped for 3 months before she can conceive.

Tamoxifen is a selective estrogen receptor modulator (SERM) and is used as a chemotherapeutic agent in receptor positive breast cancer patients to prevent recurrence after surgery. It is the drug of choice in pre-menopausal women. It has an agonistic action on the endometrium, leading to an increased risk of endometrial cancers (2 fold risk). Hence, it is not used in post-menopausal women.

It is contraindicated in pregnancy due to teratogenic effects such as craniofacial abnormalities and reproductive tract abnormalities. For the same reason, it has to be stopped for 3 months (due to long half-life) prior to conceiving. It is also stopped during lactation as it is secreted in breast milk. Other complications of tamoxifen include deep vein thrombosis, pulmonary embolism, and hot flashes.

Other breast cancer risk reduction techniques:

- Lifestyle modification: Low-calorie diet, daily exercise, no smoking, no drinking, etc.
- More frequent screening with mammogram +/- MRI
- Chemoprevention: Preferred in women $\geq$ 35 years of age in patients who are ER/PR positive. Given for 5-10 years.

- SERMS (selective estrogen receptor modulators) - Tamoxifen and raloxifene. These have adverse effects.
- Aromatase Inhibitors (AIs): They are used only in post-menopausal women.
- Risk-reduction surgeries: preferred in patients with BRCA1/2 mutations or patients with first-degree relatives with breast cancers. Provides >90% risk-reduction. preferred in patients who have completed their family.
- Bilateral mastectomy
- Salpingo-oophorectomy: Best results when done < 40 years of age.

Solution to Question 8:

Bebtelovimab is not used in the treatment of sickle cell disease. The clinical features of multiple blood transfusions, bone pain, and sickle cells on peripheral smear are suggestive of sickle cell disease.

Bebtelovimab is the new monoclonal antibody used for the treatment of COVID-19. It works by binding to the spike protein of the COVID-19 virus. It has been found useful in reducing the risk of hospitalization or death. Bebtelovimab retains the activity against the omicron variant.

Other options:

Option B: Voxelotor is an inhibitor of hemoglobin S polymerase and stabilizer of sickle cell hemoglobin (HbS) that is used in the therapy of sickle cell disease.

Option C: Hydroxyurea improves fetal haemoglobin (HbF) and is the primary drug for controlling HbS in sickle cell disease. Thus improving the oxygen-carrying capacity of the blood. Hydroxyurea also inhibits the enzyme ribonucleotide reductase (RNR), which is a rate-limiting step in the biosynthesis of DNA. This property of hydroxyurea is used in the treatment of various cancers, and myeloproliferative disorders like chronic myelogenous leukemia, and polycythemia.

Option D: L-glutamine is a naturally occurring amino acid with antioxidant properties. It increases the availability of reduced glutathione which may lessen oxidative damage in sickle red blood cells.

Solution to Question 9:

The half-life of letrozole is 45 hours.

Letrozole is a reversible aromatase inhibitor. It is a non-steroidal compound that inhibits the aromatization of estrogen thereby causing estrogen depletion. Letrozole is rapidly absorbed with 99.9% oral bioavailability and has a $t_{1/2}$ of around 41 hours.

Letrozole is used as the first-line drug for estrogen-positive breast cancer following mastectomy in post-menopausal women.

Adverse effects of letrozole are due to the depletion of estrogen. The commonly seen adverse effects are hot flashes, vaginal dryness, nausea, diarrhea, dyspepsia, accelerated bone loss, and

thinning of hair.

Solution to Question 10:

A false statement about eicosanoids is they are synthesized from arginine. Eicosanoids are synthesized from arachidonic acid (Option B)

Eicosanoids are the most universally distributed autocooids in the body. Prostaglandins, thromboxanes, and leukotrienes comprise eicosanoids. Every cell is capable of synthesizing one or more eicosanoids. Platelets primarily synthesize thromboxanes (Option D).

Eicosanoids are synthesized from arachidonic acid. Arachidonic acid is derived from membrane phospholipids. The release of arachidonic acid (AA) from the membrane phospholipids is mainly mediated by phospholipase A.

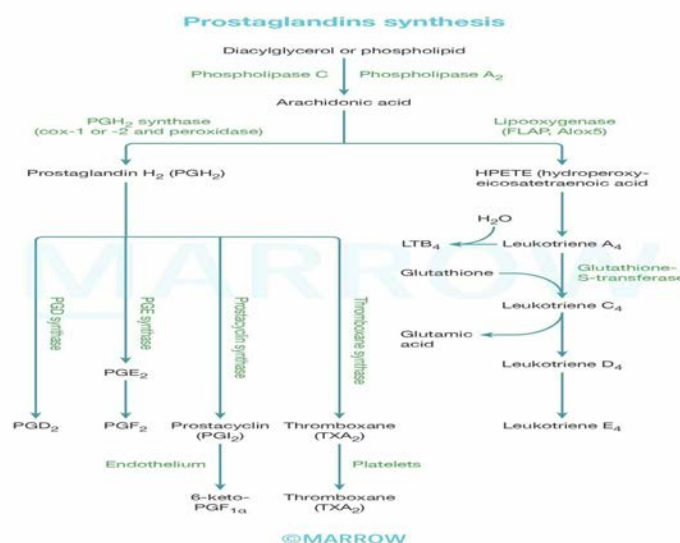
AA is metabolized by cyclooxygenases to produce prostaglandins and lipoxygenases to produce leukotrienes and lipoxins.

Inhibitors of metabolism of arachidonic acid:

- Corticosteroids cause inhibition of prostaglandin synthesis by inhibiting phospholipase A2.
- Non-steroidal anti-inflammatory (NSAIDs) drugs inhibit the synthesis of prostaglandins by inhibiting the enzyme cyclooxygenase.
- Zileuton inhibits the enzyme lipoxygenase thereby decreasing the synthesis of leukotrienes.

Chemically, all prostaglandins are considered to be derivatives of prostanoic acid, having a five-carbon ring (Option C).

The image given below shows the synthesis of various eicosanoids from arachidonic acid:



Solution to Question 11:

Beta lactams act on step 4 in the given image.

Mechanism of action of beta-lactams: They are the cell wall synthesis inhibitor. The β -lactam antibiotics inhibit this last step in peptidoglycan synthesis. They inhibit the enzyme transpeptidase which is required for cross-linking of the peptidoglycan chain. Transpeptidases cleave the terminal D-alanine of the peptide chains and form cross-linkages between them. This cross-linking provides rigidity and stability to the cell wall.

Transpeptidases and other related proteins constitute penicillin-binding proteins (PBP) and, are located in the bacterial cell membrane. β -lactam antibiotics are bactericidal. This is because bacteria swell and burst due to deficient cell walls when they divide in the presence of β -lactam antibiotics. Penicillins, cephalosporins, monobactams, and carbapenems are examples of β -lactam antibiotics.

Gram-positive bacteria are more susceptible to β -lactam antibiotics than gram-negative because their cell wall is almost entirely made of peptidoglycan.

Resistance to β -lactam antibiotics develops due to the production of the beta-lactamase enzyme and the development of altered penicillin-binding proteins like in methicillin-resistant *Staphylococcus aureus* (MRSA). β -lactamases inactivate β -lactam antibiotics by opening the β -lactam ring. This could be overcome by using drugs that inhibit this enzyme. Examples of β -lactamase inhibitors are clavulanic acid, sulbactam, and tazobactam.

Other options:

Option A: Fosfomycin inhibits step 1 in the image. It inhibits the cytoplasmic enzyme enol pyruvate transferase that catalyses the initial step in the bacterial cell wall and hence cell wall synthesis. It is used for the treatment of urinary tract infections.

Option B: Bacitracin inhibits step 2 in the image. It is a cell wall synthesis inhibitor that interferes with dephosphorylation in the cycling of the lipid carrier which transfers peptidoglycan subunits to the growing cell wall. It is only given for topical application because of its severe nephrotoxicity when given systemically.

Option C: Vancomycin inhibits step 3 in the image. It binds to the D-Ala-D-Ala terminus of the newly formed peptidoglycan pentapeptide. This prevents further elongation of the peptidoglycan and its cross-linking. It is used against MRSA and *Clostridium difficile* infection. Redman syndrome due to histamine release following infusion is an important side effect.

Note: The FDA has approved the use of cefepime-enmetazobactam, a combination of a fourth-generation cephalosporin and beta-lactamase inhibitor for use in complicated UTIs (including pyelonephritis).

Solution to Question 12:

Phospholipase C acts via the Gq type of G protein-coupled receptors.

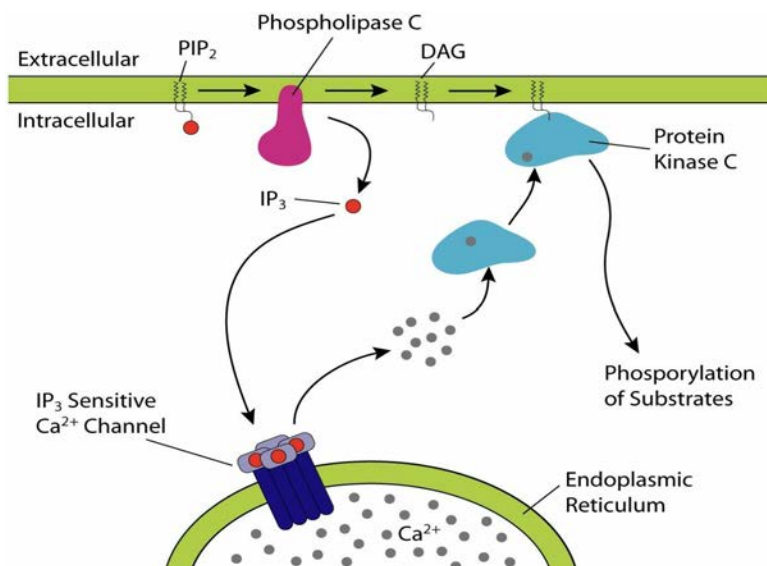
G protein-coupled receptors (GPCRs) are cell membrane receptors that span across the membrane 7 times. They also serve as receptors for various peptide hormones, neurotransmitters, amino acids, etc.

Subtypes of GPCRs are distinguished by their alpha subunits. The important subtypes of GPCRs are Gs, Gp, Gq, and Go.

GPCRs act via various second messengers. The important ones with their action on the effector are:

- Gs causes activation of adenylyl cyclase and results in Ca^{2+} channel opening.
- Gi causes inhibition of adenylyl cyclase and results in K^{+} channel opening.
- Go causes Ca^{2+} channel inhibition.
- Gq causes activation of phospholipase C which generates the second messengers inositol 1,4,5-trisphosphate (IP_3) and diacylglycerol (DAG). The water-soluble IP_3 diffuses to the cytosol and mobilizes Ca^{2+} from endoplasmic reticular depots.

The given below image shows the effects of phospholipase C after activation by Gq



Solution to Question 13:

Isoniazid is the first-line antitubercular drug that acts by inhibiting the synthesis of mycolic acid.

Isoniazid is a first-line bactericidal drug that is used for the treatment of tuberculosis. This drug enters bacilli by passive diffusion. It then gets activated to its toxic form by the enzyme catalase-peroxidase (the enzyme encoded by the Kat gene).

Normally, InhA and KasA are involved in the synthesis of mycolic acid. Upon activation, the reactive metabolite of isoniazid forms complexes with NAD that inhibit InhA and KasA. Hence, isoniazid inhibits the production of mycolic acid.

The reactive isoniazid metabolite also forms complexes with NADP. This complex inhibits mycobacterial dihydrofolate reductase which results in the interruption of DNA synthesis.

Side effects of isoniazid include:

- Peripheral neuritis due to interference with the production of active coenzyme pyridoxal phosphate from pyridoxine. Therefore, pyridoxine 10mg/day (vitamin B6) is given prophylactically along with isoniazid to prevent neurotoxicity.

- Hepatitis - isoniazid must be stopped at the first sign of hepatotoxicity.
- Rashes
- Anaemia
- Arthralgia

Toxicity of isoniazid results in metabolic acidosis, seizures, and coma.

Other options:

Option B: Rifampin acts by binding to DNA-dependent RNA polymerase to form a stable drug-enzyme complex. This complex suppresses RNA synthesis. Side effects of rifampin are hepatitis, flu-like illness, orange-red colored urine, and body secretions.

Option C: Pyrazinamide is a prodrug, it is converted into an active metabolite pyrazinoic acid which makes the intracellular environment acidic and collapses the transmembrane proton motive force, thereby killing the bacteria. Side effects include hyperuricemia and hepatotoxicity.

Option D: Ethambutol inhibits arabinose transferase III, thereby decreasing arabinogalactan biosynthesis. This results in disruption in mycobacterial cell wall synthesis. Optic neuritis is an important side effect of ethambutol. It results in decreased visual acuity and loss of red-green discrimination.

Solution to Question 14:

Bradycardia is the possible presenting feature which led to the diagnosis of muscarinic poisoning.

Muscarinic poisoning occurs due to the consumption of poisonous mushroom-like *Inocybe*. Muscarinic symptoms appear within an hour of eating the mushroom. These symptoms (along with nicotinic symptoms) can also be seen in organophosphate poisoning.

There are 5 subtypes of muscarinic receptors they are M1, M2, M3, M4, and M5. The effects exerted by the M receptors are given below:

- M1: Increased seizure activity.
- M2: Tremors, bronchoconstriction, bradycardia, abdominal cramping
- M3: Increased bladder contraction, miosis, increased salivary secretion
- M4: Cataplexy(sudden muscle weakness)

Atropine is used for the management of muscarinic poisoning. It competitively blocks the acetylcholine at the muscarinic receptors.

Solution to Question 15:

Valproate should be avoided for being highly teratogenic to the fetus.

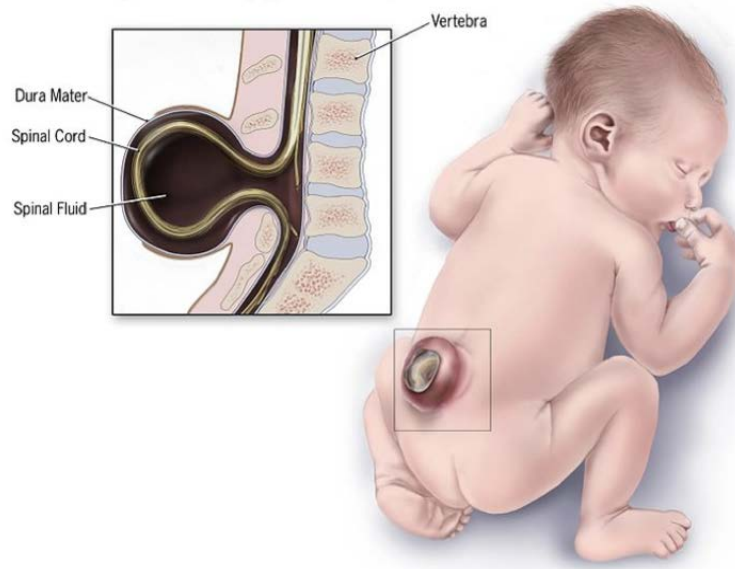
Valproate is an anti-epileptic medication. It is also used as a mood stabilizer in manic episodes as well as in bipolar disorder. The repetitive firing caused by depolarization of cortical or spinal cord neurons is inhibited by valproate. It also prolongs the sodium channel inactivation.

The side effects of valproate (mnemonic: VALPROATE)

- Vomiting
- Alopecia
- Liver toxicity
- Pancreatitis
- Rash
- Obesity
- Anorexia
- Teratogenicity
- Oedema

It causes congenital anomalies like neural tube defects (spina bifida), clefts, and cardiac anomalies. The intelligence quotient score of school-going children who were exposed in utero to valproate is found reduced. It also contributes to poor cognitive development in children. Folate supplementation reduces the chances of neural tube defects.

The images below show a baby and a USG with spinal bifida:





Other options:

Option B: Lithium is used for the treatment of bipolar disorder as well as mania. It was earlier implicated in causing Ebstein anomaly, however, recent studies suggest that lithium has a weak teratogenic effect.

Option C: Oxcarbazepine was earlier associated with teratogenicity and fetal malformations, however, recent studies suggest that oxcarbazepine monotherapy during pregnancy does not carry a significant risk of fetal malformations.

Option D: Olanzapine is an antipsychotic and an antimanic and it is not associated with teratogenicity.

Solution to Question 16:

True statements regarding the administration of 5 alpha-reductase inhibitors are 1,4. They reduce dihydrotestosterone levels and decrease prostate-specific antigen (PSA) levels.

5 alpha-reductase inhibitors are the competitive inhibitors of the enzyme 5 α -reductase which converts testosterone into the more active dihydroxy testosterone (DHT). This enzyme inhibition results in:

- Decreased DHT levels
- Decreased PSA levels
- Reduces prostate volume

However, plasma testosterone levels and LH (luteinizing hormone) levels remain unchanged

Finasteride and dutasteride are the 5 α -reductase drugs currently in use. They are used in benign prostatic hyperplasia (BPH) along with surgery and adrenergic α_1 blockers for symptomatic relief. They can also be used in male pattern of baldness and hirsutism.

Side effects of 5 α -reductase drugs are:

- Impotence
- Decreased libido
- Gynecomastia
- Skin rashes

Solution to Question 17:

This type of drug is called a misbranded drug. A drug stating that 500mg of paracetamol is present in each tablet, but contains only 200mg is a misbranded drug, according to the Drugs and Cosmetics Act, 1940 definition of misbranded drugs.

According to the Drugs and Cosmetics act, 1940, a misbranded drug is a drug if its label or container or anything accompanying the drug bears any statement, design, or device which makes any false claim for the drug or which is false or misleading in any particular.

Other options:

Option A: Spurious drugs include drugs manufactured under another drug or manufacturer's name, imitations of other drugs, and drugs with labels from a fictitious company or individual. It also includes drugs substituted partly or wholly by another drug or substance.

Option B: Adulterated drugs include those drugs that contain filthy, or decomposed substances, substances that are injurious to health, or if the drug harmful substances to reduce the quality or strength of the drug. It also includes drugs that were prepared in unsanitary conditions.

Option C: The term unethical drug is not standard terminology in pharmacology. However, the term ethical drug is sometimes used to describe those drugs that are available only by prescription of a medical practitioner.

Solution to Question 18:

Canagliflozin is associated with an increased risk of fractures.

Canagliflozin is a sodium-glucose transporter-2 (SGLT2) inhibitor. SGLT2 is a Na⁺-glucose cotransporter that is located in the proximal convoluted tubule. They reabsorb glucose and sodium from the tubular lumen back into the blood through the epithelial cells. SGLT2 inhibitors block the transport of glucose in the proximal tubule and promote glycosuria and hence, lowering blood glucose.

The currently available SGLT2 inhibitors are canagliflozin, dapagliflozin, and empagliflozin. Other than decreasing blood glucose they also decrease blood pressure and, cause weight loss of 2–4 kg. Empagliflozin and canagliflozin reduce the risk for major cardiovascular events.

The main side effect of SGLT2 inhibitors is an increase in lower urinary tract infections and genital mycotic infections. This is because of their glycosuric properties. A recent warning from the FDA says that SGLT2 inhibitors may increase the risk of fractures. They also affect circulating levels of parathyroid hormone, and 1,25-hydroxy vitamin D thereby

affecting the mineral balance in bone.

Other options:

Option A: Rosiglitazone is a thiazolidinedione that acts as a peroxisome proliferator-activated receptor γ agonist in the adipose tissue which increases insulin sensitivity and adipose differentiation. Side effects are weight gain, edema, and increased risk of fractures. Compared to rosiglitazone, canagliflozin is commonly used in clinical practice. American diabetes association recommends canagliflozin as the first-line drug along with metformin, in patients with a high risk of cardiovascular disease, renal failure, and heart failure. Even when thiazolidinediones are required, clinicians prefer pioglitazone over rosiglitazone due to the higher risk of cardiovascular-related mortality in rosiglitazone. Therefore, canagliflozin would be a better option compared to rosiglitazone.

Option B: Liraglutide is a long-acting glucagon-like peptide-1 agonist. Nausea, diarrhea, and acute pancreatitis are some frequent side effects.

Option C: Voglibose reduces the intestinal absorption of starch, dextrin, and disaccharides by inhibiting the action of α -glucosidase in the intestinal brush border. The most prominent adverse effects are malabsorption, flatulence, diarrhea, and abdominal bloating.

Solution to Question 19:

The active metabolite of cyclophosphamide is 4 - hydroxy cyclophosphamide.

Cyclophosphamide is an alkylating agent with antitumor and immunosuppressant properties. It is used for many solid tumors. The important side effects are cystitis due to metabolite acrolein and bladder carcinoma.

Cystitis caused by cyclophosphamide is treated by systemic administration of mesna and by irrigating the bladder with acetylcysteine. Both these drugs contain $-SH$ which detoxifies the toxic metabolites. Plenty of fluids and frequent bladder voiding are also helpful.

Solution to Question 20:

Alkalinization of urine will be helpful in the removal of aspirin, which is a weakly acidic drug and is susceptible to ion trapping in the urine.

Aspirin is a weak acid, with a pK_a of 3.0. Drugs that are subjected to renal clearance are excreted into the urine by glomerular filtration and active tubular secretion. Nonionized compounds are reabsorbed more rapidly than ionized polar molecules. When the pH of the urine increases, more salicylate is in its ionized form at equilibrium. Hence, salicylic acid diffuses into the tubular lumen of the kidney. Urine alkalinization is recommended as a first-line treatment only for moderately severe salicylate poisoning that does not meet the criteria for hemodialysis.

Alkalinization of the urine is achieved by administration of 100–150 mEq of sodium bicarbonate in 1 L of 5% dextrose in water (D5W). It is infused intravenously at twice the maintenance fluid requirements.

Alkalinization of urine is contraindicated in renal failure or if fluid administration may worsen pulmonary edema or congestive heart failure.

Solution to Question 21:

A hospital conducting a double-blinded study in multiple centers to check the efficacy of statins describes phase 3 of the clinical trial. A phase 2 clinical trial is generally conducted at a few centers (2-4) by physicians.

Phase 3 of a clinical trial is also known as a multisite trial. It is conducted on a large patient population (500–3000) by several physicians at multiple centers. This phase involves patients who have been diagnosed with the disease that the drug claims to treat. Its purpose is to confirm the efficacy of the newly developed drug in a larger population. Phase 3 is randomized, controlled, and may be blinded.

Solution to Question 22:

The given clinical scenario and the image showing a pin-point pupil (miosis) suggest opioid overdose. The true statement among those given is that it is caused by activation of the Edinger-Westphal nucleus.

Edinger-Westphal nucleus is a part of the light reflex pathway and is located in the midbrain. This nucleus is responsible for causing pupillary constriction when the retina is exposed to light.

Opioid overdose symptoms are a triad of coma, respiratory depression, and pinpoint pupils (miosis). Other features are hypotension, bradycardia, shallow breathing, and hypothermia.

Tolerance to miosis does not develop even in addicts who develop a high tolerance to opioids. Miosis is induced by all opioid agonists.

Naloxone is the drug of choice for opioid overdose. It is administered after adequate ventilation is established. Airway protection and ventilatory maintenance are the most important steps in the treatment of opioid overdose; this is because respiratory depression is the major cause of mortality in such cases. Single-dose activated charcoal can be administered if the opioid ingestion occurred within one hour.

Levallorphan is an opioid antagonist, with properties similar to those of naloxone. It can be used for opioid overdose. It reverses severe opioid-induced respiratory depression.

Phenylephrine is an α_1 -selective agonist that causes mydriasis, but it is not useful in the reversal of opioid-induced miosis.

Solution to Question 23:

The correct answer is option B: A-4, B-1, C-3, D-2

In a patient with head injury, morphine is contraindicated because it increases intracranial tension due to CO₂ retention, precipitates respiratory depression, and may cause vomiting, miosis, and altered mental state which interferes with assessment of mental state.

Amiodarone should not be used in patients with prolonged QT interval because it may cause marked prolongation of the QT interval, bradycardia, and torsades de pointes.

Vigabatrin, an inhibitor of GABA-transaminase, an anti-convulsant, is contraindicated in pregnancy due to its potential teratogenic effect.

Estrogen preparations can contribute to the incidence and severity of thromboembolism, particularly if other risk factors are present. The risk of serious cardiovascular side effects is particularly marked in women more than 35 years of age who smoke heavily.

A. Morphine	4. Head Injury
B. Amiodarone	1. QT prolongation
C. Vigabatrin	3. Pregnancy
D. Estrogen preparations	2. Thromboembolism

Solution to Question 24:

1 and 2 are true statements regarding memantine.

Memantine is a non-competitive NMDA receptor antagonist. It inhibits glutamate excitotoxicity. It delays the progression of dementia in Alzheimer's disease. Commonly, it is started in patients with moderate-to-severe disease.

It can be given both as monotherapy or can be given along with cholinesterase inhibitors. It can be started at a dose of 5 mg once daily and then progressively increased up to 10 mg twice daily for a minimum of 6 months.

It can cause dizziness and headache. Since it is excreted in the kidney, careful dose adjustments should be made in patients with renal failure.

Solution to Question 25:

Step 5 in the given life cycle shows the integration of the viral DNA into the host genome.

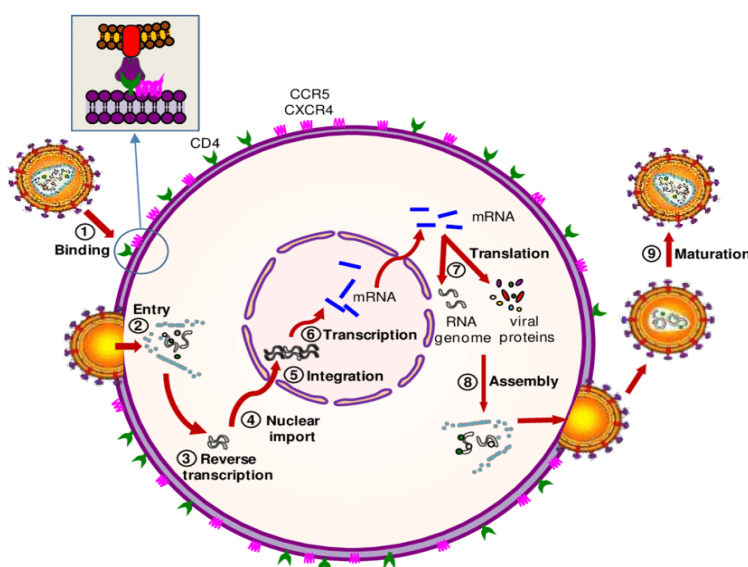
Raltegravir is an integrase inhibitor and is active against both HIV-1 and HIV-2. It can be used in adults and children >1 month old.

It is advised for patients to consume antacids and sucralfate, 6 hours before or 2 hours after raltegravir intake.

It may cause headaches, nausea, weakness and fatigue. Rarely, it can lead to drug reactions with eosinophilia and systemic symptoms (DRESS) syndrome.

The steps involved in the lifecycle of HIV as given in the image are described as follows:

- Step 1: Binding of HIV-1 to the cell membrane of a CD4⁺ cell
- Step 2: HIV-1 enters the cell via interaction with the membrane glycoproteins
- Step 3: Reverse transcription of the viral genetic material using the virus' reverse transcriptase enzyme
- Step 4: The HIV-1 genome travels into the nucleus
- Step 5: The viral genome integrates into the host's DNA through the integrase enzyme
- Step 6: The HIV-1 proviral genome undergoes transcription with the host DNA
- Step 7: The viral messenger RNA (mRNA) undergoes translation to produce new viral proteins
- Steps 8 and 9: Assembly and maturation of the virion



Other options:

Option B: Enfuvirtide acts at step 2 in the given image. It blocks gp41 of the cell membrane and blocks the entry of the virus into the cell. It prevents the free viral particles from invading new CD4⁺ cells. It is effective against HIV-1.

Option C: Tenofovir acts at step 3 in the given image. It is a nucleotide reverse transcriptase inhibitor (NRTI) that is effective against HIV-1,2 and hepatitis B infections. It is used in adults and children >2 years old. It can also be used for pre-exposure prophylaxis.

Option D: Zidovudine also acts at step 3 in the given image. It is an NRTI and is effective against HIV-1,2 and Human T-cell Lymphoma Virus (HTLV). It can be used in adults and children. It can also be used to prevent mother-to-child transmission.

Solution to Question 26:

Phase 4 clinical trials are also called post-marketing surveillance is the correct statement.

Other options:

Option A: New drug application is done after phase 3 of clinical trials.

Option C: Phase 3 trials may or may not be blinded.

Option D: Phase 2 trials are the first round of trials in patients. Phase 1 trials are done in healthy volunteers.

Solution to Question 27:

The correctly matched pairs of teratogenic drugs and their effects is 1-d, 2-c, 3-b, 4-a.

Lithium use in early pregnancy causes Ebstein anomaly in the fetus. Fetal lithium exposure is associated with neonatal goiter, CNS depression, hypotonia (floppy baby syndrome), and cardiac murmur.

Gray baby syndrome occurs in neonates, especially premature ones, exposed to high doses of chloramphenicol (100 mg/kg), which are administered prophylactically.

Fetal warfarin syndrome or warfarin embryopathy presents with CNS abnormalities, depressed nasal bridge/ saddle nose at birth, bleeding diathesis, and growth retardation. Hypoplasia of the nose, eye sockets, and long bones with stippling of epiphyses are also seen.

Fetal thalidomide exposure causes phocomelia. This disorder is associated with underdeveloped limbs and absent pelvic bones. Both upper and lower limbs are affected, and a part of the limbs may be missing. Fused fingers and missing thumbs are also seen. Thalidomide was originally used for the treatment of pregnancy-associated morning sickness but was withdrawn due to this adverse effect. It re-entered the market for the treatment of erythema nodosum leprosum. It also has antiangiogenic and immunomodulatory effects.

Solution to Question 28:

The drug used to manage this condition is prazosin as it can counteract the effects of breathlessness most probably caused by pulmonary edema.

The venom is a potent autonomic stimulator that causes the release of massive amounts of catecholamines from the adrenals. It also has a direct effect on the myocardium.

It can present as neurotoxicity with restlessness or seizure-like activity, abnormal oculomotor function, and loss of pharyngeal muscle control that leads to respiratory compromise. Tongue fasciculations and hypersalivation are also common. Cardiovascular toxicity from systemic envenomation includes tachycardia, hypertension, pulmonary edema, and cardiogenic shock.

If the scorpion has hemolytic venom, the reaction is mainly local, with edema, pain, and reddening. This usually lasts for one to two hours.

Treatment:

- Pain - Paracetamol, NSAIDs, or opioids

- Sedation for agitation, oculomotor abnormalities, uncoordinated neuromuscular activity, and muscle spasms - Benzodiazepines
- Hypertension and pulmonary edema - Prazosin counteracts the adrenergic surge that can lead to these effects
- Cardiogenic shock - Antivenom, prazosin, nitroglycerin, dobutamine
- Cholinergic effects - Atropine
- Supportive care, wound care

Solution to Question 29:

Phase 4 in the graph indicates the nicotinic action of acetylcholine at higher doses.

In phases 1 and 2, acetylcholine (ACh) acts on muscarinic receptors (at 2 micrograms and 50 micrograms) bringing down blood pressure. Atropine blocks the muscarinic receptor, so this action of acetylcholine is not seen in phase 3. However, in phase 4, higher doses of ACh (like 5 mg) act via the nicotinic receptor, shooting up the blood pressure. This phenomenon is called the reversal of ACh in blood pressure and was studied by Dale.

Nicotinic receptors are ligand-gated ion channels. It can be divided into two groups: muscle receptors (NM), which are found at the skeletal neuromuscular junction, and neuronal receptors (NN), which are found in the ganglia and CNS. When stimulated at higher doses, ACh acts on NN receptors, leading to the stimulation of sympathetic ganglia and adrenal medulla. This increases the heart rate, blood pressure, and myocardial contractility.

Muscarinic receptors are G-coupled protein receptors involved in the parasympathetic nervous system. M1, M2, M3, M4, and M5 are the receptor subtypes. They act via Gi or Gq subunits. When ACh is administered, it acts on muscarinic receptors, primarily the M3 receptors, which are located on vascular endothelial cells. It stimulates guanylyl cyclase and promotes relaxation via a cyclic GMP-dependent mechanism. At higher doses, it acts on M2 receptors, which have direct effects on the heart. It also inhibits the release of norepinephrine from sympathetic nerve terminals and decreases cyclic AMP levels.

Solution to Question 30:

The most likely drug causing hyperpigmentation of the hands and soles would be emtricitabine.

Emtricitabine is an anti-retroviral drug that acts by inhibiting nucleoside reverse transcriptase (NRTI). It is well-tolerated and has minimal toxic effects. Adverse effects include hyperpigmentation of the skin (especially sun-exposed areas, palms, and soles), headache, vomiting, and nausea.

Other options:

Option A: Nelfinavir is a protease inhibitor that causes diarrhea or loose stools. It usually resolves within a month of starting the drug.

Option C: Abacavir is an NRTI that can cause a life-threatening hypersensitivity syndrome. The syndrome presents with fever, abdominal pain, and maculopapular rash within 6 weeks of initiating abacavir. The drug must be immediately discontinued and must not be administered to the patient thereafter. Rarely, there might be respiratory and musculoskeletal side effects.

Option D: Zidovudine is an NRTI that causes malaise, fatigue, myalgia, anorexia, and insomnia. Chronic use leads to nail hyperpigmentation. It can also cause bone marrow suppression.

Solution to Question 31:

Dorzolamide is a topical ocular carbonic anhydrase inhibitor.

It acts on carbonic anhydrase in the ciliary processes of the eye and mediates the formation of HCO_3^- in aqueous humor. Inhibition of carbonic anhydrase decreases the rate of formation of aqueous humor and consequently reduces intraocular pressure.

The major indication for carbonic anhydrase inhibitors is open-angle glaucoma. Dorzolamide and brinzolamide, which are available only as ophthalmic drops, are used for this purpose.

Dorzolamide may be used for secondary glaucoma and in acute angle-closure glaucoma to lower intraocular pressure before surgery.

Solution to Question 32:

Sildenafil is a phosphodiesterase 5 inhibitor and the action of the drug is mediated through cGMP. Erectile dysfunction is treated by phosphodiesterase-5 (PDE-5) inhibitors like sildenafil, vardenafil, and tadalafil.

The corpus cavernosum produces nitric oxide during arousal that stimulates the formation of cGMP. cGMP causes relaxation of the smooth muscle of the corpus cavernosum and penile arteries, engorgement of the corpus cavernosum, and erection.

The above-mentioned drugs compete for cyclic GMP, binding at the site of cyclic GMP hydrolysis on PDE5, enhancing the accumulation of cGMP and hence used in the management of erectile dysfunction. They are useful in treating pulmonary arterial hypertension.

Adverse effects of the drug are headache, flushing, nasal congestion, nasopharyngitis, and dyspepsia. Sildenafil and Vardenafil also cause retinal PDE-6 inhibition and can result in blurring of vision and blue-green tinting of the vision.

Solution to Question 33:

The correct match is as follows:

Acetazolamide, a carbonic anhydrase inhibitor, acts on the enzyme in epithelial cells of the proximal convoluted tubule. Acetazolamide is mainly used in patients with secondary glaucoma to reduce intra-ocular pressure, provide symptomatic relief in acute mountain sickness, and as

treatment for absence seizures.

Torsemide, a loop diuretic, inhibits the Na⁺-K⁺-ATPase symporter in the thick ascending loop of Henle. It is used in acute pulmonary edema, edema associated with congestive cardiac failure, hypertension with renal insufficiency, and hyponatremia.

Chlorthalidone, a thiazide-like diuretic, inhibits the Na⁺-Cl⁻ symporter in the distal convoluted tubule. It is used in hypertension and in edema associated with congestive cardiac failure, hepatic cirrhosis, nephrotic syndrome, and chronic renal failure.

Vasopressin, also known as anti-diuretic hormone, is secreted by the posterior pituitary and mainly acts on the collecting duct, and increases the reabsorption of water (anti-diuresis). Vasopressin or its agonists (such as desmopressin) are used in cases of diabetes insipidus, type I and type II von Willebrand's disease, primary nocturnal enuresis, and postoperative ileus.

Column A	Column B
A. Acetazolamide	4. Proximal convoluted tubule
B. Torsemide	1. Thick ascending loop of Henle
C. Chlorthalidone	3. Distal convoluted tubule
D. Vasopressin	2. Collecting duct

Solution to Question 34:

Alkalinization of urine will be helpful in the removal of weakly acidic drugs as they are susceptible to ion trapping in the urine.

Drugs that are subjected to renal clearance are excreted into the urine by glomerular filtration and active tubular secretion. Nonionized compounds are reabsorbed more rapidly than ionized polar molecules. When the pH of the urine increases, more salicylate is in its ionized form at equilibrium. Hence, salicylic acid diffuses into the tubular lumen of the kidney. Urine alkalinization is recommended as a first-line treatment only for moderately severe salicylate poisoning that does not meet the criteria for hemodialysis.

Alkalinization of the urine is achieved by administration of 100–150 mEq of sodium bicarbonate in 1 L of 5% dextrose in water (D5W). It is infused intravenously at twice the maintenance fluid requirements.

Alkalinization of urine is contraindicated in renal failure or if fluid administration may worsen pulmonary edema or congestive heart failure.

Solution to Question 35:

Hyponatremia occurs more frequently in the case of oxcarbazepine than carbamazepine.

Both carbamazepine and oxcarbazepine are sodium-channel blockers.

Carbamazepine is most effective in complex partial seizures. It is also used in GTCS and simple partial seizures, trigeminal and related neuralgias, and manic depressive psychosis.

Carbamazepine potentiates the action of antidiuretic hormone (ADH). It can result in water retention with decreased osmolarity and hyponatremia, especially in elderly patients with cardiac disease. Carbamazepine and oxcarbazepine can cause leukopenia, aplastic anemia, hepato-toxicity and hyponatremia in the elderly.

Oxcarbazepine shows lesser hypersensitivity reaction. Hence, rashes are more commonly seen with carbamazepine than with oxcarbazepine.

Studies have shown that carbamazepine elevates enzyme gamma-glutamyl transpeptidases (GGT) more frequently than oxcarbazepine because of enzyme P450 induction.

Mechanism of action of some common antiepileptics:

Mechanism of action	Antiepileptics
Sodium-channel blockers	Phenytoin Carbamazepine, Oxcarbazepine Eslicarbazepine Lamotrigine Topiramate Felbamate Valproate Rufinamide Levetiracetam
Calcium-channel blockers	Valproate Lamotrigine Ethosuximide
GABA _A receptor modulators	Benzodiazepines Phenobarbital Felbamate Primidone Topiramate Carbamazepine, Oxcarbazepine Stiripentol Clobazam
GABA uptake inhibitors	Tiagabine Vigabatrin
NMDA receptor antagonist	Felbamate
AMPA/Kainate receptor antagonist	Phenobarbital Topiramate Perampanel
HCN channel activity enhancers	Lamotrigine
Inhibitors of brain carbonic anhydrase	Acetazolamide Zonisamide Topiramate
Potassium-channel openers	Retigabine Topiramate

Solution to Question 36:

The correct pairs are as follows:

Trastuzumab is a humanized monoclonal antibody against the external domain of HER2/neu. It is indicated as an adjuvant in the management of metastatic HER-2-positive breast cancer.

Infliximab binds with high affinity to TNF- α and prevents the cytokine from binding to its receptors. Infliximab is used for the treatment of rheumatoid arthritis; it is used in combination with methotrexate in patients who do not respond to methotrexate alone. Infliximab is also used for the treatment of the symptoms of moderate-to-severe Crohn's disease.

Sirolimus is an antimetabolic and antiproliferative drug. It acts by forming a complex with an immunophilin called FKBP-12, which in turn inhibits the mammalian target of rapamycin (mTOR). mTOR is a protein kinase that blocks cell cycle progression at the G1 phase to S phase transition. It is used in the prophylaxis of organ transplant rejection.

Imatinib is a first-generation tyrosine kinase inhibitor. Imatinib attacks BCR-ABL translocation and is used in chronic myelocytic leukemia. It also inhibits c-kit and produces extended control of GI stromal tumors.

A. Trastuzumab	4. HER2/neu
B. Infliximab	3. TNF α
C. Sirolimus	2. mTOR
D. Imatinib	1. BCR-ABL tyrosine kinase

Solution to Question 37:

Oxazepam can be used in patients with elevated transaminases. It is short-acting and so, can be used in patients with liver impairment or brain damage.

Oxazepam is an orally available benzodiazepine used to treat anxiety and alcohol withdrawal states. Hepatic oxidation is decreased in persons with liver disease and the elderly. Oxazepam and lorazepam undergo only hepatic glucuronidation.

Other options:

Option A: Midazolam is the benzodiazepine of choice for induction of anesthesia. The onset of anesthesia is within 30 to 60 seconds. It can also be used to control seizures, but there is no difference in the efficacy of intravenous midazolam and intravenous diazepam.

Option B: Diazepam is a long-acting benzodiazepine. It is metabolized in the liver to active metabolites like nordiazepam which are long-acting. Diazepam and its long-acting metabolites can accumulate significantly in elderly patients with liver disease and therefore should not be used in such cases.

Option C: Alprazolam is metabolized by hepatic oxidation which is significantly decreased in liver impairment and elderly patients.

Solution to Question 38:

Atorvastatin 80 mg can be used to treat this condition. The first line of therapy in controlling cholesterol in patients with atherosclerotic cardiovascular disease (ASCVD) is starting a high-intensity statin. High-intensity statins include atorvastatin (40-80 mg) and rosuvastatin (20-40 mg).

Statins are competitive inhibitors of HMG-CoA reductase. This leads to a decrease in cholesterol biosynthesis. Statins are also effective in reducing LDL and triglyceride levels.

Other options:

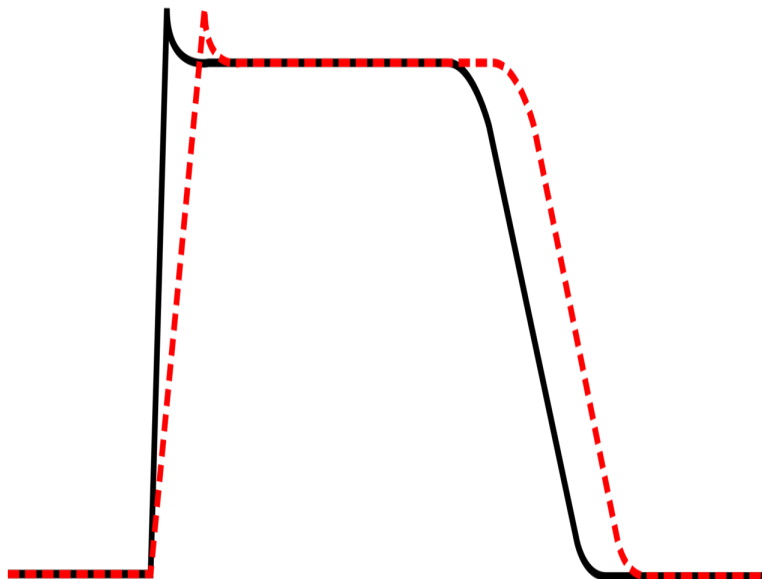
Option A. Fenofibrate reduces lipoprotein levels by interacting with Peroxisome proliferator-activated receptor (PPAR- α) and stimulating lipoprotein lipase. They are used as drugs of choice for treating hyperlipoproteinemia and severe hypertriglyceridemia (>1000 mg/dL).

Option B. Rosuvastatin and fenofibrate combination was previously used to reduce atherosclerotic cardiovascular disease risk. But recent studies have indicated no change in reduction and hence is not recommended anymore.

Option C: Rosuvastatin 10 mg is a low-intensity statin and is mainly used for primary prevention against atherosclerotic cardiovascular disease.

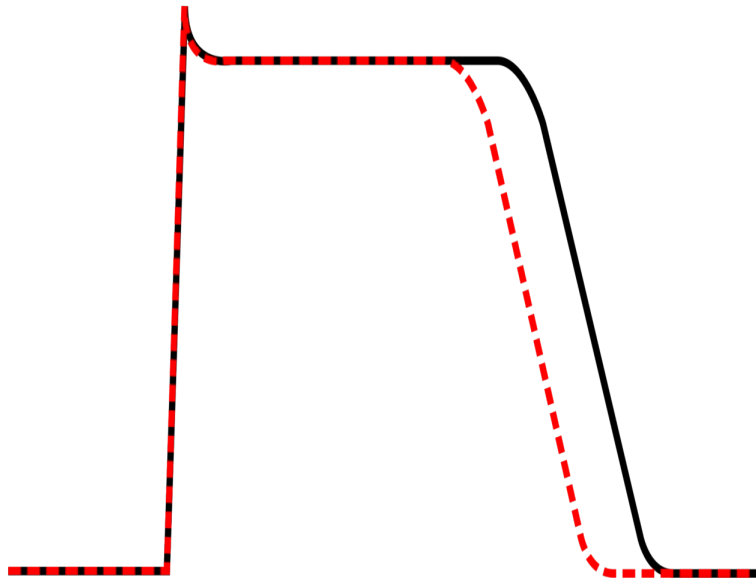
Solution to Question 39:

Quinidine is a class IA anti-arrhythmic agent. Class 1a agents are sodium channel blockers with significant potassium channel blocking properties. This slows conduction and prolongs the QRS duration and action potential duration.



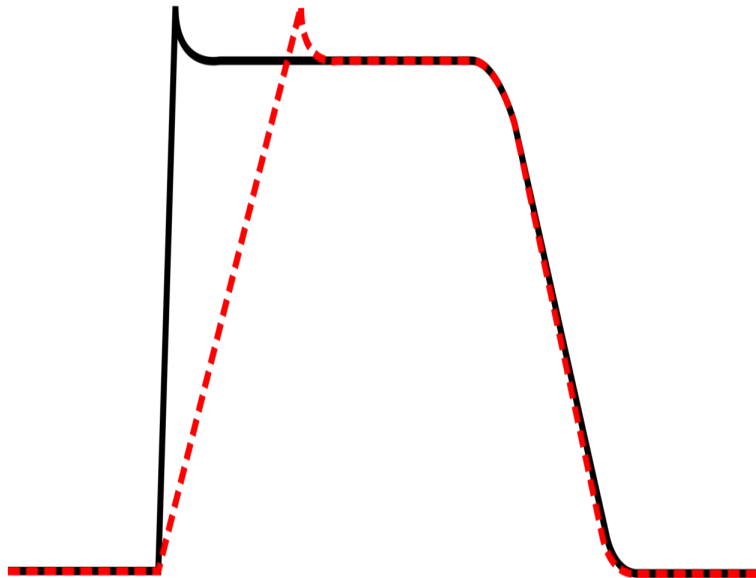
Lignocaine and tocainide (Options D and A) are class 1b anti-arrhythmic drugs. These are sodium channel blockers that inhibit both inactivated and open sodium channels and can open potassium channels. It reduces phase 4 slope and reduces automaticity. It also alters the excitation threshold.

The below image shows the changes in the action potential of a cardiac muscle when a class 1b anti-arrhythmic drug is used.



Class 1c agents like flecainide and propafenone block sodium channels with insignificant potassium channel-blocking properties.

The image below illustrates the action potential changes when a class 1c anti-arrhythmic is used.



Option B: Propranolol is a β -receptor-blocker and a class II antiarrhythmic.

Solution to Question 40:

The false statement regarding statins is that they can be given with verapamil and other enzyme inhibitors.

Statins like atorvastatin and rosuvastatin are metabolized by the cytochrome 3A4 enzymes of the liver. Verapamil is a cytochrome 3A4 inhibitor and may interfere with statin degradation. This leads to an increase in serum statin levels. The most important feature of statin toxicity is myopathy. It can present as myalgia but can sometimes also lead to fatal rhabdomyolysis.

Statins are the most commonly used hypolipidemic drugs. Statins act by inhibiting the rate-limiting HMG-CoA reductase enzyme, which in turn inhibits cholesterol synthesis. They decrease levels of low-density lipoproteins (LDL) and triglycerides and increase the levels of high-density lipoproteins (HDL) in plasma.

Other options:

Option A: Lipoprotein (lp) a is an LDL molecule that has apolipoprotein b and an additional apolipoprotein a. Studies indicate that these molecules are atherogenic and may independently cause cardiovascular disease. Although HMG-CoA reductase inhibitors substantially reduce the risk of cardiovascular events, there is a mild increase in lipoprotein a levels.

Option B: Statins can be continued during surgery. These drugs have proved to decrease overall mortality due to cardiovascular events. Statins also influence inflammatory responses and can be safely used in trauma.

Option D: Chronic administration of high statin doses also leads to an increased incidence of type 2 diabetes mellitus.

Solution to Question 41:

The drugs marked A, B and C are aspirin, eptifibatide, and vorapaxar, respectively.

Aspirin causes inhibition of cyclooxygenase-1 (COX-1), which would inhibit the formation of thromboxane A₂. Thromboxane A₂ is an inducer of platelet aggregation and is a potent vasoconstrictor. Hence, inhibition of its formation is responsible for antiplatelet action.

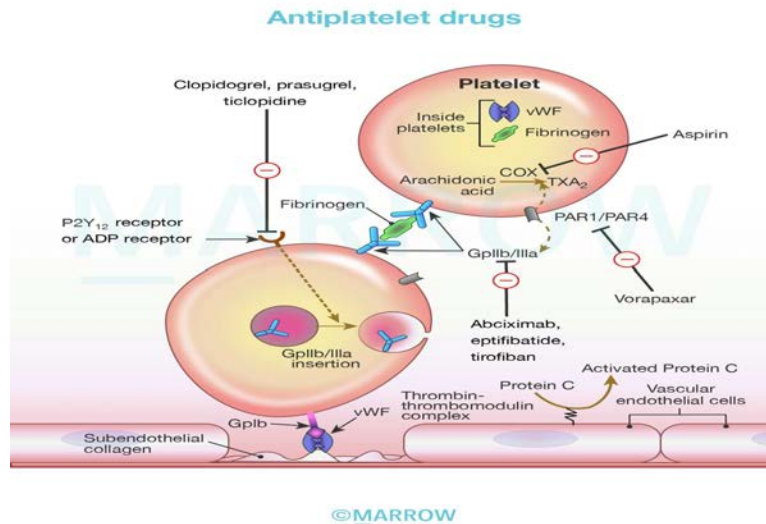
Aspirin is a non-competitive, irreversible inhibitor of the cyclooxygenase (COX) enzyme. It is a nonselective COX inhibitor and inhibits both COX-1 and COX-2.

Abciximab, tirofiban, and eptifibatide act by blocking Gp IIb/IIIa receptors. They are mainly used during percutaneous coronary intervention for acute coronary syndrome. They carry a high risk of bleeding and can cause thrombocytopenia.

Vorapaxar primarily inhibits platelet aggregation caused by thrombin. PAR-1 activation by thrombin contributes to platelet aggregation. Vorapaxar is a competitive antagonist of protease-activated receptor-1 (PAR-1). Vorapaxar is indicated in combination with aspirin and/or clopidogrel to reduce thromboembolic events in patients with MI or peripheral vascular disease.

Prasugrel is an irreversible P₂Y₁₂ inhibitor and lasts for the entire lifetime of a platelet.

Prasugrel is a prodrug that is used in patients with acute coronary syndrome who have undergone percutaneous coronary intervention. It reduces the rate of thrombotic cardiovascular events in such patients.



Solution to Question 42:

The wrong statement/s regarding disulfiram is only 3.

Disulfiram is an irreversible inhibitor of both cytosolic and mitochondrial forms of aldehyde dehydrogenase. It is used in behavioral modification for alcohol de-addiction. The principle of action is aversive conditioning (not anti-craving).

Disulfiram can be given only to highly motivated individuals. It should not be administered until the person has abstained from alcohol for at least 12 hours. The disulfiram-alcohol reaction may persist as long as 14 days after the last dose of disulfiram.

Alcohol consumption during disulfiram therapy can lead to increased concentrations of acetaldehyde. This can cause aversive reactions, resulting in flushing, pulsating headache, throbbing pain in the head and neck, nausea, vomiting, rashes, respiratory difficulties, chest pain, and autonomic manifestations.

Disulfiram is usually given in the dose of 250 mg/day (range of 125–500 mg/day).

Solution to Question 43:

G6PD deficiency will manifest with primaquine.

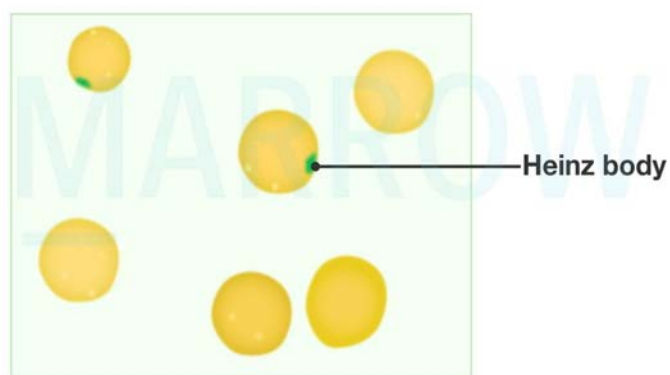
G6PD deficiency is an X-linked recessive (XLR) disorder that causes congenital hemolytic anemia due to enzyme deficiency. Hemolysis occurs in conditions of high oxidative stress that are triggered by infections, fava beans, and certain drugs.

G6PD is an enzyme in the hexose monophosphate pathway (HMP) that maintains the cellular levels of NADPH. NADPH then provides reducing equivalents needed for the conversion of oxidized glutathione to reduced glutathione. Reduced glutathione helps neutralize free radicals, protecting against oxidant injury.

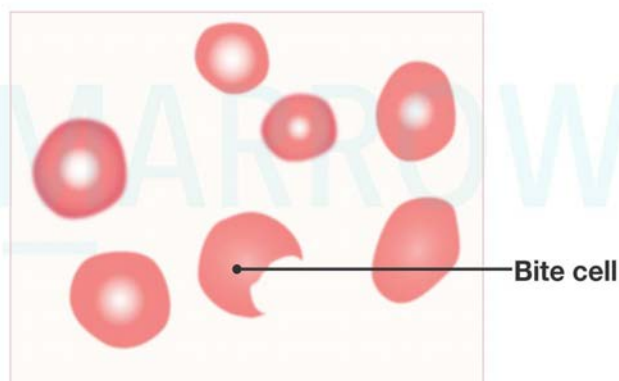
Clinically, most patients are asymptomatic. Some patient develops episodic hemolysis, causing hemolytic anemia, hemoglobinemia, and hemoglobinuria.

Both primaquine and chloroquine cause hemolysis in G6PD-deficient individuals, but hemolysis is rare with chloroquine and it can be used in patients with G6PD deficiency, whereas primaquine is contraindicated in these individuals.

Hematological findings in G6PD deficiency are shown below:



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Drugs that carry a risk of clinical hemolysis in persons with G6PD deficiency:

CLASS	DRUGS
Antimalarials	Primaquine >Chloroquine Dapsone/chlorproguanil
Sulphonamides/sulphonamides	Sulfamethoxazole Sulfasalazine, Sulfisoxazole

CLASS	DRUGS
Antibacterial/antibiotics	Cotrimoxazole, Ciprofloxacin, Chloramphenicol, Nalidixic acid, Norfloxacin, Nitrofurantoin
Others	Naphthalene, Rasburicase, Methylene blue

Solution to Question 44:

The correct option regarding the drugs used to treat schizophrenia symptoms is option C.

Drugs used to treat schizophrenia can be classified:

- Typical or 1st generation antipsychotics.
- Atypical or 2nd generation antipsychotics.

All typical antipsychotics markedly improve positive symptoms of schizophrenia, which is attributed to their antagonism at the D₂ receptor. Some of the drugs included in this class are chlorpromazine and haloperidol.

Atypical antipsychotics are antagonists or inverse agonists at the 5HT_{2A} receptor. 5HT_{2A} antagonism is responsible for reducing negative symptoms in schizophrenia. Some of the drugs that belong to this class are clozapine, aripiprazole, and brexpiprazole.

Atypical antipsychotics have lower extrapyramidal symptoms as compared to typical antipsychotics as the former has a lesser affinity to the D₂ receptor. Muscarinic antagonists can be given to decrease the extrapyramidal symptoms. For example, promethazine or hydroxyzine is given for acute muscular dystonia, which is a form of extrapyramidal symptoms.

5HT_{1A} receptor antagonism, not agonism, has been found to reduce food intake and hence reduce body weight.

Aripiprazole and brexpiprazole are atypical antipsychotics that have partial agonist effects at the 5HT_{1A} receptor. They have been approved for the treatment of schizophrenia and as an adjunctive treatment for depression.

Solution to Question 45:

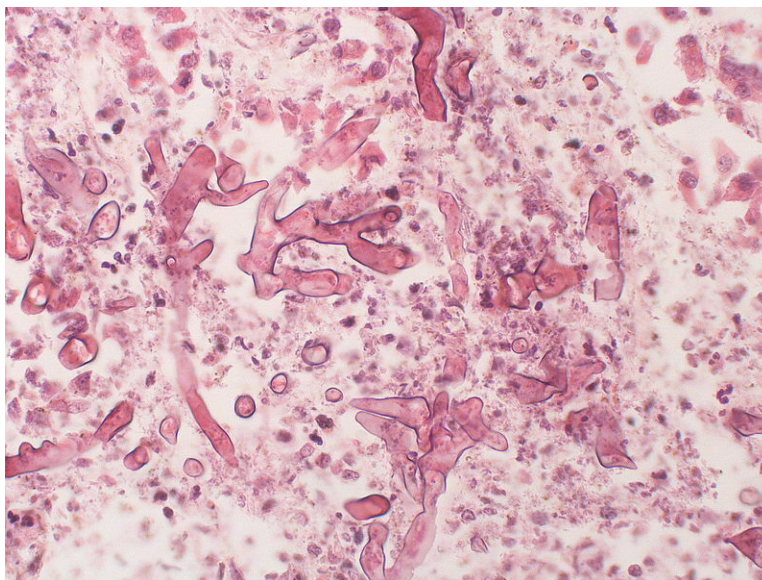
The given clinical scenario is suggestive of mucormycosis. The patient can be given injection amphotericin B for 8 weeks, followed by oral posaconazole.

Mucormycosis is an opportunistic mycosis. It is typically seen in immunocompromised patients and invades blood vessels.

The rhino-orbital-cerebral is the most common form of mucormycosis. Patients present with rhinitis, eye or facial pain, and facial numbness, followed by the onset of conjunctival suffusion and swelling along with blurry vision.

Diagnosis is made by biopsy with histopathologic examination. The biopsy shows characteristic wide, ribbon-like aseptate hyphal elements that branch at right angles.

The image below shows biopsy findings seen in mucormycosis:



Treatment of the infection includes 4 steps:

- Initiation of therapy (amphotericin B) as early as possible. Posaconazole can also be used in the management of this condition.
- Surgical debridement
- Reversal of underlying conditions and risk factors
- Treatment of underlying malignancy if present

Amphotericin B is a broad-spectrum antifungal. It binds to ergosterol in the fungal membrane and forms pores or channels in the fungal cell membrane. This allows the leakage of cytosolic molecules and ions, leading to a loss of membrane integrity.

Other options:

Option A: Itraconazole can be used for the treatment of chronic aspergillosis if there is resistance or failure of voriconazole. Ketoconazole is used for both dermatophytosis and deep mycosis, but it has been replaced by itraconazole.

Option C: Fluconazole is used for the treatment of candidiasis and cryptococcosis.

Option D: Amphotericin B in combination with 5-flucytosine is used as induction treatment for cryptococcal meningitis.

Solution to Question 46:

The antilipidemic drugs likely to cause gallstones are the peroxisome proliferator-activated receptors (PPARs) that act by activating PPAR alpha.

PPARs are a superfamily of nuclear receptors involved in the regulation of energy homeostasis and metabolic function. It has three subtypes:

- PPAR α - reduces triglyceride level
- PPAR γ - causes insulin sensitization
- PPAR δ - enhances fatty acids metabolism

Lipoprotein lipase (LPL) is a key enzyme in the degradation of VLDL. Fibrates bind to PPAR alpha and cause stimulation of lipoprotein lipase enzyme. Increased LPL synthesis enhances the clearance of triglyceride-rich lipoproteins. Additionally, fibrates also stimulate fatty acid oxidation and reduce the expression of apo C-III (inhibitor of lipolysis). All fibrates increase the lithogenicity of bile, thereby increasing the risk of gall stones

Other options:

Class	HMG- CoA reductase inhibitors (statins)	Niacin	Bile acid-binding resins
Examples	Lovastatin, atorvastatin	Niacin	Colestipol,colesevelam
Mechanism of action	Inhibits HMG-CoA reductase	Decreases the production of VLDLs and the catabolism of HDLs	Binds to bile acids in the intestinal lumen and increases excretion.
Major effects on lipoprotein levels	Reduces LDL levels	Reduces triglycerides, LDL, and lipoprotein a levelsSignificant increase HDL levels	Reduces LDL levels
Important side effects	MyopathyElevations of serum amino-transferase activityElevation of creatine kinase activity	Cutaneous vasodilationElevations in amino-transferases levels	ConstipationBloating

Solution to Question 47:

In the given image, drug A & drug B are agonists.

Drugs that bind to physiological receptors and mimic the regulatory effects of endogenous signaling compounds are agonists. Drugs that are only partially as effective as agonists are partial agonists.

The efficacy of a drug indicates the capacity of a drug to activate a receptor and generate a cellular response. A drug with high efficacy is a full agonist, eliciting a full response at some concentration. A drug with a low intrinsic efficacy will be a partial agonist. Drug A & drug B are agonists (Option C) that have the same efficacy (Option A), although, at different concentrations.

Drug C does not produce a full response at any concentration, making it a partial agonist. In the presence of a full agonist, a partial agonist behaves like a competitive antagonist (Option D).

When two drugs produce equivalent responses, the drug whose dose-response curve lies to the left is said to be the more potent since it requires a lower dose to produce the same effect. Drug A

& drug B produce the same action and level of response in the tissue. The curve of drug A on the left indicates that it produces the same response at a lower concentration, making it more potent than drug B (Option B).

Solution to Question 48:

Dipivefrine and dopexamine are β 2-adrenergic receptor agonists.

Dipivefrine causes a decrease in aqueous production and increases outflow. It was used in the treatment of open-angle glaucoma and ocular hypertension but has since been replaced.

Dopexamine is a synthetic analogue related to dopamine with activity at D1 and D2 receptors as well as at β 2 receptors. It is used in patients with severe congestive heart failure, sepsis, and shock.

Other options:

B. Fenoldopam is a peripheral D1 agonist with moderate affinity for α 2-adrenergic receptors. It is a rapidly acting vasodilator that is used in severe hypertension.

D. Mirabegron is a β 3-adrenergic receptor agonist which is used for bladder incontinence. Activation of this receptor in the bladder leads to detrusor muscle relaxation and increased bladder capacity. This action prevents voiding and provides relief for those with an overactive bladder and urinary incontinence.

Solution to Question 49:

Sildenafil acts by inhibiting phosphodiesterase-5 (PDE-5) and is used in the treatment of erectile dysfunction.

PDE-5 inhibitors cause the accumulation of cGMP in the corpus cavernosum, which leads to the relaxation of smooth muscles of penile arteries that fill the corpus cavernosum, thereby improving erectile function in patients with erectile dysfunction.

In addition to erectile dysfunction, sildenafil can be used in the treatment of pulmonary arterial hypertension.

Adverse effects include headache, flushing, nasal congestion, blue-green tinting of vision, and erections lasting more than 4 hours.

Sildenafil is contraindicated in patients using nitrate vasodilators, which could result in dangerously low blood pressure.

Solution to Question 50:

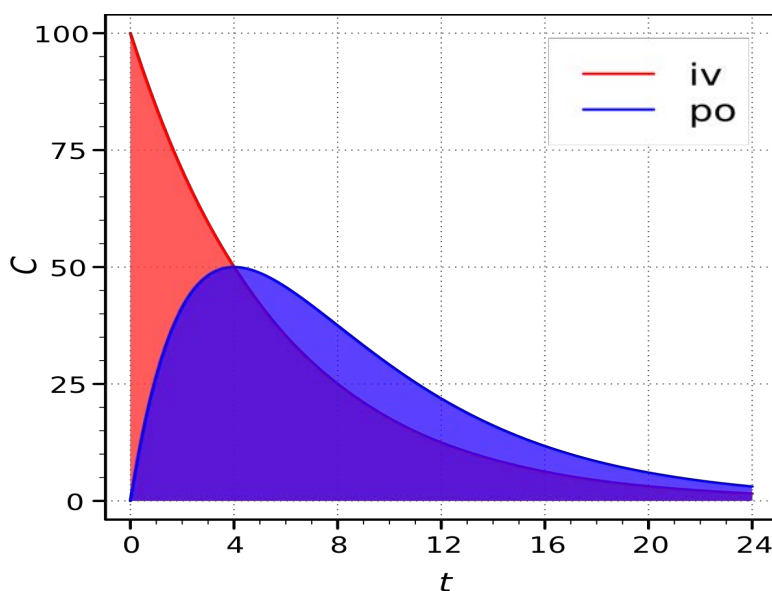
The correct option among the following is A, C, and D.

Bioavailability is defined as the fraction of a drug reaching systemic circulation in an unchanged form after administration by any route. The area under the blood concentration-time curve (AUC) is proportional to the dose and the extent of bioavailability for a drug.

Bioavailability (F) = Quantity of drug reaching systemic circulation/Quantity of drug administered

Bioavailability depends on two factors, absorption and 1st pass metabolism. The bioavailability of the drug injected intravenously is 100%. But it is lower after oral ingestion because of incomplete absorption and first-pass metabolism in the intestinal wall or liver and lies between 5% to <100%. The bioavailability of subcutaneous and intramuscular routes lies between 75% to ≤100%.

The image given below shows the difference between the bioavailability of drugs injected intravenously (IV) and administered orally (PO). Note the area under the respective curves shaded in red and blue colors which represent the absorption of the drug.



Solution to Question 51:

In this hypertensive patient with hyperkalemia, losartan must be stopped.

Losartan is competitive angiotensin I (AT₁) receptor antagonist, which binds to the AT₁ receptor with high affinity. Uses of losartan are:

- Antihypertensive
- Renoprotective drug in diabetes mellitus
- Stroke prophylaxis
- Portal hypertension in cirrhosis without compromising renal function
- Anti-aggregant as it is a competitive antagonist of the thromboxane A₂ receptor.

Hyperkalemia is an important side effect of ARBs. Losartan appears to be a safe and efficacious agent to lower serum uric acid levels in patients with hyperuricemia.

Option A: Gingival hyperplasia and ankle swelling are side effects of calcium channel blockers like amlodipine. Ankle edema occurs due to increased hydrostatic pressure in the capillaries present in the dependent parts of the body. This is due to reflex constriction of postcapillary vessels. Other common side effects of amlodipine include headache, dizziness, and abdominal pain.

Option B: Metoprolol is a cardioselective beta blocker. Cardioselective beta blockers are less likely to cause hyperkalemia as an adverse effect. The risk of metoprolol-induced hyperkalemia is increased if the patient is also diabetic.

Option D: Furosemide is a loop diuretic. It acts by inhibiting the $\text{Na}^+ - \text{K}^+ - 2\text{Cl}^-$ symporter in the thick ascending limb of the loop of Henle resulting in increased excretion of sodium, potassium, and chloride. Hypokalemia, hyperuricemia, hypocalcemia, and hypomagnesemia are the important side effects.

Solution to Question 52:

Varenicline is the most effective oral drug for smoking cessation.

Nicotine is the principal constituent of tobacco responsible for its addictive behaviour. Varenicline works by blocking the effects of nicotine on the brain. It blocks α_4 , and β_2 nicotinic acetylcholine receptor subtypes and inhibits dopaminergic activation produced by smoking through their partial agonism. It decreases the craving and withdrawal syndrome that occurs with cessation attempts. It prevents nicotine stimulation of the mesolimbic dopamine system associated with nicotine addiction, thus helping to occlude the rewarding effect of smoking.

Varenicline is prescribed only to patients 18 years or older. It is taken orally and the dose taken is 0.5 mg once daily on days 1 to 3, increased to 0.5 mg twice daily on days 4 through 7, and 1 mg twice daily after day 8 for 11 weeks. Therapy should start one week before the target quit date. It can be continued for another 12 weeks (1 mg BD) to maintain abstinence.

The adverse effects include nausea, disturbed sleep, sleepwalking, agitation, drowsiness, and constipation, and may also increase the risk of pancreatitis. Varenicline has also been implicated in specific skin syndromes such as Stevens-Johnson syndrome, erythema multiforme, and photosensitivity. Varenicline can cause renal failure and kidney stones, thus renal function must be monitored.

The FDA has issued a warning about mood and behavioural changes associated with its use, and there is some evidence of increased cardiovascular risk.

Cevimeline is a muscarinic agonist with an affinity for both M_1 and M_3 receptors on the lacrimal and salivary glands. While M_1 activation results in increased secretion, M_3 activation results in smooth muscle contraction, and expulsion of the secretions. Cevimeline enhances lacrimal secretions in Sjogren's syndrome.

Solution to Question 53:

The correct option is Option C: 1-a, 2-d, 3-b, 4-c

Quinidine is a class IA anti-arrhythmic agent. Class 1a agents are sodium channel blockers with significant potassium channel blocking properties. This slows conduction and prolongs the QRS duration and action potential duration.

Digoxin is a Na⁺ K⁺ ATPase inhibitor that works by increasing AV node refractoriness.

Esmolol is a β -receptor-blocker and a class II antiarrhythmic.

Ibutilide is a class III anti-arrhythmic agent. It works by blocking K⁺ channels.

Solution to Question 54:

The drug that increases trabecular outflow is netarsudil.

Netarsudil is a rho-kinase inhibitor used in the treatment and management of glaucoma and ocular hypertension. It lowers the intra-ocular pressure directly through the relaxation of the trabecular meshwork, leading to an increase in aqueous outflow. It also reduces aqueous humor production.

Netarsudil ophthalmic solution is instilled in one drop into the affected eye once daily in the evening. Adverse effects are conjunctival hyperemia, and corneal verticillata.

Other options:

Option A- Timolol is a non-selective β -blocker that decreases the IOP by reducing the aqueous humor production through the blockade of the β -receptors on the ciliary epithelium. Ocular adverse effects are allergy, decreased corneal sensation, and punctate keratitis.

Option C- Brimonidine is a selective alpha-2 agonist which decreases aqueous synthesis and increases uveoscleral outflow. It can cross the blood-brain barrier and, hence, causes drowsiness and CNS depression. It is contraindicated in infants and children.

Option D- Acetazolamide is a topical carbonic anhydrase inhibitor that decreases IOP by reducing aqueous formation. The main adverse effect is the bitter taste following administration. It is contraindicated in people with allergies to sulfonamide antibiotics.

Solution to Question 55:

Finasteride is used to treat male pattern baldness acts by inhibiting 5-alpha reductase.

Finasteride is a competitive inhibitor of the 5-alpha-reductase enzyme that inhibits the conversion of testosterone into its active form dihydrotestosterone (DHT).

Finasteride inhibits type 2 5-alpha reductase within the hair follicle and reduces the production of DHT. Hence, it is used in the treatment of male pattern hair loss. But, new hair growth will be lost when finasteride is discontinued, similar to minoxidil.

On long-term use (nearly 6 months), it produces an effect by reducing prostatic epithelial cell proliferation, thereby reducing the size of the prostate and improving urine flow.

Adverse effects of finasteride:

- Decreased libido

- Erectile dysfunction
- Ejaculation disorder and decreased ejaculate volume

Finasteride is highly teratogenic and pregnant women should not be exposed to the drug because of the potential for inducing genital abnormalities in male fetuses.

Solution to Question 56:

Rituximab is a monoclonal antibody that acts against CD20 B-cell antigen expressed on the surface of B-lymphocytes, plasma cells, and B-cell lymphomas.

It promotes apoptosis through transmembrane signalling as well as by antibody dependent and complement dependent cytotoxicity. It is used in the treatment of B-cell lymphoma, non-Hodgkin lymphoma, and chronic lymphocytic leukemia. It can be used both as a single agent as well as in combination with cytotoxic chemotherapy.

Adverse effects are infusion reactions like chills, fever, urticarial rashes, pruritus, dyspnea, and hypotension. Pretreatment with antihistaminics and corticosteroids prevents the reactions.

Solution to Question 57:

New drug application is filed after phase 3 of clinical trials.

Other options:

Option A: Phase 4 clinical trials are also called post-marketing surveillance.

Option C: Phase 2 trials are the first round of trials in patients.

Option D: Phase 1 trials are done in healthy volunteers.

Pharmacology NEET 2022

Question 1:

A male patient with chronic obstructive pulmonary disease (COPD) was prescribed theophylline. He noticed that his urine output had increased the following day. This action of the drug is mediated through which of the following receptors?

- a) Interleukin-10
- b) Histone deacetylase
- c) Adenosine A₁
- d) Beta 2 adrenergic receptors

Question 2:

Which of the following drugs is not likely to cause a pleural pathology?

- a) Metformin
- b) Methysergide
- c) Bleomycin
- d) Nitrofurantoin

Question 3:

A patient with deep vein thrombosis was started on a new drug. After 2 days, he presented with the given finding. Which of the following drugs is implicated in causing the above condition?



- a) Warfarin
- b) Heparin
- c) Dabigatran
- d) Rivaroxaban

Question 4:

A patient undergoing chemotherapy was given an antiemetic, after which he developed symptoms like acute dystonia, bradykinesia, and tremors. Which of the following drugs would have caused these symptoms?

- a) Ondansetron
- b) Metoclopramide
- c) Meclizine
- d) Scopolamine

Question 5:

A young patient started to take a weight loss medication that acts by inhibiting fat absorption from food. After a few weeks, she developed easy bruising and increased menstrual bleeding. Deficiency of which of the following vitamins is responsible for her condition?

- a) Vitamin E
- b) Vitamin K
- c) Vitamin B6

d) Vitamin D

Question 6:

A woman presenting with symptoms of urinary tract infection was prescribed a drug that causes tendon rupture and arthropathy. What is the mechanism of action of the drug?

- a) DNA gyrase inhibition**
- b) Ribosomal inhibition**
- c) Cell wall synthesis inhibition**
- d) Inhibition of folic acid synthesis**

Question 7:

A patient was brought to the emergency with a history of consumption of 8 tablets of digoxin. On examination, he was unstable and his heart rate was 56 bpm. ECG showed a 3rd-degree heart block. What is the next step in the management of this patient?

- a) Antibody against digoxin**
- b) Lidocaine**
- c) DC cardioversion**
- d) Phenytoin**

Question 8:

An elderly man, who is a known case of diabetic nephropathy, presented to the emergency with palpitations and chest discomfort. ECG showed tall T-waves. Laboratory investigations showed elevated potassium levels. Which of the following drugs will cause the shift of potassium back into the cell?

- a) Epinephrine**
- b) Glucagon**
- c) Atropine**
- d) Lactic acid**

Question 9:

A patient with a pituitary tumor that overproduced growth hormone underwent surgical removal of the tumor. The resection was found to be incomplete. What is the first-line treatment for this patient?

- a) Leuprolide
- b) Goserelin
- c) Nafarelin
- d) Octreotide

Question 10:

A middle-aged male patient presents with protrusion of the chin, excessive sweating, impaired glucose tolerance, and enlargement of hands and feet. Which of the following is a growth hormone receptor antagonist used to treat this condition?

- a) Octreotide
- b) Pegvisomant
- c) Cabergoline
- d) Olcegepant

Question 11:

A diabetic patient presented with rhinitis and facial swelling. An intranasal biopsy revealed the presence of broad-based aseptate hyphae with branching at right angle. Evidence of vascular invasion was present. What will be the drug of choice for this condition?

- a) Fluconazole
- b) Amphotericin B
- c) Ketoconazole
- d) Griseofulvin

Question 12:

A tuberculosis patient on anti-tubercular treatment presents with a tingling sensation and paresthesia in the lower limbs. He is not a diabetic and occasionally consumes alcohol. Which of the following vitamins must be supplemented to this patient?

- a) Vitamin B6
- b) Vitamin B12

- c) Vitamin B2
- d) Vitamin B3

Question 13:

An elderly woman received a botox injection for the treatment of wrinkles. Which of the following is the mechanism of action of botulinum toxin?

- a) Inhibits the release of acetylcholine
- b) Release of noradrenaline at synaptic cleft
- c) Selectively and irreversibly inhibits nicotinic receptors
- d) Stimulates muscarinic and nicotinic receptors

Question 14:

Which of the following is a PCSK9 inhibitor?

- a) Evolocumab
- b) Ezetimibe
- c) Bempedoic acid
- d) Clofibrate

Question 15:

An elderly man presents with rigidity and tremors. On examination, he has blank facial expressions. Which of the following drugs can be used to manage this condition?

- a) Clozapine
- b) Donepezil
- c) Selegiline
- d) Haloperidol

Answer Key

Question No.	Correct Option
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1	c
2	a
3	a
4	b
5	b
6	a
7	a
8	a
9	d
10	b
11	b
12	a
13	a
14	a
15	c

Detailed Explanations

Solution to Question 1:

An increased urine output after administering theophylline is due to the antagonism at adenosine A1 receptors.

The adenosine A1 receptors are present in the afferent arteriole of the glomeruli. When activated, they cause vasoconstriction that reduces the glomerular filtration rate (GFR) and renal blood flow. This reduces urine output. Antagonistic action on these receptors will cause an increased GFR and renal blood flow and hence increased urine output.

Methylxanthines like theophylline and aminophylline are the bronchodilators used mainly in chronic obstructive pulmonary disease (COPD) and bronchial asthma. The pharmacological actions of theophylline are:

- It inhibits phosphodiesterase (PDE) non-selectively, thereby increasing cellular cAMP and cGMP, causing bronchodilation.
- Adenosine causes bronchoconstriction by releasing histamine and leukotrienes. Theophylline antagonizes adenosine A1 receptors and causes bronchodilation.
- It increases IL-10 release, which has anti-inflammatory actions.
- It causes a decrease in gene transcription of pro-inflammatory cytokines.
- It promotes apoptosis of granulocytes and thereby, reduces the inflammatory load in the airways.

Adverse effects of theophylline:

- Headache, nausea, and vomiting (due to inhibition of PDE4)
- Increased acid secretion (due to PDE inhibition)
- Diuresis (due to inhibition of adenosine A1 receptors)
- Cardiac arrhythmias (due to inhibition of cardiac PDE3 and antagonism of cardiac A1 receptors).
- Seizures (due to central A1 receptor antagonism).

Theophylline is mainly metabolized mainly by the enzyme CYP1A2. Therefore, it has various drug interactions, which include:

- Agents that lower its plasma level are smoking, phenytoin, rifampicin, and phenobarbitone.
- Drugs that increase its plasma level are erythromycin, ciprofloxacin, oral contraceptives, and allopurinol.

Solution to Question 2:

Metformin is not associated with pleural pathology.

Metformin belongs to the biguanides class of oral hypoglycemic agents. It acts by activation of AMP-dependent protein kinase (AMPK). This results in reduced hepatic glucose production.

Side effects of metformin are lactic acidosis, gastrointestinal disturbance, and vitamin B12 deficiency.

Contraindications for metformin include :

- GFR $\leq$ 30 ml/min
- Unstable CHF
- Radiographic contrast studies
- Hospitalized patients who cannot take anything orally
- Acidosis

Other options:

Option B: Methysergide is a serotonin(5-HT) receptor antagonist. It was used for the treatment of migraine prophylaxis and carcinoid syndrome. But its use has been discontinued due to side effects like abdominal, pulmonary and endocardial fibrosis.

Option C: Bleomycin is an anticancer antibiotic used for the treatment of squamous cell cancer and Hodgkin lymphoma. It chelates copper or iron and produces superoxide ions that damage DNA. Side effects are mucocutaneous toxicity and pulmonary fibrosis.

Option D: Nitrofurantoin is a urinary antiseptic. They are used for the treatment of lower urinary tract infections (UTIs). It acts by damaging DNA and inhibiting protein synthesis in susceptible bacteria. Important side effects are gastrointestinal disturbances, liver damage and pulmonary fibrosis.

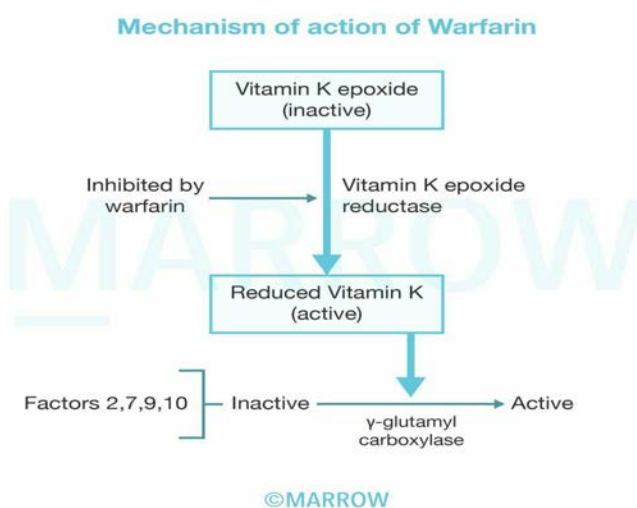
Solution to Question 3:

The clinical scenario of the development of purple toe syndrome, after starting a new drug for the treatment of deep vein thrombosis, is suggestive of the adverse effect of warfarin.

Purple toe syndrome is a rare complication of warfarin therapy. It occurs due to cholesterol microemboli released from atheromatous plaques, which cause arteriolar occlusion. Warfarin must be immediately stopped and replaced with an oral factor Xa inhibitor (e.g. apixaban).

Activation of coagulation factors II, VII, IX, and X and proteins C and S are done by carboxylation of their glutamate residues, with the help of the reduced form of vitamin K as a co-factor. This reduced form of vitamin K needs to be constantly recycled back from its oxidized form for the reaction to continue. This recycling is done by the enzyme vitamin K epoxide reductase, which is competitively inhibited by warfarin. Hence, there will be a reduction in the activation of coagulation factors.

The image below shows warfarin's mechanism of action.



The half-life of proteins C and S are 8 and 24 hours, respectively, which is relatively shorter than the half-lives of other clotting factors, particularly factor II. Therefore, the full antithrombotic effect of warfarin is not achieved for 4 to 5 days. For this reason, warfarin must be bridged with a rapidly acting parenteral anticoagulant, such as heparin, low-molecular-weight heparin (LMWH), or fondaparinux, in patients with thrombosis or at high risk for thrombosis.

Uses of warfarin are acute deep vein thrombosis, pulmonary embolism, and preventing stroke or systemic embolization in patients with a mechanical heart valve.

Adverse effects of warfarin are bleeding, skin necrosis, purple toe syndrome, and fetal warfarin syndrome (hypoplasia of the nose, eye socket, hand bones, and growth retardation).

The therapeutic dose of warfarin is monitored by an INR range of 2-3. Warfarin toxicity can occur when the INR exceeds 4 and can lead to fatal bleeding. Treatment includes withholding warfarin, with vitamin K, and blood transfusions or fresh frozen plasma to replenish clotting factors.

Warfarin, LMWH, and UFH do not accumulate in breast milk and do not have an anticoagulant effect in the newborn. They are compatible with breastfeeding.

Warfarin is mainly metabolized by the CYP2C9 enzyme. Drugs that affect the CYP2C9 enzyme can alter the anticoagulant function of warfarin and these include the following:

Reduced efficacy:

- CYP2C9 activation by drugs: barbiturates, carbamazepine, rifampin
- Oral contraceptives: increase blood levels of clotting factors

Increased efficacy:

• Drugs causing CYP2C9 inhibition:

- Amiodarone
- Azole antifungals
- Cimetidine
- Clopidogrel
- Cotrimoxazole
- Disulfiram
- Isoniazid
- Metronidazole
- Fluoxetine
- Sulfinpyrazone
- Zafirlukast

Other options:

Option B: Heparin acts by activating antithrombin III, thereby inactivating the active form of clotting factors II, IX, X, XI, XII, and XIII. Adverse effects are bleeding and heparin-induced thrombocytopenia (HIT) due to the formation of antibodies against the heparin-platelet complex.

Option C: Dabigatran is a direct thrombin inhibitor. It is used for the prevention of embolism and stroke in patients with atrial fibrillation. Bleeding and dyspepsia are the most important adverse effects.

Option D: Rivaroxaban is a direct inhibitor of factor Xa. It is used for prophylaxis and treatment of deep vein thrombosis. Side effects reported are bleeding, nausea, hypotension, tachycardia, and edema.

Solution to Question 4:

In the above scenario, the patient has developed extrapyramidal adverse effects like acute dystonia, bradykinesia, and tremors. This is likely due to the antagonist action on dopamine receptors by metoclopramide.

Metoclopramide is a prokinetic and antiemetic. It has the following pharmacological actions:

- Dopamine receptor antagonist

- Serotonin (5HT₄) receptor agonist
- Serotonin (5HT₃) antagonist
- Sensitization of muscarinic receptors on smooth muscle. This results in increased lower esophageal sphincter tone and stimulates gastric antral contractions.

Metoclopramide acts on both the gastrointestinal system and central nervous system. It is used as an upper antiemetic for many types of vomiting, especially migraine, drug-induced, and chemotherapy-induced. It is also used as a prokinetic in gastroparesis to increase gastric emptying.

Adverse effects of metoclopramide are mainly extrapyramidal effects, and these include acute dystonia, Parkinson-like symptoms (tremors, bradykinesia, rigidity), tardive dyskinesia, galactorrhea due to hyperprolactinemia, and methemoglobinemia in neonates.

Extrapyramidal symptoms can be treated by discontinuation of metoclopramide and administration of anticholinergics or antihistaminics.

Other options:

Option A: Ondansetron is a 5-HT₃ receptor antagonist. It blocks impulses at the peripheral origin in GIT. It is most commonly indicated for the treatment of postoperative and chemotherapy-induced nausea and vomiting. Headache and dizziness are the common adverse effects.

Option C: Meclizine is a long-acting H₁ antihistaminic drug. It is used for seasickness. It is less sedative and less anticholinergic.

Option D: Scopolamine (hyoscine) is a muscarinic antagonist. It blocks neural pathways from the vestibular apparatus in the inner ear to the emetic center in the brainstem. A transdermal patch of scopolamine is used as an antiemetic in motion sickness. Constipation, drowsiness, dry mouth, and blurred vision are the common adverse effects.

Solution to Question 5:

The clinical scenario of easy bruising and increased menstrual bleeding following weight loss medication that inhibits fat absorption from food is suggestive of vitamin K deficiency.

Vitamin K is a fat-soluble vitamin. Hence, drugs that inhibit the absorption of fat (eg; ezetimibe) also inhibit the absorption of fat-soluble vitamin K. The following are the vitamers of vitamin K:

- Phylloquinone (Vitamin K₁) is the normal dietary source of Vitamin K that is found in green vegetables.
- Menaquinone (Vitamin K₂) is synthesized by intestinal bacteria.
- Menadione (Vitamin K₃) is a synthetic compound that is metabolized to phylloquinone.

Vitamin K is required for γ -carboxylation of glutamate residues of clotting factors like prothrombin (factor II), factors VII, IX, and X. Warfarin inhibits γ -carboxylation by preventing the conversion of vitamin K to its active hydroquinone form.

Vitamin K deficiency in adults may be seen in patients with liver disease, obstructive jaundice, malabsorption, and prolonged antimicrobial therapy. Vitamin K deficiency is usually diagnosed

on the basis of an elevated prothrombin time or reduced clotting factors.

Vitamin K deficiency is treated with a parenteral dose of 10 mg followed by 5mg (duration depends on the severity of vitamin K deficiency).

Other options:

Option A: Vitamin E is a fat-soluble vitamin. The levels of vitamin E also will get decreased following taking weight loss medication that inhibits fat absorption from food. But the symptoms of vitamin E deficiency include hemolytic anemia and neuromuscular manifestations.

Option C: Vitamin B6 is a water-soluble vitamin. Drugs that cause vitamin B6 deficiency are isoniazid, hydralazine, cycloserine, penicillamine, and oral contraceptives. Patients usually present with peripheral neuritis, anemia, seborrheic dermatitis, glossitis, lowered seizure threshold, or convulsions (due to fall in brain GABA levels).

Option D: Vitamin D is a fat-soluble vitamin. Vitamin D levels will also decrease following weight loss medication that inhibits fat absorption from food. Patients with vitamin D deficiency will typically develop rickets in children and osteomalacia in adults.

Solution to Question 6:

In the above scenario, the antimicrobial prescribed for urinary tract infection, which has a propensity to cause tendon rupture and arthropathy, is likely to be a fluoroquinolone. The mechanism of action of the drug is DNA gyrase inhibition.

Normally, during replication, to prevent excessive positive supercoiling of the strands, DNA gyrase nicks double-stranded DNA and introduces negative supercoils. Fluoroquinolones, by inhibiting DNA gyrase (in gram-negative bacteria), result in the production of damaged DNA. This is finally digested by exonucleases, resulting in bactericidal action.

Fluoroquinolones inhibit topoisomerase IV in gram-positive bacteria.

Fluoroquinolones are used in the treatment of urinary tract infections, prostatitis, and gastrointestinal, respiratory infections, and as a second-line drug in anti-tubercular therapy. They are avoided in children and pregnant women.

Adverse effects of fluoroquinolones are arthralgia, tendon rupture, *Clostridium difficile* colitis, dysglycemia, prolonged QT interval, rashes, hallucinations, delirium, and seizures.

Other options:

Option B: Ribosomal inhibition is the mechanism of action of various antimicrobials like tetracyclines, aminoglycosides, chloramphenicol, and macrolides.

Option C: Cell wall synthesis inhibition is done by beta-lactam antibiotics.

Option D: Inhibition of folic acid synthesis is the mechanism of action of sulfonamides and co-trimoxazole.

Solution to Question 7:

Administration of digibind (antibody against digoxin) is the next step in managing this patient. An unstable patient with bradycardia and 3rd-degree heart block following digoxin toxicity should be treated with digibind.

Digibind is a nonimmunogenic antibody used to treat cardiotoxicity due to digoxin. It is nonimmunogenic because it lacks the Fc fragment.

Digoxin is the cardiac glycoside used to treat heart failure and control ventricular rate in atrial fibrillation/flutter.

It inhibits Na⁺/K⁺ ATPase, a membrane-bound transporter to increase intracellular sodium. As a result, there is less expulsion of calcium from the cell by the sodium-calcium exchanger (NCX). Increased calcium inside the myocyte leads to an increase in cardiac contractility.

The most common cardiac manifestations of digitalis toxicity are ventricular tachycardia, partial to complete A-V block, severe bradycardia, and atrial fibrillation/flutter.

Factors that increase the risk of digoxin toxicity are:

- Hypokalemia
- Drugs that decrease digoxin elimination (inhibitors of P-glycoprotein) like quinidine
- Renal/hepatic failure
- Thyrotoxicosis

Option B: Lidocaine is commonly used as a local anesthetic. It has anti-arrhythmic properties and is used to suppress ventricular tachycardia caused by digitalis toxicity.

Option C: DC cardioversion is used for tachyarrhythmias when pharmacologic treatment fails. It is not used in bradyarrhythmias.

Option D: Phenytoin is an antiepileptic used for the treatment of generalized tonic-clonic seizures and partial seizures. It can also be used for the treatment of polymorphic ventricular tachycardia due to QT prolongation.

Solution to Question 8:

The clinical scenario with ECG showing tall T-waves and elevated serum potassium levels are suggestive of hyperkalemia. Epinephrine will cause the shift of potassium back into the cell.

Epinephrine is an agonist at alpha 1, beta 1, and beta 2 receptors.

The β actions are mediated through cAMP. Epinephrine binds to Gs → ATP and is broken down to cAMP by the enzyme adenylyl cyclase. This results in:

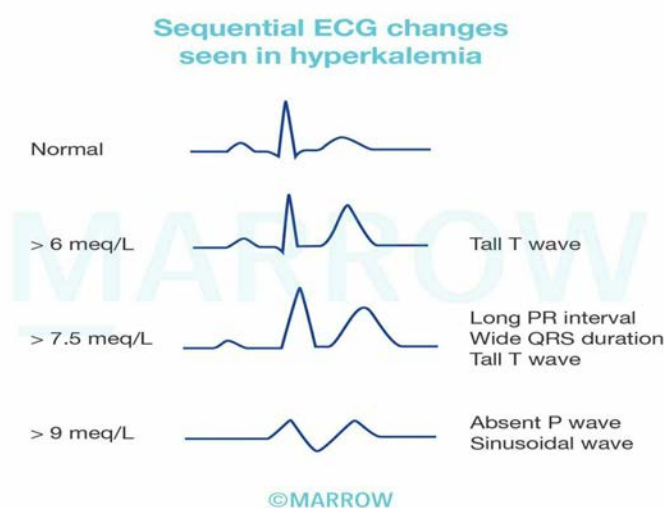
- In the liver and muscle- hyperglycemia and hypokalemia due to potassium uptake in muscle and liver
- In adipose tissue- increased plasma free fatty acids
- In the heart- an increased force of contraction reinforces the positive inotropic action
- In pancreatic islets- increases glucagon secretion

Epinephrine is used in anaphylaxis, to prolong the action of local anesthetics, cardiac arrest, and as inhalational epinephrine in the treatment of post-intubation and infectious croup.

Epinephrine is contraindicated in anesthesia with halothane due to the risk of arrhythmias and in patients receiving beta-blockers because a marked rise in BP can occur due to unopposed alpha action.

Hyperkalemia refers to an increase in the levels of serum potassium $>5.5\text{mEq/L}$.

The image given below shows sequential ECG changes seen in hyperkalemia.



Hyperkalemia is a medical emergency and has to be treated when ECG changes are present or severe hyperkalemia (serum $\text{K}^+ >6.5$) without ECG changes. Management is as follows:

- Intravenous infusion of 10 ml of 10% calcium gluconate for 2-3 minutes with cardiac monitoring. It helps to antagonize the cardiac effects.
- Intravenous regular insulin 10 units followed by 50mL of 50 % dextrose. This helps in shifting the plasma K^+ into the cells.
- Nebulisation with β_2 agonists such as albuterol or salbutamol.
- Cation exchange resins, diuretics, and/or dialysis are useful to remove the excess potassium.

Solution to Question 9:

The above scenario with a pituitary tumor producing growth hormone is likely to cause acromegaly in the patient. Octreotide is the first-line treatment for this condition.

Octreotide is a synthetic analog of somatostatin.

Octreotide can be used in the following conditions:

- Acromegaly

- Secretory diarrheas associated with carcinoid, AIDS, cancer chemotherapy, or diabetes- by suppression of hormones that enhance intestinal mucosal secretion.
- Bleeding esophageal varices- reduces hepatic blood flow and helps stop bleeding.
- Dumping syndrome in post gastric surgery patients

Adverse effects of octreotide are:

- Abdominal pain
- Nausea
- Steatorrhea
- Gall stones (due to biliary stasis)
- Hyperglycemia

Other options:

Option A: Leuprolide is a long-acting GnRH agonist. It is used for palliation of carcinoma prostate, uterine fibroids, endometriosis, and precocious puberty. Side effects include hot flashes, decreased bone mineral density, vaginal dryness, and erectile dysfunction.

Option B: Goserelin is also a GnRH agonist used for the treatment of endometriosis, dysfunctional uterine bleeding, and for prostate and breast cancer.

Option C: Nafarelin is a long-acting GnRH agonist. It is used for the treatment of endometriosis, uterine fibroids, and central precocious puberty.

Solution to Question 10:

The clinical features of protrusion of the chin, excessive sweating, impaired glucose tolerance, and enlargement of hands and feet are suggestive of acromegaly. Pegvisomant is a growth hormone receptor antagonist used to treat this condition.

Acromegaly is commonly caused due to growth hormone-secreting somatotroph adenoma. It occurs secondary to mutations of the pituitary-specific gene PIT-1.

Patients present with frontal bossing, increased hand and foot size, mandibular enlargement with prognathism, widened space between the lower incisor teeth, soft tissue swelling resulting in increased heel pad thickness, coarse facial features, and large fleshy nose.

Serum IGF-1 levels (elevated) and GH suppression test with oral glucose are useful in the diagnosis. The levels of GH do not reduce in response to the oral glucose load.

Transsphenoidal surgical resection of the tumor, octreotide, and pegvisomant will be helpful in the management of this condition.

Other options:

Option A: Octreotide is a synthetic analog of somatostatin. It is used for the treatment of acromegaly and secretory diarrhea.

Option C: Cabergoline is a dopamine D2 receptor agonist. It is used as a first-line drug for the treatment of hyperprolactinemia.

Option D: Oglcegepant is a calcitonin gene-related peptide (CGRP) antagonist. It is used for the treatment of migraine.

Solution to Question 11:

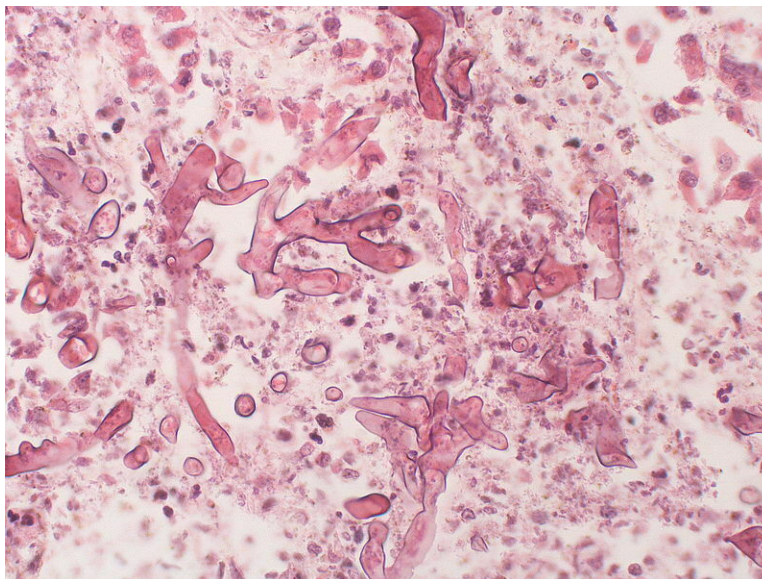
The above scenario and biopsy findings showing broad-based aseptate hyphae with branching at right angles are suggestive of mucormycosis. The drug of choice for mucormycosis is amphotericin B.

Mucormycosis is an opportunistic mycosis. It is typically seen in patients with diabetes mellitus and/or diabetic ketoacidosis and invades blood vessels.

The rhino-orbital-cerebral is the most common form of mucormycosis. Patients present with rhinitis, eye or facial pain, and facial numbness followed by the onset of conjunctival suffusion and swelling along with blurry vision.

Diagnosis is made by biopsy with histopathologic examination. The biopsy shows characteristic wide, ribbon-like aseptate hyphal elements that branch at right angles.

The below image shows biopsy findings seen in mucormycosis.



Treatment of the infection includes 4 steps:

- Initiation of therapy (amphotericin B) as early as possible.
- Surgical debridement.
- Reversal of underlying conditions and risk factors.
- Treatment of underlying malignancy if present.

Amphotericin B binds to ergosterol in the fungal membrane and forms pores or channels in the fungal cell membrane. This allows the leakage of cytosolic molecules and ions, leading to a loss of membrane integrity.

It is a broad-spectrum antifungal that can be used against *Candida albicans*, *Cryptococcus neoformans*, *Blastomyces dermatitidis*, *Histoplasma capsulatum*, *Sporothrix schenckii*, *Coccidioides immitis*, *Aspergillus*.

Amphotericin B also has antiprotozoal action along with antifungal properties and is effective against *Leishmania*.

Adverse effects of amphotericin B:

- Infusion-related fever and chills
- Azotemia/nephrotoxicity- all lipid formulations of amphotericin B are less nephrotoxic than conventional amphotericin B
- Renal tubular acidosis
- Renal wasting of potassium and magnesium
- Hypochromic normocytic anemia- due to decreased erythropoietin production from the kidney

Note: Mucormycosis recently emerged as an important superinfection in COVID-19 patients, especially in India. The surge in cases is most likely due to diabetes mellitus and steroid use.

Other options:

Option A: Fluconazole is used for the treatment of candidiasis and cryptococcosis.

Option C: Ketoconazole is used for both dermatophytosis and deep mycosis, but it has been replaced by itraconazole.

Option D: Griseofulvin is used for dermatophytosis. It is effective against the mycotic disease of the skin, hair, and nails due to *Microsporum*, *Trichophyton*, and *Epidermophyton*.

Solution to Question 12:

Vitamin B6 must be supplemented for this patient. The clinical features of tingling sensation and paresthesias in a patient on anti-tubercular treatment are suggestive of the adverse effect of isoniazid.

Isoniazid is the first-line antitubercular drug that inhibits the production of mycolic acid. It is a bactericidal drug against tubercle bacilli.

Adverse effects of isoniazid:

- Peripheral neuritis (paresthesias, numbness, tingling sensation) due to interference with the production of active coenzyme pyridoxal phosphate from pyridoxine. Therefore, pyridoxine 10 mg/day (vitamin B6) is given prophylactically along with isoniazid to prevent neurotoxicity.
- Hepatitis - isoniazid must be stopped at the first sign of hepatotoxicity.
- Rashes
- Anemia
- Arthralgia

Toxicity of isoniazid results in metabolic acidosis and seizures.

Note: Vitamin deficiency in alcoholics includes the following:

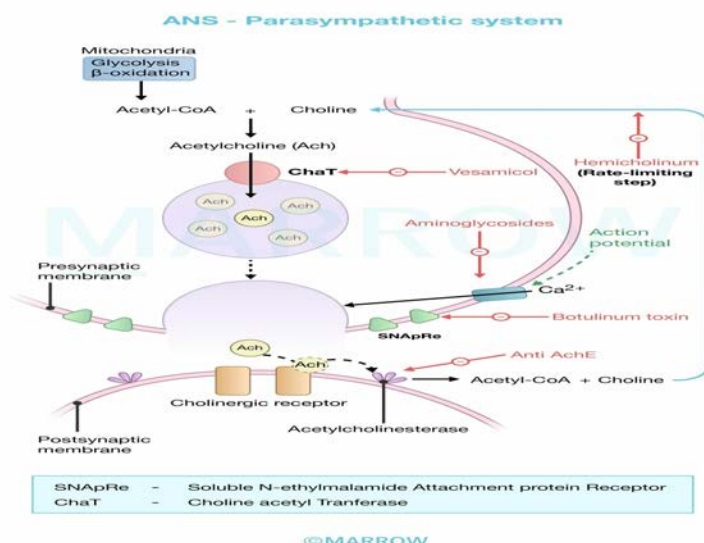
- Folate causes megaloblastic anemia. It is treated by supplementation of oral doses of 5-15 mg of folic acid.
- Vitamin B6 results in peripheral neuritis, anemia, and glossitis, and lowers the seizure threshold. It is treated by supplementation of oral doses of 2-5 mg of pyridoxine.
- Thiamine results in Wernicke's encephalopathy. It is treated with 200 mg of intravenous thiamine three times daily, followed by 10 mg/day until complete recovery.
- Vitamin A results in xerophthalmia and night blindness. It is treated with supplementation of oral vitamin A 3000-5000 IU/day.

Solution to Question 13:

The botulinum toxin acts by inhibiting the release of acetylcholine into the synaptic cleft.

Botulinum toxin cleaves SNARE protein, which is involved in the exocytosis of acetylcholine. Thus, it inhibits the release of acetylcholine.

The image given below shows the site of action of botulinum toxin:



This toxin causes a long-lasting loss of cholinergic transmission. Hence, it can be employed for the treatment of several spastic and neurological conditions caused by the overactivity of cholinergic nerves, such as age-related facial wrinkles, blepharospasm, spastic cerebral palsy, spasmodic torticollis, post-stroke spasticity, and axillary hyperhidrosis.

Adverse effects occur due to incorrect injection and overdose; they include ptosis, diplopia, facial swelling, dry mouth, muscle weakness, and respiratory paralysis.

Botulinum toxin is an exotoxin produced by the bacteria *Clostridium botulinum*. It is responsible for a type of food poisoning called botulism.

Solution to Question 14:

Evolocumab is a PCSK9 inhibitor.

PCSK9 refers to proprotein convertase subtilisin/kexin type 9. They are mainly involved in the downregulation of low-density lipoprotein (LDL) receptors in the liver.

Evolocumab is a monoclonal antibody that inhibits PCSK-9 (proprotein convertase subtilisin/kexin). Inhibition of PCSK-9 prevents the degradation of LDL receptors and thus increases LDL-cholesterol clearance by the liver. They are used in patients with atherosclerotic cardiovascular disease (ASCVD) to decrease LDL levels.

Side effects of PCSK9 inhibitors are injection-site reactions, nasopharyngitis, urinary tract infection, and upper respiratory tract infection and it does not increase the risk of myopathy.

Other options:

Option B: Ezetimibe inhibits cholesterol absorption by enterocytes in the small intestine. It lowers the total and LDL-C levels.

Option C: Bempedoic acid inhibits adenosine triphosphate-citrate lyase (ACL) and results in decreased LDL in patients not responding to statin therapy.

Option D: Clofibrate binds to PPAR-alpha and causes stimulation of fatty acid oxidation and increased LPL synthesis. This results in reduced triglycerides, LDL and increased high-density lipoprotein (HDL) levels. It causes myopathy when combined with statins and is associated with an increased risk of gallstone formation.

Solution to Question 15:

The clinical features of rigidity, tremors, and blank facial expressions are suggestive of Parkinson's disease. Monoamine oxidase-B (MAO-B) inhibitors like selegiline are used for the treatment of Parkinson disease.

Normally, MAO-B oxidizes dopamine. Selegiline is a selective and irreversible MAO-B inhibitor. Therefore, it increases dopamine levels in the brain. It is used as an adjuvant to levodopa in Parkinson's disease. It prolongs and enhances the action of levodopa.

Adverse effects of selegiline are postural hypotension, nausea, confusion, involuntary movements, and psychosis. Selegiline is contraindicated in patients with convulsive disorders.

Other options:

Option A: Clozapine is an atypical antipsychotic that selectively blocks dopamine (D4) receptors in the brain. It is used in the treatment of refractory schizophrenia. Agranulocytosis and metabolic syndrome are the important adverse effects.

Option B: Donepezil is a reversible cholinesterase inhibitor used for the treatment of Alzheimer's disease.

Option D: Haloperidol is a typical antipsychotic used for the treatment of acute schizophrenia, Huntington's disease, and Gilles de la Tourette syndrome.

Pharmacology NEET 2023

Question 1:

A bronchial asthma patient on inhalational steroids presented with white patchy lesions on the tongue and buccal mucosa. What is the drug that can be used to treat this condition?

- a) Clotrimazole
- b) Griseofulvin
- c) Terbinafine
- d) Flucytosine

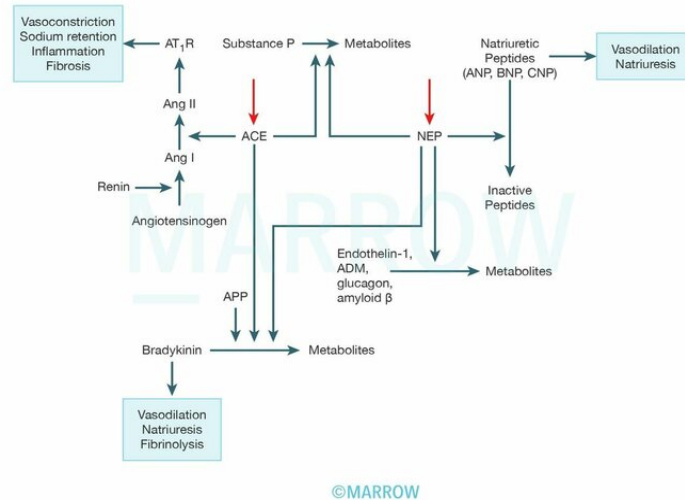
Question 2:

A pregnant woman with a history of bronchial asthma is in the third stage of labor. Which of the following drugs should be avoided in the management of postpartum hemorrhage in this mother?

- a) Carboprost
- b) Oxytocin
- c) Dinoprostone
- d) Methyl ergometrine

Question 3:

Name the drug that acts on both the marked areas.



- a) Sacubitril
- b) Omapatrilat
- c) Losartan
- d) Nesiritide

Question 4:

A patient with a previous history of myocardial infarction and ventricular arrhythmia is on treatment for a few months. He developed fatigue, dyspnea, and weight gain. He had also developed pulmonary fibrosis. Which of the following drugs is responsible for the above-mentioned side effects?

- a) Amiodarone
- b) Atenolol
- c) Aspirin
- d) Spironolactone

Question 5:

A young female comes to the emergency department after consuming 100 tablets of aspirin. What should be the next step in management?

- a) N-acetyl cysteine to replenish glutathione stores
- b) Pralidoxime
- c) Glucagon to control bradycardia and hypoglycemia

- d) Give sodium bicarbonate to alkalize urine

Question 6:

A chronic smoker was on nicotine replacement therapy and clonidine tablets for smoking de-addiction. He stopped taking clonidine tablets and now presents with a headache. What is the reason behind this condition?

- a) Postural hypotension
- b) Receptor upregulation
- c) Rebound hypertension
- d) Receptor hypersensitivity

Question 7:

A patient with recurrent attacks of gout was started on a therapy that inhibits the synthesis of uric acid. His symptoms were reduced after therapy. Which drug was he started on?

- a) Probenecid
- b) Colchicine
- c) Diclofenac
- d) Allopurinol

Question 8:

A patient with a malignancy is undergoing chemotherapy. The platelet counts were reduced after the previous cycle of chemotherapy. Which of the following drugs can be used in the treatment of this patient?

- a) Oprelvekin (IL-11)
- b) Filgrastim
- c) Erythropoietin
- d) Amifostine

Question 9:

The true statement among the following is _____

- a) The dose of telmisartan should be reduced in renal failure but not in hepatic failure.
- b) The dose of irbesartan should be reduced in case of mild-moderate hepatic failure and renal failure.
- c) The dose of candesartan should be reduced in mild-moderate liver failure but not in renal failure.
- d) Losartan acts as thromboxane A₂ antagonist and inhibits platelet aggregation.

Question 10:

A patient given digoxin started having side effects like nausea and vomiting. The serum concentration of digoxin was 4 mg/dL. The plasma therapeutic range is 1 mg/dL. If the half-life of digoxin is 40 hours, how long one should wait before resuming the treatment?

- a) 40 hours
- b) 80 hours
- c) 120 hours
- d) 140-180 hours

Question 11:

A patient comes to the casualty with organophosphate poisoning. He was started on atropine infusion and pralidoxime. After 2 hours, the patient had a sudden rise in temperature. What is the likely cause of fever?

- a) Atropine toxicity
- b) Side effect of pralidoxime
- c) Due to organophosphate poisoning
- d) Idiopathic

Question 12:

The mechanism of action of tissue plasminogen activator is _____

- a) Inhibit extrinsic pathway
- b) Inhibits platelet aggregation
- c) Enhance fibrin degradation
- d) Inhibit clot formation

Answer Key

Question No.	Correct Option
1	a
2	a
3	b
4	a
5	d
6	c
7	d
8	a
9	d
10	b
11	a
12	c

Detailed Explanations

Solution to Question 1:

The given clinical scenario of white patchy lesions on the tongue and buccal mucosa in a patient using inhalational steroids is oral candidiasis. The drug that can be used for the treatment is clotrimazole.

MDI results in the deposition of steroids in the oropharyngeal space, which can cause cough, dysphonia, and oropharyngeal candidiasis.

Clotrimazole is a topical imidazole used in the treatment of tinea infections like ringworm. It is particularly used for the treatment of vaginitis as a once-daily application because of its long-lasting residual effect.

For oral candidiasis, 10 mg troche of clotrimazole is allowed to dissolve in the mouth 3-4 times a day. Alternatively, lotion or gel is applied and swirled in the mouth for as long as possible. Side effects are local irritation and a burning sensation.

Other options:

Option B: Griseofulvin acts by inhibiting microtubule assembly, disrupting the fungal cell division. It is mainly used against dermatophytes and is the drug of choice for tinea capitis in children. It is not effective against Candida.

Option C: Terbinafine acts by inhibiting epoxidation of fungal squalene that reduces ergosterol biosynthesis. It is helpful for treating nail onychomycosis and tinea capitis.

Option D: Flucytosine acts by inhibiting fungal DNA synthesis. It is used exclusively in combination with amphotericin B for cryptococcal meningitis.

Solution to Question 2:

The drug that should be avoided in the management of postpartum hemorrhage in this mother with a history of bronchial asthma is carboprost.

Carboprost is an analog of PGF₂α. It is used in obstetrics to induce second-trimester abortions and to control postpartum hemorrhage. Vomiting, diarrhea, and bronchoconstriction can occur.

Contraindications of carboprost:

Absolute	Relative
Asthma Pulmonary hypertension Suspected amniotic fluid embolism	Pre-eclampsia Renal disease Liver disease Heart disease

Other options :

Option B: Oxytocin is the first-line drug to prevent and treat PPH according to WHO guidelines. It is administered by intravenous route and may lead to hypotension and reflex tachycardia.

Option C: Dinoprostone is a prostaglandin E₂ analog used for cervical ripening, induction of labor, and in atony treatment. It should be avoided in patients with a history of asthma, glaucoma, or myocardial infarction. One major adverse effect is uterine hyperstimulation.

Option D: Ergot derivatives are used in the control of postpartum uterine bleeding. It is given intramuscularly and is effective within 1–5 minutes. It is given at the time of delivery of the placenta or immediately after if bleeding is significant. Side effects are gastrointestinal disturbances like diarrhea, nausea, and vomiting.

Solution to Question 3:

The drug that acts on both the marked areas is omapatrilat.

Omapatrilat is a dual inhibitor of neutral endopeptidase and Angiotensin-converting enzyme (ACE). It decreases angiotensin II and increases atrial and brain natriuretic peptide, promoting vasodilation, diuresis, and anti-hypertrophic effect.

Other options:

Option A - Sacubitril is a neprilysin inhibitor. It reduces the breakdown of natriuretic peptides such as ANP, BNP, and CNP, thereby increasing their levels. Sacubitril combined with valsartan reduces mortality in heart failure with reduced ejection fraction (HFrEF).

Option C - Losartan is the angiotensin receptor blocker used in the treatment of hypertension. Hyperkalemia is an important side effect of ARBs. It appears to be a safe and efficacious agent to lower serum uric acid levels in patients with hyperuricemia.

Option D - Nesiritide is a recombinant human brain natriuretic peptide (BNP). It causes vasodilatation and natriuresis. It is administered intravenously and used in acute decompensated heart failure.

Solution to Question 4:

The drug that is responsible for the above-mentioned side effects is amiodarone.

Amiodarone is a class III antiarrhythmic drug. The most serious adverse effect during chronic amiodarone therapy is pulmonary fibrosis, which is rapidly progressive and fatal. The side effects of amiodarone include:

- Pulmonary fibrosis
- Hypotension, depressed myocardial function
- Corneal microdeposits
- Hepatotoxicity
- Neuropathies
- Photosensitivity
- Hypo- or hyperthyroidism

Other options:

Option B: Atenolol is a beta-1 selective antagonist. Bradycardia, fatigue, and cold extremities are common adverse effects. It is used in combination with a diuretic, in elderly patients with isolated systolic hypertension. CNS side effects like depression and nightmares are seen.

Option C: Aspirin is a nonsteroidal anti-inflammatory drug (NSAID) that blocks the formation of thromboxane A₂ by acetylation of COX₁. Through this mechanism, it shows antipyretic, analgesic, and anti-inflammatory actions. Higher doses can result in bleeding.

Option D: Spironolactone is a potassium-sparing diuretic. It is a competitive inhibitor of the mineralocorticoid receptor in the late distal tubule and collecting duct of the kidneys. It promotes sodium diuresis but maintains the body's potassium levels. It is contraindicated in patients with hyperkalemia. Side effects are gynecomastia, impotence, decreased libido, and menstrual irregularities.

Solution to Question 5:

The next step in the management of this patient who consumed 100 tablets of aspirin is to give sodium bicarbonate to alkalinize the urine.

Alkalinization of urine will be helpful in the removal of aspirin, which is a weakly acidic drug and is susceptible to ion trapping in the urine. It is done by administration of sodium bicarbonate in 1

L of 5% dextrose in water (D5W). It is infused intravenously at twice the maintenance fluid requirements.

Alkalinization of urine is contraindicated in renal failure or if fluid administration may worsen pulmonary edema or congestive heart failure.

Other options:

Option A: N-acetyl cysteine is used as an antidote for the treatment of paracetamol poisoning which replenishes glutathione stores in the liver.

Option B: Pralidoxime is also used in organophosphate poisoning. It regenerates active acetylcholinesterase and relieves skeletal muscle end-plate block. It is usually used as an adjuvant to atropine.

Option D: Glucagon is used in beta-blocker toxicity.

Solution to Question 6:

The mechanism that causes headache in this patient, who stopped taking clonidine is rebound hypertension.

Clonidine is an alpha-2 receptor agonist and a centrally acting antihypertensive agent. It treats high blood pressure by stimulating presynaptic α_2 receptors in the vasomotor center in the brainstem. This binding has a sympatholytic effect, by suppressing the release of norepinephrine and other sympathomimetics.

Abrupt withdrawal after long-term therapy can result in rebound hypertension, restlessness, and sympathetic overactivity. Dry mouth and sedation are the major side effects of clonidine. It can also cause impotency and constipation.

Phentolamine is used for the control of hypertension due to clonidine withdrawal.

Solution to Question 7:

Allopurinol which inhibits uric acid synthesis is the likely drug that was added to the regimen.

It is a purine antimetabolite and a competitive inhibitor of the xanthine oxidase enzyme. It prevents the conversion of hypoxanthine and xanthine to uric acid reducing plasma uric acid levels. The metabolite of allopurinol called oxypurinol is a non-competitive inhibitor of xanthine oxidase.

It is used in the treatment of hyperuricemia in patients with gout and also in patients with hematological malignancies about to undergo chemotherapy.

Other options:

Option A: Probenecid is a uricosuric agent that promotes the excretion of uric acids by inhibiting the organic anion transporter, particularly URAT-1 in the renal tubule. It is used to decrease the body pool of urate in patients with chronic gout. It is used as an adjuvant to allopurinol in chronic gout and hyperuricemia. Plenty of fluids are given to avoid urate

calculi formation.

Option B: Colchicine is an antimitotic drug that is used in the treatment of acute gout. It acts by inhibiting microtubule and spindle formation. Side effects are nausea, vomiting, myelosuppression, aplastic anemia, and rhabdomyolysis.

Option C: Diclofenac is a commonly used NSAID that has analgesic, antipyretic, and anti-inflammatory activities. It acts by inhibiting the cyclooxygenase-2 enzyme. Side effects are rashes, fluid retention, edema, and renal function impairment.

Solution to Question 8:

The most probable diagnosis for this patient is chemotherapy-induced thrombocytopenia. Oprelvekin can be used in the treatment of this patient.

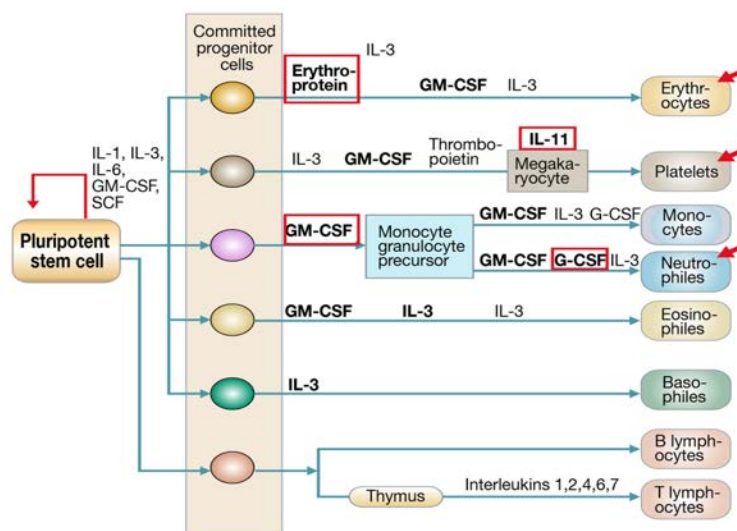
Oprelvekin is an IL-11 analog used in the treatment of chemotherapy-induced thrombocytopenia. It increases thrombopoiesis. It is used in the treatment of thrombocytopenia due to non-myeloid malignancies. Important adverse effects are fluid retention, tachycardia, palpitation, edema, and shortness of breath.

Other options:

Option B: Filgrastim is a granulocyte-colony-stimulating factor (G-CSF) that is used for the treatment of severe neutropenia after high-dose cancer chemotherapy.

Option C: Epoetin alfa and darbepoetin alfa are recombinant human erythropoietin. They are used in the treatment of anemias associated with surgery, AIDS, cancer chemotherapy, and certain chronic inflammatory conditions.

Option D: Amifostine is used in the reduction of nephrotoxicity associated with cisplatin.



Solution to Question 9:

The true statement is losartan acts as a thromboxane A₂ antagonist and inhibits platelet aggregation.

Losartan is a competitive angiotensin receptor antagonist. It binds to the AT₁ receptor with high affinity. It is also a competitive antagonist of the thromboxane A₂ receptor and attenuates platelet aggregation.

Losartan is the angiotensin receptor blocker used in the treatment of hypertension. Hyperkalemia is an important side effect of ARBs. It appears to be a safe and efficacious agent to lower serum uric acid levels in patients with hyperuricemia.

Other options:

Option A: The clearance of telmisartan is mainly by biliary secretion. The clearance is affected by hepatic insufficiency. The dose of telmisartan should be reduced in hepatic failure but not in renal failure.

Option B: Irbesartan is metabolized to glucuronide conjugate, and is cleared mainly by biliary excretion. The plasma clearance is unaffected by either renal or mild-to-moderate hepatic insufficiency. Hence, the dose need not be reduced in either of the cases.

Option C: The plasma clearance of candesartan is by biliary excretion and renal elimination. The clearance is affected by renal insufficiency. Hence, the dose should be reduced in case of renal failure and not in mild-moderate liver failure.

Solution to Question 10:

The treatment can be resumed after 80 hours.

The half-life ($T_{1/2}$) is the time it takes for the plasma concentration of the drug to be reduced by 50%. In the given scenario, the half-life of digoxin is given as 40 hours. The serum concentration of digoxin is given as 4 mg/dL.

After 40 hours, the serum concentration of digoxin becomes 2 mg/dL (half of the initial concentration), and after 80 hours, it will be reduced to 1 mg/dL. So, to reach a serum concentration of 1 mg/dL (therapeutic range), it will take 80 hours.

Solution to Question 11:

The most likely cause of fever in this patient is atropine toxicity.

The clinical scenario is suggestive of organophosphate consumption and atropine is used for its management. Atropine is a muscarinic receptor antagonist, and it competitively blocks the acetylcholine at muscarinic receptors.

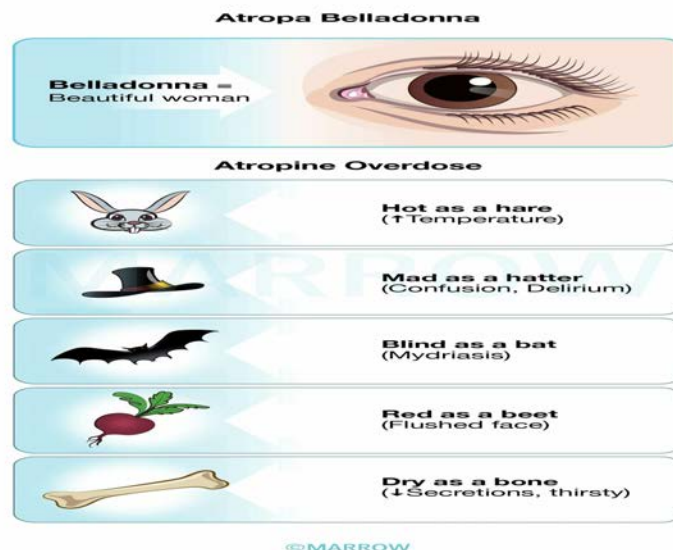
Organophosphates are powerful acetylcholinesterase inhibitors. It is absorbed by inhalation or through the skin. Mucous membranes and the gastrointestinal tract are also the sites for absorption.

The predominant initial signs are of muscarinic excess which includes: (DUMBBELSS)

- Diarrhoea
- Urination
- Miosis
- Bronchospasm
- Bradycardia
- Emesis
- Lacrimation
- Salivation
- Sweating

Nicotinic manifestations include twitching of eyelids, tongue, and facial muscles which then leads to paralysis. CNS stimulation results in headache, tremors, ataxia, coma, and death.

Atropine in larger doses can lead to atropine toxicity. The features are depicted in the following image:



Solution to Question 12:

The mechanism of action of the tissue plasminogen activator is to enhance fibrin degradation.

Tissue plasminogen activators activate plasminogen bound to fibrin, and catalyze the conversion of plasminogen to plasmin, resulting in increased formation of plasmin. This results in fibrinolysis of formed thrombus (thrombolysis).

It is indicated in pulmonary embolism with hemodynamic instability, acute ischemic stroke, and acute myocardial infarction. Recombinant variants of t-PA like reteplase and tenecteplase have a longer half-life.

All thrombolytics can cause bleeding, which may be counteracted by the administration of antifibrinolytics, such as aminocaproic acid and tranexamic acid.

Pharmacology INI-CET 2023

Question 1:

A patient underwent a hysterectomy and was given morphine in the post-operative period and in the ward for pain. She now presents with respiratory depression and pin point pupils. What is the drug used to treat this condition?

- a) Naloxone
- b) Pethidine
- c) Atropine
- d) Buprenorphine

Question 2:

Drug used for trastuzumab-resistant her 2 positive breast cancer is:

- a) Vismodegib
- b) Lapatinib
- c) Vemurafenib
- d) Erlotinib

Question 3:

Nomogram used for vancomycin dosing is?

- a) Salazar
- b) Halsted
- c) Rummack-mathew
- d) Matzke

Question 4:

Which of the following is true statement regarding lithium?

- a) It is not absorbed from the stomach
- b) It can cause fine tremors in the therapeutic range

- c) Approved for the treatment of absent seizures
- d) It is not teratogenic

Question 5:

Match the following

- a) A-2, B-3, C-4, D-1
- b) A-3, B-4, C-1, D-2
- c) A-4, B-2, C-3, D-1
- d) A-1, B-2, C-3, D-4

Question 6:

True about therapeutic index of a drug is:

- a) Denotes safety and efficacy after experiments on humans
- b) Denotes safety and efficacy in the population
- c) Denotes safety and efficacy in an individual
- d) Ratio of median lethal dose in 50% of population to median effective dose in 50% of population

Question 7:

Osimertinib is used in NSCLC with which of the following mutation?

- a) T790M mutation
- b) L858R mutation
- c) T890M mutation
- d) M790T mutation

Question 8:

A patient was started on a drug for mood disorder and he developed priapism. The causative drug is:

- a) Trazodone
- b) Venlafaxine

- c) Bupropion
- d) Mirtazapine

Question 9:

Cyclosporine-induced cholestasis is due to which receptor?

- a) MDR 1
- b) BRCA
- c) MRP 1
- d) OAT1B1

Question 10:

A diabetic patient was started on a drug that decreased her HbA1C from 7.6 to 6.7. She started complaining of itching around vulvovaginal area . What was the most likely drug she was started on?

- a) Acarbose
- b) Canagliflozin
- c) Linagliptin
- d) Liraglutide

Question 11:

An anti-diabetic drug causes a decrease in HbA1C by causing an increase in GIP and GLP-1 levels and a decrease in glucagon levels. It is an oral drug causing a reduction in both pre and post-prandial glucose. The drug is:

- a) Pramlintide
- b) Alogliptin
- c) Liraglutide
- d) Metformin

Question 12:

What is true about keynote 189 trial?

- a) Pembrolizumab plus chemotherapy used in NSCLC
- b) Pembrolizumab used in NSCLC
- c) Nivolumab plus chemotherapy used in NSCLC
- d) Nivolumab is used in NSCLC

Question 13:

Which of the following is not a mechanism for thalidomide teratogenicity?

- a) Release of reactive oxygen species and free radicals
- b) Suppression of angiogenesis
- c) Cereblon binding
- d) Inhibition of IL2, IL4, IL5

Question 14:

Which is a rare but very severe side effect of clozapine that requires regular monitoring?

- a) Suicidal tendency
- b) Agranulocytosis
- c) Myocarditis
- d) Cerebral bleeding

Question 15:

Which of the following newly approved anti-diabetic drugs can be given for heart failure with reduced ejection fraction and impaired renal function?

- a) Empaglifozin
- b) Degludec
- c) Linagliptin
- d) Rosiglitazone

Question 16:

A patient diagnosed with ulcerative colitis presents with 2-3 episodes of bloody stools per day. What should be given to manage this patient ?

- a) Budesonide
- b) Azathioprine
- c) 6-mercaptopurine
- d) 5-aminosalicylic acid

Question 17:

A 25 year old male patient presents with several purpuric skin lesions and is diagnosed with idiopathic thrombocytopenic purpura. Which of the following tyrosine kinase inhibitors can be used in his condition?

- a) Eltrombopag
- b) Romiplostim
- c) Bosutinib
- d) Fostamatinib

Question 18:

A 40-year-old woman with multidrug-resistant tuberculosis presents with fatigue, constipation, weight gain, and neck swelling. Which of the following drugs is implicated in this case?

- a) Pyrazinamide
- b) Ethionamide
- c) Cycloserine
- d) Streptomycin

Question 19:

A patient on amitriptyline was found in an unconscious state. Which of the following does not suggest amitriptyline toxicity?

- a) Arrhythmias
- b) Confusion
- c) Hypothermia
- d) Mydriasis

Question 20:

Which of the following is beneficial in reducing the painful crisis of sickle cell anemia?

- a) Hydroxyurea
- b) Methotrexate + leukovorin
- c) Methionine
- d) Eltrombopag

Question 21:

What is the most useful drug to treat doxorubicin induced cardiotoxicity?

- a) Dexrazoxane
- b) Dexamethasone
- c) Aprepitant
- d) Amifostine

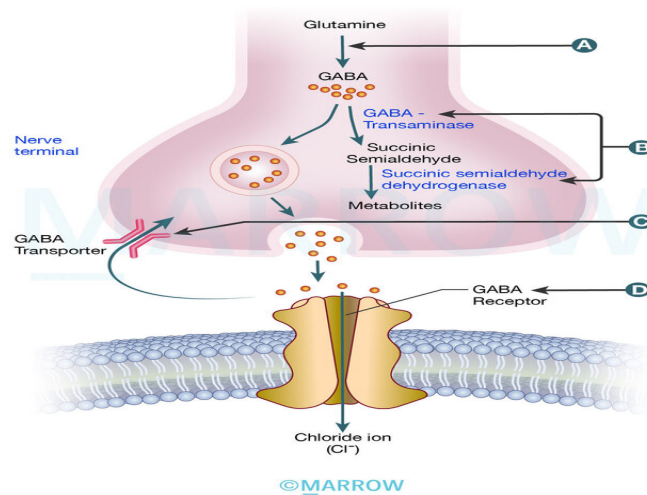
Question 22:

An elderly patient with dementia is not compliant with oral medications. Which of the following will be beneficial in this case?

- a) Donepezil
- b) Rivastigmine
- c) Galantamine
- d) Memantine

Question 23:

Sodium valproate has an inhibitory action at which of the following sites?



- a) A
- b) B
- c) C
- d) D

Question 24:

Which of the following drugs acts via the ATP-binding cassette transporter?

- a) Verapamil
- b) Diltiazem
- c) Nifedipine
- d) Amlodipine

Question 25:

A 12 year old boy presents with complaints of profuse watery stools and an elevated WBC count of 16000 cells/mm³. He has been on treatment with amoxicillin for the past 1 week. What is the treatment of choice for his condition?

- a) Fidaxomycin
- b) Clindamycin
- c) Zanamivir
- d) Cilastatin

Question 26:

Which of the following mood stabilisers causes hepatotoxicity and hyperammonemia?

- a) Lithium
- b) Valproate
- c) Carbamazepine
- d) Topiramate

Question 27:

Which of the following pathogens is correctly matched with its intrinsic drug resistance?

- a) *Aspergillus niger* - Voriconazole
- b) *Candida krusei* - Fluconazole
- c) *Candida Glabrata* - Amphotericin B
- d) *Sporothrix* - Itraconazole

Question 28:

A child was receiving a blood transfusion. At what serum iron level will desferrioxamine be administered to him?

- a) 350 mcg/dL
- b) 400 mcg/dL
- c) 450 mcg/dL
- d) 500 mcg/dL

Question 29:

A patient on fluoxetine was started on a pain control drug for neuropathic pain. He developed an episode of syncope after 2 weeks. ECG reveals QT interval of 477 ms. Which of the following is the causative drug?

- a) Methadone
- b) Morphine
- c) Meperidine

d) Propoxyphene

Question 30:

All of the following side effects are more commonly seen with carbamazepine than oxcarbazepine except

- a) Rashes**
- b) Hyponatremia**
- c) Blood dyscrasia**
- d) Hepatitis**

Question 31:

A patient with hemophilia on antiretroviral therapy is brought in altered mental status following complaints of nausea, vomiting and headache. Which of the following medications was he taking?

- a) Lopinavir**
- b) Tipranavir**
- c) Darunavir**
- d) Ritonavir**

Question 32:

A 60-year-old patient with chronic kidney disease suddenly develops chest pain and palpitations. He had previously complained about feeling weak for the past week. Serum potassium was found to be 6.5mEq/L. Which of the following is likely to be found in this patient's ECG?

- a) Tall peaked T waves**
- b) Flat T waves**
- c) Tall QRS complexes**
- d) Prolonged PR intervals**

Question 33:

Which of the following drugs can be used in diabetic neuropathy, post herpetic neuralgia and spinal cord injury?

- a) Amitriptyline
- b) Phenytoin
- c) Carbamazepine
- d) Pregabalin

Question 34:

Which of the following antibiotics acts by inhibiting the 50s ribosome?

- a) Aztreonam
- b) Daptomycin
- c) Linezolid
- d) Rifampicin

Answer Key

Question No.	Correct Option
1	a
2	b
3	d
4	b
5	b
6	b
7	a
8	a
9	a
10	b
11	b
12	a
13	d
14	b
15	a

16	d
17	d
18	b
19	c
20	a
21	a
22	b
23	b
24	a
25	a
26	b
27	b
28	d
29	a
30	b
31	b
32	a
33	d
34	c

Detailed Explanations

Solution to Question 1:

The drug of choice for the reversal of respiratory depression is naloxone. Naloxone is a competitive antagonist at μ -opioid receptors and rapidly reverses the effects of opioid toxicity when administered.

Opioid overdose symptoms:

- Triad of coma, respiratory depression, and pinpoint pupils (miosis).
- Hypotension, bradycardia
- Shallow breathing
- Hypothermia

Naloxone is a pure opioid antagonist which reverses all effects, except sedation. Naloxone is also used for reversing neonatal asphyxia due to opioid usage in labor.

Other options

Option B - Meperidine (pethidine) is a mu-opioid receptor (MOR) agonist. It has a strong analgesic effect on the central nervous system. It is mainly used as an analgesic but can also be

used in post-anesthetic shivering. It has a greater ability to enter the CNS and produce analgesia at lower systemic concentrations compared to morphine. Therefore, constipation, urinary retention, and nausea are less frequent compared to morphine.

Meperidine can cross the placenta and cause significant respiratory depression in the fetus. In patients or addicts who are tolerant to the depressant effects of meperidine, large doses repeated at short intervals may produce an excitatory syndrome that includes hallucinations, tremors, muscle twitches, etc.

Option C -Atropine is an anticholinergic drug. It acts through reversible, competitive inhibition of the postsynaptic muscarinic acetylcholine receptors. It is indicated in the treatment of organophosphate/muscarinic poisoning, in bradycardia, and to reduce salivary secretions.

The adverse effects of atropine are due to the antimuscarinic effects, including xerostomia, blurring of vision, photophobia, tachycardia, flushing, warm, dry skin, constipation, urinary retention, anhidrosis, delirium, and coma. Intoxication with atropine results in all the above symptoms along with convulsions, and circulatory collapse due to respiratory failure. Death may occur secondary to respiratory paralysis.

Option D - Buprenorphine is a partial μ agonist and an antagonist at δ and κ receptors. It is used as an analgesic and in the management of opioid dependence.

Solution to Question 2:

Lapatinib is used for trastuzumab-resistant HER-2 neu-positive breast carcinoma.

Lapatinib is an orally bioavailable, small-molecule inhibitor of the EGFR and HER2 tyrosine kinases. Lapatinib is metabolized by CYP3A4.

Lapatinib in combination with capecitabine is approved for the treatment of patients with metastatic HER2-positive trastuzumab-refractory breast cancer. Lapatinib is also used in combination with the aromatase inhibitor letrozole to treat postmenopausal women with hormone receptor-positive, metastatic breast cancer that overexpresses HER2.

Frequent adverse effects include acneiform rash, diarrhea, cramping, and exacerbation of gastroesophageal reflux. Cardiac toxicity appears less pronounced than with trastuzumab, although lapatinib should be used with caution in combination with cardiotoxic drugs. A dose-dependent prolongation of the QT interval has been reported; thus, careful monitoring of patients with heart disease is recommended. Hepatotoxicity that may be severe has been observed.

Trastuzumab is a monoclonal antibody that inhibits the HER2 receptor. It is used in the treatment of HER2-positive breast cancer.

Other options:

Option A: Vismodegib is a hedgehog pathway inhibitor used to treat metastatic and locally advanced BCCs.

Option C: Vemurafenib is a BRAF inhibitors used in metastatic melanoma and thyroid cancers with BRAF V600E mutation.

Option D: Erlotinib is a protein kinase antagonists which is the first line treatment of non-small cell lung carcinoma with EGFR deletion of exon 19 or L858R mutation

Solution to Question 3:

The nomogram used for vancomycin dosing is the Matzke nomogram.

The Matzke nomogram includes dosing intervals ranging from 0.5 to 12 days on a continuum of CL_{cr}(Creatinine clearance) values from 0 to 120 mL/min. It is designed to guide the determination of the initial empiric dose and interval of intravenous (IV) vancomycin in hospitalized adults who are not receiving hemodialysis. The nomogram does not apply to patients who exhibit highly variable vancomycin pharmacokinetics. These patients include the critically ill, those with burn injuries affecting more than 20 percent of their body surface, patients with rapidly changing renal function, pregnant women, those with liver failure/ascites, individuals at the extremes of weight, patients who are acutely post-transplant, and those with cystic fibrosis

The Rumack-Matthew nomogram uses the serum acetaminophen concentration, in relation to the time after ingestion, to assess potential hepatotoxicity after a single, acute ingestion of acetaminophen.

Solution to Question 4:

The true statement about lithium is it can cause fine tremors in the therapeutic range.

Lithium has a very narrow therapeutic index of 0.8-1.2 mEq/L. The severity of lithium toxicity is often divided into the following three grades:

A fine tremor is common in patients taking lithium at therapeutic doses. Therapeutic lithium tremor is generally symmetric, limited to the arms, and nonprogressive. The tremor from lithium toxicity is coarser and more disabling, may also affect the legs, and occurs with other symptoms of toxicity. Other adverse effects include mental confusion, hypothyroidism, nephrogenic diabetes insipidus, and edema. Lithium use in early pregnancy causes Ebstein anomaly in the fetus(Option D). Fetal lithium exposure is associated with neonatal goiter, CNS depression, hypotonia (floppy baby syndrome), and cardiac murmur.

As lithium is readily dialyzed, lithium overdose or toxicity is treated with hemodialysis.

Lithium is a mood stabilizer with a very narrow therapeutic index. Lithium is administered orally. Lithium is almost completely absorbed from the GI tract(Option A). It is the first line of management in bipolar disorder.

Lithium is also used in the management of mania and has anti-suicidal properties. (Option C) It has a narrow therapeutic index, and blood levels in excess of this may cause symptoms of toxicity. Therapeutic drug monitoring is done by measuring lithium levels 12 hours after the last dose.

Grade	Sr. lithium concentration(in mEq/L)	Symptoms
Mild symptoms	1.5-2.5	Nausea, vomiting, tremor, and fatigue
Moderate intoxication	2.5-3.5	Confusion, agitation, delirium, tachycardia, and hypertension
Severe intoxication	>3.5	Coma, seizures, hyperthermia, and hypotension

Solution to Question 5:

Mirabegron is a β_3 -adrenergic receptor agonist which is used for bladder incontinence. Activation of this receptor in the bladder leads to detrusor muscle relaxation and increased bladder capacity. This action prevents voiding and provides relief for those with an overactive bladder and urinary incontinence.

Indacaterol is a very long acting beta 2 agonists (V-LABA) and can be used once daily in COPD maintenance.

Betaxolol is a β_1 selective third-generation β -blocker with additional calcium entry blockade activity and a slight membrane-stabilizing property. It does not have agonist activity.

Isoproterenol is a non-selective beta agonist ($\beta_1=\beta_2$) with no α action. The direct action on cardiac β_1 receptors causes an increased risk of tachycardia and arrhythmias. The decrease in peripheral vascular resistance is due to β_2 action causing a fall in diastolic blood pressure. Isoproterenol currently may be used as a cardiac stimulant in patients with bradycardia or heart block, particularly in anticipation of inserting an artificial cardiac pacemaker.

A. Mirabegron	3. Beta 3
B. Indacaterol	4. Beta 2
C. Betaxolol	1. Beta 1
D. Isoproterenol	2. Beta

Solution to Question 6:

The therapeutic index of a drug denotes the safety and efficacy of the drug in the population (option B), not the individual (option C).

The therapeutic index is calculated with the following equations:

LD₅₀/ED₅₀ or TD₅₀/ED₅₀

where LD₅₀ is the median lethal dose, TD₅₀ is the mean toxic dose, and ED₅₀ is the median effective dose

The therapeutic index (TI) describes a relationship between the doses of a drug that causes lethal or toxic effects with the dose that causes therapeutic effects. It is also called the therapeutic ratio. The therapeutic index is used to assess the safety of the drug administered.

Median lethal dose (LD₅₀) and median toxic dose (TD₅₀)

- LD₅₀ is the dose of the drug at which 50% of the experimental animals die
- In human trials, TD₅₀ can be considered. It is the dose of the drug that causes adverse effects in 50% of the population (Option D)
- The larger the LD₅₀ or TD₅₀, the safer the drug

Median effective dose (ED₅₀) is the quantity of a drug that can produce the desired therapeutic effects in 50% of the population.

Therapeutic window refers to a range of concentrations at which the likelihood of efficacy is high and the probability of adverse effects is low.

Note: Option D is incorrect as it does not make the necessary distinction that the therapeutic index is the ratio of the median lethal dose in 50% of animals (as the human population cannot be used for lethal studies) to the median effective dose in 50% of the population. LD₅₀ is measured in animals, not in humans. Option B is a better choice.

Solution to Question 7:

The T790M mutation in the EGFR gene is known to cause resistance to first and second-generation EGFR inhibitors. Osimertinib, as a third-generation EGFR inhibitor, is effective against this resistance, hence it is often used for patients with the T790M mutation after they've progressed on first or second-generation EGFR inhibitors.

The question seems to be asking specifically for the mutation that osimertinib is particularly known for being used against, and that would be the T790M mutation.

Osimertinib is a kinase inhibitor that is approved for

- Patients with metastatic non small cell lung carcinoma(NSCLC) that has progressed after prior EGFR TKI treatment and is positive for the EGFR T790M mutation.
- First-line treatment of patients with metastatic NSCLC and adjuvant therapy after resection of NSCLC with EGFR exon 19 deletions or exon 21 L858R mutations

So, osimertinib is used both for the L858R mutation and T790M mutation in different settings. But T790M mutation is often highlighted because of its association with resistance to first and second-generation EGFR inhibitors, and because osimertinib can overcome this resistance.

Osimertinib is administered orally and is a substrate of CYP3A. The drug is primarily eliminated in the feces and to a lesser extent in the urine. Concurrent administration of inducers of CYP3A4 may necessitate an increase in the dose.

The adverse effects after osimertinib treatment are leukopenia, diarrhea, skin rash, interstitial lung disease, QTc prolongation, and left ventricular dysfunction.

Solution to Question 8:

Trazodone causes priapism.

Trazodone is an atypical antidepressant drug. It is a serotonin receptor antagonist. Other drugs which belong to this include nefazodone, mirtazapine, and mianserin. Low doses of trazodone (50–200 mg) have been used widely, both alone and concurrently with SSRIs or SNRIs, to treat insomnia. Trazodone use is associated with priapism in rare instances. Intracavernous injection of phenylephrine can be used to treat priapism.

Other options:

Option B: Venlafaxine is a second-line drug used in generalized anxiety disorder. It belongs to the class of serotonin-norepinephrine reuptake inhibitors (SNRIs). Venlafaxine can induce sustained diastolic hypertension (diastolic blood pressure ≥ 90 mmHg at consecutive weekly visits). Venlafaxine also is associated with an increased risk of perinatal complications.

Option C: Bupropion is a first-line drug used for nicotine cessation and has the added benefit of causing weight loss. Weight gain is commonly seen after stopping nicotine use. Hence this drug can be used in this patient as it treats both of these.

Option D: Mirtazapine is used in major depressive disorder. It increases appetite and causes significant weight gain.

Solution to Question 9:

Cyclosporine induced cholestasis is due to MDR1 transporter.

ABC transporters such as MRP2, MDR1, BCRP, BSEP, and MDR3 in the bile canalicular membrane of hepatocytes mediate the efflux (excretion) of drugs and their metabolites, bile salts, and phospholipids against a steep concentration gradient from liver to bile. This primary active transport is driven by ATP hydrolysis.

Cyclosporine, an MDR1 substrate can cause cholestatic liver injury through the following mechanisms:

- Competitive inhibition of ATP dependent transporters.
- Inhibition of intrahepatic vesicle transport and targeting of ATP dependent transporters to the canalicular membrane.
- Impairment of bile secretion partly by increasing canalicular membrane fluidity without affecting the expression of canalicular transporters.

Cyclosporine is an immunomodulatory drug that inhibits calcineurin activity by forming a complex with cyclophilin, a cytoplasmic-receptor protein present in target cells, to form a cyclosporine-cyclophilin complex. Calcineurin is a calcium/calmodulin-dependent phosphatase required for the induction of T cell lymphokine production and proliferation.

This drug-receptor complex specifically and competitively binds to and inhibits calcineurin, a calcium and calmodulin-dependent phosphatase. This, in turn, inhibits the translocation of a family of transcription factors (NF-AT), leading to reduced transcriptional activation of cytokine genes for interleukin (IL)-2, tumor necrosis factor (TNF)-alpha, IL-3, IL-4, CD40L, granulocyte-macrophage colony-stimulating factor, and interferon-gamma. Ultimately, the proliferation of T lymphocytes is reduced, resulting in immunomodulation.

Its uses are as follows:

- In solid organ transplantation, to prevent organ rejection in kidney, liver, and heart allogeneic transplants
- In patients with rheumatoid arthritis, when the disease has not adequately responded to methotrexate
- Cyclosporine reportedly is effective in Behçet's disease, acute ocular syndrome, endogenous uveitis, atopic dermatitis, inflammatory bowel disease, and nephrotic syndrome.

Adverse effects of cyclosporine include a decrease in glomerular filtration rate (GFR) and metabolic disturbances like hypomagnesemia, hyperkalemia, dyslipidemia, gynecomastia, hypertrichosis; neurotoxicity in the form of convulsions. There is an increased risk of developing skin and lymphoproliferative malignancies in cyclosporine-treated psoriasis patients.

Solution to Question 10:

In the above clinical scenario, the most likely cause of itching around the vulvovaginal area is canagliflozin. The side effects associated with canagliflozin are urinary tract infections, genital mycotic infections due to the high glucose content of the urine, increased risk of fractures, and lower extremity amputations.

Canagliflozin is a sodium-glucose transporter 2 (SGLT-2) inhibitor. Other SGLT-2 inhibitors include dapagliflozin and empagliflozin. They act by inhibiting the SGLT-2 transporter which is present on the proximal segment of the renal tubule. It is responsible for the reabsorption of 80-90% of filtered glucose. It inhibits glucose reabsorption resulting in glycosuria and a change in the renal threshold.

SGLT2 inhibitors can cause mild weight loss and reduction in blood pressure. Drugs like empagliflozin and canagliflozin are found to reduce major cardiovascular events.

The common side effects include urinary tract infections and genital mycotic infections due to the high glucose content of the urine. There is a risk of the development of diabetic ketoacidosis. It increases the risk of fractures and lower limb amputations due to the creation of an imbalance with the minerals and circulating parathormone and vitamin D. Since SGLT2 inhibitors depend on the rate of glucose filtration to be effective, potency decreases if given in stage 3 kidney disease (GFR 60–30 mL/min).

Other options:

Option A: Acarbose is an alpha-glucosidase inhibitor. It reduces the intestinal absorption of glucose by inhibiting the intestinal brush border enzyme, alpha-glucosidase. These drugs also increase GLP-1 levels in circulation. They reduce post-prandial glucose levels.

Option C: Linagliptin is a dipeptidyl-peptidase-4 (DPP-4) inhibitor. It is used in type 2 diabetes as monotherapy or in combination therapy when metformin is not tolerated or contraindicated. It binds extensively to plasma proteins and is cleared primarily by the hepatobiliary system. Hence, no dosage adjustment is needed in renal failure. Dipeptidyl peptidase-4 inhibitors enhance the release of insulin postprandially, decrease glucagon, and thereby lower mealtime as well as fasting glucose. They are weight neutral.

Option D: Liraglutide is also a GLP-1 receptor agonist and is given only in the subcutaneous injectable form. It was recently approved in the treatment of obesity.

Solution to Question 11:

Alogliptin is a dipeptidyl peptidase-4 (DPP) inhibitor. Glucagon-like peptide (GLP) and glucose-dependent insulintropic polypeptide (GIP) are both incretins that promote the exocytosis of insulin into circulation. Both these incretins are rapidly inactivated by dipeptidyl peptidase. So, inhibition of dipeptidyl peptidase enhances the action of incretins and facilitates insulin release. Dipeptidyl peptidase-4 inhibitors enhance the release of insulin postprandially, decrease glucagon, and thereby lower mealtime as well as fasting glucose.

Other DPP-4 inhibitors include saxagliptin, linagliptin, alogliptin, and vildagliptin. They are weight neutral.

DPP-4 inhibitors are associated with severe joint pain which reduces on discontinuation of the drug. Linagliptin is cleared primarily by hepatobiliary system, with little renal clearance. Only saxagliptin is metabolized by the liver and must be used in lower doses if used along with a cytochrome p450 enzyme inhibitor drug. Other drugs are renally excreted and hence must be used with caution in renal insufficiency.

Other options:

Option A - Pramlintide is an amylin agonist. It reduces postprandial glycemia and leads to weight loss.

Option C - Liraglutide is also a GLP-1 receptor agonist and is given only in the subcutaneous injectable form. It was recently approved for the treatment of obesity.

Option D - Metformin belongs to the biguanides class of oral hypoglycemic agents. It acts by activation of AMP-dependent protein kinase (AMPK). This results in reduced hepatic glucose production. Side effects of metformin are lactic acidosis, gastrointestinal disturbance, and vitamin B12 deficiency.

Solution to Question 12:

Keynote 189 trial is a efficacy and safety study of pembrolizumab and chemotherapy in metastatic non-small cell carcinoma of lung.

Keynote 189 trial is an efficacy and safety study of pembrolizumab (MK-3475) combined with pemetrexed/platinum chemotherapy versus pemetrexed/platinum chemotherapy alone in participants with advanced or metastatic nonsquamous non-small cell lung cancer (NSCLC) who

have not previously received systemic therapy for advanced disease. Participants will be randomly assigned to receive pembrolizumab combined with pemetrexed/platinum.

Methods

In this double-blind, phase 3 trial, patients were randomly assigned (in a 2:1 ratio). 616 patients with metastatic nonsquamous NSCLC without sensitizing EGFR or ALK mutations who had received no previous treatment for metastatic disease to receive pemetrexed and a platinum-based drug plus either 200 mg of pembrolizumab or placebo every 3 weeks for 4 cycles, followed by pembrolizumab or placebo for up to a total of 35 cycles plus pemetrexed maintenance therapy. Crossover to pembrolizumab monotherapy was permitted among the patients in the placebo-combination group who had verified disease progression. The primary endpoints were overall survival and progression-free survival, as assessed by a blinded, independent central radiologic review.

Conclusions

In patients with previously untreated metastatic nonsquamous NSCLC without EGFR or ALK mutations, the addition of pembrolizumab to standard chemotherapy of pemetrexed and a platinum-based drug resulted in significantly longer overall survival and progression-free survival than chemotherapy alone.

Solution to Question 13:

Thalidomide does not cause teratogenicity by inhibition of IL2, IL4, and IL5.

The mechanism of thalidomide teratogenicity include oxidative stress, anti-angiogenesis, and thalidomide-binding protein, cereblon which has a role in the development of cerebral tissues and because its expression in the hippocampus among other areas is associated with memory and learning processes (CRBN). Fetal thalidomide exposure causes phocomelia. This disorder is associated with underdeveloped limbs and absent pelvic bones. Both upper and lower limbs are affected, and a part of the limbs may be missing. Fused fingers and missing thumbs are also seen. Thalidomide was originally used for the treatment of pregnancy-associated morning sickness but was withdrawn due to this adverse effect. It is now used for the treatment of erythema nodosum leprosum.

Thalidomide is an immunomodulator. Thalidomide absorption from the GI tract is slow and highly variable. It distributes throughout most tissues and organs, without significant binding to plasma proteins. Thalidomide and its metabolites are excreted in the urine, while the nonabsorbed portion of the drug is excreted unchanged in feces.

Common adverse effects of thalidomide include sedation and constipation. However, one of the most serious adverse effects is peripheral sensory neuropathy, characterized by asymmetrical, painful, peripheral paresthesia with sensory loss. Symptoms often present as numbness of the toes and feet, muscle cramps, weakness, signs of pyramidal tract involvement, and carpal tunnel syndrome.

Solution to Question 14:

Agranulocytosis is a rare but very severe side effect of clozapine that requires regular monitoring. A patient who is on clozapine should undergo regular hematological monitoring due to the risk of agranulocytosis. Avoid combining clozapine with other agranulocytosis-causing drugs like carbamazepine, phenytoin, propylthiouracil, and sulfonamides.

Adverse effects of clozapine are:

- Sedation and weight gain- H₁ antagonism
- Syncope, tachycardia, hypotension- α_1 antagonism
- Hypersalivation/sialorrhea- M₄ agonism
- Seizures- dose-dependant
- Agranulocytosis- dose-independent

Clozapine reduces the seizure threshold thus increasing the risk of seizures. Clozapine, an atypical antipsychotic, is an antagonist of 5HT_{2A}, D₁, D₃, D₄, and α -receptors. Clozapine is used in treatment-resistant schizophrenia, suicidal patients with schizophrenia, severe psychotic depression, and Huntington's disease.

Solution to Question 15:

Empagliflozin, which is a sodium-glucose transporter-2 (SGLT-2) inhibitor has been recently approved for use in patients with Heart Failure with Reduced Ejection Fraction, and can be initiated in patients with impaired renal function with estimated glomerular filtration rate ≥ 20 mL/min/1.73 m².

Empagliflozin works by inhibiting the sodium-glucose co-transporter-2 (SGLT-2) in the proximal renal tubules resulting in reduced reabsorption and increased urinary excretion of glucose. The glucose-lowering effect is independent of insulin. Empagliflozin acts as a diuretic and natriuretic by reducing sodium and volume load, causing intravascular contraction. Empagliflozin is also associated with weight loss, with reductions in blood pressure without increasing heart rate. Empagliflozin has been found to decrease the risk of mortality from cardiovascular causes in diabetic patients.

SGLT-2 inhibitors have a cardioprotective effect resulting from improved cardiac cell metabolism, improvement of ventricular loading mechanism, reduction of cardiac cell necrosis and fibrosis. It also has a renoprotective property.

Except for empagliflozin which can be used in chronic kidney disease, use of other SGLT-2 inhibitors might lead to acute kidney injury and should be avoided in patients with renal failure. Side effects of empagliflozin includes dehydration, euglycemic diabetic ketoacidosis, urinary tract infections, genital mycotic infections, hypoglycemia and necrotizing fasciitis.

Side effects of SGLT-2 inhibitors include dehydration, hyperkalemia, worsening of diabetic ketoacidosis, increased risk of fractures, leg and foot amputation with canagliflozin, and a possible increased risk of bladder cancer with dapagliflozin. Dapagliflozin has also been recently approved by the FDA for treatment of Heart failure with Reduced Ejection Fraction in both diabetic and non-diabetic patients, it has been associated with increased risk of bladder cancer and breast cancer.

Other options :

Option B : Degludec is a form of long acting insulin, and is not indicated for the treatment of heart failure.

Option C : Linagliptin is a dipeptidyl peptidase-4 (DPP-4) inhibitor, which enhances the action of incretins that promote the release of insulin into circulation. Dipeptidyl peptidase-4 inhibitors enhance the release of insulin postprandially, decrease glucagon, and thereby lower mealtime as well as fasting glucose. Linagliptin is safe to use in chronic kidney disease as it is cleared primarily by hepatobiliary system, but is not approved for heart failure.

Option D : Rosiglitazone is a thiazolidinedione, which activates PPAR- γ receptors in adipose tissue and skeletal muscles. These are insulin sensitizers and increase insulin-mediated glucose uptake in patients with type 2 diabetes. Rosiglitazone is currently banned in India as it has shown an increased risk of developing myocardial infarction, stroke, congestive heart failure, and death.

Solution to Question 16:

5-aminosalicylic acid is effective in inducing and maintaining remission and is considered to be the first line agents for mild to moderate ulcerative colitis, seen in this patient with 2-3 bloody stools per day.

Treatment of ulcerative colitis differs based on the severity of the disease and involves medical and surgical management. Patients with mild to moderate disease are treated with topical or systemic 5-aminosalicylate drugs such as Salazopyrine/Sulfasalazine and Mesalamine (5 ASA) which are first-line therapy. Glucocorticoids (option A), immunosuppressants, and biologics are effective in patients with moderate to severe disease.

Azathioprine and cyclosporin are used as steroid-sparing agents. Anti-TNF alpha inhibitors (infliximab, adalimumab) can be used as anti-inflammatories.

Surgery is recommended for patients who do not improve after medical treatment or with toxic megacolon, and perforation. Indications for elective surgery include failure of medical therapy, growth retardation, extraintestinal disease, and malignant change.

Ulcerative colitis is an inflammatory bowel disease resulting in superficial erosions confined to the mucosa and submucosa of the colon. It starts from the rectum and extends proximally in a continuous manner. The clinical features include abdominal pain, diarrhea, rectal bleeding, tenesmus, and passage of mucus. Other symptoms include anorexia, nausea, vomiting, fever, and weight loss.

Ulcerative colitis is graded based on the frequency of bowel movements and systemic involvement.

- Mild disease - ≤ 4 stools per day with/without rectal bleeding;
- Moderate disease - ≥ 4 stools per day with few systemic signs of illness;
- Severe disease - ≥ 6 blood stools per day with fever, tachycardia (≥ 90 beats/min), anemia (hemoglobin < 10.5 g/dL), and elevated levels of inflammatory markers (ESR ≥ 30 mm/h)
- Fulminant disease - > 10 stools per day with severe anemia requiring blood transfusion and progressive colonic dilation (toxic megacolon)

Investigations show elevated C-reactive protein, platelet count, erythrocyte sedimentation rate, fecal calprotectin, and fecal lactoferrin. A colonoscopy reveals loss of typical vascular pattern, erythema, friability, regenerative mucosal nodules, pseudopolyps, and ulcerations.

Extraintestinal manifestations associated with IBD include ocular features (episcleritis, and uveitis), dermatological manifestations (erythema nodosum, pyoderma gangrenosum), rheumatological (arthropathies, sacroiliitis, ankylosing spondylitis) and primary sclerosing cholangitis. Complications include toxic megacolon, hemorrhage, strictures, and perforations. These patients are also at increased risk of colon cancer.

Other options

Option B & C : Azathioprine and 6-mercaptopurine are not first line agents for ulcerative colitis.

Solution to Question 17:

Fostamatinib is a tyrosine kinase inhibitor used in the treatment of chronic immune thrombocytopenia (idiopathic thrombocytopenic purpura).

Fostamatinib has activity against spleen tyrosine kinase (SYK). The major metabolite of fostamatinib, R406, inhibits signal transduction of Fc-activating receptors and B-cell receptors. The fostamatinib metabolite R406 reduces antibody-mediated destruction of platelets.

Idiopathic thrombocytopenic purpura (ITP) is an acquired disorder characterized by the immune-mediated destruction of platelets and possibly inhibition of platelet release from the megakaryocyte. It is an autoimmune disorder caused by antibodies directed against platelet membrane glycoproteins IIb-IIIa or Ib-IX. Secondary causes of ITP include systemic lupus erythematosus (SLE), HIV-AIDS, and Hepatitis C.

ITP usually presents with mucocutaneous bleeding, very low platelet count, and normal peripheral blood cells. Patients typically exhibit ecchymoses and petechiae or they may incidentally discover thrombocytopenia during a routine complete blood count. Rarely, life-threatening bleeding may occur. The presence of wet purpura (blood blisters in the mouth) and retinal hemorrhages can serve as warning signs of potentially life-threatening bleeding.

The treatment of ITP is as follows:

Other options :

Option A and B: Eltrombopag and Romiplostim are thrombopoietin (TPO) receptor agonists used in the treatment of immune thrombocytopenia which have failed to respond to conventional treatments. Both drugs act via the Mpl receptor to increase platelet counts and not via tyrosine kinase inhibition.

Option C: Bosutinib is a BCR-ABL tyrosine kinase inhibitor indicated in CML.

Clinical features		Management
1.	Minor bleeding manifestations (and/or) Platelet count > 5000/ μ L	Prednisone (OR) Anti-D immunoglobulin therapy - effective only in Rh-positive patients (OR) Intravenous immunoglobulin (IVIG)
2.	Severe ITP (any of the following):- Significant GI / intracranial bleeding- Platelet count < 5000/ μ L- Signs of impending bleeding (retinal or large oral mucosal bleeds)	Platelet transfusion (AND) High-dose glucocorticoids (AND) IVIG / Anti-D immunoglobulins + Rituximab
3.	Relapse after first-line therapy	Splenectomy
4.	Unresponsive to all other therapy or relapse after splenectomy	Thrombopoietin receptor agonists - romiplostim (subcutaneous), eltrombopag (oral)

Solution to Question 18:

The given clinical features are suggestive of hypothyroidism with goiter, which is a known adverse effect of ethionamide.

Multidrug-resistant tuberculosis (MDR-TB) is treated with 2nd line agents that include ethionamide (ETH). ETH is not a first-line agent due to its gastrointestinal side effects. Ethionamide inhibits activity of the *InhA* gene product, resulting in inhibition of mycolic acid synthesis. It is structurally related to thionamides (propylthiouracil and methimazole) and therefore causes hypothyroidism by preventing organification of iodine in thyroid hormone synthesis. Another antitubercular drug that can cause hypothyroidism is Para-amino salicylic acid (PAS).

ETH/PAS-induced hypothyroidism should be treated with thyroxine replacement therapy, instead of changing the regimen. Thyroid function usually optimizes within 2 months after completing MDR therapy.

Ethionamide also causes hepatotoxicity and neurological adverse effects and may require pyridoxine supplementation.

MDR-TB is when there is resistance to at least isoniazid and rifampicin which are the most potent first-line drugs. The treatment of MDR-TB is curated with drugs grouped into different classes by the World Health Organization (WHO).

Group A: Levofloxacin, bedaquiline, linezolid

Group B: Clofazimine, cycloserine

Group C: Ethambutol, pyrazinamide, delamanid, ethionamide

In patients treated previously or with additional drug resistance, a longer MDR-TB regimen of 18-20 months is recommended. Drugs belonging to group A are preferred along with one drug from group B whenever possible. A shorter regimen of 6 months including bedaquiline with other

drugs can be used in patients not treated before and without extensive disease. In pregnant women with MRD-TB, drugs such as bedaquiline and delamanid should be avoided.

Other options :

Option A : Pyrazinamide inhibits fatty acid synthase-I leading to inhibition of cell membrane synthesis. Pyrazinamide causes hyperuricemia and hepatotoxicity.

Option C : Adverse effects of cycloserine are mainly neurological. Half of the patients on cycloserine (1g/day) experience neuropsychiatric symptoms. It ranges from headaches to psychosis, seizures, and suicidal ideas.

Option D : Streptomycin is an aminoglycoside which binds to the 30S ribosomal subunit and interferes with the initiation of protein synthesis. It is a second-line agent for the treatment of active tuberculosis. Streptomycin produces ototoxicity, vestibulotoxicity, optic nerve dysfunction, including scotomas, and peripheral neuritis.

Solution to Question 19:

Hypothermia is not associated with amitriptyline use or tricyclic antidepressant toxicity.

Tricyclic antidepressants (TCAs) such as amitriptyline are used in the treatment of psychotic depression, insomnia, and several pain conditions. Their mechanism of action is inhibition of norepinephrine and serotonin reuptake. They have a quinidine-like sodium channel blockade effect on cardiac conduction. Hence, they cause QT prolongation. Other cardiotoxic effects include atrioventricular (AV) block and hypotension. Rightward-axis deviation is also an associated finding.

TCA toxicity can result in ventricular tachyarrhythmias (Option A), status epilepticus, hyperthermia, and coma. It can also result in anticholinergic syndrome due to muscarinic receptor antagonism. Symptoms include dilated pupils (Option D), high temperature, confusion (option B), dry mouth, urinary retention, tachycardia, and myoclonic movements. It is managed with intravenous sodium bicarbonate bolus (50-100 mEq). This helps in the reversal of sodium channel blockade. Gastric lavage and administration of activated charcoal can be carried out in cases of recent ingestion.

Other side effects of TCAs include postural hypotension from anti-alpha 1 adrenergic action and sedation from anti-histaminic action. ECG changes such as T wave inversion or suppression are also common.

In some cases, it can result in polymorphic ventricular tachycardias with QT prolongation known as torsades de pointes. Recurrent episodes can be treated with intravenous magnesium sulfate.

Solution to Question 20:

Hydroxyurea is beneficial in reducing the painful vasoocclusive crisis of sickle cell anemia.

Hydroxyurea is a myelosuppressive agent that is effective in reducing the frequency of painful episodes like dactylitis and acute chest syndrome, decreasing the rate of hospitalization, and requirement for blood transfusion. It is safe and well-tolerated in children and infants.

Hydroxyurea is approved for long-term maintenance of sickle cell disease, and is recommended for patients of all ages regardless of symptoms.

Hydroxyurea increases fetal hemoglobin levels by an unknown mechanism and interferes with the polymerization of abnormal HbS.

Newer drugs for management of acute painful crisis of sickle cell anemia include L-glutamine and crizanlizumab. For individuals >5 years of age with sickle cell disease who have pain despite hydroxyurea (or who cannot tolerate hydroxyurea), L-glutamine is recommended, regardless of whether they are receiving hydroxyurea. The mechanism of action of this drug, is that it reduces oxidative stress in sickle erythrocytes. Crizanlizumab is a P-selectin-blocking monoclonal antibody which is given intravenously every month to reduce acute painful episodes.

Sickle cell anemia is an autosomal recessive disorder that occurs due to the point mutation in the β -globin gene which results in the replacement of glutamate residue with valine. This causes the production of sickle-shaped red blood cells due to the polymerization of deoxygenated hemoglobin (HbS). Factors that predispose to sickling are infection, dehydration, and acidosis. Sickle cell disease causes hemolytic anemia due to increased erythrocyte destruction. The anemia is generally well compensated. Complications occur due to abnormal erythrocyte configurations that increase hemoglobin polymerization leading to a painful vaso-occlusive state.

Clinical features of sickle cell anemia :

- Moderately severe hemolytic anemia with jaundice.
- Painful bone crisis : hand-foot syndrome or dactylitis of the bones of hands and feet.
- Acute chest syndrome typically presents with fever, cough, chest pain, and pulmonary infiltrates.
- Hyposthenuria : inability to concentrate urine.
- Aplastic crisis stems from the infection of red cell progenitors by parvovirus B19, which causes a transient cessation of erythropoiesis and a sudden worsening of the anemia.
- Spleen : In children, splenic sequestration crises occur which results in splenomegaly due to red pulp congestion. In adolescence or early adulthood, autosplenectomy occurs due to splenic infarction, fibrosis, and progressive shrinkage.
- Increased susceptibility to infection with encapsulated organisms like pneumococcal, meningococcal, and Haemophilus influenza septicemia, but can be reduced by vaccination and prophylactic antibiotics.
- Other clinical features include priapism, leg ulcers, stroke, retinopathy, pigment gallstones, and growth retardation.

Other options :

Option B : Methotrexate interferes with folate metabolism by acting as a competitive inhibitor of the enzyme dihydrofolate reductase. This starves cells for tetrahydrofolate and makes them unable to synthesize purines and thymidine monophosphate. It does not treat painful crisis of sickle cell anemia.

Option C : Methionine is an essential amino acid involved in the synthesis of folate and vitamin B12. It is not used in the management of sickle cell anemia.

Option D : Eltrombopag is a thrombopoietin analog used to treat immune thrombocytopenia. It is not used in the treatment of painful sickle cell crisis.

Solution to Question 21:

Dexrazoxane is the drug of choice to treat doxorubicin induced cardiomyopathy.

Doxorubicin is an anthracycline group of anticancer antibiotics that acts by inhibiting topoisomerase II, and is known to cause cardiotoxicity. Dexrazoxane is a cardioprotective iron-chelating agent that is indicated for use along with high dose doxorubicin to avoid cardiotoxicity.

Dexrazoxane is currently approved to prevent or reduce anthracycline-induced cardiotoxicity in women with metastatic breast cancer who have received a total cumulative dose of doxorubicin of 300 mg/m².

Doxorubicin-induced cardiomyopathy takes two forms:

- Acute: pericarditis-myocarditis syndrome, arrhythmias, ST and T wave changes in ECG. Occurs within the first 2–3 days, is transient and in most cases asymptomatic.
- Chronic: Dose dependent dilated cardiomyopathy associated with heart failure.

Doxorubicin, daunorubicin and epirubicin share similar cardiotoxicity profile. Idarubicin has least cardiotoxicity when compared to other anthracyclines.

Endomyocardial biopsy showing vacuolar degeneration and myofibrillar loss is seen in cardiotoxicity due to anthracyclines.

Difference between cardiotoxicity due to anthracyclins and trastuzumab:

Anthracyclines like doxorubicin can result in a radiation recall reaction, where erythema and desquamation of skin are observed at sites of previous radiation injury. The risk of cardiotoxicity also increases with previous irradiation. Therefore, careful monitoring and dosage adjustments are required for doxorubicin.

Other options :

Option B, C and D : Dexamethasone, aprepitant and amifostine are not used in the management of anthracycline induced cardiotoxicity.

Anthracyclins	Trastuzumab
Irreversible	Reversible
Dose-dependent	Not dose-dependent
Biopsy shows myocyte destruction	Biopsy does not show myocyte destruction

Solution to Question 22:

Rivastigmine transdermal patch can be prescribed to patients with dementia who are non compliant on oral medications.

Rivastigmine is a cholinesterase inhibitor that is available as a transdermal patch for use in Alzheimer's disease and dementia. It can be applied for over 24 hours. Transdermal delivery reduces the fluctuation and avoids direct gastric irritation effects associated with oral formulations.

Alzheimer's disease is the most common cause of dementia in the elderly. The characteristic feature of Alzheimer's is the degeneration of cholinergic neurons leading to reduced levels of acetylcholine.

Treatment includes cholinesterase inhibitors like Donepezil, Rivastigmine and Galantamine and NMDA receptor antagonist like memantine. The cholinesterase inhibitors cause significant side effects such as nausea, vomiting, diarrhea and peripheral cholinergic effects, which can be avoided by using a transdermal formulation.

Anticholinesterase drugs

Other options :

Options A, C and D: Donepezil, galantamine and memantine are currently available in the oral form and transdermal formulations are under development and yet to be available in the market.

Reversible inhibitors	
Carbamates	Non-Carbamates
Physostigmine, Neostigmine, Pyridostigmine, Rivastigmine	Edrophonium, Tacrine, Donepezil, Galantamine
Irreversible inhibitors	
Carbamates	Organophosphates
Carbaryl, Propoxur	Dyflos, Echthiophate, Malathion, Diazinon, Tabun, Sarin, Soman

Solution to Question 23:

Sodium valproate has an inhibitory action on the metabolism and degradation of GABA, and also increases its synthesis.

Valproate is an anti-epileptic medication and also used as a mood stabilizer in bipolar disorder. Valproate acts via multiple mechanisms. It blocks both sodium and calcium channels. It prolongs sodium channel inactivation and attenuates calcium-mediated 'T' current. It inhibits metabolism and degradation of the inhibitory neurotransmitter GABA and increases its synthesis from glutamic acid (Option A). It also inhibits excitatory NMDA glutamate receptors. The repetitive firing caused by the depolarization of cortical or spinal cord neurons is inhibited by valproate.

Valproate is a broad spectrum antiseizure drug used in partial seizures, generalised tonic-clonic seizures (GTCS), absence seizures, myoclonic seizures and atonic seizures. It can be used in absence seizures, but it is preferred when the child also suffers from GTCS. It is the drug of choice for juvenile myoclonic epilepsy.

The side effects of valproate (mnemonic: VALPROATE)

Vomiting

Alopecia

Liver toxicity

Pancreatitis

Rash

Obesity

Anorexia

Teratogenicity

Edema

It causes congenital anomalies like neural tube defects (Spina Bifida), clefts, and cardiac anomalies. The intelligence quotient score of school-going children who were exposed in utero to valproate is found reduced. It also contributes to poor cognitive development in children. Folate supplementation reduces the chances of neural tube defects.

Other options :

Option C : Drugs such as vigabatrine and tiagabine inhibit GABA reuptake by inhibiting the GABA reuptake transporter. Vigabatrine is the drug of choice for infantile spasms (West syndrome) seen in tuberous sclerosis.

Option D : Flumazenil acts as an antagonist at the GABA receptor and is used as an antidote for benzodiazepine overdose. Benzodiazepines act by promoting GABA inhibition by binding to the GABA receptor itself and inhibiting nerve transmission.

Solution to Question 24:

Verapamil is a potent inhibitor of P-glycoprotein drug transporter which belongs to the family of ATP-binding cassette (ABC) transporters.

A group of primary active transporters, the ATP-binding cassette (ABC) family, hydrolyze ATP to export substrates across membranes. P-glycoprotein (P-gp) is an ATP-binding cassette (ABC) transporter, which acts as an efflux membrane transporter. It is responsible for limiting cellular uptake and the distribution of xenobiotics and toxic substances. P-glycoprotein is also called ABCB1 or MDR1, and it exports bulky neutral or cationic compounds from cells. MDR1 exports many drugs as well. It is associated with multidrug resistance (MDR) to cancer chemotherapeutics through the multidrug resistance protein 1 or MDR1.

Verapamil can cause interactions with drugs whose disposition is dependent upon P-glycoprotein. Drugs that inhibit P-glycoprotein are -

- Clarithromycin
- Erythromycin
- Amiodarone

- Quinidine
- Verapamil
- Diltiazem
- Cyclosporine
- Itraconazole
- Propafenone
- Flecainide.

P-glycoprotein pumps play a major role in the elimination of certain drugs such as digoxin. Drugs that inhibit P-glycoprotein can thereby decrease their elimination and lead to toxicity. For eg, digoxin is metabolized by P-glycoprotein, and drugs that inhibit it increase the plasma levels of digoxin and may result in toxicity.

Increased P-gp expression: Reduces the absorption of drugs that are substrates for P-gp.

Decreased P-gp expression: Leads to supratherapeutic plasma concentrations and drug toxicity.

Other options :

Option B, C, and D: Although diltiazem, nifedipine, and amlodipine have been reported in the literature to act as p-glycoprotein inhibitors, verapamil is more potent and commonly used for p-glycoprotein inhibition in trials, and would be the best answer.

Solution to Question 25:

Oral Fidaxomicin is the antibiotic indicated for this patient's condition which is suggestive of *Clostridium difficile* infection, characterized by profuse watery diarrhea and leukocytosis following long term antibiotic use such as ampicillin.

C difficile infection is most commonly associated with prolonged antimicrobial use which interferes with the commensal colonic microflora. The most commonly implicated antibiotics are cephalosporins, ampicillin, clindamycin, and fluoroquinolones. Spores of toxigenic *C. difficile* when ingested, survive gastric acidity and colonize the lower intestinal tract. Here they produce two toxins: toxin A (an enterotoxin) and toxin B (a cytotoxin). These toxins initiate processes that disrupt the epithelial-cell barrier function, diarrhea, and pseudomembrane formation. The patient presents with non bloody diarrhea, fever, abdominal pain, and leukocytosis.

Clinical features can range from asymptomatic carriage to fulminant colitis.

- Nonsevere disease - Watery diarrhea (≥ 3 loose stools in 24 hours) is the cardinal symptom. Other manifestations include lower abdominal pain and cramping, low-grade fever, nausea, anorexia, mucus, or occult blood in the stool. Physical examination may demonstrate lower abdominal tenderness. Unexplained leukocytosis in hospitalized patients (even in the absence of diarrhea) may reflect underlying CDI
- Severe colitis - Clinical manifestations of severe colitis include diarrhea, lower quadrant or diffuse abdominal pain, abdominal distention, fever, hypovolemia, lactic acidosis, hypoalbuminemia, elevated creatinine, and marked leukocytosis.

- Fulminant colitis - Characterized by hypotension or shock, ileus, or megacolon.
- Recurrent disease - resolution of CDI followed by recurrence of symptoms within two to eight weeks after treatment has been stopped

Nucleic acid amplification tests including polymerase chain reaction (PCR) assays, for C. difficile toxin A or B gene in stool, are widely used diagnostically. Glutamate dehydrogenase is an enzyme immunoassay to detect C. difficile in stool. Confirmation of the presence of toxin in the stool in addition to PCR or glutamate dehydrogenase positivity is recommended.

Oral vancomycin, fidaxomicin, and metronidazole are used in the management of this condition. Fidaxomicin acts by inhibiting protein synthesis through the blockade of RNA polymerase.

Management of nonsevere to severe initial disease:

Management of recurrent disease:

Management of fulminant disease -

Other options:

Option B : Clindamycin is implicated as a risk factor for development of clostridium difficile colitis and is not used to treat it.

Option C : Zanamavir is a neuraminidase inhibitor used against influenza A and B virus.

Option D : Cilastatin is a reversible dehydropeptidase inhibitor and used adjunctively with imipenem to prolong its action as it rapidly hydrolysed by renal dehydropeptidase. It is not used in the treatment of clostridium difficile colitis.

Nonsevere diseaseSupportive clinical data: White blood cell count $\leq 15,000$ cells/mL and serum creatinine < 1.5 mg/dL	Vancomycin 125 mg orally 4 times daily for 10 days, OR Fidaxomicin 200 mg orally twice daily for 10 days If above agents are unavailable: Metronidazole 500 mg orally 3 times daily for 10 days
Severe diseaseSupportive clinical data: White blood cell count $> 15,000$ cells/mL and/or serum creatinine ≥ 1.5 mg/dL	Vancomycin 125 mg orally 4 times daily for 10 days, OR Fidaxomicin 200 mg orally twice daily for 10 days

First recurrence	If vancomycin was used for the initial episode: Vancomycin pulsed-tapered regimen: 125 mg orally 4 times daily for 10 to 14 days, then 125 mg orally twice daily for 7 days, then 125 mg orally once daily for 7 days, then 125 mg orally every 2 or 3 days for 2 to 8 weeks, OR Fidaxomicin 200 mg orally twice daily for 10 days. If fidaxomicin or metronidazole was used for the initial episode: Vancomycin 125 mg orally 4 times daily for 10 days
Second or subsequent recurrence	Vancomycin pulsed-tapered regimen (outlined above), OR Fidaxomicin 200 mg orally twice daily for 10 days, OR Vancomycin followed by rifaximin: Vancomycin 125 mg orally 4 times per day for 10 days, then Rifaximin 400 mg 3 times daily for 20 days, OR FMT

Initial episode	Enteric vancomycin plus parenteral metronidazole: Vancomycin 500 mg orally or via nasogastric tube 4 times daily, AND Metronidazole 500 mg intravenously every 8 hours. If ileus is present, additional considerations include: FMT (administered rectally) Rectal vancomycin (administered as a retention enema 500 mg in 100 mL normal saline per rectum; retained for as long as possible and readministered every 6 hours)
Recurrent episode	Antibiotics as for initial fulminant episode above. Some favor FMT in the context of a first or second recurrence that presents as a fulminant episode

Solution to Question 26:

Valproate is a mood stabilizer with the side effect of hepatotoxicity and hyperammonemia.

Valproate is an anti-epileptic medication and also used as a mood stabilizer in manic episodes as well as in bipolar disorder. The repetitive firing caused by the depolarization of cortical or spinal cord neurons is inhibited by valproate. It also prolongs the sodium channel inactivation. Valproate can cause an idiosyncratic hepatic toxicity and interfere with conversion of ammonia to urea leading to hyperammonemia. Valproate is contraindicated in patients with genetic defects in urea metabolism.

Valproate therapy reduces carnitine levels in tissues, potentially impacting mitochondrial function and leading to hyperammonemia and microvesicular steatosis. Genetic factors also play a role, as individuals who are heterozygous for mutations in the gamma polymerase gene, which is primarily present in mitochondria, are more susceptible to valproate-induced liver damage. Intravenous administration of carnitine can be used to manage this condition.

The side effects of valproate (mnemonic: VALPROATE)

Vomiting

Alopecia

Liver toxicity

Pancreatitis

Rash

Obesity

Anorexia

Teratogenicity

Edema

It causes congenital anomalies like neural tube defects (Spina Bifida), clefts, and cardiac anomalies. The intelligence quotient score of school-going children who were exposed in utero to valproate is found reduced. It also contributes to poor cognitive development in children. Folate supplementation reduces the chances of neural tube defects.

Other options :

Option A : Lithium is a mood stabilizer with a very narrow therapeutic index and is the first line of management in bipolar disorder. Its adverse effects can be acute such as tremors, ataxia, weakness, polyuria, polydipsia or chronic such as nephrogenic diabetes insipidus, thyroid dysfunction, and hyperparathyroidism. It is not associated with hepatotoxicity or hyperammonemia.

Option C : Carbamazepine is a sodium channel blocker used in the management of complex partial seizures, GTCS, simple partial seizures and trigeminal neuralgia. It is known to be hepatotoxic but is not associated with hyperammonemia. Other adverse effects include hypersensitivity reactions, eosinophilia, agranulocytosis, aplastic anemia, hyponatremia and stevens-johnson syndrome.

Option D : Topiramate is an anti epileptic and non-opioid pain medication which is associated with significant weight loss of up to 7% as a side effect.

Solution to Question 27:

Among the given options, *Candida krusei* shows an intrinsic resistance to fluconazole.

The primary or intrinsic resistance of *C. krusei* to azoles (fluconazole and itraconazole) is mediated through reduced susceptibility of the target enzyme (lanosterol 14 α -demethylase) to inhibition by these drugs. However, improved susceptibility is seen with newer generation azoles such as voriconazole.

Life-threatening systemic infections caused by *Candida* and *Aspergillus* species are usually seen in severely ill and immunocompromised patients such as HIV, chemotherapy-induced neutropenia, and transplant recipients on immunosuppressants. *C. albicans* and *C. dubliniensis* are most susceptible to currently available azoles, *C. glabrata* is less susceptible, whereas *C. krusei* is intrinsically resistant to fluconazole. Resistance occurs due to overexpression or point mutation in the gene encoding the target enzyme lanosterol 14 α -demethylase. Voriconazole is the preferred therapy for multi-drug resistant infection with *C. krusei*.

C. albicans and *C. glabrata* (Option C) showing resistance to Amphotericin B has been increasingly reported over the past several years. But this is a secondary resistance due to alteration or a decrease in the amount of ergosterol in the cell membrane. Amphotericin B is indicated to treat *C. glabrata* that has developed resistance to azoles.

Aspergillus fumigatus, the most common species to cause *Aspergillus* infection worldwide, also shows an intrinsic resistance to fluconazole. It is susceptible to echinocandins like micafungin, which acts by inhibiting beta-(1,3)-D-glucan synthase, an enzyme necessary for the synthesis of the cell wall.

Other options

Option A : *Aspergillus niger* is a common cause of otomycosis and is susceptible to azoles like voriconazole, amphotericin B, and echinocandins.

Option D : The drug of choice for treatment of sporotrichosis is itraconazole.

Solution to Question 28:

IV desferrioxamine (also known as deferoxamine) is indicated when serum iron levels are greater than 500 mcg/dL in cases of iron toxicity.

Desferrioxamine is selectively used as a chelating agent to manage cases of acute iron poisoning, hemochromatosis, and patients with iron overload from repetitive blood transfusions. Desferrioxamine can be administered intramuscularly, intravenously, or subcutaneously but not orally.

Deferiprone and deferasirox are both oral drugs and can be given in transfusion related siderosis but are less effective than desferrioxamine. Although, they can be preferred because of the ease of administration through oral route and multiple transfusions required for beta thalassemia patients.

Iron exhibits toxic effects by hydration of the ferric (Fe³⁺) ion which generates H⁺. Non transferrin bound iron catalyses free radical formation which causes mitochondrial injury, lipid peroxidation, increased capillary permeability, vasodilation, and organ toxicity.

Clinical features of iron toxicity include initial nausea, vomiting, abdominal pain and diarrhea. In severe cases there is high anion gap metabolic acidosis, cardiovascular and CNS depression, hepatitis, coagulopathy, and seizures.

In the case of toxic iron ingestion, radiopaque iron tablets may be seen on abdominal x-ray.

Management of iron toxicity includes the following :

- Whole-bowel irrigation for large ingestions, endoscopy and gastrostomy if clinical toxicity and large number of tablets are still visible on x-ray.
- IV hydration.
- Sodium bicarbonate for acidemia.
- IV desferrioxamine for systemic toxicity, iron level $>90 \mu\text{mol/L}$ ($500 \mu\text{g/dL}$).

Solution to Question 29:

Methadone is the causative drug for this patient's presentation of syncope and prolonged QT interval after coadministration other QT prolonging drugs such as fluoxetine.

Unlike other opioids, QT prolongation is primarily a side effect of methadone, which is additive with other agents known to cause QT prolongation such as fluoxetine. Serious cardiac arrhythmias including torsades de pointes have been noted with methadone use. Normal QTc is about 440 ms. Values of more than 440 ms in males and 460 ms in females are considered to be prolonged. QT prolongation can present as syncope or cardiac arrest.

Methadone is a synthetic opioid that is chemically different but has similar pharmacological action to that of morphine (Option B). It is a long-acting mu-opioid receptor agonist. Although an effective analgesic, the primary use of methadone hydrochloride is detoxification and maintenance therapy treatment for opioid use disorder. Methadone is also used for the management of chronic pain.

Side effects of methadone are similar to that of morphine such as lethargy, constipation, respiratory depression, and hypotension. Like other opioids, methadone will produce tolerance and dependence. Development of physical dependence during the long-term administration of methadone can be demonstrated following abrupt drug withdrawal or administration of an opioid antagonist.

Other options :

Option B : Morphine is not associated with QT prolongation.

Option C : Meperidine, also known as pethidine, though chemically unrelated to morphine interacts with μ opioid receptors and its actions are blocked by naloxone. Apart from usual opioid side effects, pethidine produces some atropinic effects like dry mouth, blurred vision, tachycardia. Pethidine injected in patients receiving a selective serotonin reuptake inhibitor (SSRI) may produce serotonin syndrome. It is occasionally used to control shivering during recovery from anaesthesia.

Option D : Propoxyphene is a methadone analogue that was used to treat mild-to-moderate pain. Due to reports of cardiotoxicity, it was withdrawn in UK, Europe, and USA and it was banned in India in 2013.

Solution to Question 30:

Hyponatremia occurs more frequently in the case of oxcarbazepine than carbamazepine.

Both carbamazepine and oxcarbazepine are sodium-channel blockers that stabilize neuronal membranes. Oxcarbazepine is the 10-keto analog of carbamazepine and cannot form an epoxide metabolite. Carbamazepine potentiates the action of antidiuretic hormone (ADH). It can result in water retention with decreased osmolarity and hyponatremia, especially in elderly patients with cardiac disease. Although hyponatremia may occur more commonly with oxcarbazepine than with carbamazepine, most adverse effects of oxcarbazepine are similar to those of carbamazepine.

Carbamazepine and oxcarbazepine can cause leukopenia, aplastic anemia, hepato-toxicity in the elderly.

Oxcarbazepine shows lesser hypersensitivity reaction. Hence, rashes (option A) are more commonly seen with carbamazepine than with oxcarbazepine.

Studies have shown that carbamazepine elevates enzyme gamma-glutamyl transpeptidases (GGT) more frequently than oxcarbazepine because of enzyme P450 induction. Hepatotoxicity from oxcarbazepine occurs less commonly compared to carbamazepine (Option D).

Side effects of carbamazepine therapy include: mnemonic: HEADS

Hypersensitivity reactions

Eosinophilia

Agranulocytosis: report symptoms like sore throat/fever

Aplastic anemia

Ataxia

Dilutional hyponatremia

Diplopia, blurred vision

Splenomegaly

Stevens-Johnson syndrome

Carbamazepine is most effective in complex partial seizures. It is also used in GTCS and simple partial seizures, trigeminal and related neuralgias, and manic depressive psychosis.

Oxcarbazepine has been used to treat intractable trigeminal neuralgia refractory to other anticonvulsants.

Mechanism of action of some common antiepileptics:

Mechanism of action	Antiepileptics
Sodium-channel blockers	Phenytoin Carbamazepine, Oxcarbamazepine Eslcarbazepine Lamotrigine Topiramate Felbamate Valproate Rufinamide Levetiracetam

Mechanism of action	Antiepileptics
Calcium-channel blockers	ValproateLamotrigineEthosuximide
GABA A receptor modulators	BenzodiazepinesPhenobarbitalFelbamatePrimidoneTopiramateCarbamazepine, OxcarbamazepineStiripentolClobazam
GABA uptake inhibitors	TiagabineVigabatrin
NMDA receptor antagonist	Felbamate
AMPA/Kainate receptor antagonist	PhenobarbitalTopiramatePerampanel
HCN channel activity enhancers	Lamotrigine
Inhibitors of brain carbonic anhydrase	AcetazolamideZonisamideTopiramate
Potassium-channel openers	RetigabineTopiramate

Solution to Question 31:

This patient was most likely taking tipranavir, as it is known to be associated with an increased risk of intracranial hemorrhage and hepatotoxicity which is the likely cause of altered mental status in this patient with hemophilia following complaints of nausea, vomiting and headache.

Tipranavir is a protease inhibitor (PI) used in patients who are resistant to other PIs. It is often combined with ritonavir, to achieve better therapeutic outcomes. It should be avoided in patients with head trauma or bleeding diathesis, such as hemophilia as it is associated with an increased risk of intracranial hemorrhage. Other common adverse effects of tipranavir include nausea, vomiting, diarrhea, abdominal pain, urticarial or maculopapular rash, and liver toxicity.

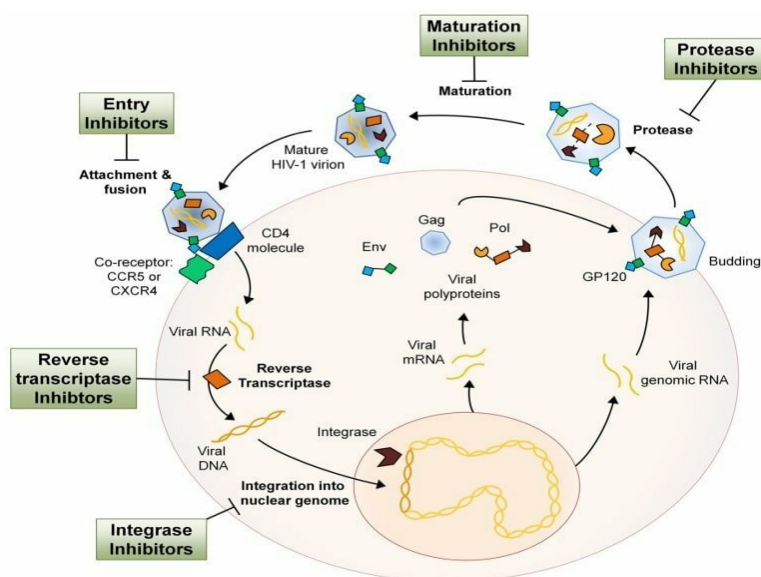
Other Options :

Option A, C and D : Lopinavir, Darunavir and Ritonavir are protease inhibitors that can cause nausea and vomiting, although they are not contraindicated in patients with bleeding diathesis and not associated with an increased risk of intracranial hemorrhage.

The illustration below shows the mechanism of action of various antiretroviral drugs.

Antiretroviral drug	Mechanism of action
Nucleotide/ nucleoside reverse transcriptase inhibitors	Inhibits reverse transcriptase
Non-nucleoside reverse transcriptase inhibitors	Inhibits reverse transcriptase by binding to an allosteric site of the enzyme

Antiretroviral drug	Mechanism of action
Protease inhibitors	Inhibits the viral protease enzyme necessary to produce mature virions upon budding from the host membrane.
Entry inhibitors	Interferes with binding, fusion, and entry of HIV -1 into the cell by blocking a target. Maraviroc works by targeting CCR5, a coreceptor located on human helper T-cells
Integrase inhibitors	Inhibits the viral enzyme integrase, which is responsible for the integration of viral DNA into the DNA of the infected cell.

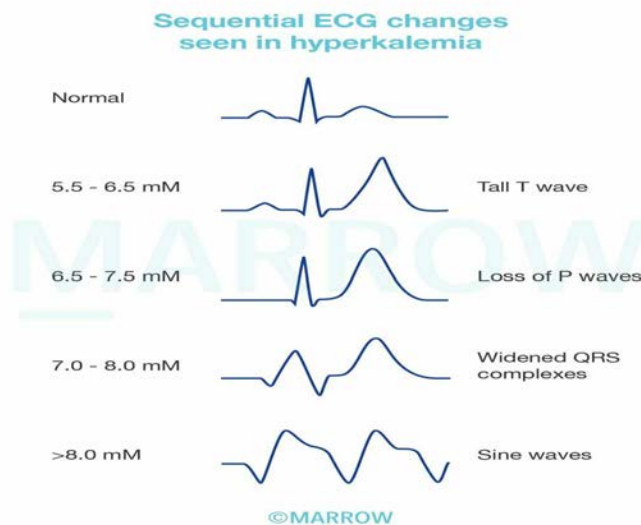


Solution to Question 32:

A chronic kidney disease (CKD) patient with chest pain, palpitations, muscle weakness, and serum potassium of 6.4mEq/L indicates hyperkalemia.

Hyperkalemia can be defined as potassium levels >5.5 mM. In normal patients, hyperkalemia is prevented by aldosterone-dependent secretion and potassium excretion via the gastrointestinal tract. But in patients with CKD, certain situations overcome these preventive mechanisms and cause hyperkalemia. These include high potassium intake, hemolysis, red blood cell transfusion, metabolic acidosis, and drugs that inhibit potassium excretion. The most common drugs that can precipitate hyperkalemia in CKD patients are renin-angiotensin system (RAS) inhibitors, spironolactone, and other potassium-sparing diuretics like triamterene, eplerenone, and amiloride.

ECG shows tall and peaked T waves, widening of QRS complexes, loss of P waves, and sine wave pattern.



Management of hyperkalemia involves an immediate reversal of cardiac effects by administration of 10mL of 10% calcium gluconate intravenously. This leads to the reversal of the depolarization blockade due to hyperkalemia. Redistributing the excess potassium by administering 10U of intravenous insulin followed by 50mL of 50% dextrose can also be given, as insulin facilitates the intracellular shift of potassium. Beta-2-agonists can also be used in the management of hyperkalemia. But patients with end-stage renal disease are resistant to beta-2-agonists and hence must be used with caution. Removal of potassium can be done using cation exchange resins like patiromer and sodium polystyrene sulfonate, diuretics, and dialysis.

Solution to Question 33:

Pregabalin can be used for diabetic neuropathy, post herpetic neuralgia and spinal cord injury. Although amitriptyline can also be used for these conditions, pregabalin is found to be recommended over amitriptyline considering safety and efficacy and hence is the best option.

Pregabalin is an anti-epileptic drug. Its analgesic effect is mediated by binding to $\alpha 2\delta$, a protein that is a subunit of voltage-gated calcium channels and causes a decrease in glutamate release at excitatory synapses. Decreased neurotransmitter release is associated with reduced neuronal hyperexcitability. It is used in the management of neuropathic pain associated with diabetic peripheral neuropathy, spinal cord injury or postherpetic neuralgia.

Currently, only duloxetine and pregabalin are approved by the U.S. Food and Drug Administration and the European Medicines Agency to treat painful diabetic neuropathy. Diabetic Neuropathy is the most common cause of peripheral neuropathy in developed countries. Risk factors include long-standing, poorly controlled DM and the presence of retinopathy and nephropathy.

Side effects of pregabalin include sedation and peripheral edema.

Other options :

Option A : Tricyclic antidepressants (TCAs) such as amitriptyline are used in the treatment of several pain conditions such as diabetic neuropathy, post herpetic neuralgia and following spinal cord injury. Their mechanism of action is inhibition of norepinephrine and serotonin reuptake. Although both pregabalin and amitriptyline are among the first line drugs for neuropathic pain, amitriptyline would be the next best option in this question, owing to its side effect profile and good yet controversial efficacy in pain control. It is a non FDA approved drug for diabetic neuropathy (off label indication). Side effects associated with amitriptyline include anticholinergic side effects (dry mouth, tachycardia, hyperthermia), cardiac effects (arrhythmias) and sedation.

Option B : Phenytoin is an antiepileptic drug that is not used in the treatment of pain.

Option C : Carbamazepine is an iminostilbene derivative. It blocks voltage-activated sodium channels, slowing their rate of recovery and limiting the repetitive firing of action potentials. It is used in the treatment of generalized, focal-to-bilateral and idiopathic tonic-clonic seizures, focal seizures, neuropathic pain associated with trigeminal and glossopharyngeal neuralgias.

Solution to Question 34:

Linezolid is a protein synthesis inhibitor that acts by inhibiting the 50s ribosomal subunit.

Linezolid binds to the P site of the 50s ribosomal subunit and inhibits bacterial protein synthesis. It is an oxazolidinone that is active against various resistant organisms including MRSA. It has excellent oral bioavailability. Linezolid is used to treat infections caused by gram-positive bacteria, particularly vancomycin-resistant enterococcus and methicillin-resistant *Staphylococcus aureus*. Linezolid is also approved for the treatment of community-acquired pneumonia due to *S. pneumoniae* and nosocomial pneumonia due to *S. aureus*.

The intrinsic resistance of gram-negative bacteria to linezolid is mediated by endogenous efflux pumps.

Linezolid is associated with the side effect of myelosuppression (anemia, leukopenia, pancytopenia, and thrombocytopenia). Thrombocytopenia is the most common side effect. Linezolid has MAOI (monoamine oxidase inhibitors) activity and therefore can precipitate serotonin syndrome with concomitant use of SSRI (paroxetine) or with other serotonergic medications.

Other adverse effects include peripheral neuropathy, optic neuropathy, lactic acidosis.

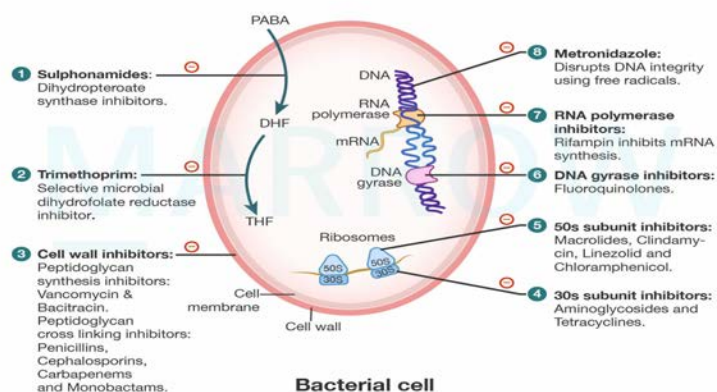
Other options :

Option A : Aztreonam is a monobactam, and acts similar to penicillin by inhibiting peptidoglycan crosslinking and inhibiting bacterial cell wall synthesis.

Option B : Daptomycin is a cyclic lipopeptide antibiotic. It acts by depolarising bacterial membranes resulting in loss of membrane potential and cell death.

Option D : Rifampicin is an anti tubercular antibiotic that acts by inhibiting bacterial DNA-dependent RNA polymerase.

Mechanism of action of antibiotics



Pharmacology INI-CET 2024

Question 1:

Which of the following drugs used in the treatment of diabetes is known to cause urinary tract infections?

- a) Dapagliflozin
- b) Sitagliptin
- c) Pioglitazone
- d) Glimeperide

Question 2:

Which of the following medications used in treating leprosy causes skin pigmentation?

- a) Ofloxacin
- b) Dapsone
- c) Clofazamine
- d) Rifampicin

Question 3:

A patient is set to receive intermittent treatment with teriparatide. Which of the following complications does this drug predispose him to?

- a) Osteonecrosis of the jaw bone
- b) Subtrochanteric fractures
- c) Net loss of bone mass
- d) Osteosarcoma

Question 4:

Which of the following monoclonal antibodies is not a HER - 1 inhibitor?

- a) Pertuzumab
- b) Cetuximab

- c) Panitumumab
- d) Necitumumab

Question 5:

A G2P1 delivered a preterm baby at 36 weeks. The baby was admitted in the NICU for a week and was given expressed breast milk. Unfortunately the baby did not survive. Which of the following drugs can be used to reduce the engorgement and pain of the lactating breasts?

- a) Cabergoline
- b) Sumatriptan
- c) Chlorpromazine
- d) Pramipexole

Question 6:

Coagulation profile monitoring is required for which of the following drugs?

- a) Lepirudin
- b) Enoxaparin
- c) Fondaparinux
- d) Dabigatran

Question 7:

A patient known case of diabetes, shows microalbuminuria on urine analysis, done twice at an interval of 6 months. Renal function tests are normal. Which of the following drugs is not useful in preventing the progression of renal damage?

- a) Finerenone
- b) Hydrochlorothiazide
- c) Angiotensin receptor blockers
- d) SGLT2 inhibitor

Question 8:

Which of the following causes a decrease in serum parathyroid hormone?

- a) Calcitriol
- b) Risedronate
- c) Furosemide
- d) Lithium

Question 9:

Which of the following acts by suicide inhibition?

- a) 1,2
- b) 3,4
- c) 1,2,4
- d) 1,3,4

Question 10:

Which of the following diuretics is matched correctly with its location of action?

- a) 1,2,3
- b) 3,4
- c) 2, 4
- d) 1,2

Question 11:

Which of the following drugs acts as an inhibitor of PCSK-9?

- a) Bempedoic acid
- b) Lomitapide
- c) Ezetemibe
- d) Evolocumab

Question 12:

Inhaled nitric oxide (iNO) is used for?

- a) Persistent pulmonary hypertension of newborn

- b) Unstable angina
- c) Obstructive sleep apnea
- d) Erectile dysfunction

Question 13:

A patient presents with complaints of a cough producing tenacious white sputum. Which of the following statements is true about the drug that could be prescribed for this condition?

- a) Codeine as it is a potent drug and quickly relieves symptoms.
- b) Bromhexine and Dextromethorphan syrup as they liquify sputum and reduce cough due to antitussive effect
- c) Dextromethorphan alone as it does not have addictive potential like codeine
- d) Bromhexine as it liquifies sputum

Question 14:

Which of the following antidiabetic drugs is known to cause unexplained diarrhea?

- a) Sitagliptin
- b) Metformin
- c) Glimepiride
- d) Pioglitazone

Question 15:

A young female weighing 46 kg is on digoxin with $vd=6$ L/kg and clearance 4.6 L/h. Calculate the half-life of digoxin.

- a) 4h
- b) 68h
- c) 42h
- d) 0.9h

Question 16:

Choose the correct option regarding phenytoin.

- a) Acute oral overdose results primarily in cerebellar and vestibular symptoms and high doses have been associated with marked cerebellar atrophy
- b) High dose causes cardiotoxicity in old and nephrotoxicity in young patients
- c) These complications can be minimized by administering fosphenytoin at a rate of less than 250 mg of phenytoin sodium equivalents per minute
- d) Given in all patients with seizures regardless of cardiac arrhythmias, hypotension and/or CNS depression

Question 17:

True statement about Iron Isomaltoside 1000 is

- a) Iron requirement can be fulfilled in single dose
- b) More immunogenic than iron dextran
- c) Given intravenously and gets deposited in adipose tissue therefore less toxic
- d) Cannot be given with erythropoietin

Question 18:

The drug that binds to CD4 and has action on both CCR5 and CXCR4 in tropic HIV infections is

- a) Maraviroc
- b) Ibalizumab
- c) Elvitegravir
- d) Fosamprenavir

Question 19:

Inhibitors of Poly ADP ribose polymerase (PARP) are approved for BRCA-positive breast cancers. Which among the following drugs is used?

- a) Pablociclib
- b) Rucaparib
- c) Regorafenib
- d) Vandetanib

Question 20:

A 60-year-old patient who is a known case of hypertension and diabetes underwent PCI stent placement for NSTEMI last year. He is currently on aspirin, metoprolol and atorvastatin. He comes for a follow-up, and his lipid profile is as follows- Total cholesterol 252mg/dl LDL: 156mg/dl, triglyceride 158mg/dl and HDL- 40 mg/dl. Which drugs can be added based on his lipid profile?

- a) 1, 2 and 4
- b) Only 1 and 2
- c) 1,3 and 4
- d) Only 1 and 3

Question 21:

Which of the following is a true statement regarding Pretomanid?

- a) Given in XDR TB with Bedaquiline and Linezolid
- b) Contraindicated in patients on enzyme inducer drugs
- c) Acts by reactive oxygen species
- d) For resistant cases of malaria

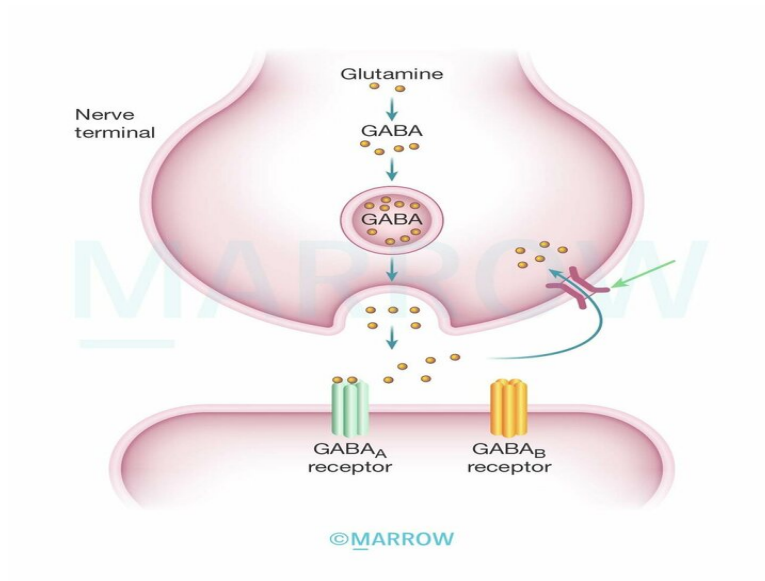
Question 22:

Which of the following drugs act on sodium channels and potentiate slow sodium inactivation?

- a) Phenytoin
- b) Lamotrigine
- c) Lacosamide
- d) Topiramate

Question 23:

Which drug acts on the marked receptor?



- a) Tiagabine
- b) Ganaxolone
- c) Felbamate
- d) Stiripentol

Question 24:

Match the treatment with its respective condition.

- a) a-i, b-ii, c-iii, d-iv
- b) a-ii, b-iii, c-i, d-iv
- c) a-iii, b-i, c-iv, d-ii
- d) a-iv, b-ii, c-iii, d-i

Answer Key

Question No.	Correct Option
1	a
2	c
3	d
4	a
5	a

6	a
7	b
8	a
9	b
10	b
11	d
12	a
13	d
14	b
15	c
16	a
17	a
18	b
19	b
20	a
21	b
22	c
23	a
24	b

Detailed Explanations

Solution to Question 1:

Dapagliflozin is known to cause urinary tract infections.

Dapagliflozin is a sodium-glucose transporter 2 (SGLT-2) inhibitor. It acts by inhibiting the SGLT-2 transporter which is present on the proximal segment of the renal tubule and is responsible for the reabsorption of 80-90% of filtered glucose. This results in glycosuria. Other SGLT-2 inhibitors include canagliflozin and empagliflozin.

Apart from their effect on lowering blood glucose levels, SGLT2 inhibitors can cause mild weight loss and reduction in blood pressure. They have also been shown to reduce the risk of hospitalization in patients with heart failure.

The common side effects include urinary tract infections and genital mycotic infections due to the high glucose content of the urine. There is a risk of the development of diabetic ketoacidosis.

Other options :

Option B - Sitagliptin is a dipeptidyl peptidase-4 inhibitor. Its use is associated with an increased risk of heart failure.

Option C - Pioglitazone is an activator of the peroxisome proliferator-activated receptor γ (PPAR γ), which is involved in the regulation of genes related to glucose and lipid metabolism. Its adverse effects include increased incidence of heart failure, peripheral edema, weight gain, macular edema, and osteoporosis.

Option D - Glimepiride belongs to the class of sulfonylureas. It stimulates insulin release by inhibiting the activity of the β cell KATP channel complex. Its side effects include hypoglycemia and weight gain.

Solution to Question 2:

Clofazimine is a dye with leprostatic and antiinflammatory properties. The mechanism of antimicrobial activity of clofazimine against mycobacteria appears to be multifactorial and includes membrane disruption, inhibition of mycobacterial phospholipase A₂, inhibition of microbial K⁺ transport, generation of hydrogen peroxide, interference with the bacterial electron transport chain, and efflux pump inhibition.

Clofazimine is orally active. It accumulates in macrophages and gets deposited as needle-shaped crystals in many tissues including subcutaneous fat.

Clofazimine is used as a component of multidrug therapy (MDT) for leprosy. Because of its anti-inflammatory properties, it is valuable in lepra reactions. Occasionally, it is used as a component of a combination regimen for extensive drug resistant TB (XDR-TB).

Adverse effects -

- Skin - The major disadvantage is reddish-black discoloration of skin, especially on exposed parts. Discoloration of hair and body secretions may also occur. Dryness of the skin, itching and scaling are often troublesome. Acneform eruptions and phototoxicity have been noted. Conjunctival pigmentation may create cosmetic problems.
- GI symptoms - Nausea, anorexia, abdominal pain, weight loss and enteritis with intermittent loose stools can occur, particularly when higher doses are used to control lepra reaction. Deposition of clofazimine crystals in the intestinal submucosa may produce gastrointestinal symptoms.

Clofazimine is to be avoided during early pregnancy and in patients with liver or kidney damage.

Other options :

Option A- Ofloxacin is a fluoroquinolone. As a component of MDT, it has been found to hasten the bacteriological and clinical response, where it is cidal to *M. leprae*. Its side effects include nausea, vomiting, and abdominal discomfort. CNS side effects like dizziness, anxiety and insomnia are also seen. Tendinitis is another adverse effect seen with the use of fluoroquinolones.

Option B- Dapsone exerts its antimicrobial effect by the inhibition of PABA incorporation into folic acid by folate synthase. Apart from being used in leprosy, dapsone is also used for the prophylaxis of *T. gondii*. The anti-inflammatory effects are the basis for therapy of dermatitis herpetiformis, linear immunoglobulin A bullous disease, and relapsing chondritis. Mild hemolytic anemia is common, especially in those with G-6-PD deficiency. Isolated instances of reversible peripheral neuropathy, drug fever, hematuria, pruritus, psychosis, and a variety of skin rashes have been reported.

Option D - Rifampicin interrupts RNA synthesis by binding to a subunit of mycobacterial DNA-dependent RNA polymerase. Apart from being used to treat leprosy, it is used in the management of tuberculosis, brucellosis, and prophylaxis of Meningococcal meningitis. Hepatitis is a major adverse effect. Other side effects include pruritus with rash, redness and watering of eyes,

flu-like symptoms and abdominal cramps. Urine and secretions may become orange-red.

Solution to Question 3:

The use of teriparatide predisposes an individual to the development of osteosarcoma.

Teriparatide is a parathormone analog, which is given as subcutaneous injections.

Pharmacokinetics and systemic actions of teriparatide on mineral metabolism are the same as for PTH. Its main target of action are the osteoblasts.

Teriparatide is administered by once-daily subcutaneous injection of 20 µg into the thigh or abdomen. The site of administration should be rotated, but the time at which the injection is made should be the same each day.

Teriparatide and abaloparatide (PTHrP analog) are the only anabolic agents currently available that increase new bone formation (option C). Candidates for treatment include women who have a history of osteoporotic fracture, who have multiple risk factors for fracture, or who failed or are intolerant of previous osteoporosis therapy. Men with primary or hypogonadal osteoporosis are also candidates for treatment with these agents.

Adverse effects include hypercalcemia, exacerbation of nephrolithiasis, and elevation of serum uric acid levels. The development of osteosarcoma has been a serious concern in patients treated with teriparatide. It comes with a black-box warning, and use should be limited to no more than 2 years and avoided in patients who are at increased baseline risk for osteosarcoma (including those with Paget disease of bone, unexplained elevations of alkaline phosphatase, open epiphyses, or prior radiation therapy involving the skeleton).

Other options :

Options A and B - Osteonecrosis of the jaw bone and subtrochanteric fractures are the adverse effects of bisphosphonates.

Solution to Question 4:

Pertuzumab is a humanized IgG1 monoclonal antibody directed against HER2.

Pertuzumab is approved for the treatment of patients with HER2-positive, locally advanced, inflammatory, or early-stage breast cancer in combination with trastuzumab and docetaxel. Its adverse effect includes cardiotoxicity.

The other three antibodies act against epidermal growth factor receptor (EGFR)/ HER-1.

Option B - Cetuximab is used to treat patients with metastatic colon cancers and head and neck squamous cell carcinoma (HNSCC).

Option C - Panitumumab is approved for the treatment of EGFR-expressing metastatic colorectal cancer.

Option D - Necitumumab is approved for first-line treatment of patients with metastatic, squamous non-small cell lung cancer in combination with gemcitabine and cisplatin.

Solution to Question 5:

In the given scenario, the woman requires suppression of lactation, for which cabergoline is indicated.

In the postpartum period, prolactin promotes lactation. Dopamine(DA) is the primary regulator of prolactin secretion from the pituitary gland. DA released from the hypothalamus into the hypophyseal portal blood supply acts on lactotroph D2 receptors to decrease prolactin secretion. The DA agonists like bromocriptine and cabergoline are used in conditions requiring suppression of prolactin.

Cabergoline is an ergot derivative with a longer t_{1/2} (~65 h), higher affinity, and greater selectivity for the DA D2 receptor compared to bromocriptine. Cabergoline is the preferred drug for the treatment of hyperprolactinemia because of greater efficacy and lower adverse effects. Therapy is initiated at a dose of 0.25 mg twice a week or 0.5 mg once a week. Cabergoline induces remission in a significant number of patients with prolactinomas. At higher doses, cabergoline is used in some patients with acromegaly alone or in conjunction with somatostatin analogs.

Adverse Effects -

Compared to bromocriptine, cabergoline has a much lower tendency to induce nausea, although it still may cause hypotension and dizziness. Cabergoline has been linked to valvular heart disease, an effect proposed to reflect agonist activity at the serotonin 5HT_{2B} receptor. However, this effect is seen primarily at the high doses used in patients being treated for Parkinson's disease and is not seen in the conventionally used doses (≤ 2 mg/week) for patients with prolactinomas.

Other options:

Option B - Sumatriptan is a 5HT_{1B/1D} receptor agonist used in the treatment of acute attacks of migraine.

Option C - Chlorpromazine is a typical antipsychotic, which has D2 antagonistic activity. It is used in the treatment of schizophrenia, acute mania, and refractory nausea and vomiting. Hyperprolactinemia is an adverse effect associated with its use.

Option D - Pramipexole is a non-ergot alkaloid with a broader DA receptor agonist activity, with the highest affinity to D₃ receptors. It is used in the treatment of Parkinson's disease and restless leg syndrome. It is not indicated for the treatment of breast engorgement.

Solution to Question 6:

Coagulation profile monitoring is required while using lepirudin.

Lepirudin is a parenteral direct thrombin inhibitor, which is a recombinant form of hirudin extracted from the saliva of leech. Its action is independent of anti-thrombin, which means it can reach and inactivate fibrin-bound thrombin in thrombi. It is excreted by the kidney and should be used with great caution in patients with renal insufficiency as no antidote exists.

Like heparin, lepirudin must be administered parenterally and is monitored by aPTT. Lepirudin was approved for use in patients with thrombosis related to heparin-induced thrombocytopenia (HIT). Its production was discontinued by the manufacturer in 2012.

Other options :

Option B - Enoxaparin is a low molecular weight heparin (LMWH) that inhibits activated factor X with less effect on thrombin. Hence, it has a smaller effect on the coagulation profile and does not need monitoring.

Option C - Fondaparinux avidly binds antithrombin, resulting in efficient inactivation of factor Xa. It is effective in the prevention and treatment of venous thromboembolism and does not appear to cross-react with pathologic HIT antibodies in most individuals. Like LMWH, it does not require monitoring of the coagulation profile.

Option D - Dabigatran is an oral direct thrombin inhibitor approved for the reduction in risk of stroke and systemic embolism with nonvalvular atrial fibrillation, treatment of venous thromboembolism (VTE) and VTE prophylaxis following hip or knee replacement surgery. It does not require monitoring of coagulation profile.

Solution to Question 7:

The given clinical scenario suggests chronic kidney disease secondary to diabetes, and hydrochlorothiazide is not useful in preventing the progression of renal damage.

Chronic kidney disease (CKD) is defined as abnormalities of kidney structure or function, present for a minimum of 3 months, with implications for health. CKD is classified based on cause, glomerular filtration rate (GFR) category (G1–G5), and albuminuria category (A1–A3). Some of the leading causes of CKD are diabetes, hypertension, glomerulonephritis and polycystic kidney disease.

In patients with diabetes mellitus, often there exists a state of glomerular hyperfiltration due to decreased vascular resistance of afferent arterioles and increased vascular resistance of efferent arterioles resulting in increased glomerular hydrostatic pressure. This contributes to the progression of the CKD.

The following drugs have been approved in the management of CKD :

- Angiotensin-converting enzyme inhibitors / Angiotensin receptor blockers - These medications appear to slow the rate of decline of kidney function in a manner that extends beyond the reduction of systemic arterial pressure and that involves a reduction in the intraglomerular hyperfiltration and hypertension.
- Finerenone - It is a non steroidal mineralocorticoid receptor antagonist which decreases proteinuria. It has been approved to reduce the risk of end-stage kidney disease, cardiovascular death, and heart failure in diabetic patients with CKD.

- SGLT-2 inhibitors - These drugs inhibit sodium-glucose cotransporter type 2 in the proximal convoluted tubule. By increasing NaCl delivery to the thick ascending limb, gliflozins trigger tubuloglomerular feedback and thereby increase preglomerular vascular resistance and decrease glomerular hypertension. This contributes to the effectiveness of gliflozins to prevent diabetic kidney disease.

Solution to Question 8:

The active form of vitamin D, 1,25-dihydroxyvitamin D (calcitriol), directly suppresses PTH gene expression

Calcitriol is the active form of vitamin D. It increases calcium absorption from the intestines and helps maintain adequate serum calcium levels. When calcium levels rise, this triggers a negative feedback mechanism on the parathyroid glands, leading to decreased secretion of parathyroid hormone (PTH). PTH's primary role is to raise blood calcium levels, so when calcium is sufficient, the need for PTH decreases, resulting in lower PTH levels.

Vitamin D and its derivatives are the primary treatment for hypoparathyroidism.

Dihydroxyvitamin D₂ (DHT), a reduced form of vitamin D₂, has a quicker onset, shorter duration, and stronger impact on bone mobilization than traditional vitamin D, making it a preferred option. While most hypoparathyroid patients respond well to any form of vitamin D, calcitriol may be preferred for temporary treatment of hypocalcemia while waiting for the effects of a slower-acting vitamin D form. Calcium is used to address calcium deficiency and as a dietary supplement. Vitamin D is the primary treatment for hypoparathyroidism.

Severe hypermagnesemia or hypomagnesemia can also inhibit PTH secretion.

Other options:

Option B: Risedronate is a bisphosphonate that inhibits bone resorption, but it doesn't directly affect PTH secretion.

Option C: Furosemide is a diuretic that can increase calcium excretion, potentially raising PTH as the body tries to conserve calcium.

Option D: Lithium can actually increase PTH secretion by affecting calcium-sensing receptors in the parathyroid glands.

Solution to Question 9:

Ritonavir and benzbromarone are both suicide inhibitors.

Suicide inhibition is when an inhibitor binds to an enzyme and then undergoes a reaction to form an irreversible complex with the enzyme, inactivating it permanently. Ritonavir is a protease inhibitor used in HIV treatment and benzbromarone is a uricosuric agent used to treat gout by increasing uric acid excretion. It also inhibits CYP2C9 via mechanism-based (suicide) inhibition.

Other options:

Ethinyl estradiol is a synthetic estrogen used in oral contraceptives and hormone replacement therapy. It acts through receptor-mediated mechanisms, not through suicide inhibition.

Spirolactone is a potassium-sparing diuretic and aldosterone antagonist. It works by competitively inhibiting aldosterone receptors in the distal tubules of the kidney, not through suicide inhibition.

Solution to Question 10:

The osmotic diuretics act at the proximal convoluted tubule and the potassium-sparing diuretics act on the collecting tubule. Hence C and D are the right statements.

A - Osmotic diuretics, like mannitol, retain water isoosmotically in the proximal convoluted tubule (PCT) and descending loop of Henle which are freely permeable to water. Luminal fluid is diluted and opposes NaCl reabsorption. They are indicated in raised intracranial tension and in dialysis disequilibrium.

B - Thiazides act on the distal convoluted tubule (DCT). They block the Na⁺/Cl⁻ co-transporter in the DCT. This results in the loss of sodium, chloride, and water contributing to the diuretic effect. Some of the examples are hydrochlorothiazide, chlorthalidone and metolazone. Their indications include hypertension, edema, diabetes insipidus and hypercalciuria.

C - Carbonic anhydrase inhibitors act on the proximal convoluted tubule (PCT). By inhibiting carbonic anhydrase, HCO₃⁻ reabsorption is blocked and diuresis occurs. Examples are acetazolamide, methazolamide, dichlorphenamide, and dorzolamide. They are used in the treatment of glaucoma, urinary alkalization in cystinuria, management of metabolic alkalosis, and acute mountain sickness.

D - Potassium-sparing diuretics include aldosterone antagonists (spironolactone, eplerenone) and inhibitors of renal epithelial Na⁺ channel (triamterene, amiloride). They act on the late distal tubule and the collecting duct. Aldosterone antagonists are used in the treatment of edema, hypertension, heart failure and Conn's syndrome. Amiloride has a role in lithium-induced diabetes insipidus.

Solution to Question 11:

Evolocumab is a monoclonal antibody that acts by inhibiting PCSK9.

PCSK9 (proprotein convertase subtilisin/kexin type) is a serine protease that causes lysosomal degradation of the low-density lipoprotein (LDL) receptors present on hepatocytes. Monoclonal antibodies against PCSK9 such as alirocumab and evolocumab counter this and cause persistence of LDL receptors on hepatocytes. This leads to more LDL uptake from the circulation, bringing down the LDL-cholesterol levels.

Evolocumab is administered as a 140-mg injection every 2 weeks or as a 420-mg injection once a month. PCSK9 inhibitors should not be used in pregnancy because transmission across the placenta is expected.

PCSK9 inhibitors are indicated only as adjunctive therapy in patients with familial hypercholesterolemia(FH) or established atherosclerotic cardiovascular disease(ASCVD) who have elevated LDL-cholesterol levels despite being on maximum statin therapy.

Other options :

Option A - Bempedoic acid acts by inhibiting de novo hepatocyte cholesterol biosynthesis, by blocking the enzyme adenosine triphosphate citrate lyase. The drug is approved as an adjunct to diet and maximally tolerated statin therapy to further lower LDL-C in patients with FH or established ASCVD.

Option B - Lomitapide acts by inhibiting microsomal triglyceride transfer protein (MTP), which is essential for the intracellular transfer of triglycerides into triglyceride-rich lipoproteins, thus inhibiting the formation of VLDLs in hepatocytes and chylomicrons in intestinal epithelial cells. It is approved as an adjunct to diet for lowering LDL-C levels, total cholesterol, apo B, and non-HDL lipoproteins in patients with FH.

Option C - Ezetimibe inhibits luminal cholesterol uptake by jejunal enterocytes, by inhibiting Niemann-Pick C1-like 1 transport protein. The role of ezetimibe as monotherapy for patients with elevated LDL-C levels is generally limited to a small group of statin-intolerant patients.

Solution to Question 12:

Inhaled nitric oxide(iNO) is used for persistent pulmonary hypertension in newborns. (Option A)

Inhaled NO selectively dilates the pulmonary vasculature, hence used in diseases associated with increased pulmonary vascular resistance. Inhaled NO is an FDA-approved drug for persistent pulmonary hypertension in the newborn.

Inhaled NO can also be used during cardiac catheterization to evaluate the pulmonary vasodilating capacity of patients with heart failure and infants with congenital heart disease. Another use of inhaled NO is to determine the diffusion capacity (DL) across the alveolar-capillary membrane. It is more effective than CO₂ because of its greater affinity for hemoglobin and its higher water solubility at body temperature.

NO is produced from the nasal passages and the lungs of normal human subjects and can be detected in exhaled gas. The measurement of fractional exhaled NO is a noninvasive marker for airway inflammation in the assessment of respiratory tract diseases, including asthma, respiratory tract infection, and chronic lung disease.

Inhaled NO is administered at low concentrations (0.1–50 ppm) since high concentrations of 50 to 100 ppm can cause significant methemoglobinemia that can cause death.

Other options:

Option B: Unstable angina is treated by anti-platelet drugs such as aspirin and clopidogrel, and anticoagulants such as heparin.

Option C: Obstructive sleep apnoea is mainly treated by using a BiPAP machine during sleep and CNS stimulants such as armodafinil for daytime sleepiness associated with the condition.

Option D: Erectile dysfunction is treated by PDE5 inhibitors such as sildenafil, tadalafil, etc.

Solution to Question 13:

Bromhexine should be prescribed for a patient with a productive cough, as it liquifies the sputum.

Bromhexine belongs to a class of drugs called mucolytics, capable of inducing thin copious bronchial secretion. It depolymerizes mucopolysaccharides directly as well as by liberating lysosomal enzymes- the network of fibers in tenacious sputum is broken. It is particularly useful if mucus plugs are present.

Other examples of mucolytics are ambroxol, carbocisteine, and N-acetyl cysteine.

Other options :

Codeine and dextromethorphan belong to the class of drugs called antitussives. These are drugs that act in the CNS to raise the threshold of the cough center or act peripherally in the respiratory tract to reduce tussal impulses, or both these actions. Because they aim to control rather than eliminate cough, antitussive drugs should be used only for dry nonproductive cough or if it is unduly tiring, disturbs sleep, or is hazardous (hernia, piles, cardiac disease, ocular surgery).

Cough is a defensive reflex and its suppression may be inappropriate in bacterial lung infection. Hence, before treatment with antitussives, it is important to identify underlying causal mechanisms that may require therapy.

Solution to Question 14:

Metformin is known to cause unexplained diarrhoea as its adverse effect.

Metformin is a member of the biguanide class of oral hypoglycemic drugs. It acts by primarily by inhibiting hepatic gluconeogenesis. It activates AMP-dependent protein kinase, leading to stimulation of hepatic fatty acid oxidation, glucose uptake, and nonoxidative glucose metabolism and reduction of lipogenesis and gluconeogenesis. It also appears to act in the gut.

Metformin is accepted as the first-line treatment of type 2 diabetes and is the most commonly used oral agent for this condition. It has been used as a treatment of infertility in women with polycystic ovarian syndrome.

The most common side effects of metformin are gastrointestinal: nausea, indigestion, abdominal cramps, bloating, and diarrhea. The use of metformin is also associated with lower blood levels of vitamin B12.

Most adverse GI effects of metformin abate over time and can be minimized by starting at a low dose and gradually titrating to a target dose over several weeks, as well as by having patients take the drug with meals. Extended-release metformin has decreased GI side effects and can be substituted for immediate-release metformin. Lactic acidosis has been linked with the use of metformin.

Other options:

Option A - Sitagliptin is a dipeptidyl peptidase-4 inhibitor. Its use is associated with an increased risk of heart failure.

Option C - Glimepiride belongs to the class of sulfonylureas. It stimulates insulin release by inhibiting the activity of the β cell KATP channel complex. Its side effects include hypoglycemia, and weight gain.

Option D - Pioglitazone is an activator of the peroxisome proliferator-activated receptor γ (PPAR γ), which is involved in the regulation of genes related to glucose and lipid metabolism. Its adverse effects include increased incidence of heart failure, peripheral edema, weight gain, macular edema, and osteoporosis.

Solution to Question 15:

The half-life of Digoxin is 42 hours.

In first-order kinetics:

$$\text{Plasma half-life (t}_{1/2}\text{)} = 0.693 \times \frac{\text{Volume of distribution (V}_d\text{)}}{\text{Clearance (CL)}}$$

Therefore,

$$t_{1/2} \sim 0.7 \times 46 \times 6 / 4.6 = 0.7 \times 60$$

$$t_{1/2} \sim 42 \text{ hours}$$

Where,

$t_{1/2}$ - Plasma half-life

V_d - Volume of distribution

CL - Clearance

Solution to Question 16:

Acute oral overdose results primarily in cerebellar and vestibular symptoms and high doses have been associated with marked cerebellar atrophy.

Dose-related toxicity of phenytoin causes ataxia, vertigo, diplopia, and nystagmus.

Other options:

Option B: High dose causes cardiotoxicity in both old and young, healthy patients. However, cardiotoxicity is more pronounced and frequent in older patients and those with known cardiac diseases.

Option C: These complications can be minimised by administering fosphenytoin at a rate of less than 150 mg of phenytoin sodium equivalents per minute, not 250 mg.

Option D: Phenytoin is not effective in generalised absent seizures.

Solution to Question 17:

Iron requirement can be fulfilled in a single dose is true regarding Isomaltoside 1000.

Iron isomaltose 1000 is a new IV iron formulation that can supply a complete replacement dose in a short, single visit. It is used in adult patients with an intolerance to oral iron or for patients with CKD not requiring hemodialysis.

Other options:

Option B: Isomaltoside 1000 is not more immunogenic than iron dextran.

Option C: The strength of iron binding causes a controlled slow release of bioavailable iron, decreasing the risk of toxicity from labile-free iron. Its decreased toxicity cannot be attributed to deposition in adipose tissue.

Option D: Isomaltoside 1000 can be given with erythropoietin.

Solution to Question 18:

The drug that binds to CD4 and has an action on both CCR5 and CXCR4 in tropic HIV infections is Ibalizumab.

Ibalizumab is an anti-CD4 monoclonal antibody approved for use in the treatment of multidrug-resistant HIV-1. It has action on both CCR5 and CXCR4 in tropic HIV infections and is administered intravenously.

It acts as an entry inhibitor. It blocks HIV-1 from infecting CD4+ T cells by binding to domain 2 of CD4 cells. It also interferes with post-attachment steps required for the entry of HIV-1 virus particles into host cells and thus prevents the viral transmission that occurs via cell-cell fusion.

Other options:

Option A: Maraviroc binds to the host cell CCR5 receptor to block the binding of viral gp120. It prevents the entry of the viral particles. It does not bind to CD4.

Option C: Elvitegravir is an integrase inhibitor that inhibits HIV 1 viral integrase enzyme and prevents proviral DNA integration into host cells.

Option D: Fosamprenavir is a protease inhibitor that blocks the viral protease enzyme necessary to produce mature virions upon budding from the host membrane.

Solution to Question 19:

Rucaparib is an inhibitor of Poly ADP ribose polymerase (PARP) used for BRCA-positive breast cancers.

Rucaparib is a poly ADP-ribose polymerase/ PARP inhibitor approved for the treatment of breast and advanced ovarian cancer.

PARP is a nuclear protein that transfers ADP-ribose from NAD⁺ to target proteins and this plays an important role in the DNA damage response. Rucaparib and olaparib are PARP inhibitors.

Inhibitors of PARP (Poly ADP Ribosyl Polymerase) approved for breast cancer:

- Olaparib
- Rucaparib
- Talazoparib

Other options:

Option A: Palbociclib is a selective inhibitor of the cyclin-dependent kinases CDK4 and CDK6. It is used in the treatment of metastatic breast cancer.

Option C: Regorafenib is an angiogenesis inhibitor that is indicated as a second-line treatment for unresectable hepatocellular carcinoma. (The first line is Sorafenib).

Option D: Vandetanib is an oral tyrosine kinase inhibitor recently FDA-approved for unresectable locally advanced or metastatic medullary thyroid carcinoma.

Solution to Question 20:

1,2 and 4 can be added based on his lipid profile.

A patient with hypertension and diabetes who underwent PCI stent placement for NSTEMI has atherosclerotic cardiovascular disease (ASCVD). The treatment goal in patients with ASCVD is to decrease LDL levels.

Ezetimibe, Evolocumab, Bemepedoic acid decrease LDL. Gemfibrozil mainly decreases triglycerides and not LDL.

1. Evolocumab is a monoclonal antibody that inhibits PCSK-9 (proprotein convertase subtilisin/kexin). Inhibition of PCSK-9 prevents the degradation of LDL receptors and thus increases LDL-cholesterol clearance by the liver. They are used in patients with atherosclerotic cardiovascular disease (ASCVD) to decrease LDL levels.

2. Ezetimibe inhibits cholesterol absorption by enterocytes in the small intestine. It lowers the total and LDL-C levels.

3. Gemfibrozil belongs to the fibrate group of drugs that binds to peroxisome proliferator-activated receptors (PPAR α), which stimulate fatty acid oxidation and thereby reduce triglyceride levels mainly.
4. Bempedoic acid inhibits adenosine triphosphate-citrate lyase (ACL) and results in decreased LDL in patients not responding to statin therapy.

Note - Ezetimibe, colestipol, bempedoic acid and PCSK-9 inhibitors like alirocumab have been proven to reduce atherosclerosis risk when added to statins, whereas the fibrate group of drugs and niacin show no significant effect in addition to statins.

Solution to Question 21:

Contraindicated in patients on enzyme inducer drugs is the true statement regarding Pretomanid.

Pretomanid is metabolized in part by CYP3A. So co-administration with strong or moderate CYP3A inducers should be avoided. Hence, it is contraindicated in patients on enzyme-inducer drugs.

Pretomanid is a bicyclic nitroimidazole that acts by inhibiting mycolic acid (a component of the mycobacterial cell wall) and protein synthesis in the mycobacteria.

Pretomanid in combination with Bedaquiline and Linezolid, was approved by the FDA in 2019 for the treatment of adult patients with Pre (XDR) TB (resistant to rifampicin and any fluoroquinolones) and for patients who are treatment-intolerant or nonresponsive to multidrug-resistant pulmonary (MDR) TB treatment.

Other options:

Option A: Pretomanid is given for Pre -XDR TB (not XDR TB) in combination with Bedaquiline and Linezolid (BPaL regimen). Pre-XDR TB is resistant to rifampicin and any fluoroquinolone, and therefore, BPaL is given to such patients.

Option C: Pretomanid acts by reactive nitrogen species like NO and not by reactive oxygen species.

Option D: Pretomanid is not used for resistant cases of malaria.

*Note: For people with MDR/RR-TB, the regimen comprises bedaquiline (B), pretomanid (Pa), linezolid (L) and moxifloxacin (M), and is referred to as BPaLM. For people who have pre-XDR-TB, the regimen can be used without moxifloxacin (BPaL).

Solution to Question 22:

Lacosamide is a drug acting on sodium channel and potentiating slow sodium inactivation.

Lacosamide acts by blocking the voltage-gated sodium channels. It is used as adjunctive therapy for focal-onset seizures in patients older than 17 years.

It is the first anti-seizure medication that limits persistent repetitive firing, the neuronal firing pattern typical of focal seizures, and prolongs the gradual inactivation of voltage-gated Na⁺

channels. Additionally, lacosamide binds to the phosphoprotein collapsing response mediator protein-2 (CRMP-2), which is mostly expressed in the nervous system and has a role in controlling axonal outgrowth and neuronal differentiation.

The role of CRMP-2 binding in seizure control is unknown.

Therapeutic Uses:

- It is used for both monotherapy and add-on therapy for focal-onset seizures in patients 17 years and older.
- It is mostly beneficial in hospitalized patients because it is available in an intravenous formulation, has minimal hepatic metabolism, and has no adverse respiratory effects.

Antiepileptic Drugs	Mechanism of Action
Phenytoin (Option A) Carbamazepine Lamotrigine (Option B)	Inhibit Na channels
Phenobarbital Vigabatrin Tiagabine	Enhance GABA inhibition Vigabatrin – block GABA aminotransferase irreversibly Tiagabine – block GABA reuptake
Ethosuximide Zonisamide Gabapentin Pregabalin	Block Ca channels Ethosuximide, Zonisamide –block T-type Ca channels Gabapentin, Pregabalin –block voltage sensitive Ca channels
Levetiracetam	Binds synaptic vesicle protein 2A
Valproate	Multiple MOA: GABA related action NMDA receptor antagonist Histone deacetylase inhibitor
Topiramate (Option D)	Multiple MOA: GABAergic action AMPA receptor antagonist Acts on Na, Ca and K channels
Drugs acting on glutamate receptors	NMDA-type glutamate receptors: Valproate, Felbamate AMPA-type glutamate receptors: Topiramate, Perampanel
Newer drugs: Rufinamide Lacosamide Ezogabine Perampanel	Prolongs inactivation of voltage-gated Na channels Enhances inactivation of voltage-gated Na channels Inhibits collapsing response mediator protein 2 Potassium channel opener AMPA-type glutamate receptor antagonist

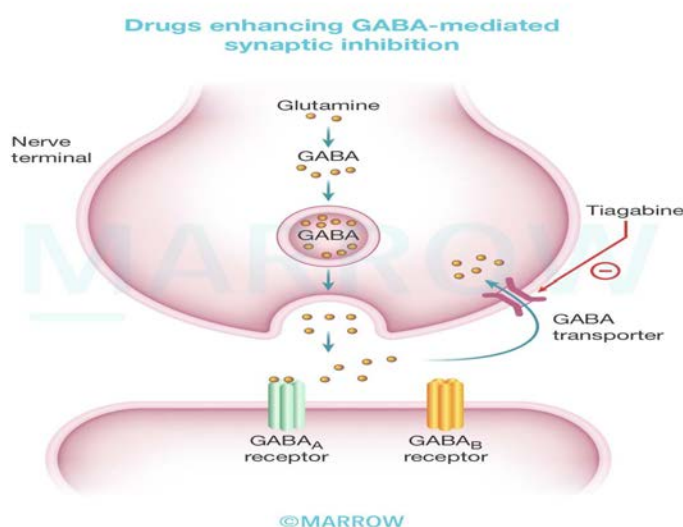
Solution to Question 23:

The marked receptor is GABA Transporter 1 Receptor (GAT-1), and Tiagabine acts on this receptor.

Tiagabine inhibits the neuronal uptake of GABA and prolongs its time in the synaptic cleft, where the GABA molecules can activate GABA receptors.

Other drugs which act on the GABAergic pathway are:

- Activation of GABA A receptors - Barbiturates, Phenobarbitone
- Activation of GABA B receptors - Baclofen
- Prevents the uptake of GABA - Tiagabine
- Prevents the degradation of GABA (inhibits GABA transaminase) - Vigabatrin



Other options:

Option B: Ganaxolone is a recently approved drug for the treatment of seizures associated with CDKL5 deficiency. The exact mechanism of action of Ganaxolone is unknown, but its anticonvulsant effects are thought to result from positive allosteric modulation of the GABA-A receptor in the CNS. It can be used in adults as well as children older than 2 years of age. The most common adverse reactions associated with the drug are somnolence, pyrexia, salivary hypersecretion and seasonal allergy.

Option C: Felbamate is the NMDA receptor antagonist prescribed to this patient. It also potentiates GABA action. It is indicated in refractory primary or secondary generalised seizures and Lennox-Gastaut syndrome. Felbamate is associated with aplastic anemia and liver failure.

Option D: Stiripentol is a GABA A receptor modulator with GABA-enhancing activity. It is clinically used in conjunction with valproate and clobazam for refractory generalized tonic-clonic seizures with severe myoclonic epilepsy in infants (Dravet syndrome).

Solution to Question 24:

The correct set of matching is a-ii, b-iii, c-i, d-iv.

Naloxone is an opioid antagonist used to reverse the effects of opioid overdose, such as respiratory depression.

Oxazepam is a benzodiazepine used for the management of delirium tremens, a severe alcohol withdrawal syndrome.

Varenicline is a partial agonist at the nicotinic acetylcholine receptors, used for smoking cessation by reducing withdrawal symptoms and cravings.

Acamprosate is a modulator of glutamate and GABA neurotransmission, used for the treatment of alcoholism by reducing cravings and the risk of relapse in alcohol abstinence.

a. Naloxone	ii. Opioid overdose
b. Oxazepam	iii. Delirium tremens
c. Varenicline	i. Smoking cessation
d. Acamprosate	iv. Treatment of alcoholism

Pharmacology NEET 2024

Question 1:

A man presents with mixed infection of *Plasmodium vivax* and *Plasmodium falciparum* in a health centre in Rajasthan. What is the dosage of the treatment given on the second day?

- a) 4 tablets of Artesunate 50mg + 6 tablets of Primaquine 2.5mg
- b) 4 tablets of Artesunate 50mg + 6 tablets of Primaquine 7.5mg
- c) 4 tablets of Artesunate 50mg + Sulphadoxine 1500mg + 6 tablets of Primaquine 2.5mg
- d) 4 tablets of Artesunate 45mg + Sulphadoxine 1500mg + 6 tablets of Primaquine 7.5mg

Question 2:

Which drug is used to treat multiple myeloma that is known to cause reactivation of Herpes Zoster?

- a) Bortezomib
- b) Lenalidomide
- c) Daratumumab
- d) Elotuzumab

Question 3:

A patient presented with breathing difficulty and nasal polyposis when he took aspirin for fever. He faced this issue previously too when he took aspirin. Which among the following is not true about this?

- a) There is no allergic sensitization to aspirin and its metabolites
- b) Responds to Monteleukast
- c) Occur on exposure to only aspirin
- d) Occurs with exposure to other NSAIDs too that block COX 1

Question 4:

A 75-year-old female presented with urinary incontinence. She also has cognition decline and dementia. Which of the following drugs is useful for treating urinary incontinence?

- a) Tiotropium
- b) Ipratropium
- c) Tolterodine
- d) Telenzipine

Question 5:

A 7-year-old boy presents with a two-week history of persistent fever, fatigue, and pallor. His parents also report easy bruising and gum bleeding. On examination, he is pale with petechiae on his legs, hepatosplenomegaly, and generalized lymphadenopathy. Blood tests show severe anemia, thrombocytopenia, and elevated white blood cell count with blast cells. What is the most appropriate initial treatment regimen for this patient?

- a) IVIG
- b) Doxorubicin + Idarubicin
- c) Prednisolone + vincristine
- d) Radiotherapy of lymph nodes

Question 6:

Lignocaine is used in a loading dose for the treatment of arrhythmias. The loading dose of this drug depends upon which of the following factors?

- a) Clearance
- b) Volume of distribution
- c) Plasma half life
- d) Bioavailability

Question 7:

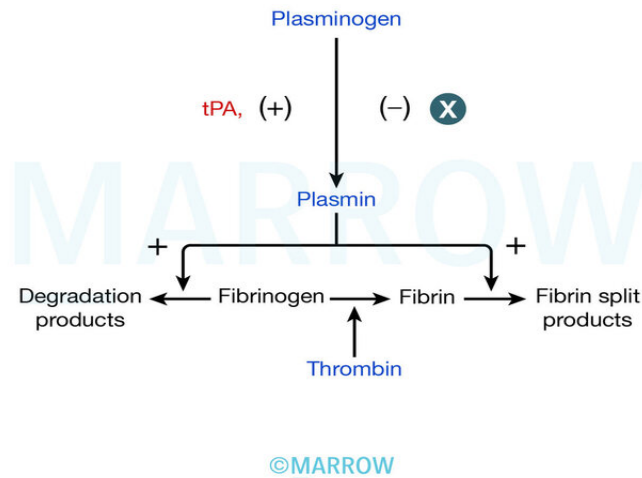
A hypertensive patient with chronic kidney disease was having frequent attacks of renal colic. Which of the following diuretics is used?

- a) Furosemide
- b) Mannitol

- c) Hydrochlorothiazide
- d) Spironolactone

Question 8:

Identify the drug marked as 'X' in the image.



- a) Clopidogrel
- b) Amino caproic acid
- c) Tenecteplase
- d) Heparin

Question 9:

Which of the following beta-adrenergic blockers are beneficial for patients with reduced ejection fraction and can reduce mortality in chronic heart failure without causing exacerbations over a few years?

- a) Propanolol
- b) Pindolol
- c) Esmolol
- d) Bisoprolol

Question 10:

Which drug is used for maintenance therapy in opium addiction?

- a) Buprenorphine
- b) Naloxone
- c) Butarphanol
- d) Clonidine

Question 11:

A patient who recently underwent surgery has complaints of nausea and vomiting. He was prescribed a drug called aprepitant. What is its mechanism of action?

- a) NK-3 antagonist
- b) NK-2 agonist
- c) NK-1 antagonist
- d) NK-1 agonist

Question 12:

An HIV patient developed a fever and dry cough. He was diagnosed with TB. He is currently on Lamivudine, Zidovudine and Nevirapine. Which Anti -TB drug should be changed in this patient?

- a) Isoniazid
- b) Rifampicin
- c) Streptomycin
- d) Ethambutol

Question 13:

A 32-year-old woman diagnosed with diarrhoea-predominant irritable bowel syndrome presented to the casualty with severe abdominal pain. She was prescribed a drug that exerts a direct spasmolytic effect on the smooth muscles of the gastrointestinal tract. Which drug could it be?

- a) Dicyclomine
- b) Scopolamine
- c) Galantamine

d) Rivastigmine

Question 14:

A patient is started on a 3 HP TPT regimen. Which of the following is true regarding the regimen?

- a) There is more compliance in 3HP than 9H and 6H**
- b) No significant difference in 3HP, 6H, 9H**
- c) Chance of abortion in 3 HP is same as 6H**
- d) 3HP regimen is less effective than 6H and 9H in preventing active TB**

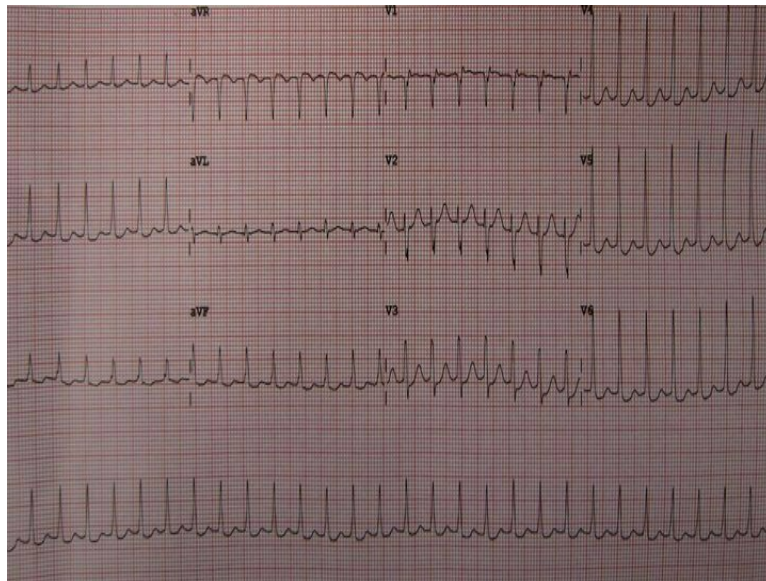
Question 15:

A female suffered a head injury from a road traffic accident. On examination, there is elevated intracranial pressure. Which diuretic among the following can be used in her condition?

- a) Mannitol**
- b) Hydrochlorothiazide**
- c) Spironolactone**
- d) Acetazolamide**

Question 16:

A 2-year-old child presents to the emergency department with lethargy, difficulty feeding, and dehydration. His vitals are stable and his ECG is shown below. What is the next step in his management?



- a) Synchronized cardioversion with 2J of current
- b) Adenosine 0.1 mg/kg
- c) Adenosine 0.2 mg/kg
- d) Synchronized cardioversion with 1J of current

Question 17:

Which of the following drugs when added with erythromycin can lead to torsades de pointes?

- a) Doxylamine
- b) Loratidine
- c) Promethazine
- d) Astemizole

Question 18:

A child was started on ciclesonide inhalational therapy for bronchial asthma. Which of the following statements is incorrect regarding his treatment?

- a) It has fewer side effects compared to other inhalational steroids
- b) It is a prodrug metabolised by bronchial esterase
- c) The occurrence of oral candidiasis is more compared to other inhaled steroids
- d) Oral bioavailability is negligible

Question 19:

A patient diagnosed with hypertriglyceridemia was started on a medication, following which, he developed flushing, hyperglycemia, and elevated liver enzymes. Identify the drug consumed by the patient.

- a) Nicotinamide
- b) Rosuvastatin
- c) Ezetemibe
- d) Fenofibrate

Question 20:

Which of the following reasons best explains the combination of Atropine (0.025 mg) and Diphenoxylate (2.5 mg) for treating secretory diarrhoea?

- a) Atropine potentiates the action of smooth muscle relaxation
- b) It prevents overdose and abuse of Diphenoxylate
- c) It antagonizes the side effect of Diphenoxylate
- d) Atropine increases the ant motility action of diphenoxylate

Question 21:

A hypertensive patient is started on a drug to control his BP. He had complaints of dry mouth, drowsiness, and confusion after taking this drug, but he still continued to take the drug for 1 year. He had to travel abroad for work, and he forgot to take his medications. He now presents with a blood pressure of 200/120mmHg. Identify the drug he was on?

- a) Amlodipine
- b) Atenolol
- c) Clonidine
- d) Enalapril

Question 22:

In which phase of clinical trials is drug dosing done?

- a) Phase I
- b) Phase II

- c) Phase III
- d) Phase IV

Question 23:

A patient with a gynecological malignancy is started on Methotrexate. Later she develops oral and gastric ulcers. Which nucleotide synthesis is affected in her?

- a) UDP
- b) CTP
- c) TMP
- d) GMP

Question 24:

The drug 'X' is a GHRH analogue that is FDA-approved for the treatment of HIV-associated lipodystrophy and not used for Growth hormone deficiency. Identify the drug 'X'.

- a) Octreotide
- b) Somatropin
- c) Tesamorelin
- d) Mecasermin

Answer Key

Question No.	Correct Option
1	a
2	a
3	c
4	c
5	c
6	b
7	c
8	b
9	d

10	a
11	c
12	b
13	a
14	a
15	a
16	b
17	d
18	c
19	a
20	b
21	c
22	a
23	c
24	c

Detailed Explanations

Solution to Question 1:

The dosage of the treatment given on the second day for mixed infection with *Plasmodium vivax* and *Plasmodium falciparum* is 4 tablets of Artesunate 50mg + 6 tablets of Primaquine 2.5mg.

For treating mixed infections of *Plasmodium vivax* and *Plasmodium falciparum*, the following treatment regimen is recommended. This approach ensures comprehensive treatment by addressing both the blood stages of *Plasmodium falciparum* with ACT and the liver stages of *Plasmodium vivax* with Primaquine.

**Dosage chart for treatment of mixed
(vivax and falciparum) malaria with ACT-SP**

Age	Day 1			Day 2		Day 3		Day 4-14
	AS tablet (50 mg)	SP tablet	PQ (2.5 mg)	AS tablet (50 mg)	PQ (2.5 mg)	AS tablet (50 mg)	PQ (2.5 mg)	PQ (2.5 mg)
Less than 1 year	½	½	0	½	0	½	0	0
1-4 years	1	1	1	1	1	1	1	1
5-8 years	2	1½	2	2	2	2	2	2
9-14 years	3	2	4	3	4	3	4	4
15 years or more	4	3	6	4	6	4	6	6

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Solution to Question 2:

The drug used to treat multiple myeloma and that has the potential to cause reactivation of Herpes Zoster is bortezomib.

Bortezomib is an inhibitor of proteasome-mediated protein degradation. It is central to the treatment of multiple myeloma, both untreated and relapsed. Bortezomib is given intravenously or subcutaneously and dosage depends upon the disease, route of administration, and the patient's history. It is known to predispose patients to the reactivation of herpesvirus, including Herpes Zoster.

Adverse effects include thrombocytopenia, fatigue, peripheral neuropathy, neutropenia, anaemia, anorexia, nausea, limb pain, dizziness, etc. Other adverse events like cardiac tamponade, disseminated intravascular coagulation, toxic epidermal necrolysis, etc. have been reported.

Other options:

Option B: Lenalidomide is the drug of choice in patients with myelodysplastic syndrome (MDS) associated with 5q deletion, and is an immunomodulatory analogue of thalidomide. It suppresses bone marrow and is often associated with significant leukopenia.

Option C: Daratumumab is a monoclonal antibody against CD-38.

Option D: Elotuzumab is a monoclonal antibody against SLAMF7, a glycoprotein expressed in multiple myeloma cells and natural killer cells. It is used in combination with lenalidomide and dexamethasone for the treatment of multiple myeloma.

Solution to Question 3:

The diagnosis for this patient with breathing difficulty and nasal polyposis, triggered by aspirin, a COX-1 inhibitor is aspirin-induced asthma, also known as Samter's triad. The statement that is

not true is that "it occurs on exposure to only aspirin", as this condition can also be triggered by other NSAIDs that inhibit COX-1, and is not restricted to just aspirin alone.

Samter's triad is also known as aspirin-induced asthma or aspirin-sensitive asthma. It consists of:

- Nasal polyposis
- Bronchial asthma
- Aspirin intolerance

A patient with Samter's triad experiences severe asthma and nasal polyposis triggered by aspirin or other COX-1 inhibitors. This condition involves nasal congestion, rhinorrhea, anosmia, and recurrent chronic rhinosinusitis, with acute asthma attacks often occurring a few hours after ingestion of these medications.

These patients overproduce leukotrienes due to COX-1 inhibition and decreased prostaglandin E₂ (PGE₂) levels. Therefore, they should avoid COX-1 inhibitors like aspirin and most NSAIDs but can generally tolerate COX-2 inhibitors and acetaminophen.

Treatment typically includes leukotriene modifiers, and in some cases, aspirin desensitization may be used to manage symptoms and allow the use of aspirin or NSAIDs if necessary for other conditions. Recent advances in treatment include biologics such as Dupilumab and IL-5 inhibitors, which are increasingly effective in managing aspirin-induced asthma and may replace aspirin desensitization except when ongoing aspirin or NSAID therapy is required.

Other options:

Option A: The reaction in aspirin-induced asthma is not due to an allergic sensitization but rather due to the inhibition of COX-1 and subsequent overproduction of leukotrienes.

Option B: Montelukast, a leukotriene receptor antagonist, is effective in managing symptoms of aspirin-induced asthma by counteracting the elevated levels of leukotrienes that contribute to inflammation and respiratory issues.

Option D: It occurs with exposure to other NSAIDs too, that block COX-1. Patients with aspirin-induced asthma can react to other NSAIDs that inhibit COX-1, not just aspirin.

Solution to Question 4:

For managing urinary incontinence in a 75-year-old female with cognitive decline and dementia, tolterodine is useful.

Tolterodine is a muscarinic receptor antagonist specifically effective for treating urinary incontinence. It exhibits selectivity for the urinary bladder, which helps to minimize side effects compared to other antimuscarinic agents. This selectivity is beneficial for patients with cognitive decline or dementia, as it reduces the risk of adverse effects. Tolterodine is metabolized by CYP2D6, so its dosage may need adjustment based on individual CYP2D6 activity, making it a suitable option for managing urinary incontinence while minimizing potential side effects.

Other muscarinic antagonists for urinary incontinence include oxybutynin, trospium chloride, darifenacin, solifenacin, and fesoterodine. Oxybutynin, though effective, often causes antimuscarinic side effects such as xerostomia. Trospium chloride, being a quaternary amine, is better tolerated and excreted primarily via the kidneys, requiring dosage adjustments in renal

impairment. Darifenacin and solifenacin have selectivity for M₃ receptors, which helps balance efficacy and side effects, while fesoterodine, a prodrug, provides a stable pharmacokinetic profile due to its conversion to tolterodine's active metabolite.

Other Options:

Option A: Tiotropium and Option C: Ipratropium are muscarinic antagonists used primarily for respiratory conditions such as chronic obstructive pulmonary disease (COPD) and asthma, rather than urinary incontinence. Tiotropium is a long-acting agent with selectivity for M₁ and M₃ receptors, while Ipratropium is used more frequently for its short-acting effects.

Option D: Telenzepine, an analogue of Pirenzepine, displays higher potency and selectivity for M₁ receptors compared to pirenzepine and is used in treating acid-peptic disease, not urinary incontinence.

Solution to Question 5:

The given clinical scenario suggests a diagnosis of Acute Lymphoblastic Leukemia (ALL), and standard initial treatment in children includes a chemotherapy regimen with drugs like prednisolone and vincristine during the induction phase.

Patients diagnosed with ALL present with symptoms of impaired marrow function like anaemia, infections secondary to neutropenia, and bleeding due to thrombocytopenia. Other symptoms include bone pain, generalized lymphadenopathy, splenomegaly, hepatomegaly, testicular enlargement, headache, vomiting, and nerve palsies.

Treatment of ALL

Induction: L-Asparaginase, vincristine, prednisolone, and anthracycline

CNS prophylaxis with intrathecal methotrexate

Consolidation: Methotrexate, cytarabine

Maintenance: 6-Mercaptopurine, methotrexate plus intrathecal therapy

Other Options:

Option A: IVIG: Intravenous immunoglobulin (IVIG) is not used for treating ALL.

option B: Both doxorubicin and idarubicin are anthracyclines used in chemotherapy. While doxorubicin can be part of certain ALL treatment protocols, idarubicin is more commonly used in Acute Myeloid Leukemia (AML) rather than ALL. This combination is not typically used in the induction phase for ALL children.

Option D: Radiotherapy is not a standard initial treatment for ALL. It may be considered in specific cases, such as central nervous system involvement or certain high-risk cases, but the primary treatment for ALL is systemic chemotherapy, not localized radiotherapy.

Solution to Question 6:

The volume of distribution of the drug determines the loading dose of the drug.

Loading Dose = Volume of distribution x Desired steady-state plasma concentration.

When the time to reach a steady-state is long, as it is for drugs with long half-lives, it may be desirable to administer a loading dose that promptly raises the concentration of the drug in plasma to the target concentration.

Other options:

Option A: Clearance is used to determine the maintenance dose.

Maintenance dose = clearance x desired steady-state plasma concentration.

Options C and D: Plasma half-life primarily determines the frequency of dosing, while bioavailability determines the route of administration. Though bioavailability may affect the dose necessary, it will not be the primary factor considered during the calculation of the loading dose.

Solution to Question 7:

For a hypertensive patient with chronic kidney disease experiencing frequent attacks of renal colic, hydrochlorothiazide, a thiazide diuretic, is often used.

Thiazide diuretics are particularly effective in cases of calcium oxalate kidney stones because they reduce urinary calcium excretion. They work by acting on the distal convoluted tubule (DCT) of the nephron, where they inhibit the Na⁺/Cl⁻ co-transporter. This inhibition decreases the intracellular sodium concentration, which, through compensatory mechanisms, leads to increased calcium reabsorption into the bloodstream.

While this mechanism helps in decreasing urinary calcium and managing calcium oxalate stones, thiazide diuretics can also cause adverse effects such as hypokalemia, hyperuricemia, and hypercalcemia. The increased reabsorption of uric acid due to the diuretic effect can exacerbate conditions like gout and should be monitored carefully.

In addition to their use in treating calcium oxalate kidney stones, thiazide diuretics are commonly prescribed for managing uncomplicated hypertension and treating edema associated with heart, liver, and kidney diseases. They are also used for conditions like osteoporosis and nephrogenic diabetes insipidus.

Other options:

Option A: Furosemide is a diuretic that can increase calcium excretion, potentially raising PTH as the body tries to conserve calcium.

Option B: Osmotic diuretics, like mannitol, retain water isoosmotically in the proximal convoluted tubule (PCT) and descending loop of Henle which are freely permeable to water. Luminal fluid is diluted and opposes NaCl reabsorption and promotes calcium excretion. They are indicated in raised intracranial tension and in dialysis disequilibrium.

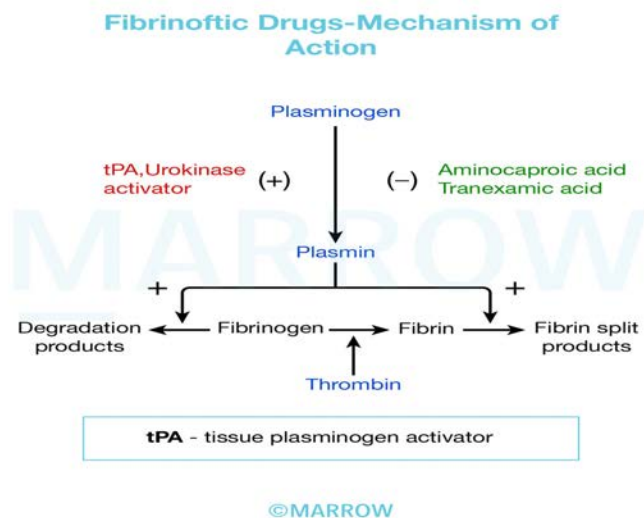
Option D: Spironolactone is a potassium-sparing diuretic and aldosterone antagonist. It works by competitively inhibiting aldosterone receptors in the distal tubules of the kidney.

Solution to Question 8:

The drug marked as 'X' in the image is aminocaproic acid.

ϵ -Aminocaproic acid, a lysine analogue that competes with plasminogen for binding sites on fibrin. By blocking this interaction, ϵ -Aminocaproic acid prevents the conversion of plasminogen into plasmin, thereby inhibiting fibrinolysis. This action makes it a useful agent for reversing the effects of fibrinolytic therapy.

However, ϵ -Aminocaproic acid should be administered with caution as it can suppress intrinsic fibrinolytic activity, potentially leading to complications such as disseminated intravascular coagulation (DIC). To address this, fresh frozen plasma and cryoprecipitate may be used to restore fibrin and clotting factors.



Solution to Question 9:

The beta-adrenergic blocker that is useful in patients with reduced ejection fraction and can reduce mortality in chronic heart failure without causing exacerbations over a few years is bisoprolol.

Bisoprolol is preferred for use in patients having heart failure with reduced ejection fraction (HFrEF).

The exacerbation mentioned in the context of beta-blockers for heart failure refers to the potential worsening of heart failure symptoms in some patients, despite the overall benefits of these medications. They can sometimes lead to a decline in left ventricular function, particularly in individuals with limited cardiac reserve who rely heavily on adrenergic stimulation. This deterioration can manifest as increased heart failure symptoms or a decrease in cardiac function. The only beta-blockers that have been proven to improve the survival of patients with HFrEF or chronic HF are -

- Bisoprolol
- Metoprolol
- Carvedilol

HFrEF is a complex clinical syndrome where the ventricle fails to adequately fill with or eject blood, typically defined by an ejection fraction (EF) less than or equal to 40%. Key etiologies include coronary artery disease, valvular heart disease, congenital heart diseases, and autoimmune diseases like giant cell myocarditis. Patients with HFrEF commonly experience symptoms like dyspnea on exertion, orthopnea, and paroxysmal nocturnal dyspnea, and may present with pedal edema, elevated jugular venous pressure (JVP), ascites, and hepatomegaly on examination.

The pharmacological management of HFrEF involves drugs with both symptomatic and mortality benefits, including Angiotensin receptor–neprilysin inhibitors (ARNIs) such as sacubitril/valsartan, angiotensin-converting enzyme (ACE) inhibitors like ramipril and enalapril, β -blockers such as metoprolol and carvedilol, mineralocorticoid receptor antagonists like spironolactone and eplerenone, sodium-glucose cotransporter 2 (SGLT2) inhibitors like dapagliflozin, and hydralazine plus isosorbide dinitrate (noting a mortality benefit specifically in African-American patients). Drugs used primarily for symptomatic relief, such as loop diuretics (e.g., furosemide), and digoxin, are reserved for specific indications or during decompensation. Non-pharmacological therapies for HFrEF include implantable cardioverter-defibrillators (ICD) and cardiac resynchronization therapy.

Solution to Question 10:

The drug used among the following options for maintenance therapy in opioid addiction is buprenorphine.

Buprenorphine is a partial μ -opioid agonist and an antagonist at δ and κ receptors. It is widely used both as an analgesic for chronic pain conditions, such as cancer pain, and as a treatment for opioid dependence, serving as an alternative to methadone.

It effectively controls cravings and blocks the effects of higher-efficacy opioids. It is often administered in combination with naloxone to prevent misuse, although it may be given alone to pregnant women or individuals with naloxone allergies. Initial dosing typically starts at 2-4 mg daily, with adjustments made to stabilize the dose at approximately 12 to 16 mg/day.

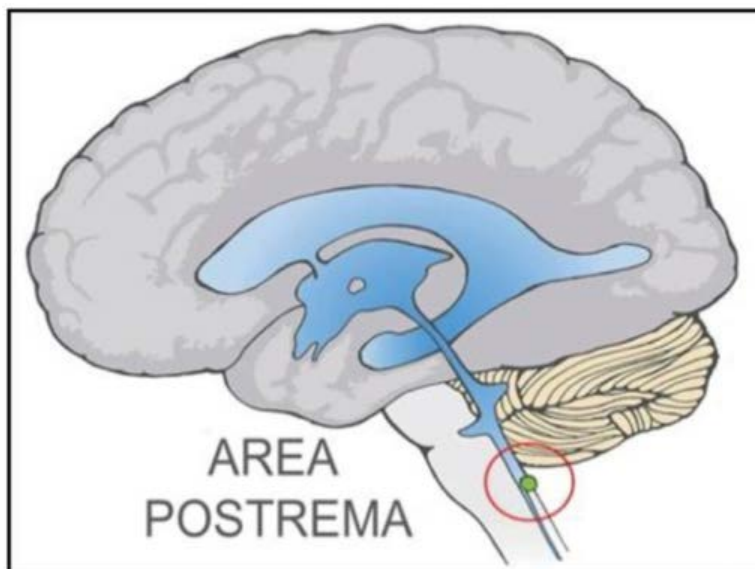
For opioid use disorder, effective management includes both withdrawal and long-term treatment strategies. Withdrawal and detoxification generally start with managing physical dependence using medications like methadone or buprenorphine. Methadone, a full agonist, is used to stabilize and reduce withdrawal symptoms, while buprenorphine helps with cravings and has a lower risk of overdose.

Clonidine and lofexidine, α -2 agonists, are used to ameliorate withdrawal symptoms. Lofexidine, approved by the FDA, is the first non-opioid treatment specifically for opioid withdrawal in adults. Naltrexone, a long-acting opioid antagonist, is used as an adjunct in patients who are already opioid-free; however, it will precipitate acute withdrawal symptoms if administered to those still dependent on opioids.

Solution to Question 11:

The class of drugs most preferred in patients suffering from postoperative nausea and vomiting (PONV), following surgery are the NK1 antagonists like aprepitant.

Substance P is a neuropeptide that interacts with neurokinin-1 (NK-1) receptors. NK1 antagonists inhibit substance P at central and peripheral receptors. Aprepitant, fosaprepitant are examples of NK1 antagonists. It also acts on the chemotrigger zone whose location is shown below.



PONV Physiology: Afferents from the chemoreceptor trigger zone (CTZ), vagal neurons in the gastrointestinal system, neurons from the vestibular system and midbrain afferents relay within the vomiting centre in the reticular formation of the brainstem which then project into the nucleus tractus solitarius.

Risk factors for PONV are divided into:

- Patient factors
 - Female > male
 - Age < 50 years
 - Prior history of motion sickness
 - Non-smokers > smokers (CTZ is desensitized)
- Preoperative factors
 - (Fasting and anxiety, while possibly contributing to PONV, have yet to be demonstrated)
- Intraoperative factors
 - Surgical - Cholecystectomy, gynaecological and laparoscopic procedures have higher incidences
 - Drugs - Nitrous oxide, Opioids, Cyclopropane,
 - General anaesthesia has a times more risk of developing PONV than regional anaesthesia.
- Postoperative factors
 - Pain and early ambulation contribute to a higher risk of PONV.

Scoring systems used to grade PONV are the Apfel score and the Koivuranta score.

Drugs used in the prophylaxis and treatment of PONV:

- 5-HT₃ antagonists - ondansetron, palonosetron
- Butyrophenones- droperidol
- Dexamethasone
- Transdermal scopolamine
- Aprepitant

Note: Double-blind studies have found no difference in the efficacy between aprepitant (NK₁ antagonists) and ondansetron (5HT₃ antagonist) in the first 24 hours following surgery. But after 24 hours aprepitant has proven to be more productive than ondansetron in the management of PONV.

Solution to Question 12:

Rifampicin should be avoided as it is a potent inducer of CYP enzymes including CYPs 1A₂, 3A₄, 2C₉, and 2C₁₉, which in turn can induce the metabolism of nevirapine and results in therapeutic failure of ART.

Rifabutin causes the least induction of cytochrome enzymes and is therefore preferred in HIV patients who are on protease inhibitors/NNRTIs.

Cytochrome enzyme-inducing potency of rifamycins is rifampicin > rifapentine > rifabutin.

Note:

Cytochrome p-450 inducers	Cytochrome p-450 inhibitors
Barbiturates Carbamazepine Rifampin Phenytoin Aminoglutethimide Cigarette smoking Chronic alcoholism	Macrolide antibiotics Azole antifungals Chloramphenicol Omeprazole Selective serotonin reuptake inhibitors (SSRIs) HIV protease inhibitors Cimetidine Ciprofloxacin Metronidazole

Solution to Question 13:

Dicyclomine is a weak muscarinic receptor antagonist with non-specific direct spasmolytic effects on the smooth muscle of the gastrointestinal tract.

Other options:

Option B: Scopolamine exerts spasmolytic activity indirectly through its antimuscarinic action. It is not preferred due to its adverse effects like drowsiness, dryness, and blurred vision.

Option C and D: Galantamine and rivastigmine are central cholinesterase inhibitors and do not possess spasmolytic activity.

Solution to Question 14:

The 3HP regimen, which involves once-weekly doses of Isoniazid and Rifapentine for 12 weeks, has been found to have higher patient compliance than the 9-month regimen of Isoniazid (9H) and the 6-month regimen (6H).

The shorter duration and reduced number of doses in the 3HP regimen make it more convenient and thus lead to better adherence. Studies have shown that patients are more likely to complete the 3HP regimen due to the shorter treatment duration and lower pill burden.

TPT stands for Tuberculosis Preventive Therapy. It is a treatment given to individuals who are at high risk of developing active tuberculosis (TB) to prevent the progression from latent TB infection (where the person is infected with the TB bacteria but does not have symptoms) to active TB disease.

There are different regimens for TPT, including:

- 3HP: A regimen consisting of once-weekly doses of Isoniazid (H) and Rifapentine (P) for 12 weeks.
- 6H: A regimen consisting of daily Isoniazid for 6 months.
- 9H: A regimen consisting of daily Isoniazid for 9 months.

TPT is particularly important for individuals with latent TB who have a higher risk of developing active TB, such as those with HIV, close contacts of TB patients, or those with certain medical conditions that weaken the immune system.

Other options:

Option B: Research has demonstrated that the 3HP regimen offers better compliance and similar effectiveness compared to the traditional 6H and 9H regimens. The shorter duration of 3HP (once weekly for 12 weeks) makes it more convenient and tolerable for patients, which leads to higher adherence rates. Thus, there is a significant difference in patient compliance between these regimens, with 3HP showing clear advantages.

Option C: Although 3HP is generally considered safe, data suggests that the risk profile for adverse effects, including potential reproductive impacts like abortion, differs between regimens. Specifically, 6H may have a different side effect profile due to the prolonged exposure to the drug over six months, compared to the shorter and intermittent dosing of 3HP.

Option D: Studies have confirmed that the 3HP regimen is as effective as 6H and 9H in preventing active tuberculosis (TB). Both regimens aim to prevent latent TB from progressing to active disease, and 3HP has been shown to achieve similar outcomes with fewer doses, enhancing patient compliance.

Solution to Question 15:

In this patient with raised intracranial pressure after a road traffic accident, mannitol is the diuretic of choice in her management to reduce intracranial pressure.

Mannitol is an osmotic diuretic that draws fluid out of the brain tissue and into the bloodstream as it forms a hypertonic solution when it reaches the bloodstream, thereby reducing intracranial pressure (ICP). It is commonly used in cases of traumatic brain injury or other conditions that cause elevated ICP. This effect helps to relieve pressure on the brain, preventing further damage.

Other options:

The other diuretics listed Hydrochlorothiazide, Spironolactone, and Acetazolamide are not typically used for managing elevated ICP. They are not hypertonic solutes and thus do not lead to a rapid reduction of the ICP as mannitol does.

Solution to Question 16:

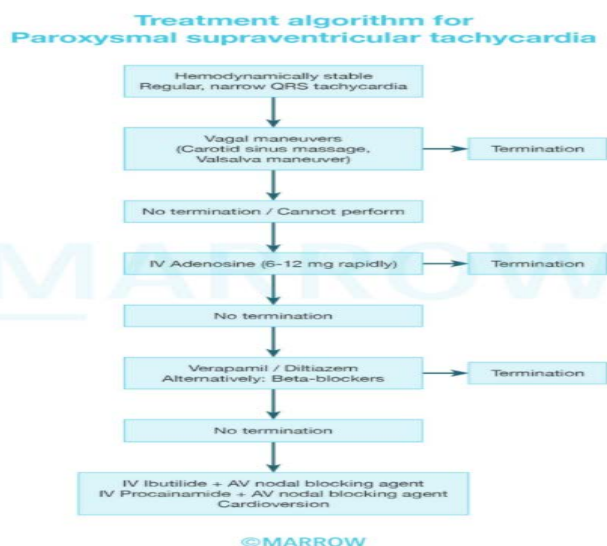
This clinical history along the EKG showing narrow QRS complexes without P waves and pseudo R waves in aVR lead suggests paroxysmal supraventricular tachycardia (PSVT) and the next step in his management would be an initial dose of adenosine at 0.1 mg/kg.

PSVT is a regular, narrow QRS complex tachycardia, most often caused due to an atrioventricular nodal re-entrant circuit. Patients may present with sudden onset palpitations, dizziness, and diaphoresis.

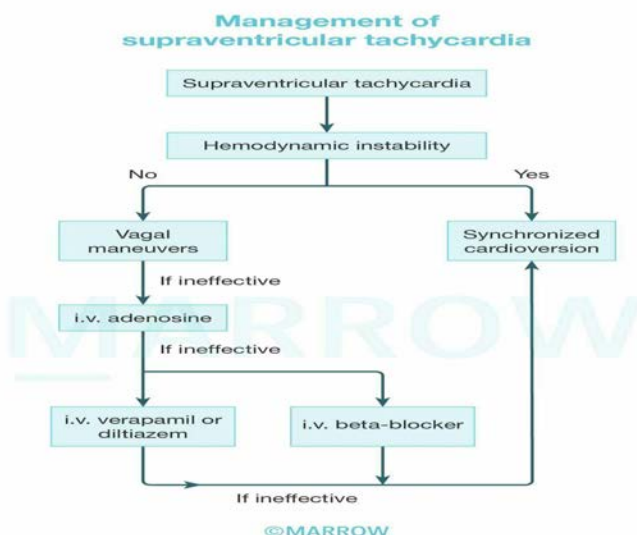
Management involves slowing AV node conduction and vagal maneuvers (e.g., Valsalva maneuver, splashing cold water on the face) are the first step. If this does not work, intravenous (IV) adenosine is the drug of choice used to terminate the majority of PSVT episodes.

Adenosine is a very short-acting endogenous purine signal molecule. It is administered as a 6 mg IV bolus over 2-3 seconds followed by a rapid flush with 20 mL NS. It can be followed up with a 12 mg IV bolus dose if required. It causes membrane hyperpolarization of cardiac muscle cells to terminate the re-entrant circuit through the AV node.

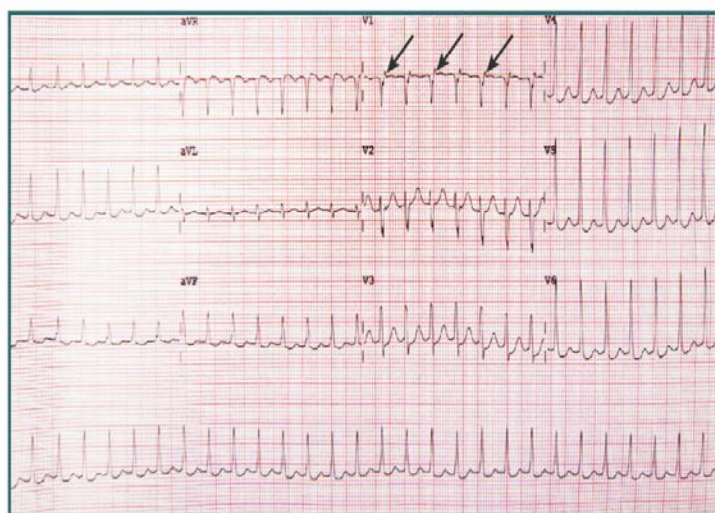
Since the action of the drug lasts for less than a minute, adverse effects are short-lived. These include transient shortness of breath, chest pain, hypotension, and cutaneous flushing due to vasodilation. It can however precipitate bronchospasm in asthmatics and verapamil is preferred in such patients.



When the patient is hemodynamically unstable, i.e., hypotension with unconsciousness or respiratory distress, QRS-synchronous direct current cardioversion is done.



The given ECG shows pseudo R' waves:



Solution to Question 17:

Among the given drugs, astemizole increases the risk of Torsades de pointes when used concomitantly with erythromycin.

Drugs that block potassium channels and prolong QT increase arrhythmogenic potential. Astemizole is a second-generation antihistamine that blocks delayed potassium channels and prolongs the QT interval.

Other options:

Option A: Doxylamine is an antihistamine commonly used as a short-term treatment for insomnia. It is combined with pyridoxine (vitamin B6) to treat nausea and vomiting during pregnancy (morning sickness). It does not prolong the QT interval.

Option B: Loratadine is a second-generation antihistamine used to treat allergic rhinitis (hay fever) and urticaria (hives). It is known for being non-sedating due to its minimal penetration into the central nervous system. It does not prolong the QT interval.

Option C: Promethazine is a first-generation antihistamine with multiple uses, including the treatment of allergies, motion sickness, nausea and vomiting, and as a sedative or sleep aid. It also has antiemetic properties and is sometimes used in the management of postoperative nausea and vomiting.

Solution to Question 18:

The incorrect statement is option C as ciclesonide has a lower risk of oral candidiasis as compared to other inhaled steroids.

Ciclesonide is an inhaled corticosteroid that is unique because it is a prodrug. It is metabolised by bronchial esterases in the lungs to its active form, which allows for targeted action within the airways. This feature reduces systemic side effects and the risk of local side effects, such as oral candidiasis (thrush), compared to other inhaled steroids.

Other options:

Option A: Ciclesonide's activation in the lungs minimises systemic absorption, leading to fewer side effects.

Option B: Ciclesonide is inactive until it is converted to its active form in the lungs.

Option D: Due to its prodrug nature and metabolism in the lungs, ciclesonide has low oral bioavailability, reducing systemic exposure.

Solution to Question 19:

The drug most likely responsible for causing flushing, hyperglycemia, and elevated liver enzymes in a patient being treated for hypertriglyceridemia is Nicotinamide (Niacin).

Niacin (nicotinic acid), is commonly used to treat hypertriglyceridemia and hypercholesterolemia. However, it is associated with side effects such as flushing, which is due to prostaglandin-mediated vasodilation. Niacin can also lead to hyperglycemia by impairing insulin sensitivity and may cause elevated liver enzymes, indicating potential hepatotoxicity.

Other options:

Option B: Rosuvastatin is a statin, primarily used to lower LDL cholesterol, and while it may elevate liver enzymes, it is less commonly associated with flushing and hyperglycemia. It can cause myopathy in patients.

Option C: Ezetimibe inhibits cholesterol absorption in the intestines and is not typically associated with these side effects.

Option D: Fenofibrate is used to lower triglycerides can elevate liver enzymes and can increase the risk of developing myopathy if used concomitantly with statins, but it does not commonly cause flushing or hyperglycemia.

Solution to Question 20:

Atropine is included in combination with diphenoxylate to prevent abuse and overdose of diphenoxylate.

Diphenoxylate is structurally similar to opioids and has the potential for misuse at higher doses. Additionally, it crosses the blood-brain barrier, making it more likely to cause dependence. The anticholinergic effects of atropine (e.g., dry mouth, tachycardia, blurred vision) become prominent at higher doses, discouraging overdose.

Diphenoxylate is an opioid agonist primarily used as an antidiarrheal agent. It works by slowing down intestinal motility, allowing more time for water and electrolytes to be absorbed from the intestines, which helps reduce diarrhea. However, because of its structural similarity to opioids, diphenoxylate has the potential for abuse and dependence if taken in large doses.

Solution to Question 21:

The adverse effects seen in this patient and the presentation of rebound hypertension on sudden drug withdrawal indicate that the drug is clonidine.

Clonidine is an alpha-2 receptor agonist and a centrally-acting antihypertensive agent. It treats high blood pressure by stimulating presynaptic α_2 receptors in the vasomotor center in the brainstem. This binding has a sympatholytic effect, by suppressing the release of norepinephrine and other sympathomimetics.

Abrupt withdrawal after long-term therapy can result in rebound hypertension, restlessness, and sympathetic overactivity. Dry mouth and sedation are the major side effects of clonidine. It can also cause impotency and constipation. Rebound hypertension is a potentially serious complication and occurs if the drug is missed for 1–2 days. This is associated with an alarming rise in blood pressure which may be more than the pretreatment level.

Phentolamine is used for the control of hypertension due to clonidine withdrawal.

Solution to Question 22:

The dosing of a new drug is tested in phase I of a clinical trial.

Phase I trials are usually performed on healthy volunteers to check the safety and pharmacokinetic properties of a drug. The participants in this phase do not gain any direct health benefits from taking part. They are generally paid a fee for their time, with the payments being

regulated and not related to the risk involved.

In clinical trials, drug dosing typically occurs in Phase I, where researchers establish the optimal dose to achieve desired effects while minimizing side effects. In these phases, they focus on safety, tolerability, and pharmacokinetics.

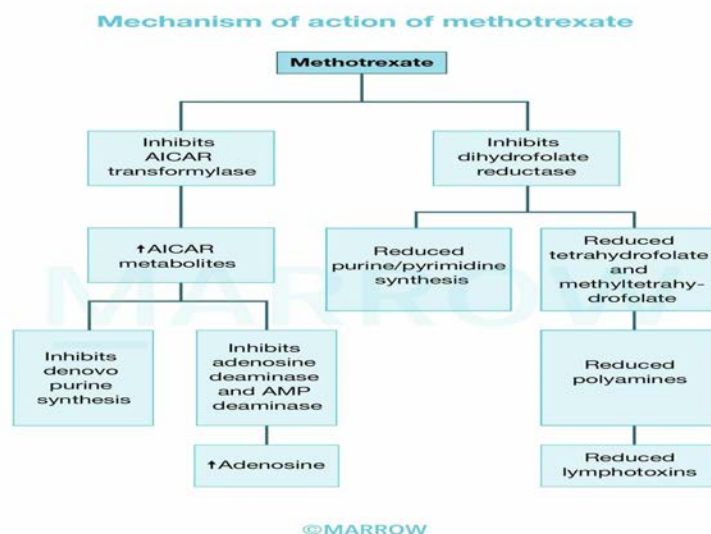
Dose-ranging, primarily conducted in Phase II trials, involves testing multiple doses to identify the most effective and safe dose range. This phase helps determine the minimum effective dose and maximum tolerated dose, providing critical data for subsequent studies in Phase 3, where the chosen dose is tested for efficacy and safety in a larger population.

Solution to Question 23:

TMP synthesis is affected due to methotrexate.

Methotrexate inhibits dihydrofolate reductase (DHFR), which is required to synthesize tetrahydrofolate. Tetrahydrofolate is essential for converting dUMP to dTMP (thymidine monophosphate) in the nucleotide synthesis pathway. Therefore, the nucleotide synthesis affected in this patient is TMP (thymidine monophosphate).

By inhibiting DHFR, methotrexate reduces the availability of THF, thereby disrupting the production of thymidylate (TMP), which is necessary for DNA replication and repair. This disruption leads to impaired DNA synthesis in rapidly dividing cells, such as cancer cells. However, it also affects other rapidly dividing cells, including those in the gastrointestinal tract and oral mucosa, leading to side effects like oral and gastric ulcers.



Solution to Question 24:

Tesamorelin is approved for use in HIV-associated lipodystrophy but not for GH deficiency. It is a synthetic N-terminal modified form of human GHRH. It does not get degraded by dipeptidyl peptidase 4 and therefore has a longer duration of action.

It increases the levels of growth hormone (GH) and insulin-like growth factor (IGF-1) but reduces visceral fat accumulation.

Tesamorelin is approved for use in HIV-associated lipodystrophy but not for GH deficiency.

Other options:

Option A: Octreotide is a somatostatin analogue that decreases growth hormone levels. It is used in the treatment of acromegaly, variceal bleeding, diarrhoea due to adenomas secreting VIP (vasoactive intestinal peptide), and sulfonylurea toxicity. One of its adverse effects is gallstone formation.

Option B: Somatropin is a growth hormone analogue. It is used in growth hormone deficiency and other short-stature conditions like Turner's syndrome, Noonan syndrome, and Prader-Willi syndrome. Their use is contraindicated in acute critical illness, active malignancy, and proliferative retinopathy.

Option D: Mecasermin is a recombinant human IGF-1. It is used in IGF-1 deficiency, growth hormone receptor defects, and antibodies against growth hormone receptors. Adverse effects include hypoglycemia and lipohypertrophy. Another recombinant human IGF-1 with its IGFBP-3, mecasermin rinfabate has been FDA-approved for HIV-associated adipose redistribution syndrome.